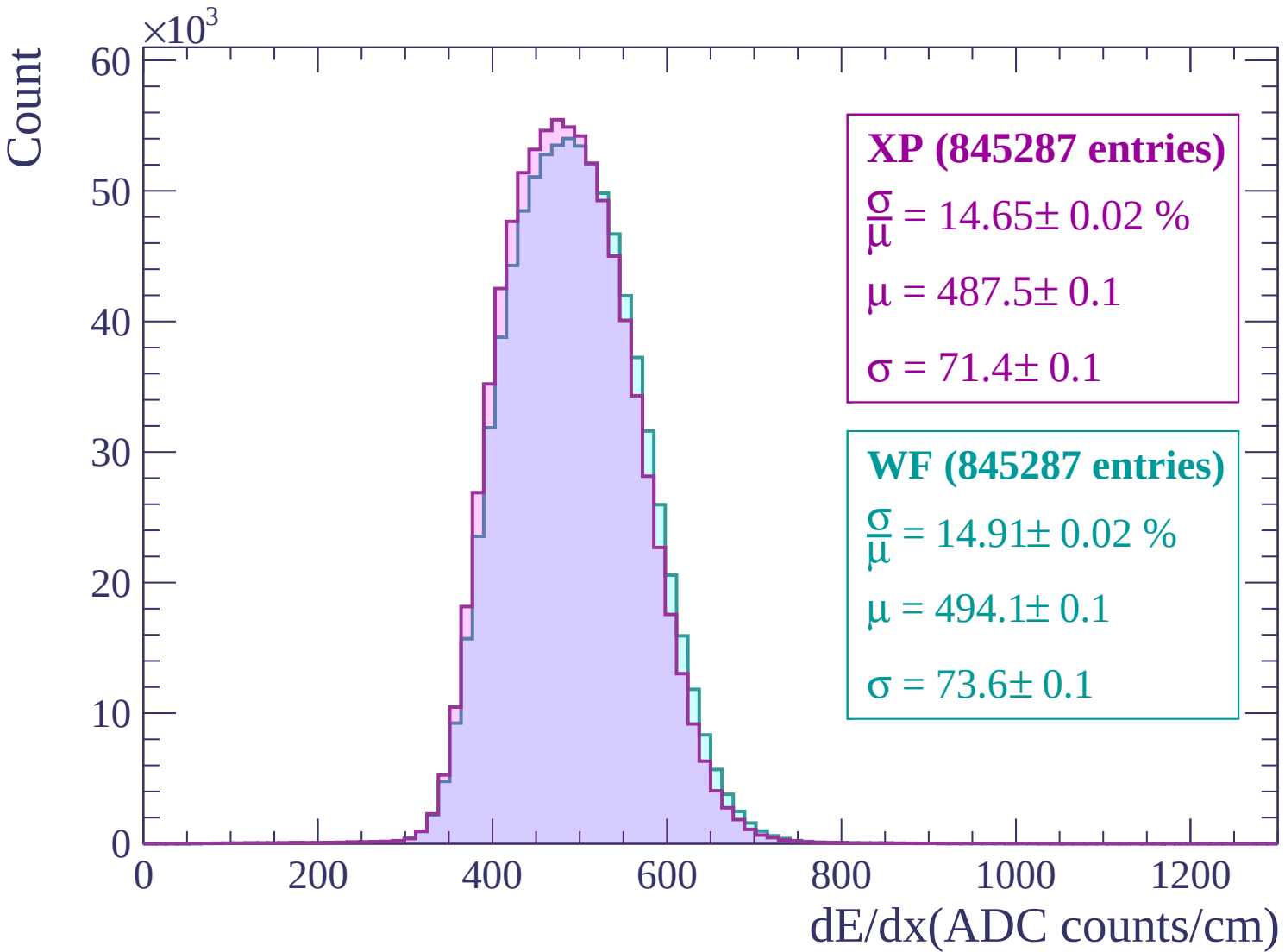
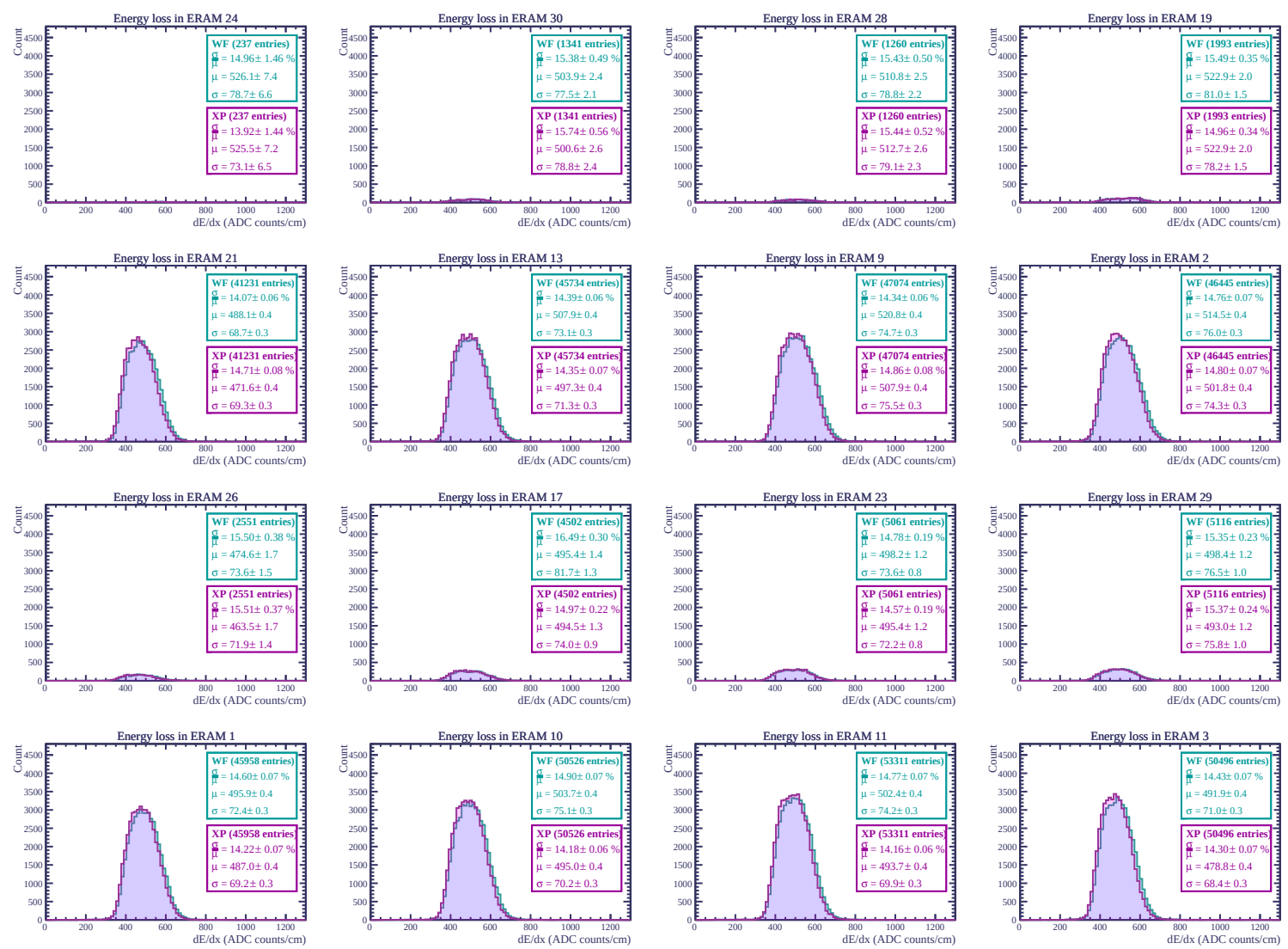
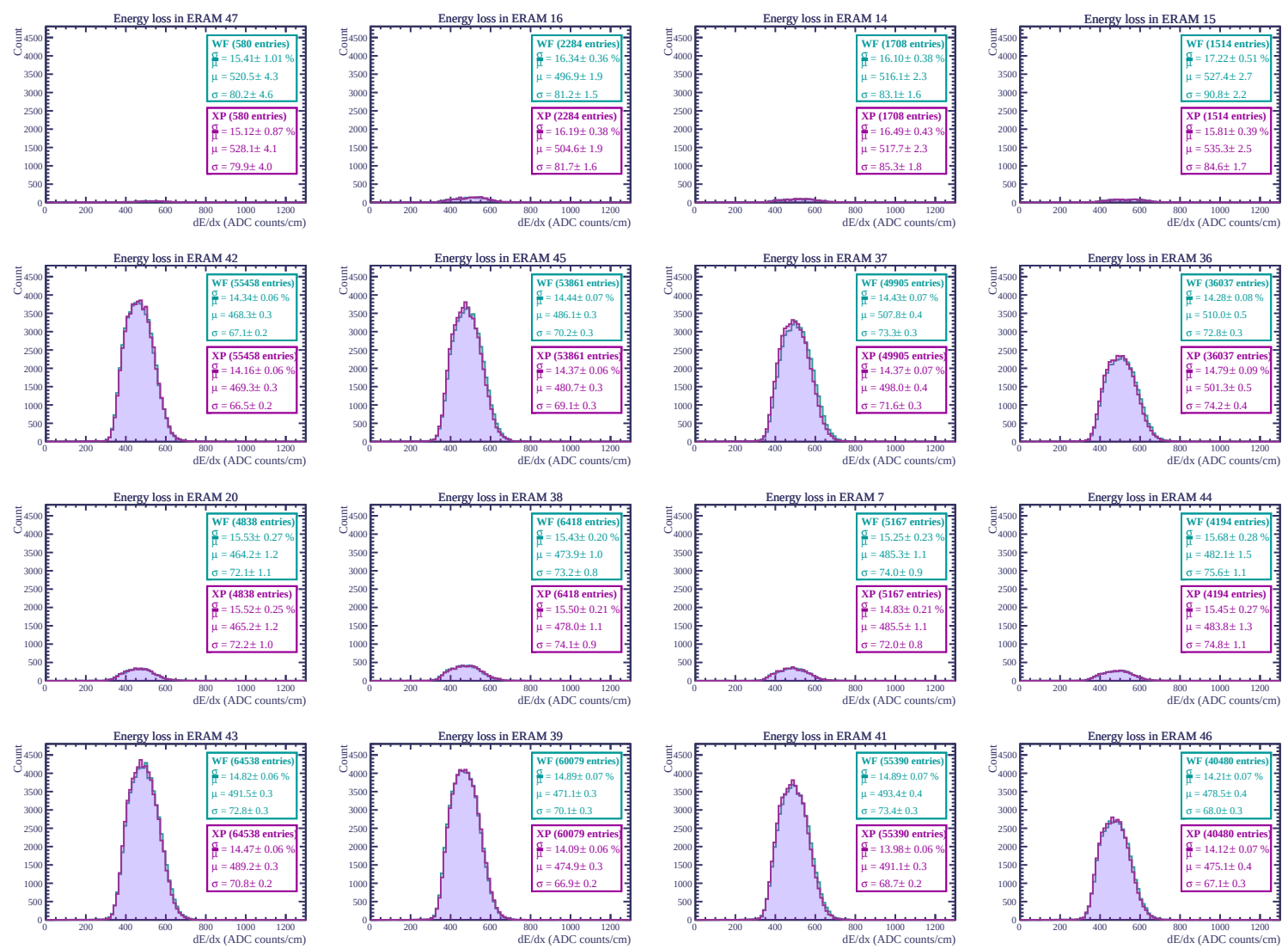
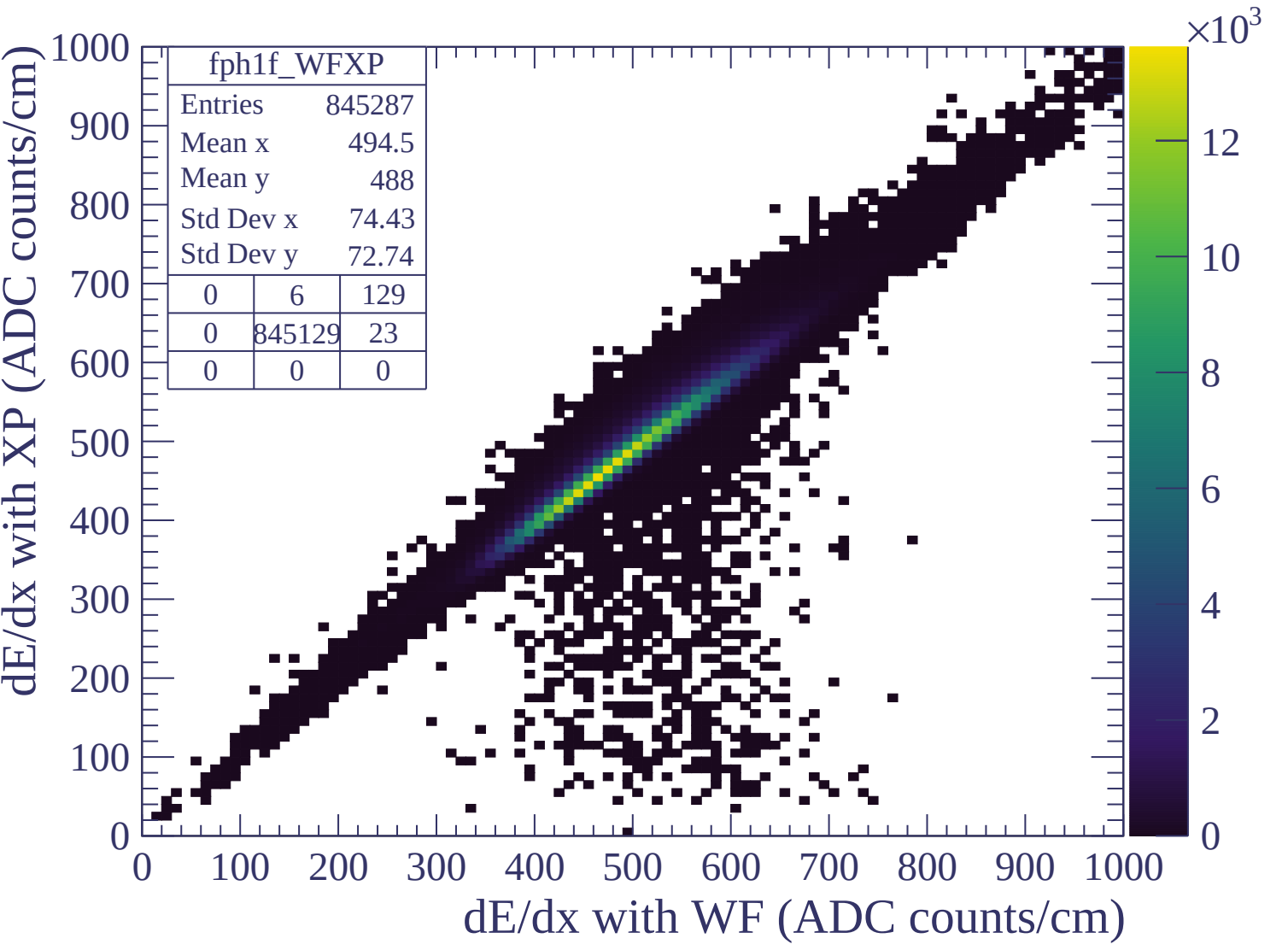
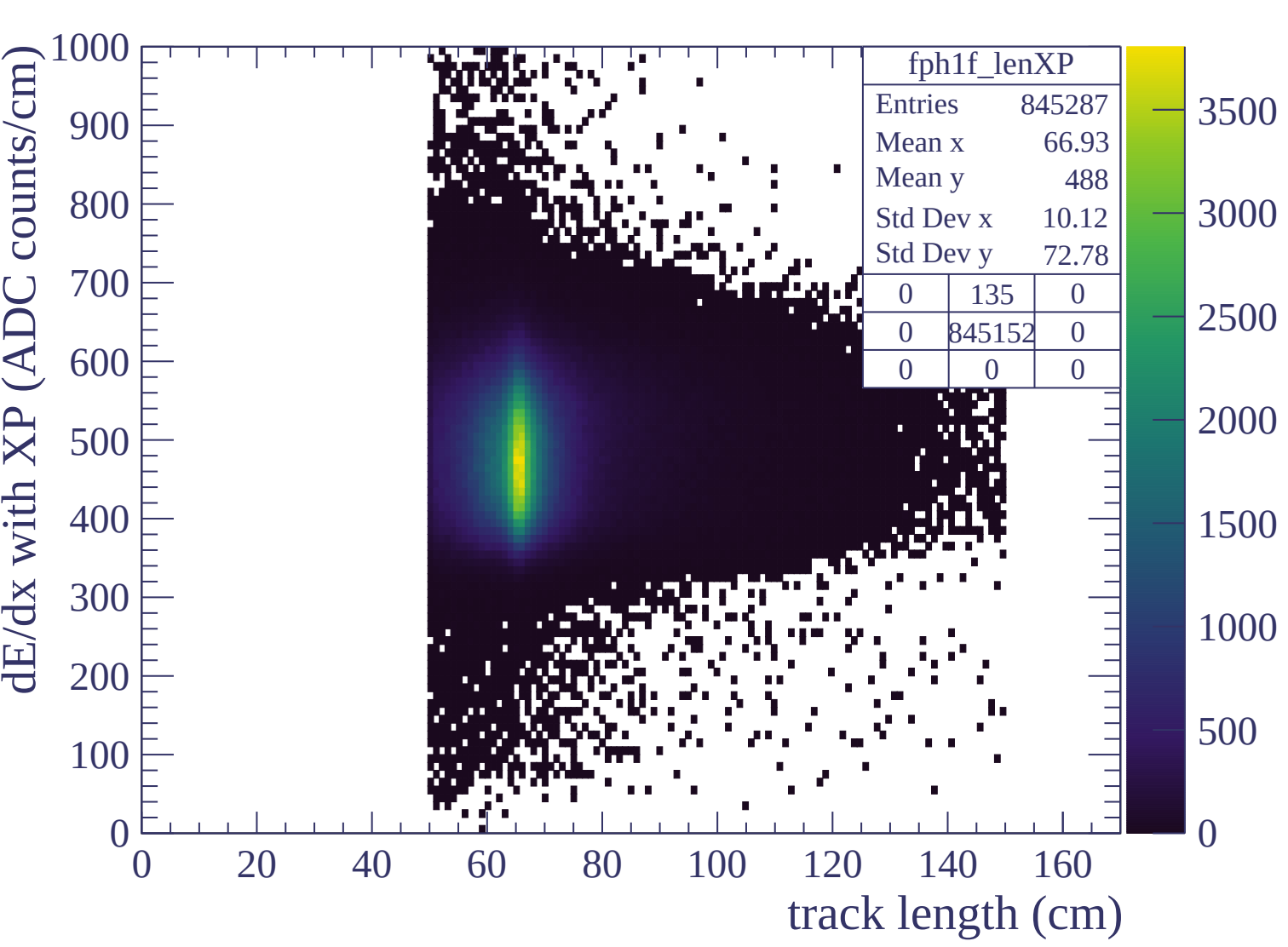


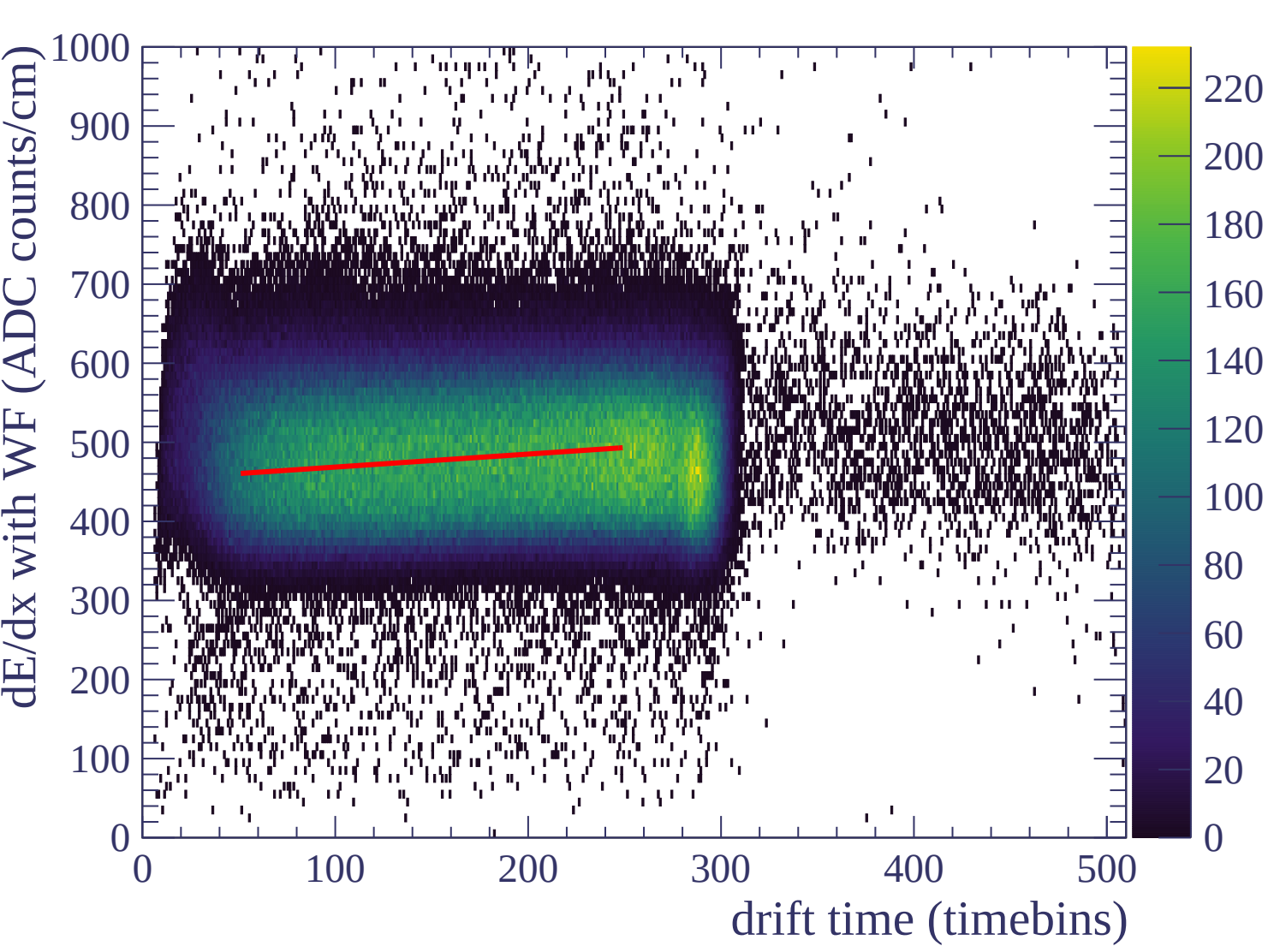


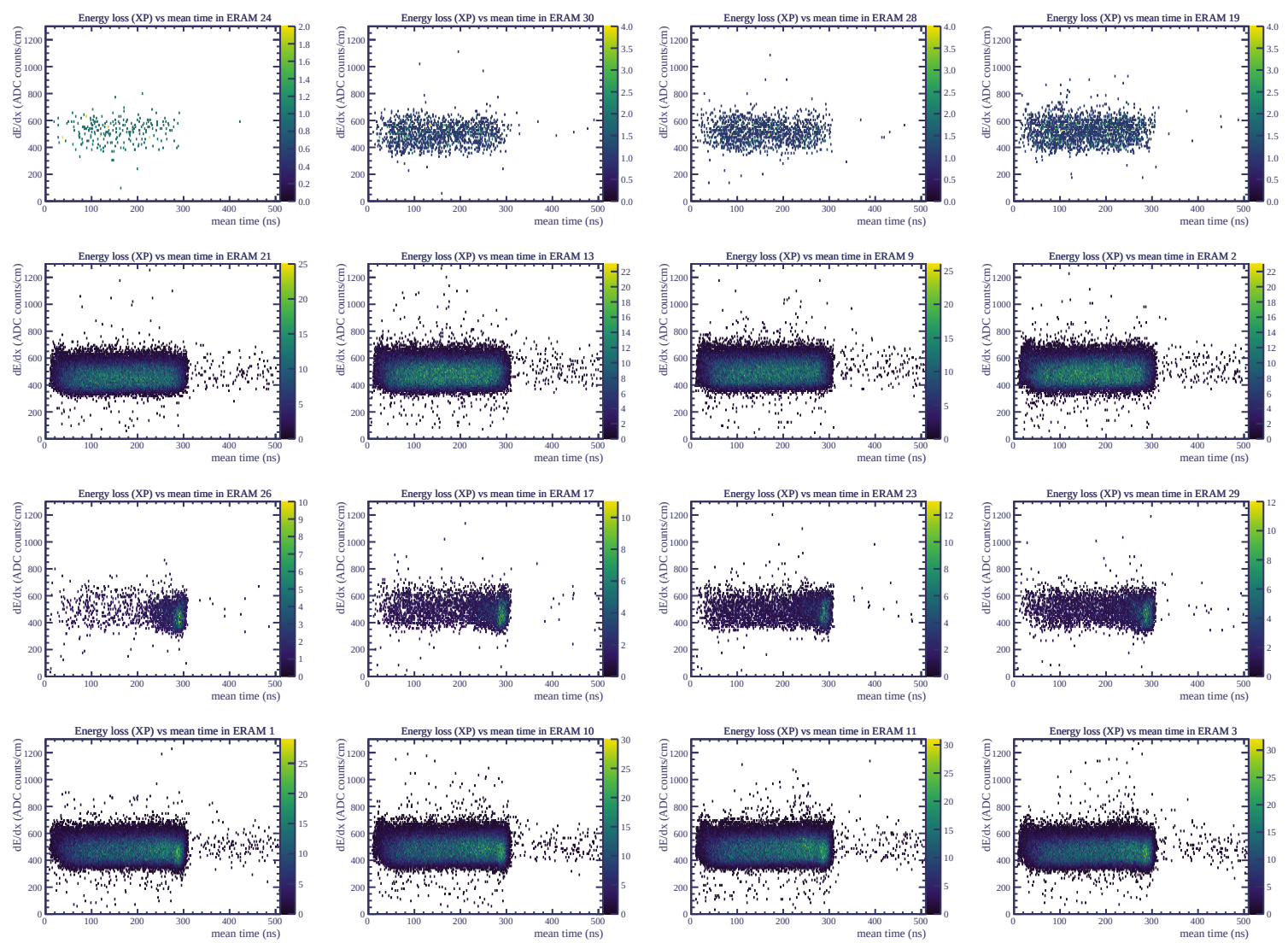


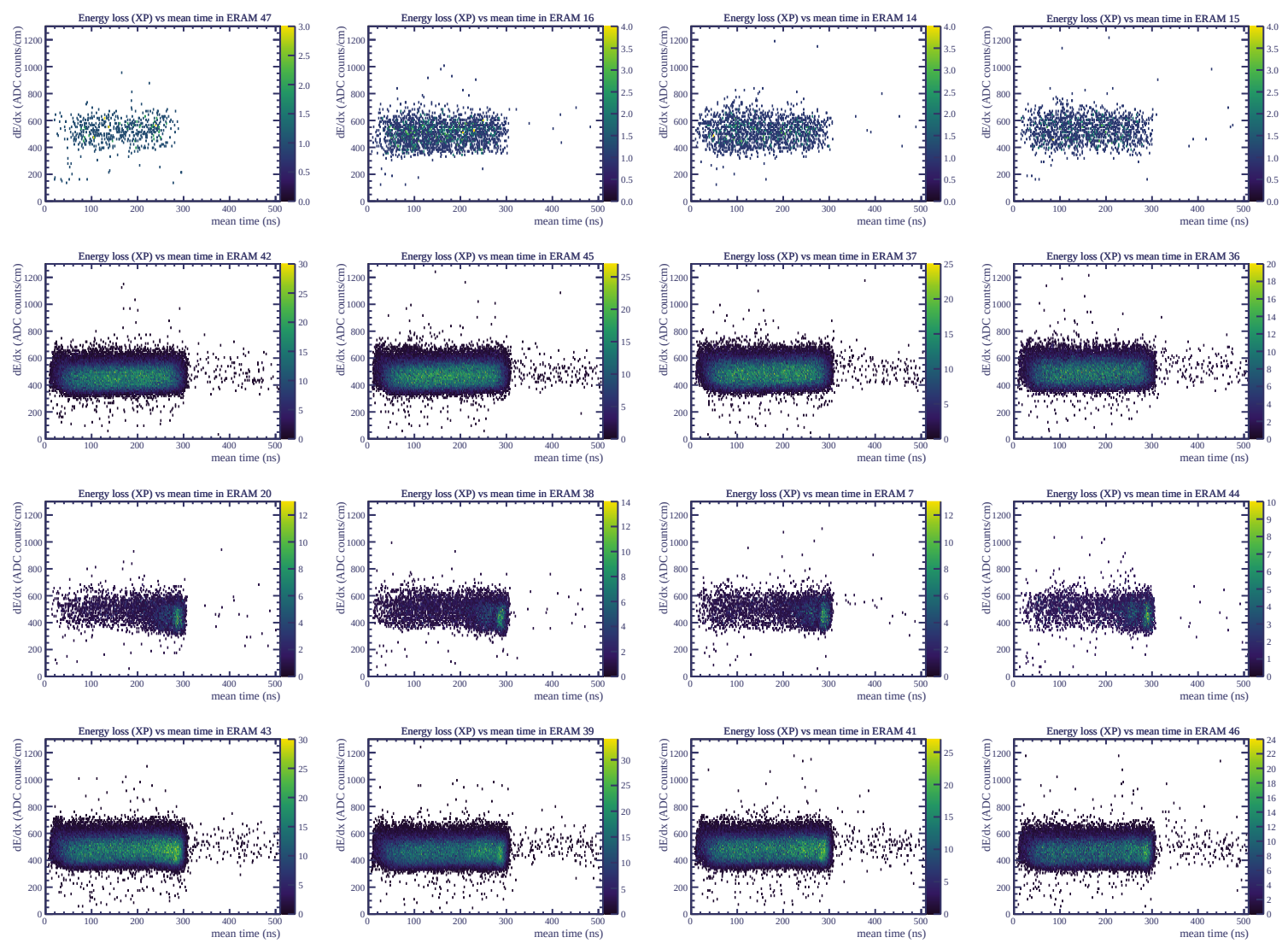




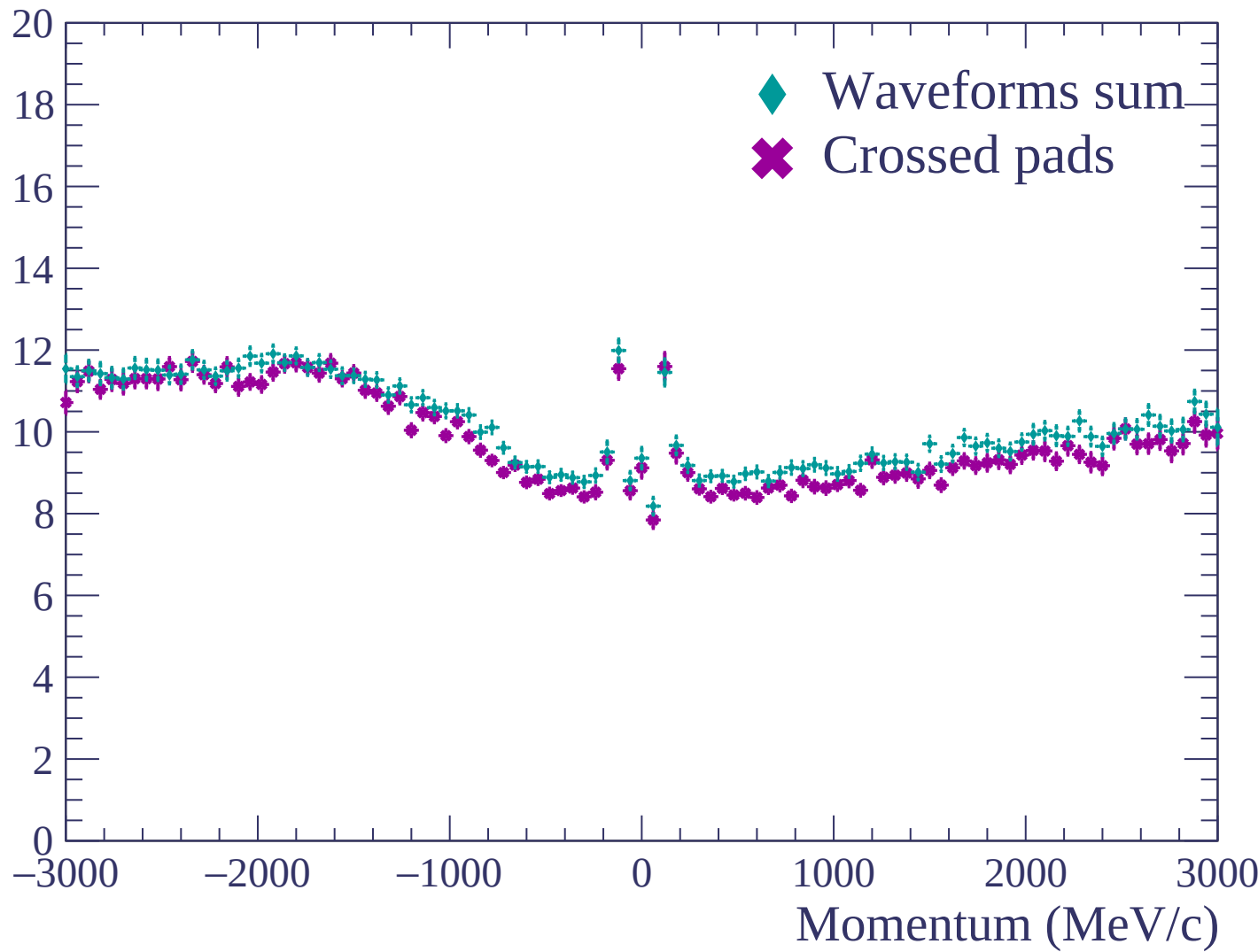


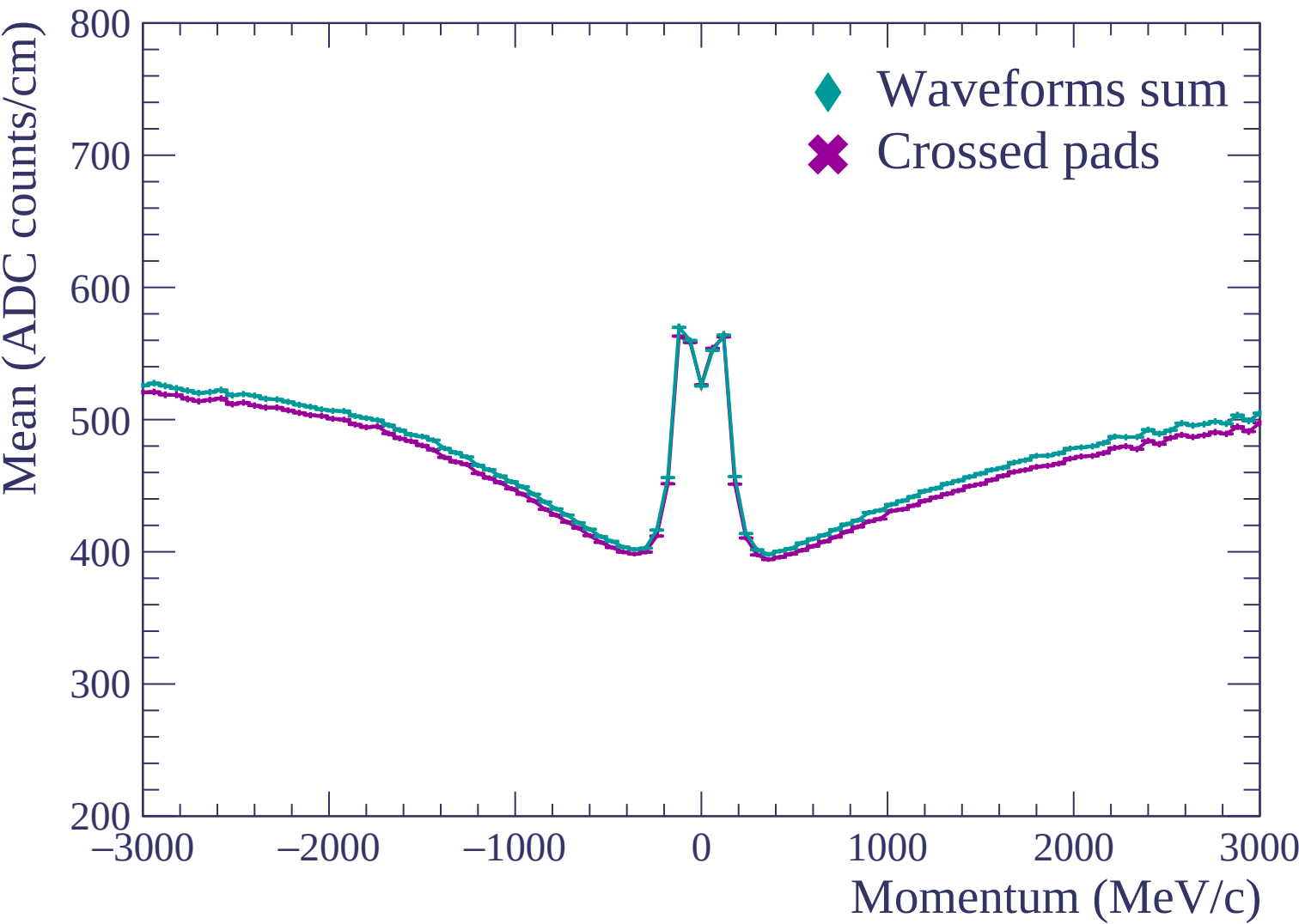


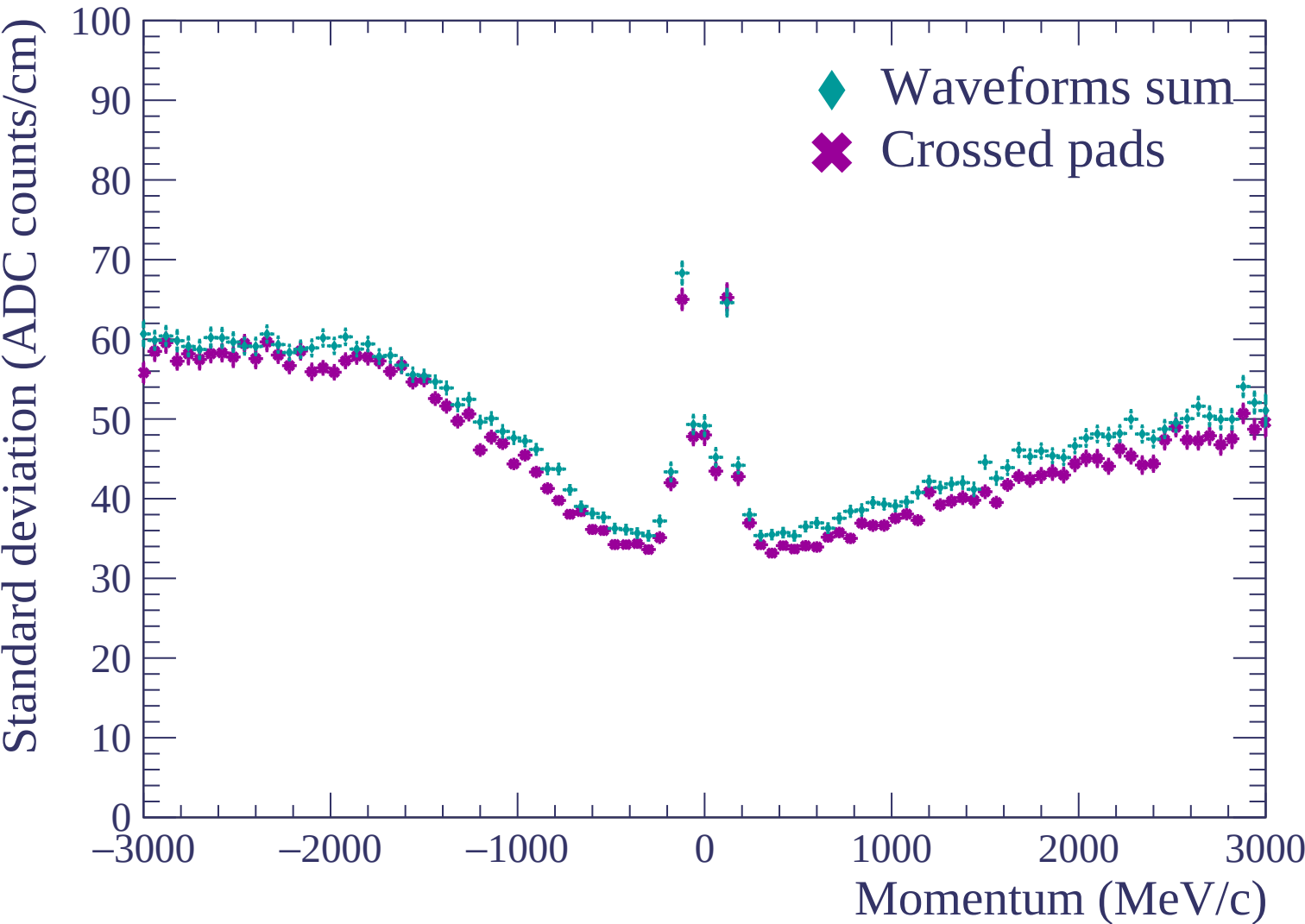


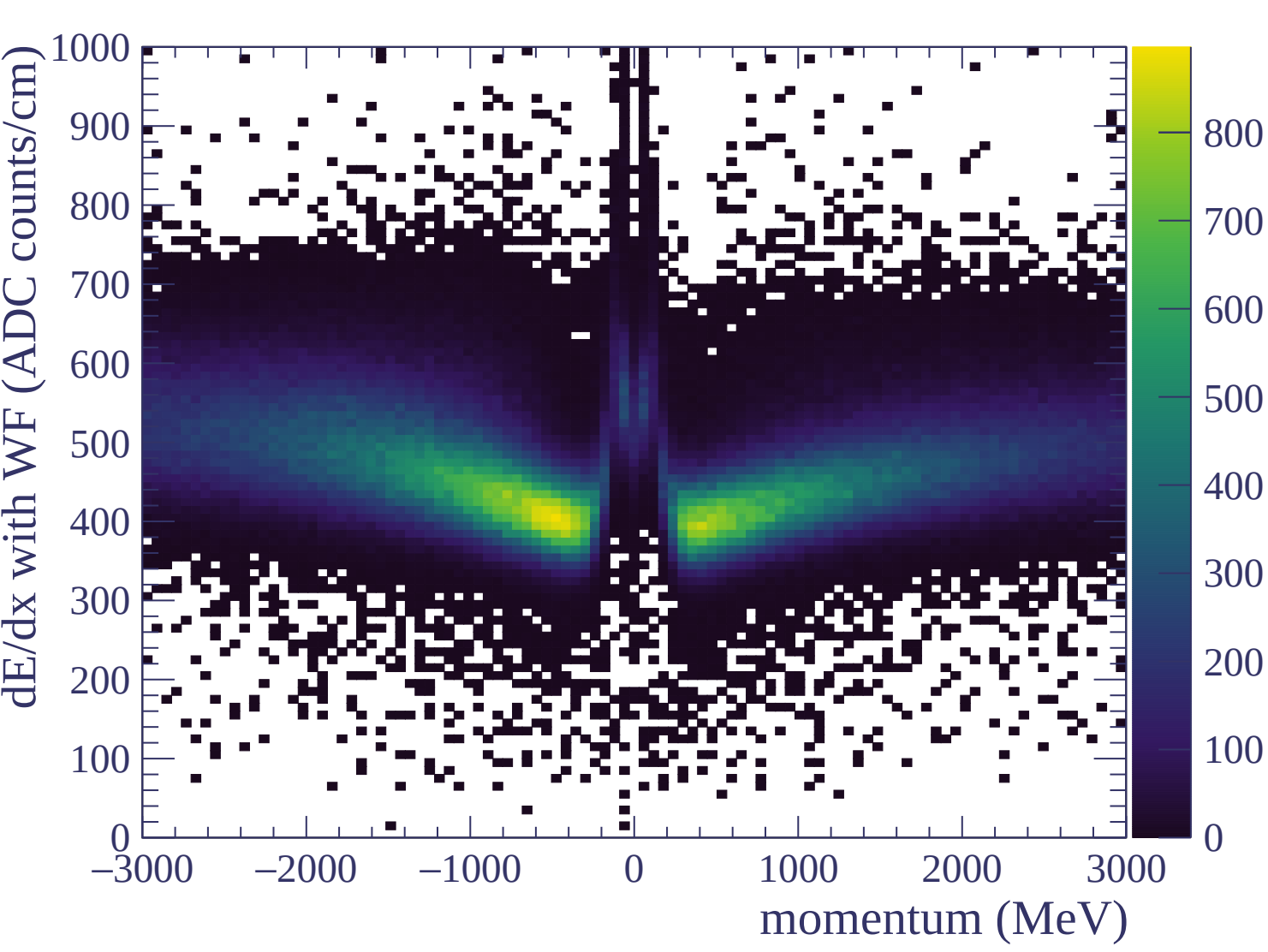


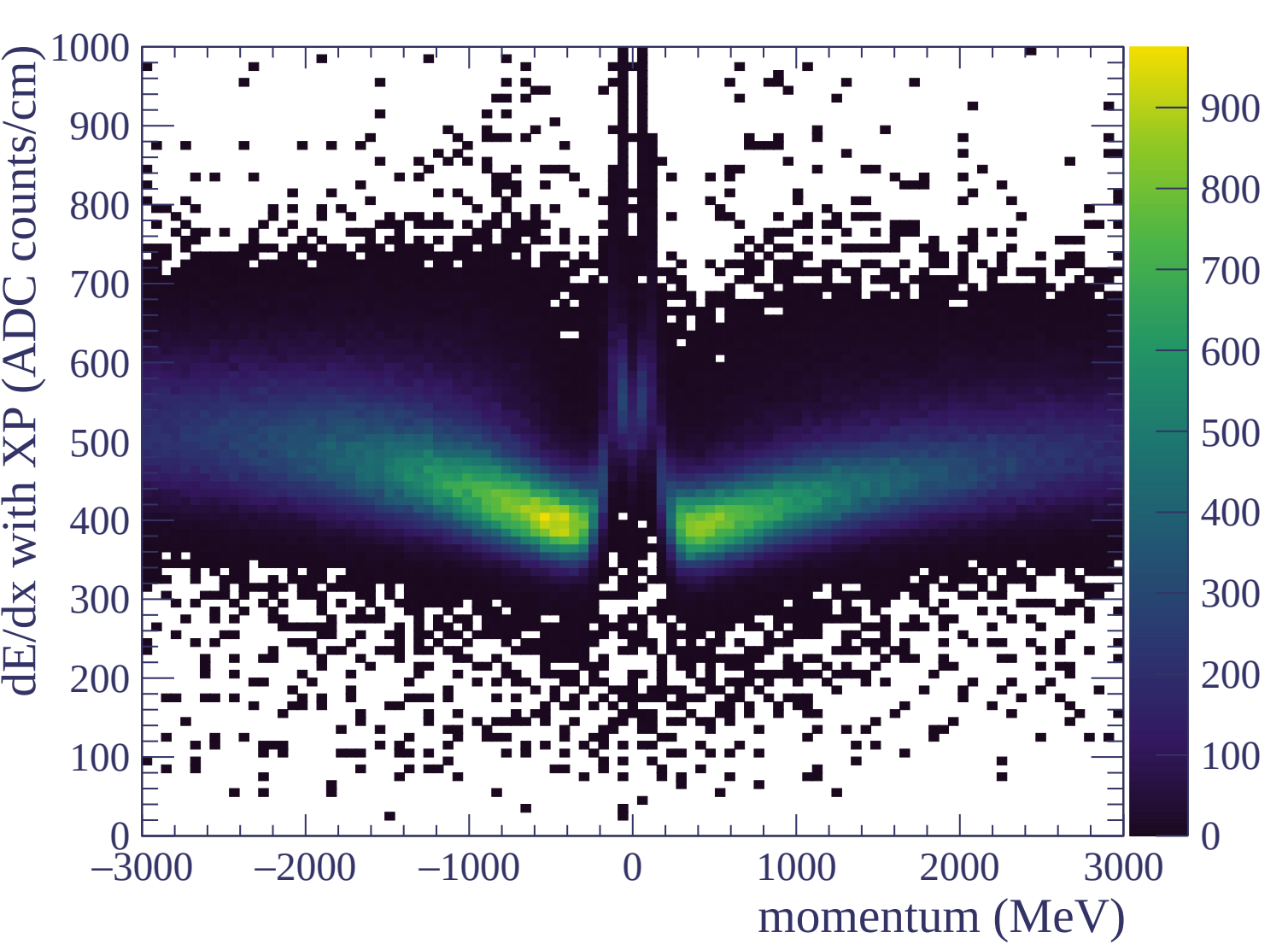
Resolution (%)

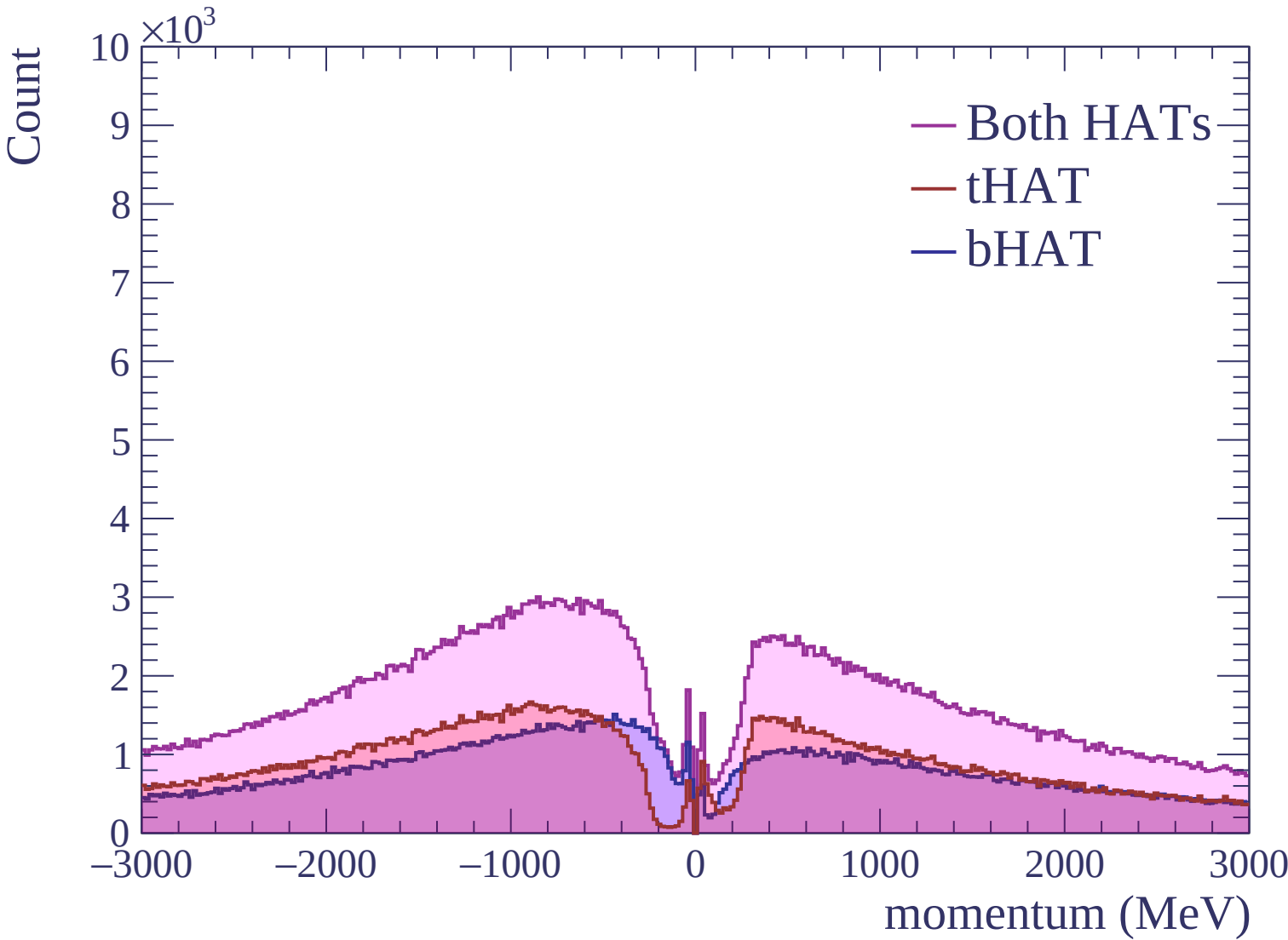






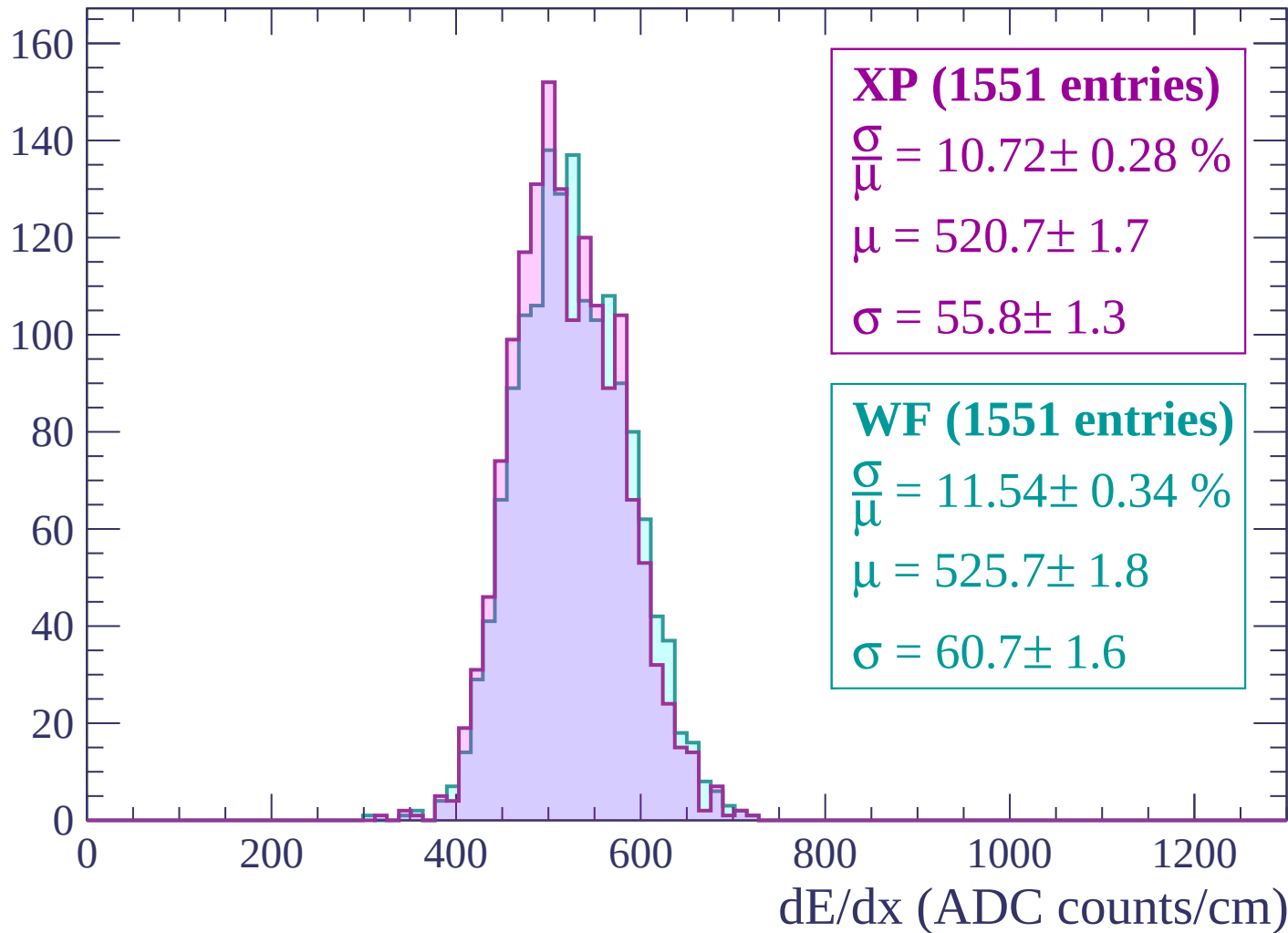






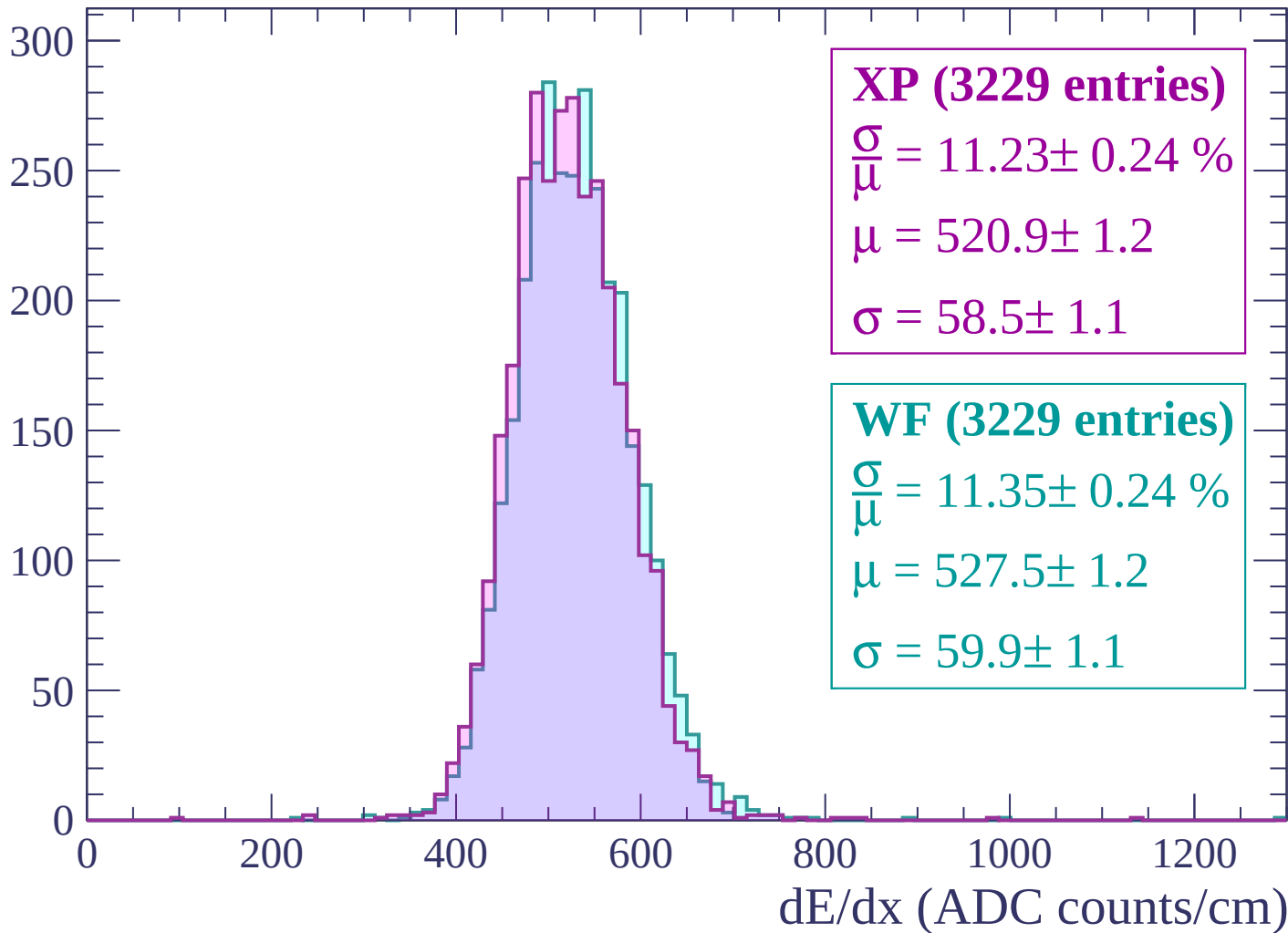
# Energy loss | -3000 < p < -2940

Count



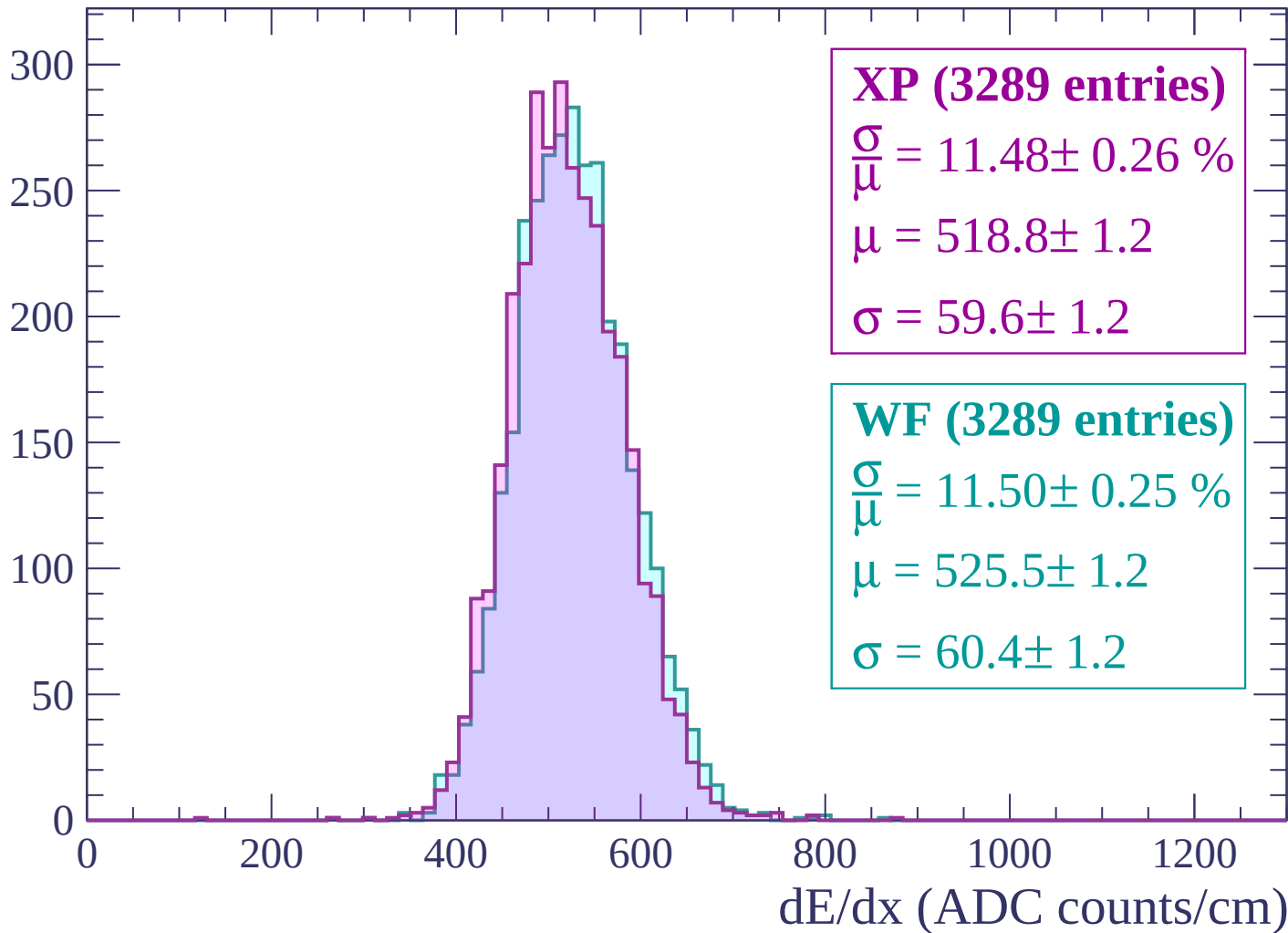
Energy loss |  $-2940 < p < -2880$

Count



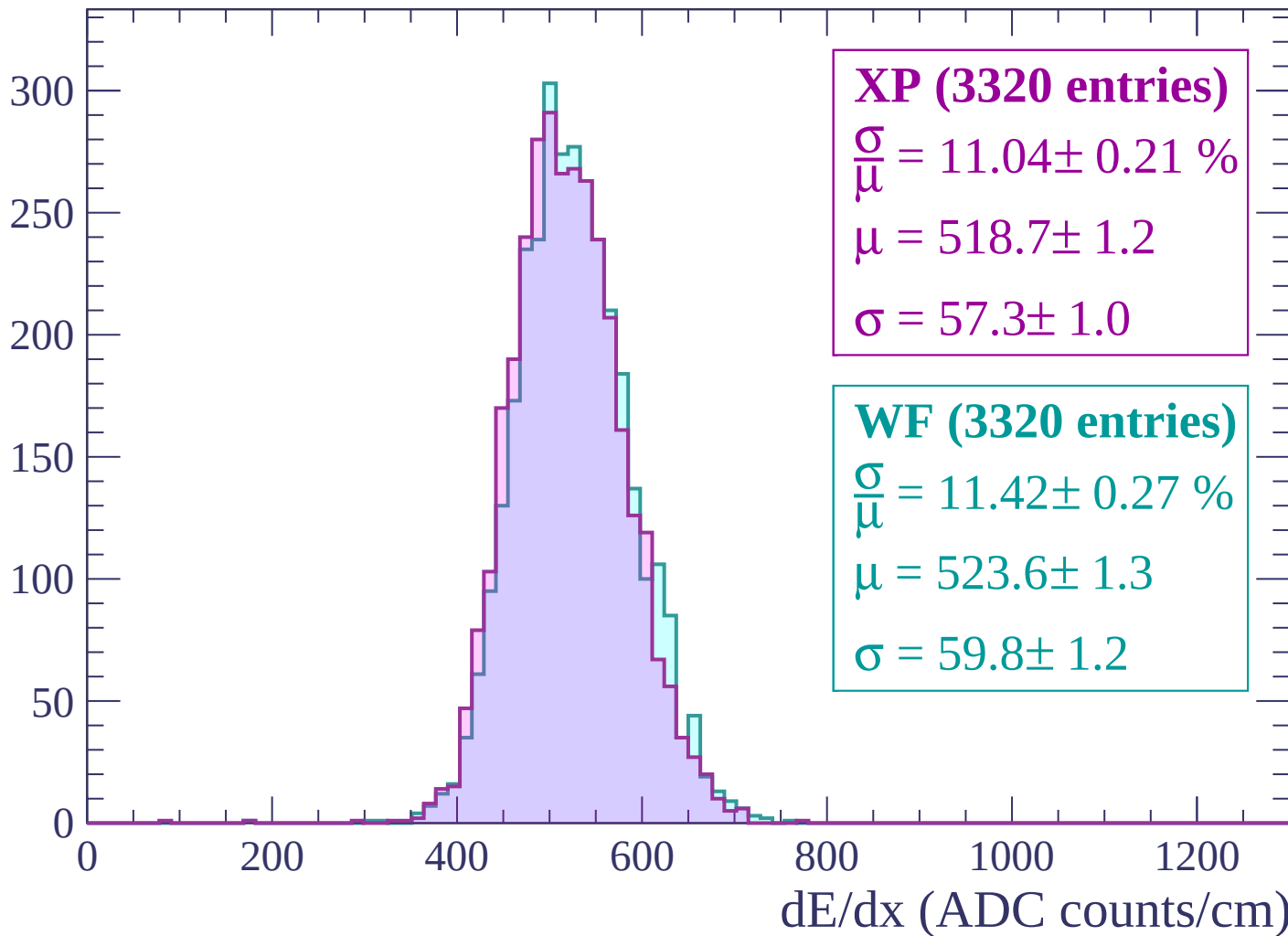
Energy loss |  $-2880 < p < -2820$

Count



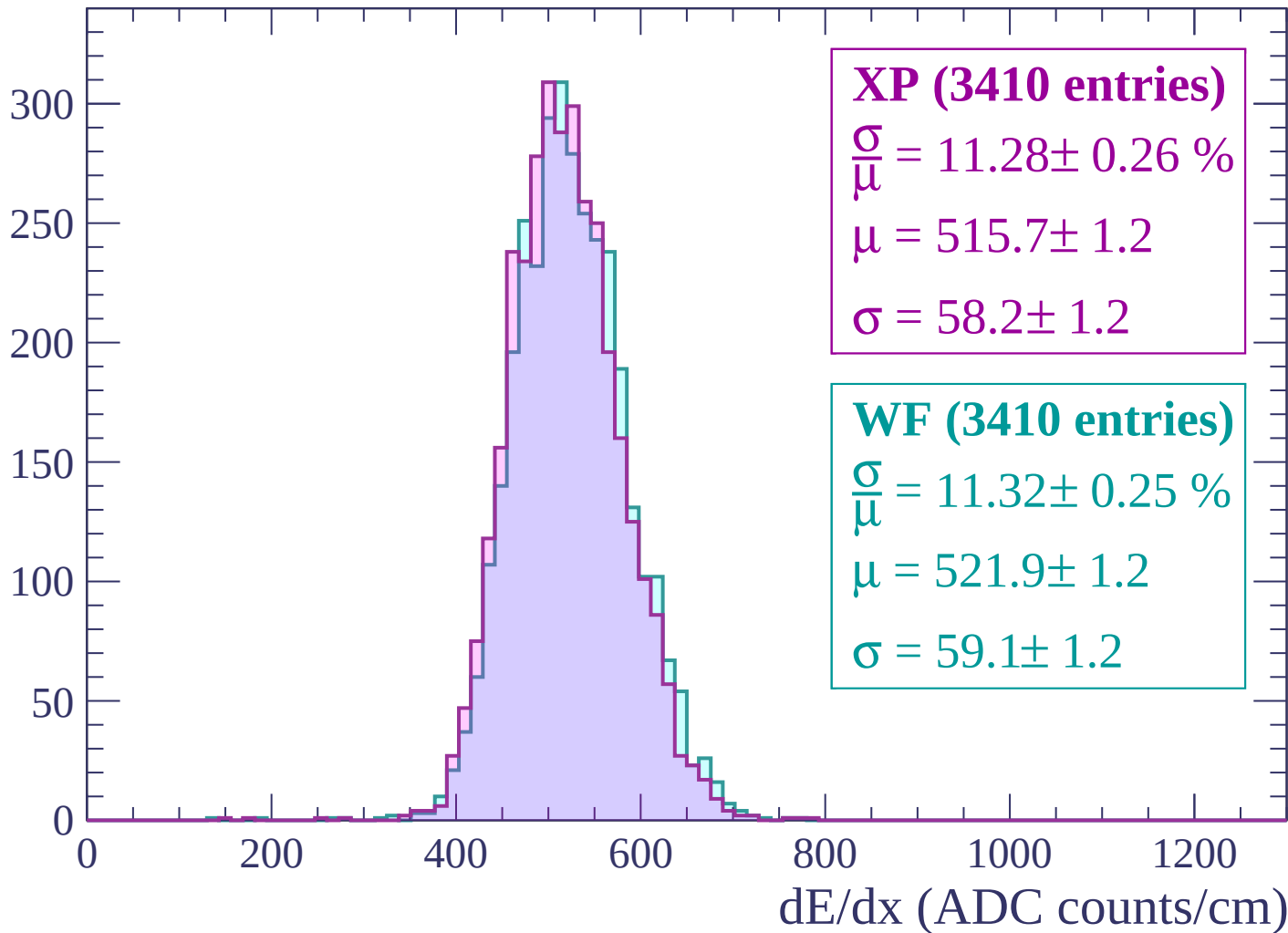
Energy loss |  $-2820 < p < -2760$

Count



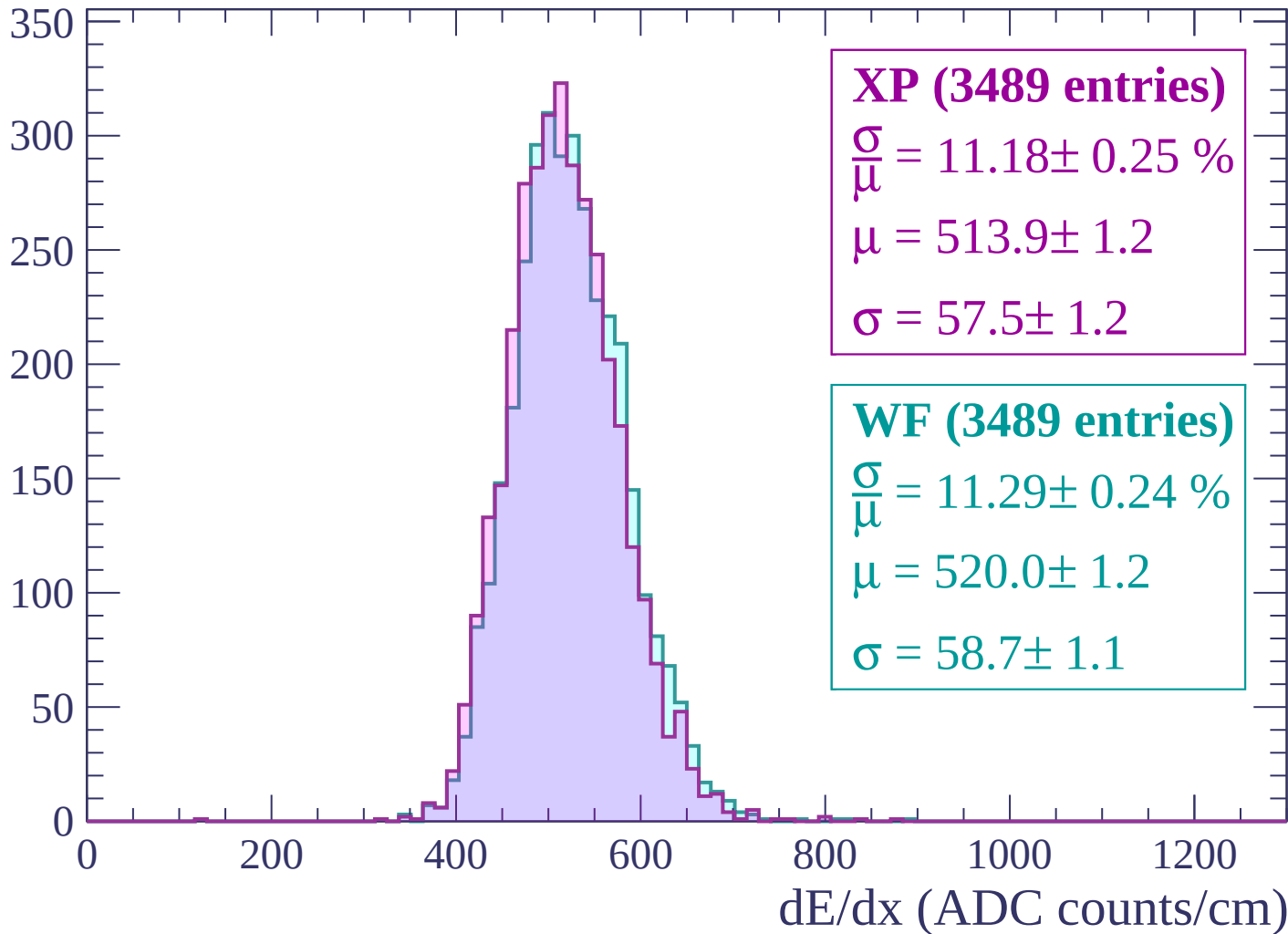
# Energy loss | $-2760 < p < -2700$

Count



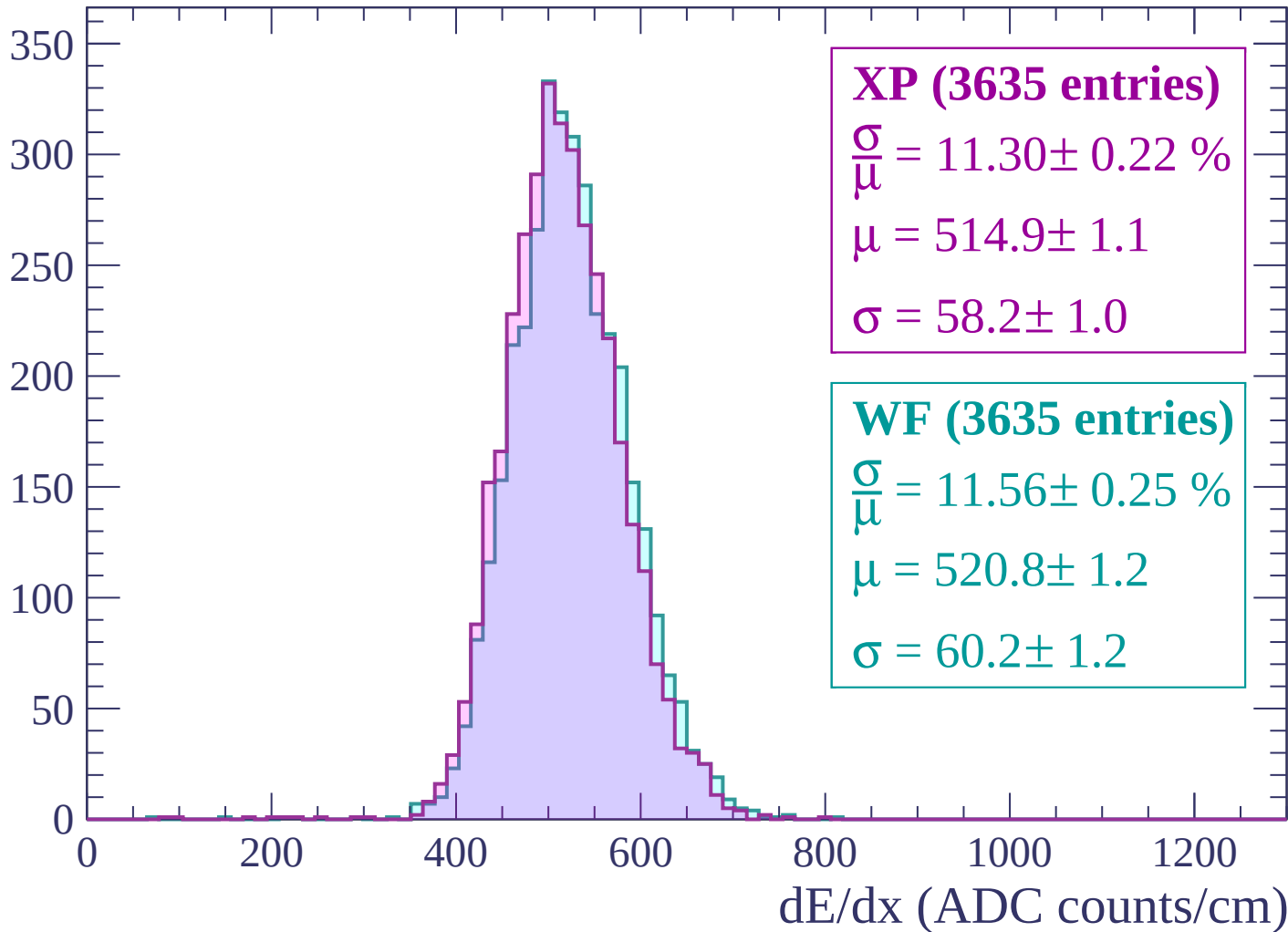
# Energy loss | -2700 < p < -2640

Count



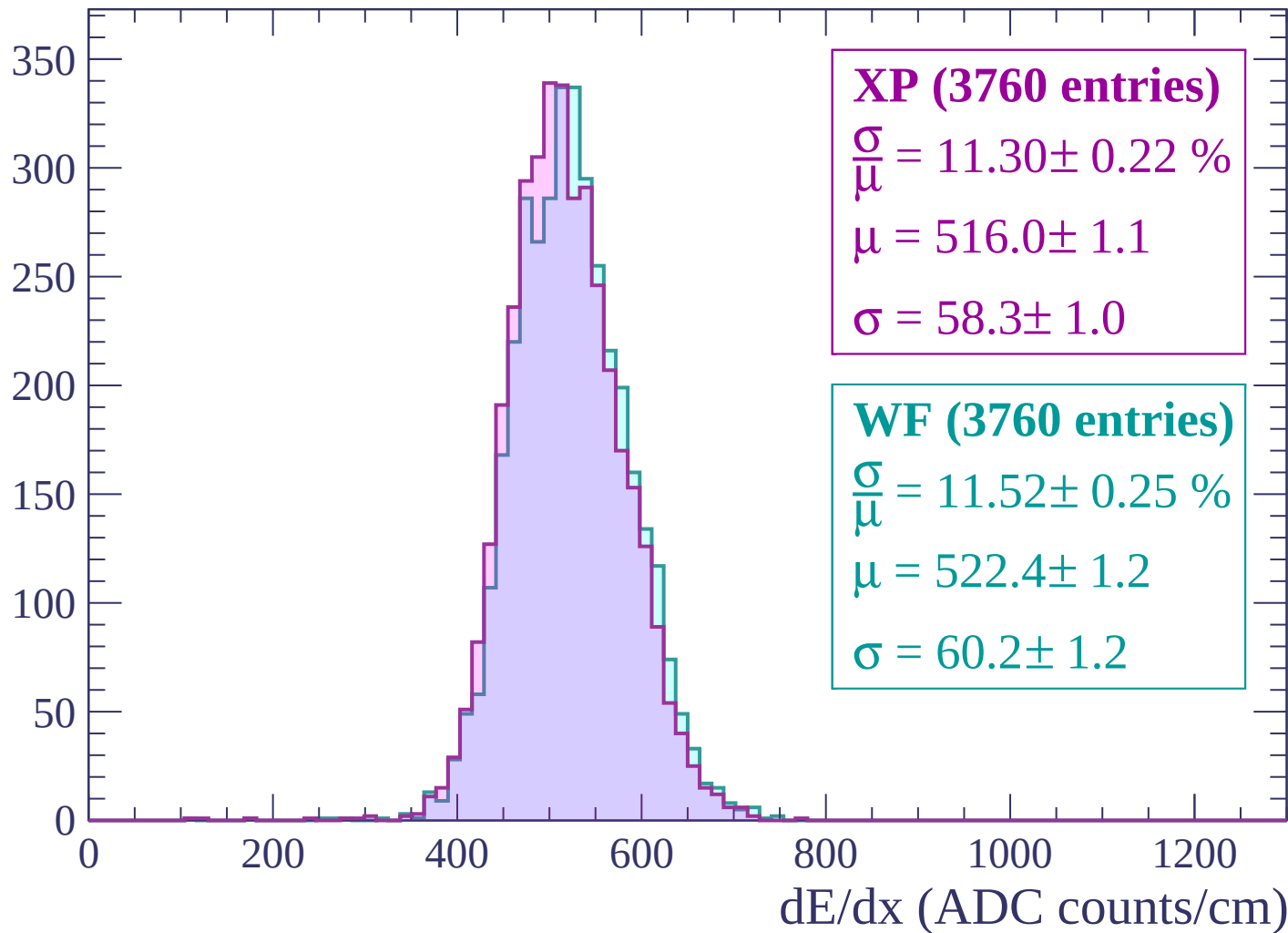
Energy loss |  $-2640 < p < -2580$

Count



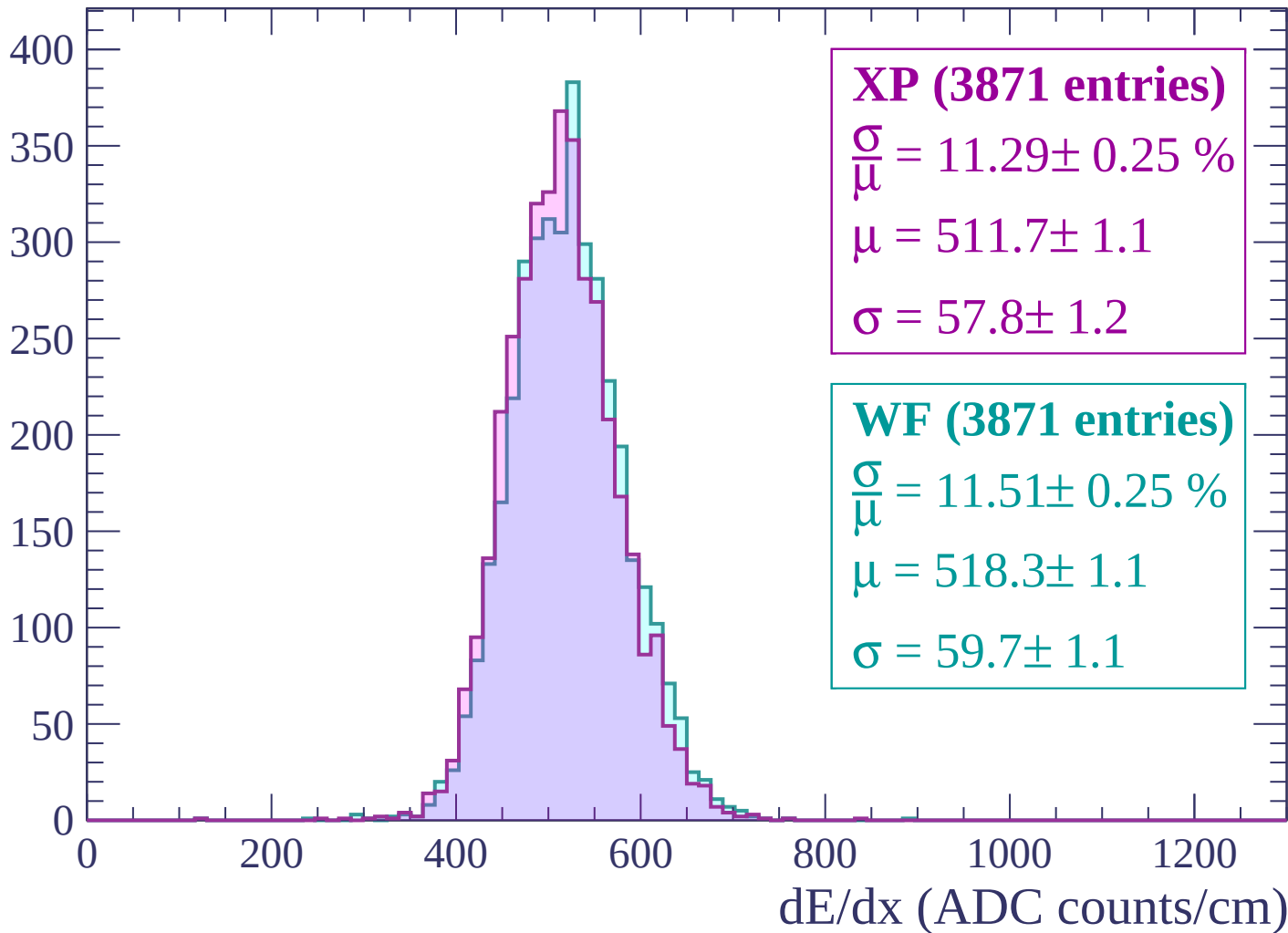
Energy loss |  $-2580 < p < -2520$

Count



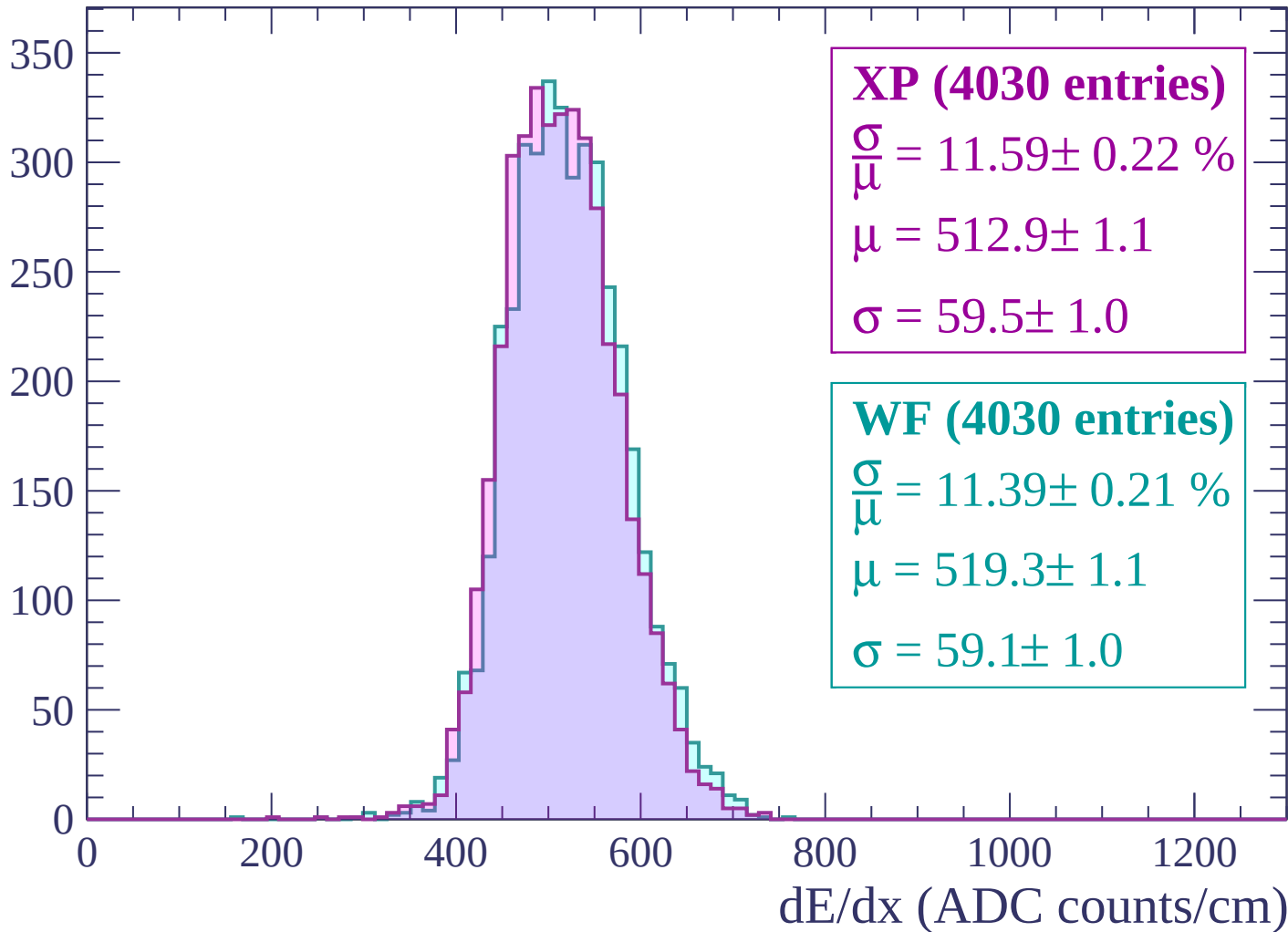
# Energy loss | -2520 < p < -2460

Count



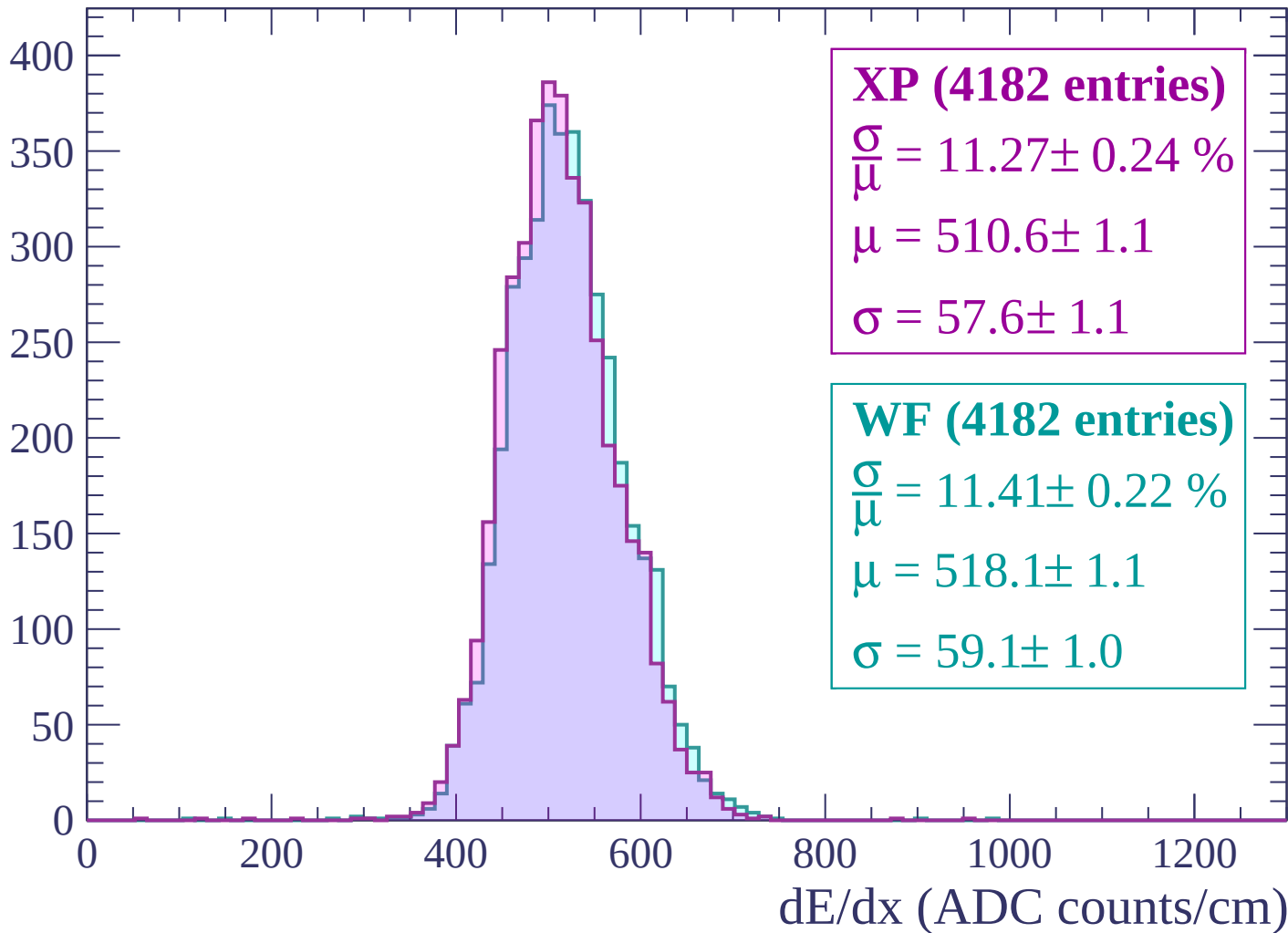
Energy loss |  $-2460 < p < -2400$

Count



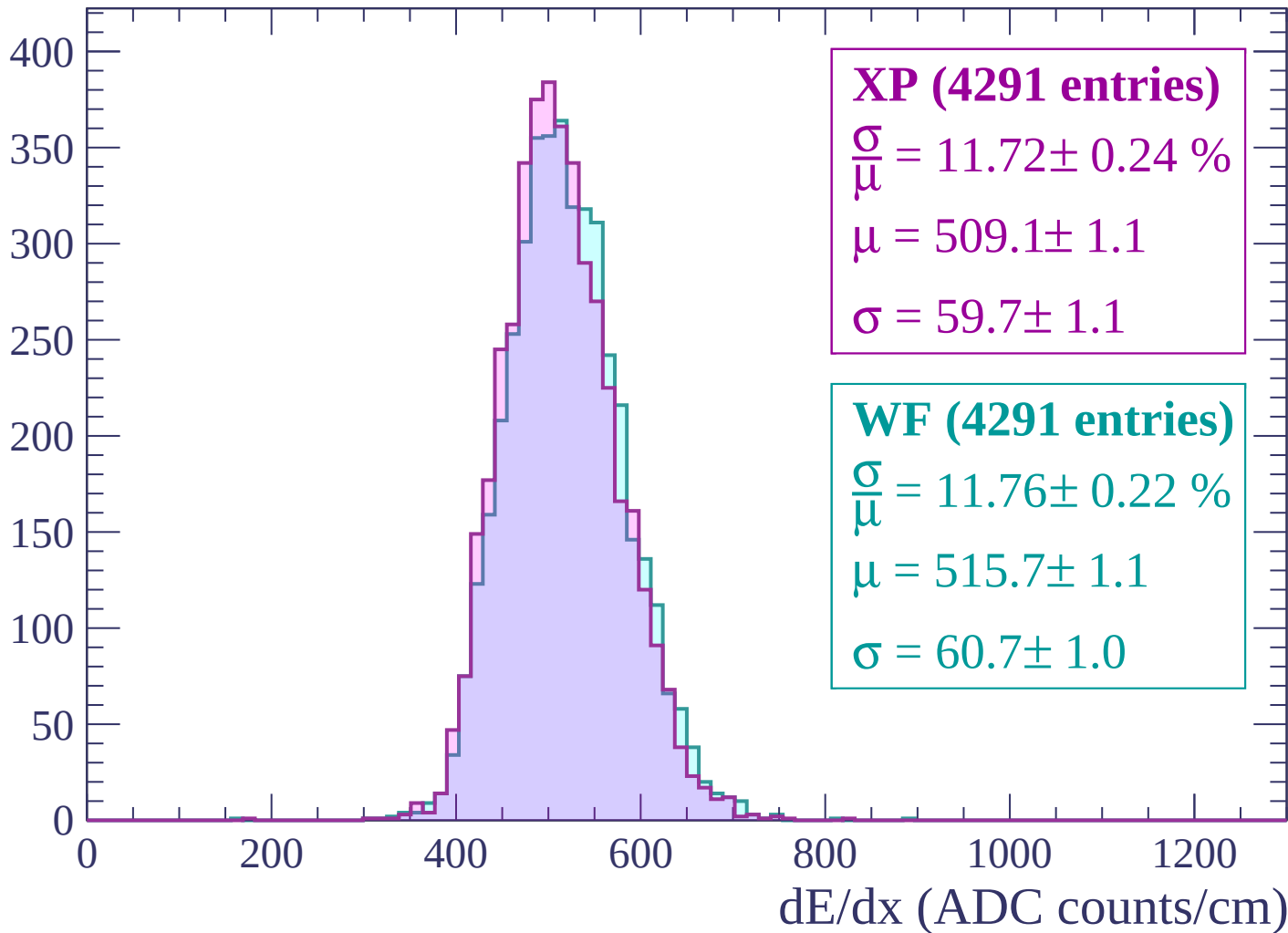
# Energy loss | -2400 < p < -2340

Count



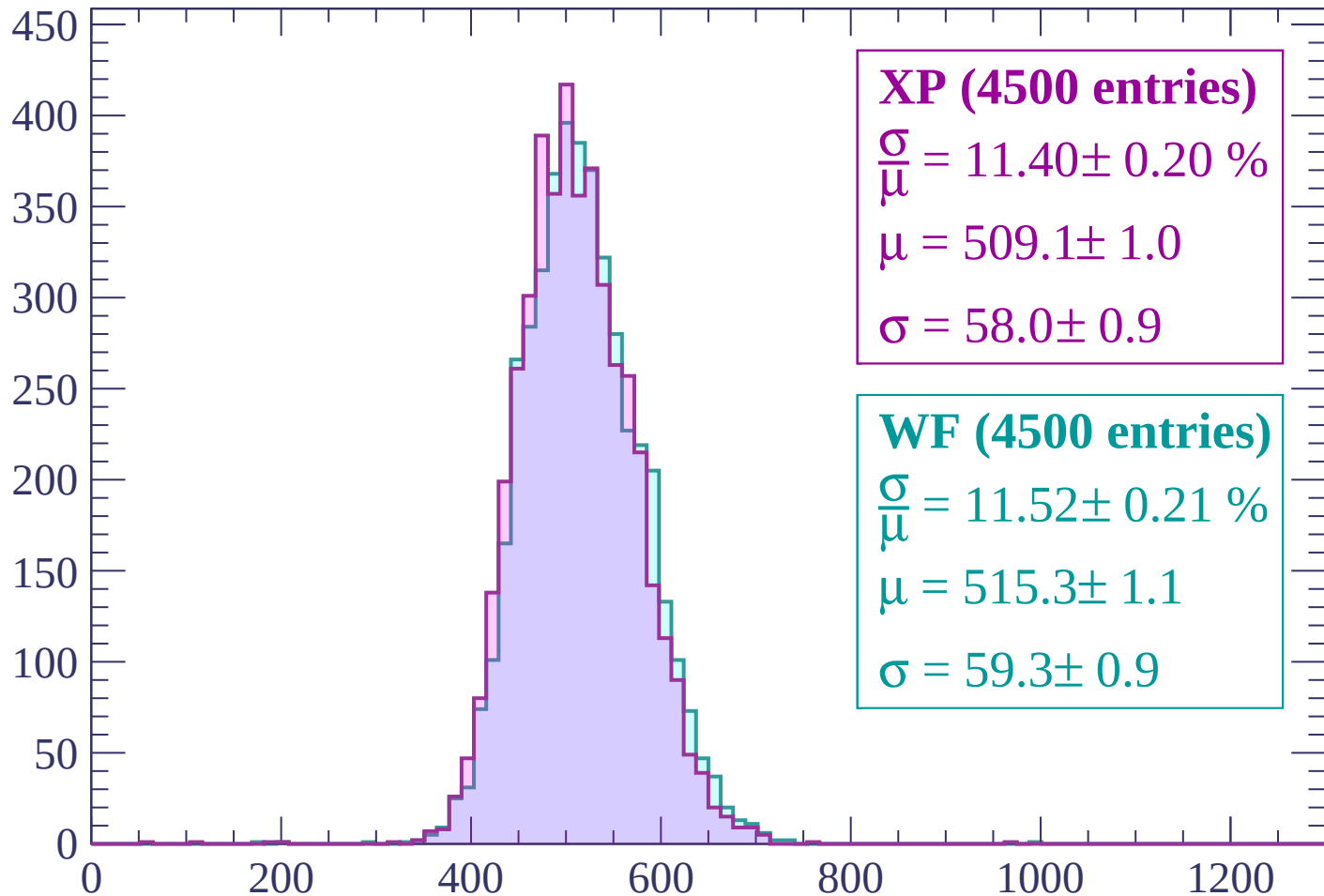
Energy loss |  $-2340 < p < -2280$

Count



# Energy loss | -2280 < p < -2220

Count



**XP (4500 entries)**

$$\frac{\sigma}{\mu} = 11.40 \pm 0.20 \%$$

$$\mu = 509.1 \pm 1.0$$

$$\sigma = 58.0 \pm 0.9$$

**WF (4500 entries)**

$$\frac{\sigma}{\mu} = 11.52 \pm 0.21 \%$$

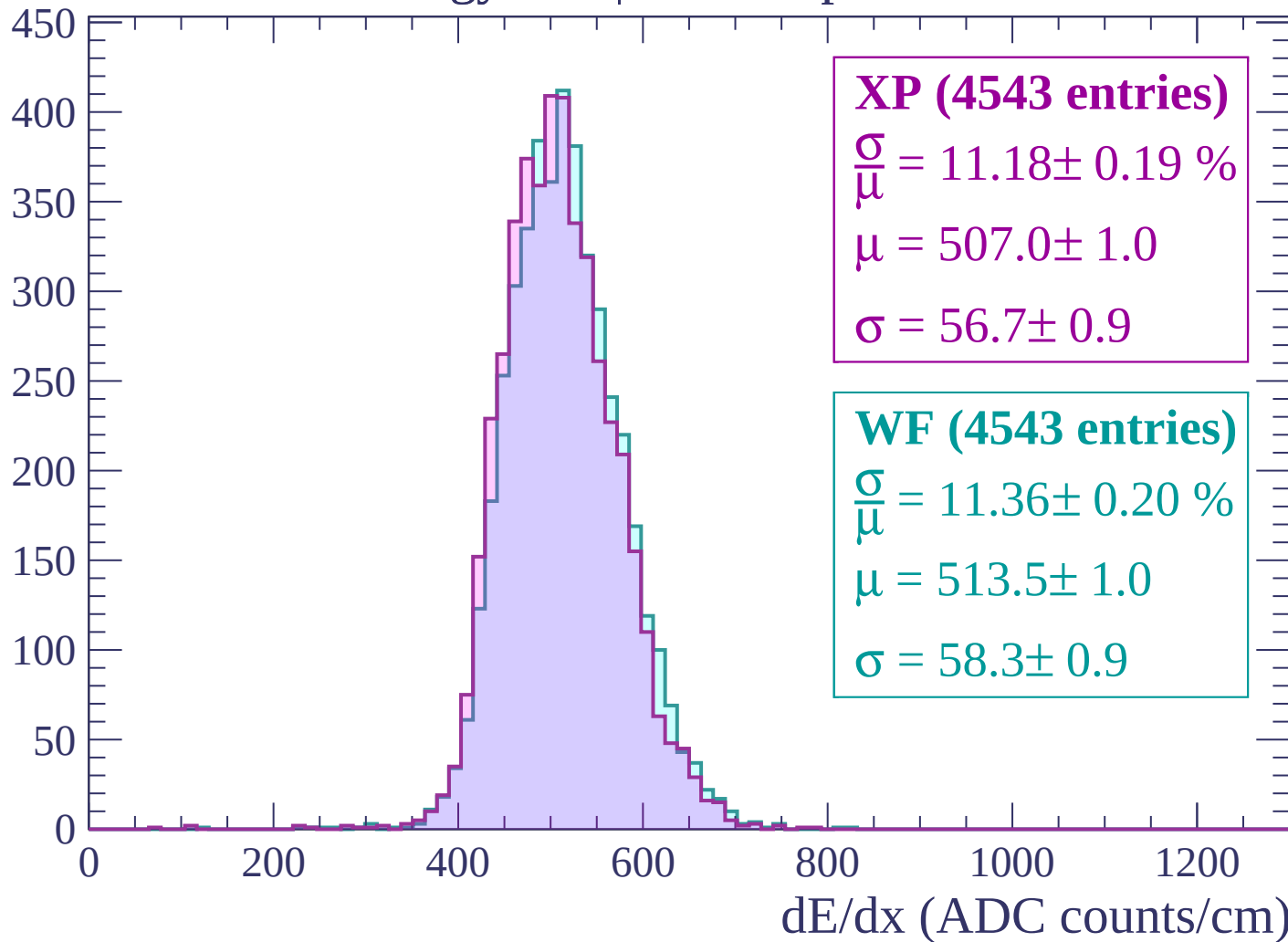
$$\mu = 515.3 \pm 1.1$$

$$\sigma = 59.3 \pm 0.9$$

dE/dx (ADC counts/cm)

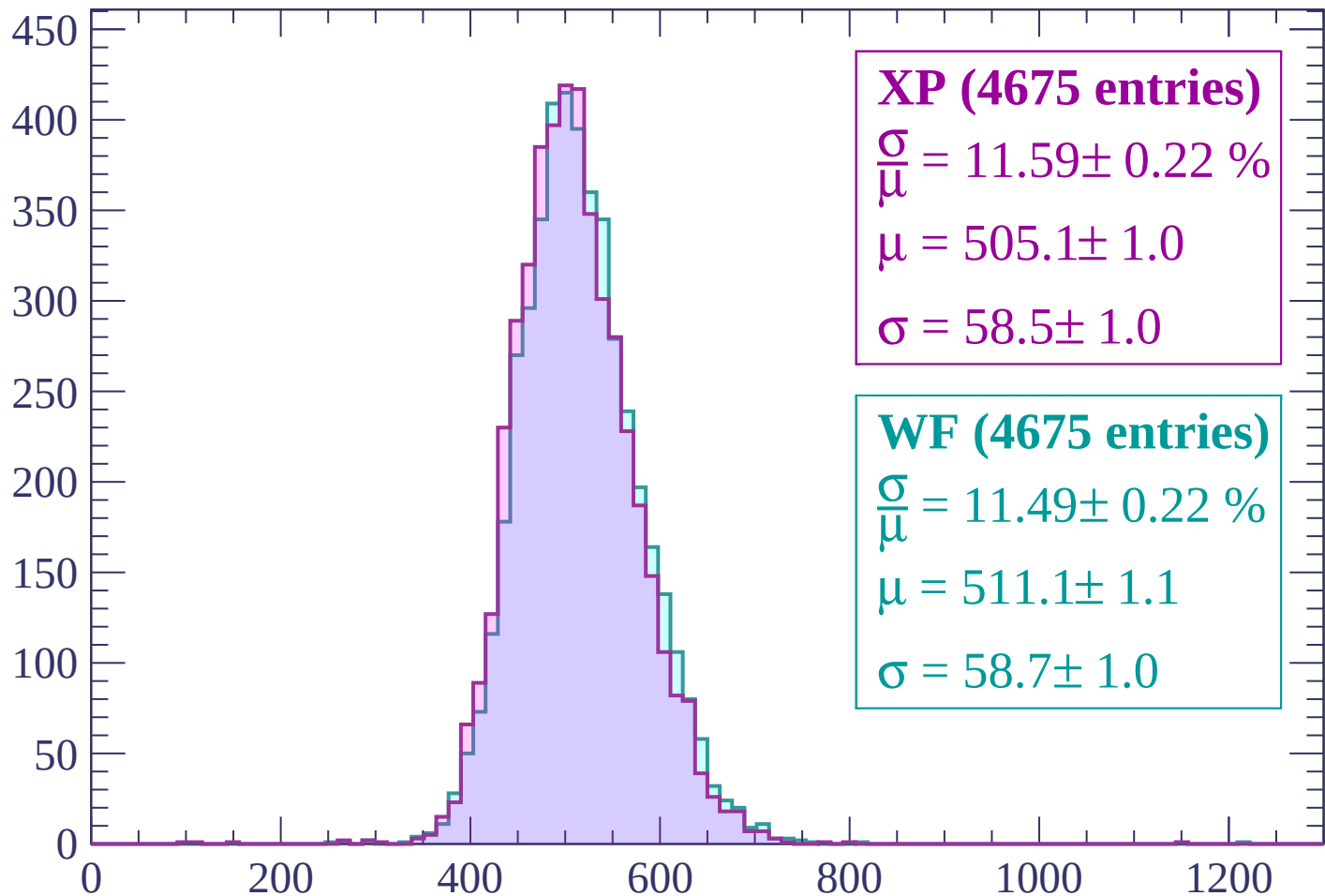
# Energy loss | -2220 < p < -2160

Count



# Energy loss | -2160 < p < -2100

Count



**XP (4675 entries)**

$$\frac{\sigma}{\mu} = 11.59 \pm 0.22 \%$$

$$\mu = 505.1 \pm 1.0$$

$$\sigma = 58.5 \pm 1.0$$

**WF (4675 entries)**

$$\frac{\sigma}{\mu} = 11.49 \pm 0.22 \%$$

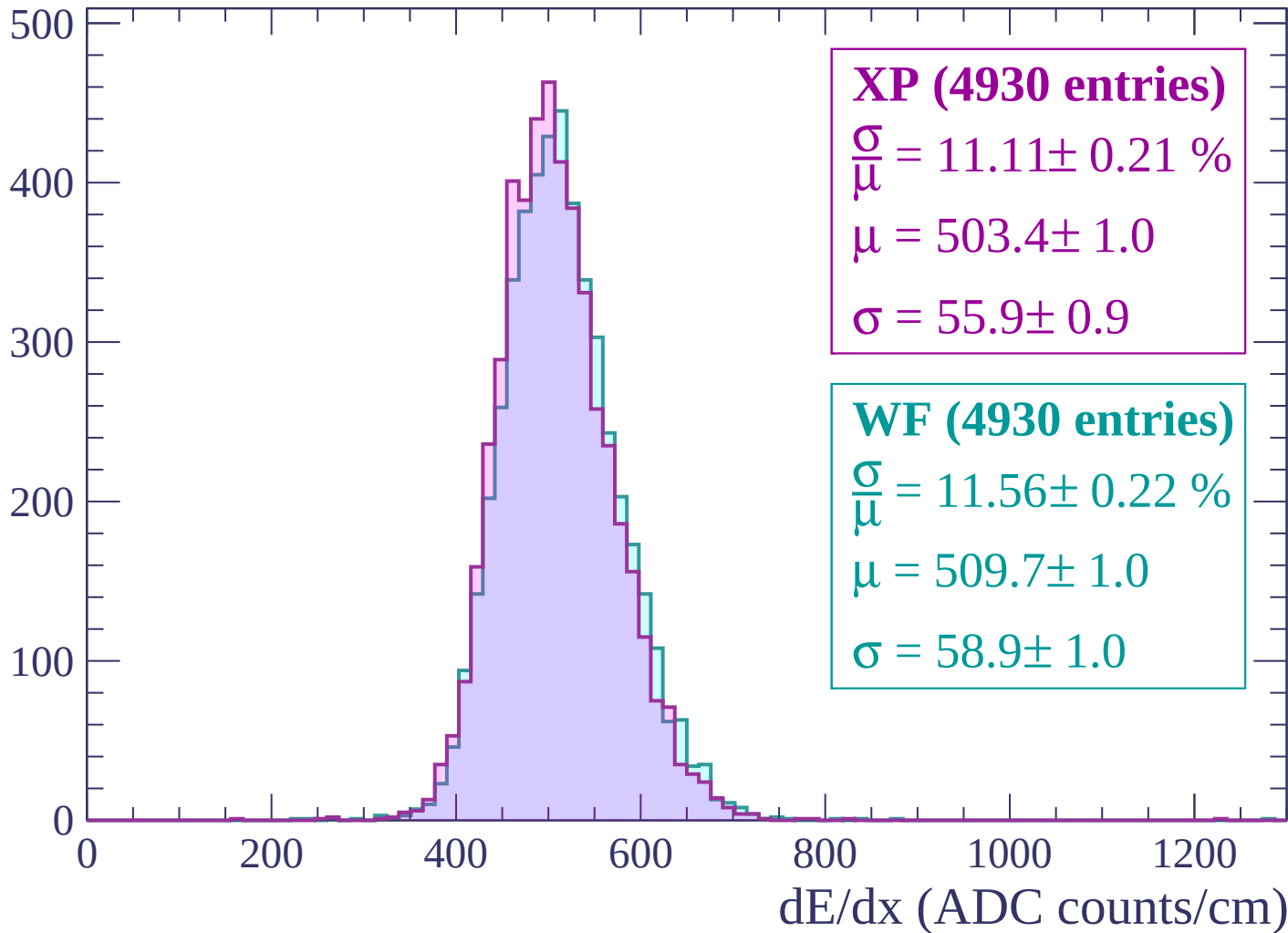
$$\mu = 511.1 \pm 1.1$$

$$\sigma = 58.7 \pm 1.0$$

dE/dx (ADC counts/cm)

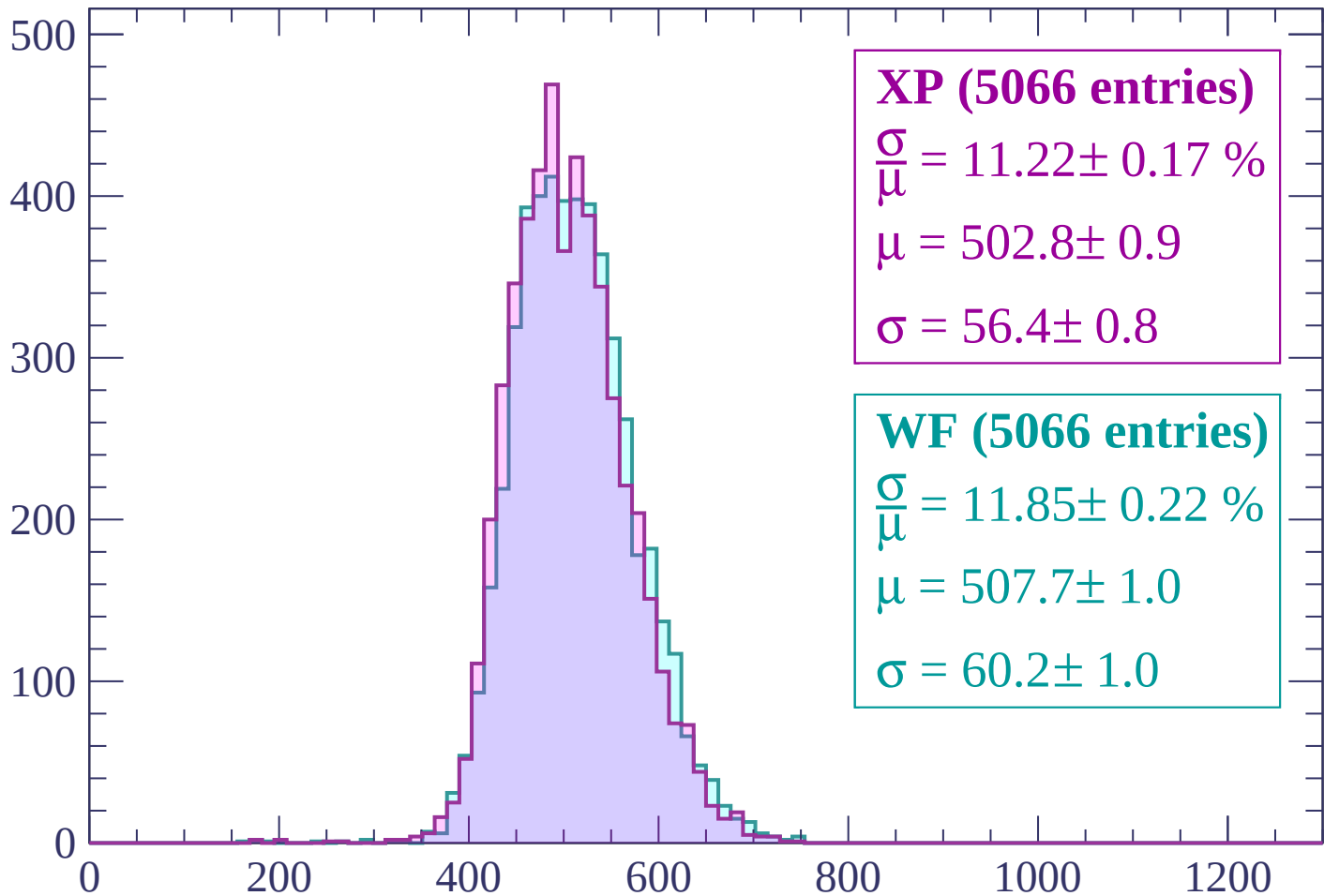
# Energy loss | -2100 < p < -2040

Count



# Energy loss | -2040 < p < -1980

Count



**XP (5066 entries)**

$$\frac{\sigma}{\mu} = 11.22 \pm 0.17 \%$$

$$\mu = 502.8 \pm 0.9$$

$$\sigma = 56.4 \pm 0.8$$

**WF (5066 entries)**

$$\frac{\sigma}{\mu} = 11.85 \pm 0.22 \%$$

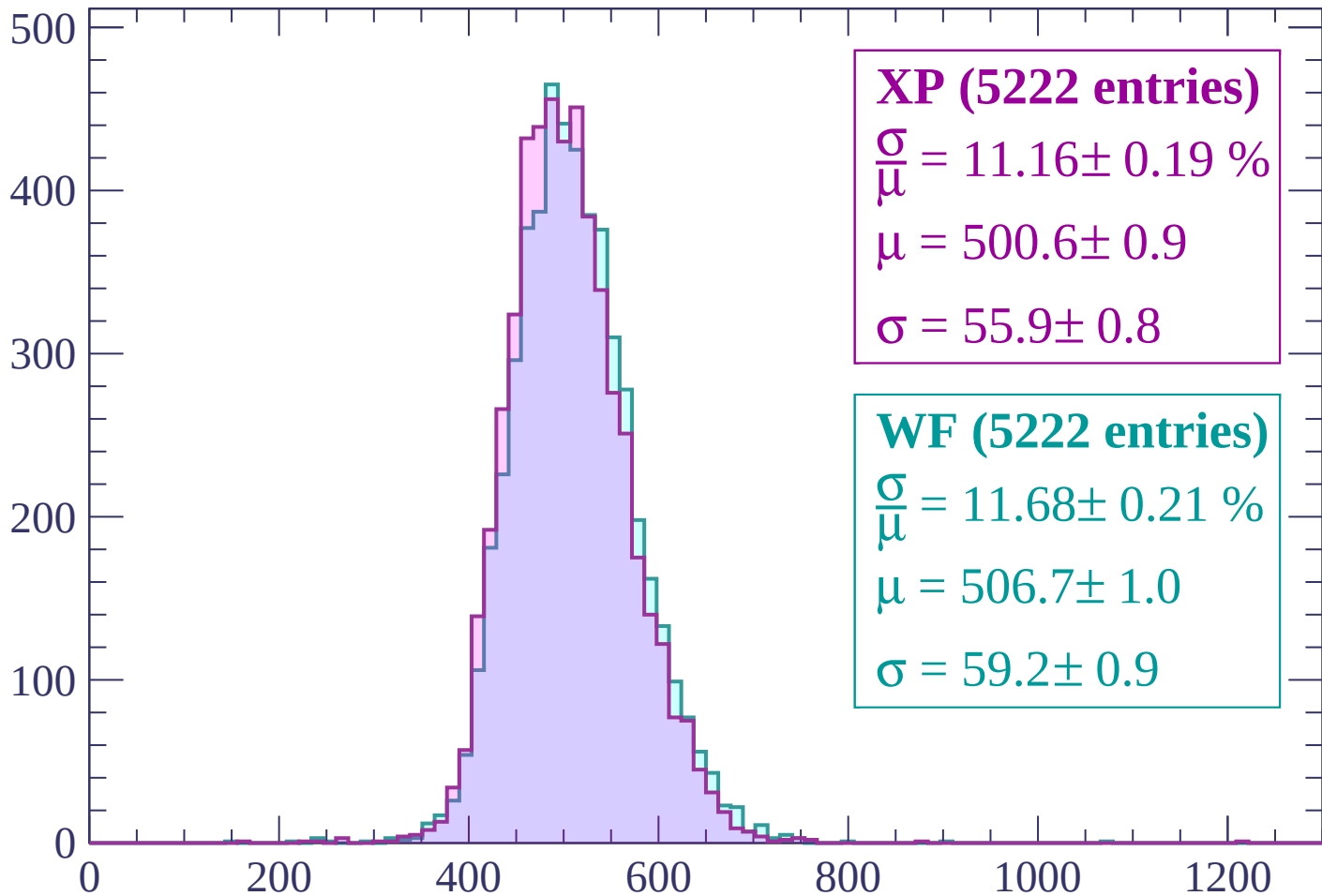
$$\mu = 507.7 \pm 1.0$$

$$\sigma = 60.2 \pm 1.0$$

dE/dx (ADC counts/cm)

# Energy loss | -1980 < p < -1920

Count



**XP (5222 entries)**

$$\frac{\sigma}{\mu} = 11.16 \pm 0.19 \%$$

$$\mu = 500.6 \pm 0.9$$

$$\sigma = 55.9 \pm 0.8$$

**WF (5222 entries)**

$$\frac{\sigma}{\mu} = 11.68 \pm 0.21 \%$$

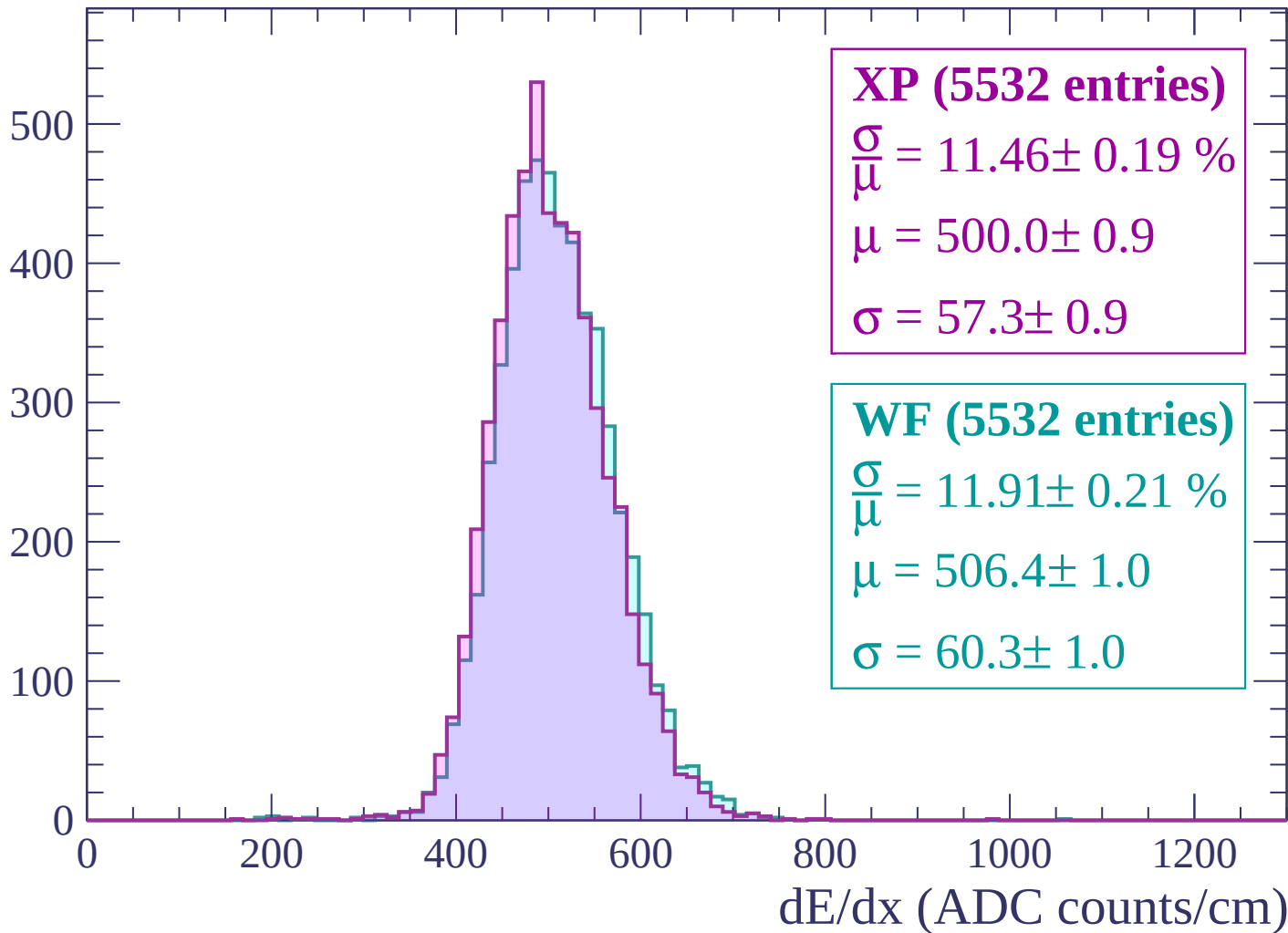
$$\mu = 506.7 \pm 1.0$$

$$\sigma = 59.2 \pm 0.9$$

dE/dx (ADC counts/cm)

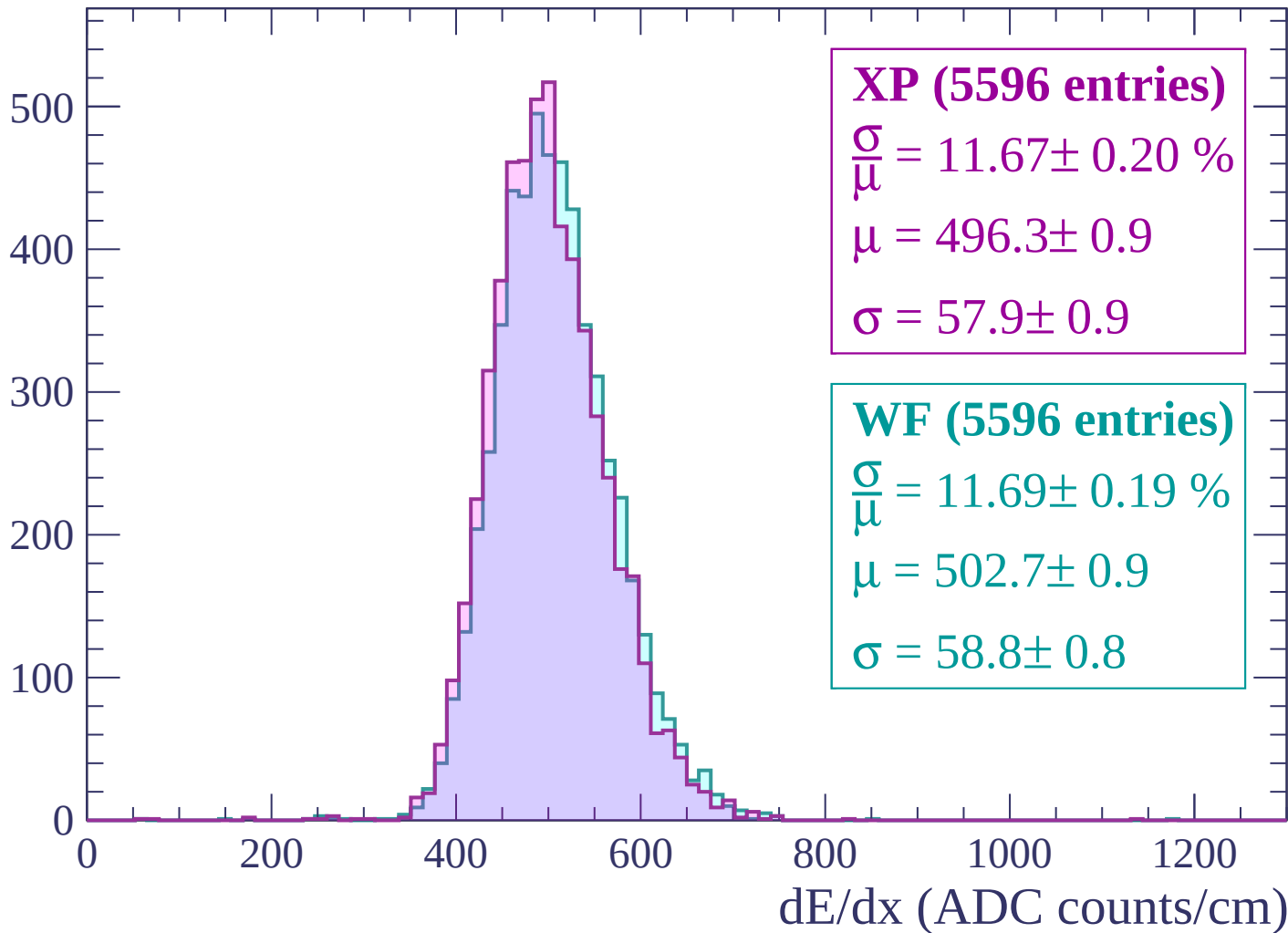
# Energy loss | -1920 < p < -1860

Count



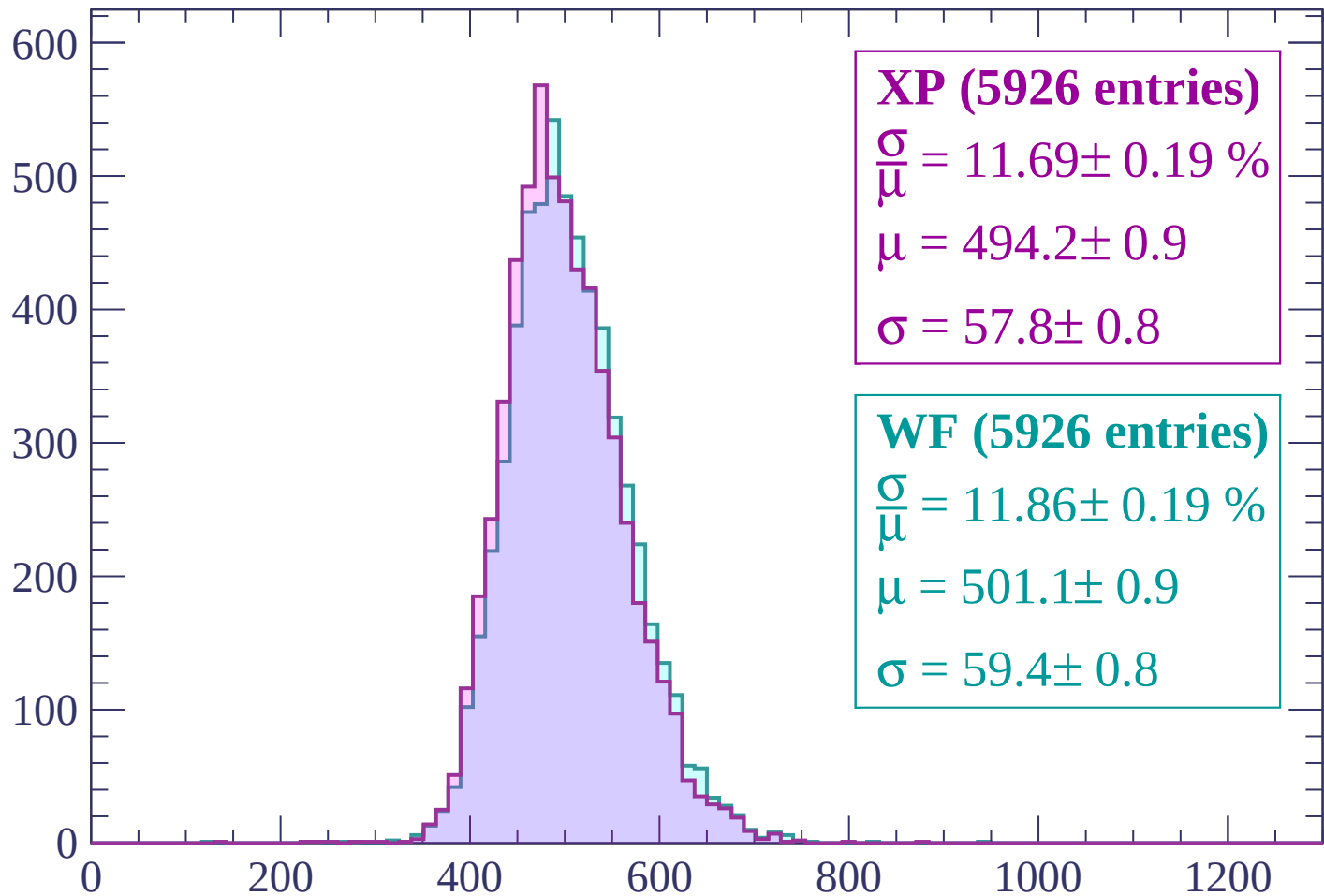
# Energy loss | -1860 < p < -1800

Count



# Energy loss | -1800 < p < -1740

Count



**XP (5926 entries)**

$$\frac{\sigma}{\mu} = 11.69 \pm 0.19 \%$$

$$\mu = 494.2 \pm 0.9$$

$$\sigma = 57.8 \pm 0.8$$

**WF (5926 entries)**

$$\frac{\sigma}{\mu} = 11.86 \pm 0.19 \%$$

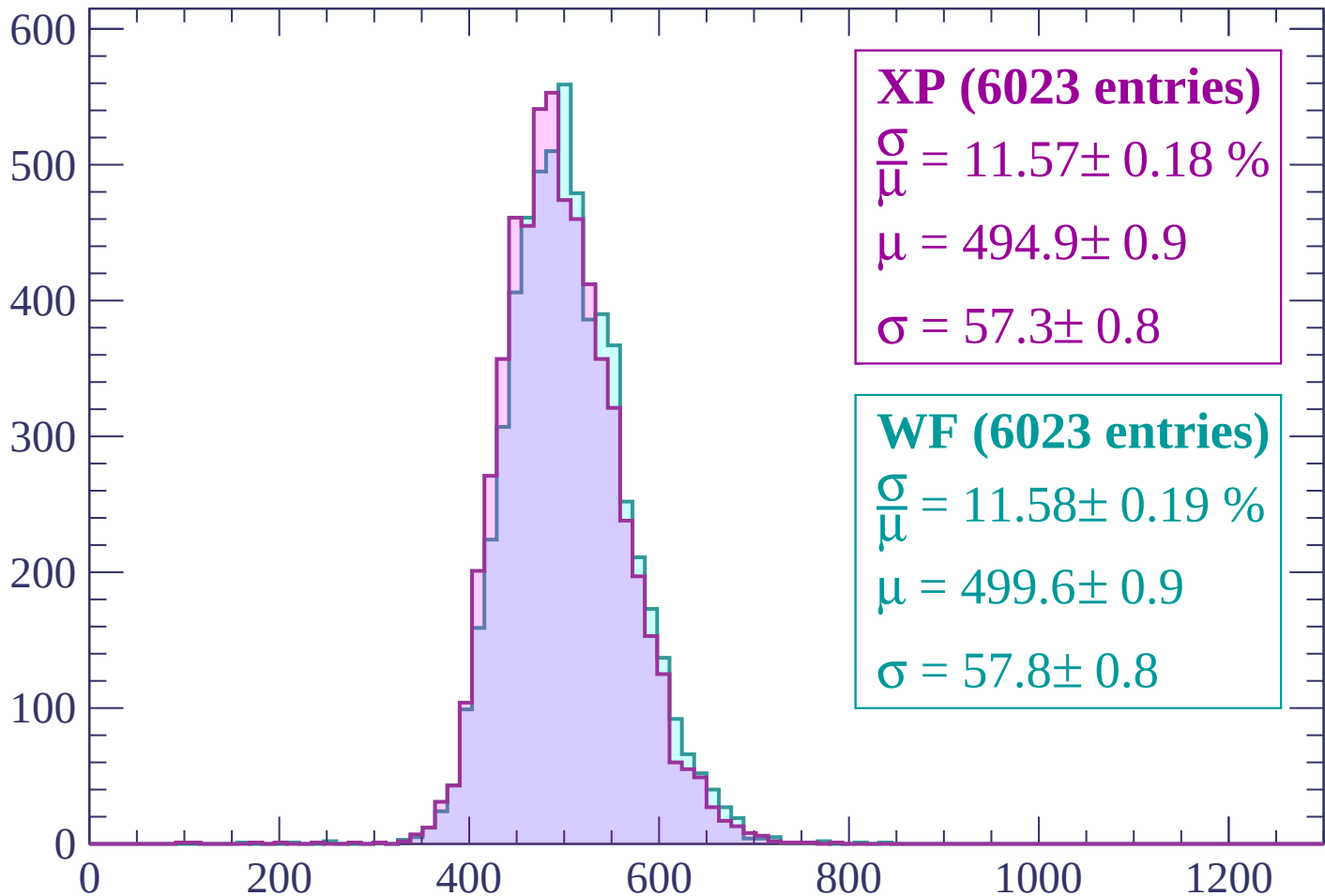
$$\mu = 501.1 \pm 0.9$$

$$\sigma = 59.4 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $-1740 < p < -1680$

Count



**XP (6023 entries)**

$$\frac{\sigma}{\mu} = 11.57 \pm 0.18 \%$$

$$\mu = 494.9 \pm 0.9$$

$$\sigma = 57.3 \pm 0.8$$

**WF (6023 entries)**

$$\frac{\sigma}{\mu} = 11.58 \pm 0.19 \%$$

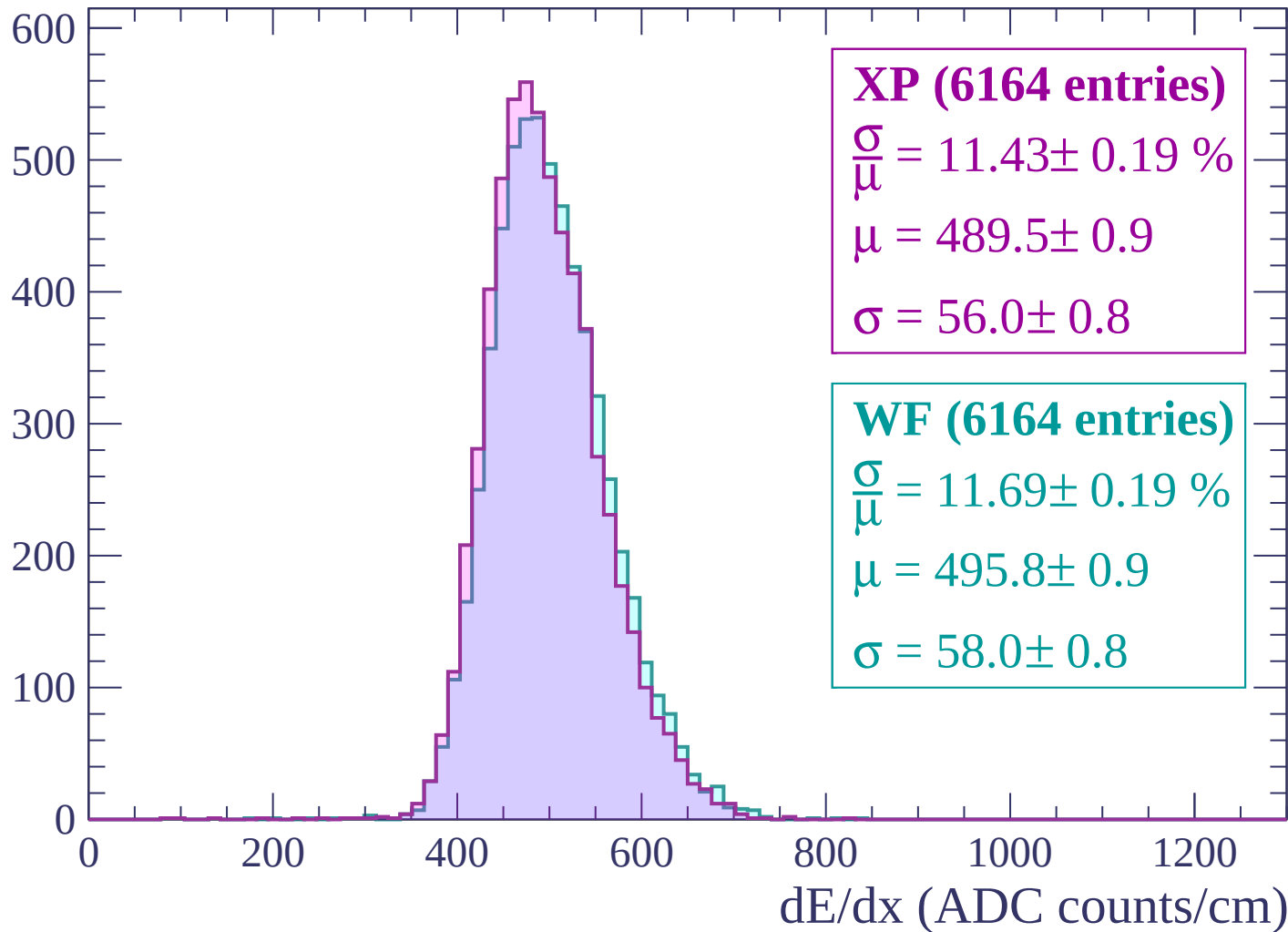
$$\mu = 499.6 \pm 0.9$$

$$\sigma = 57.8 \pm 0.8$$

dE/dx (ADC counts/cm)

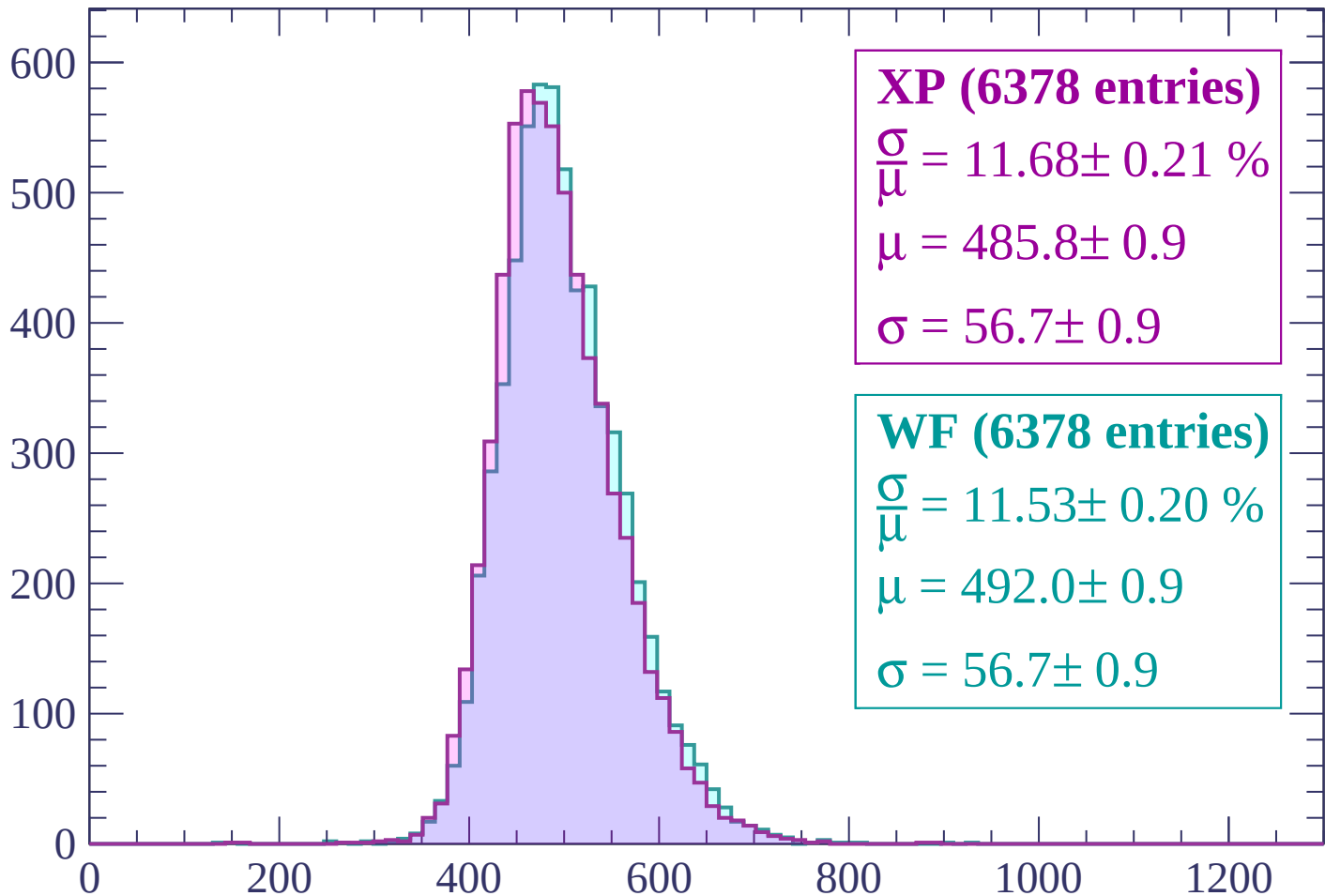
# Energy loss | -1680 < p < -1620

Count



Energy loss |  $-1620 < p < -1560$

Count



**XP (6378 entries)**

$$\frac{\sigma}{\mu} = 11.68 \pm 0.21 \%$$

$$\mu = 485.8 \pm 0.9$$

$$\sigma = 56.7 \pm 0.9$$

**WF (6378 entries)**

$$\frac{\sigma}{\mu} = 11.53 \pm 0.20 \%$$

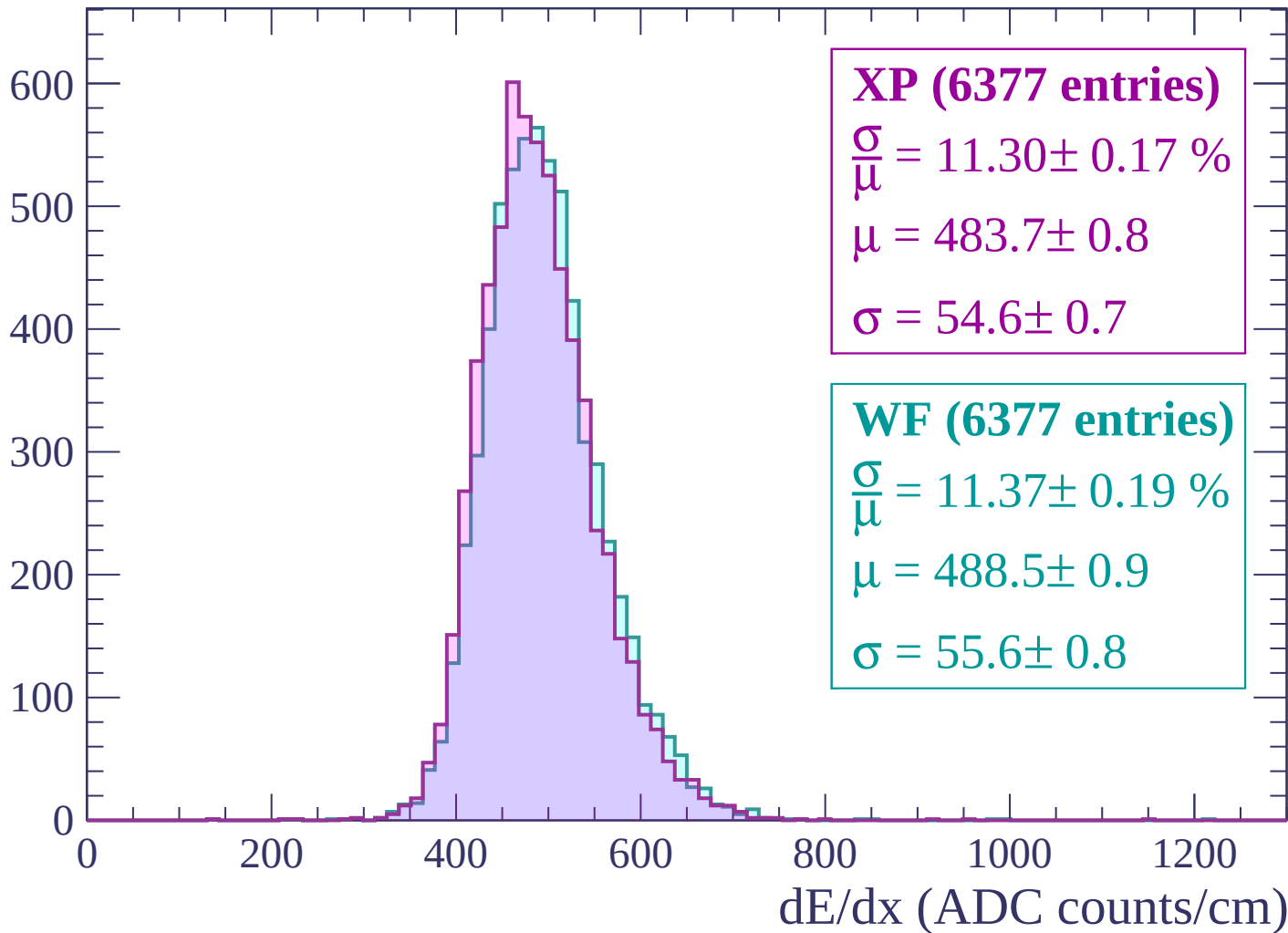
$$\mu = 492.0 \pm 0.9$$

$$\sigma = 56.7 \pm 0.9$$

dE/dx (ADC counts/cm)

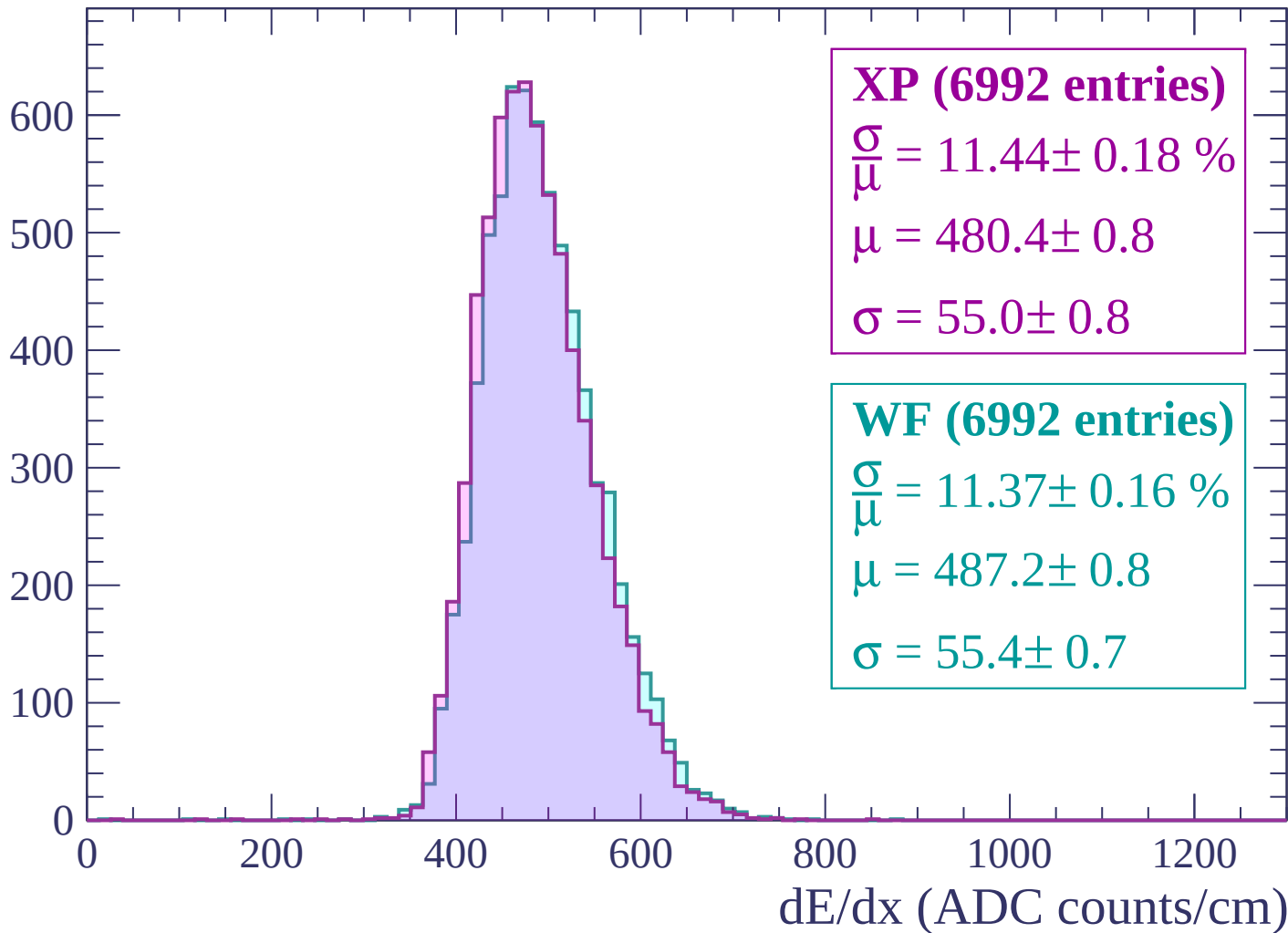
# Energy loss | -1560 < p < -1500

Count



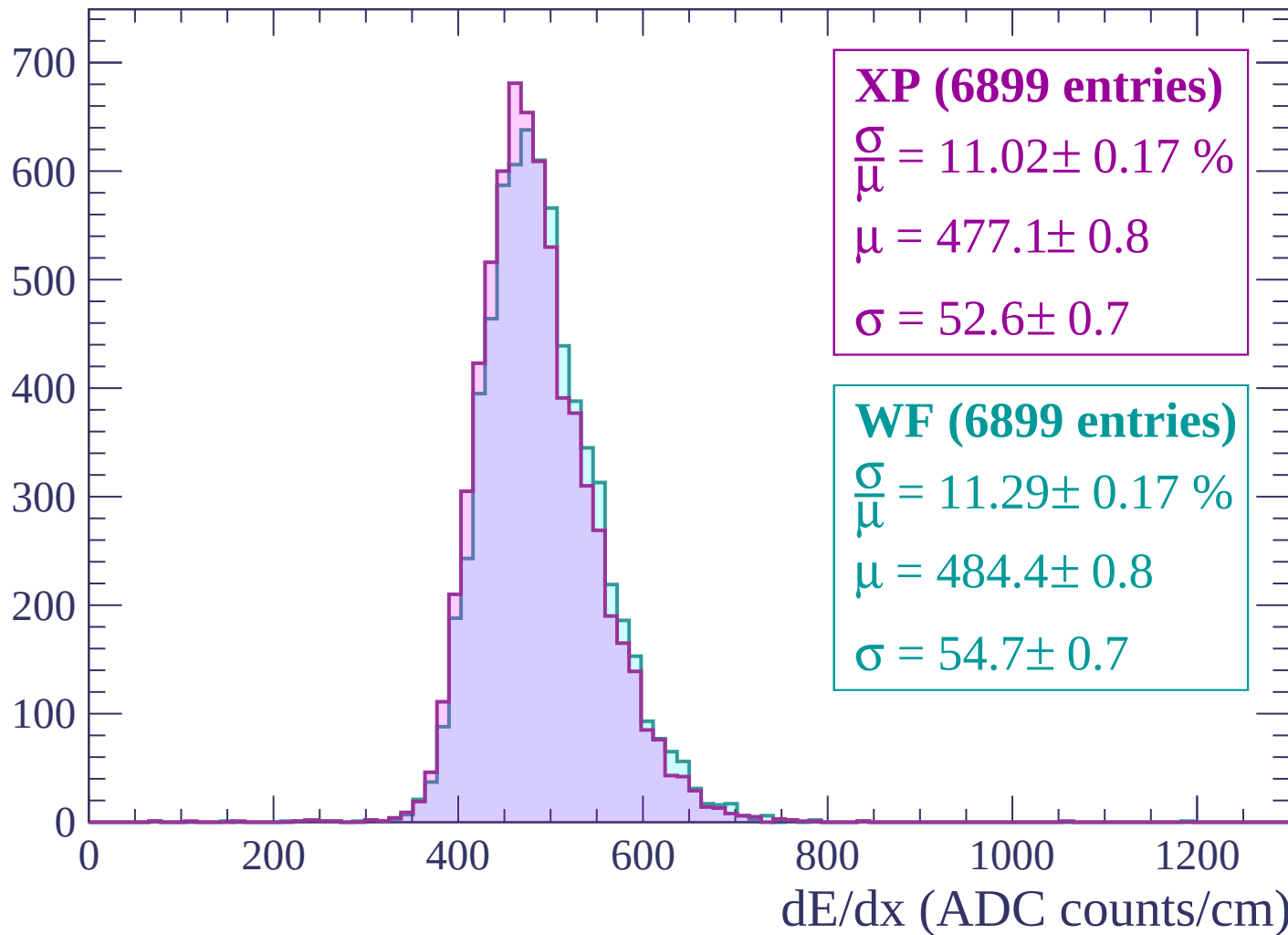
# Energy loss | $-1500 < p < -1440$

Count



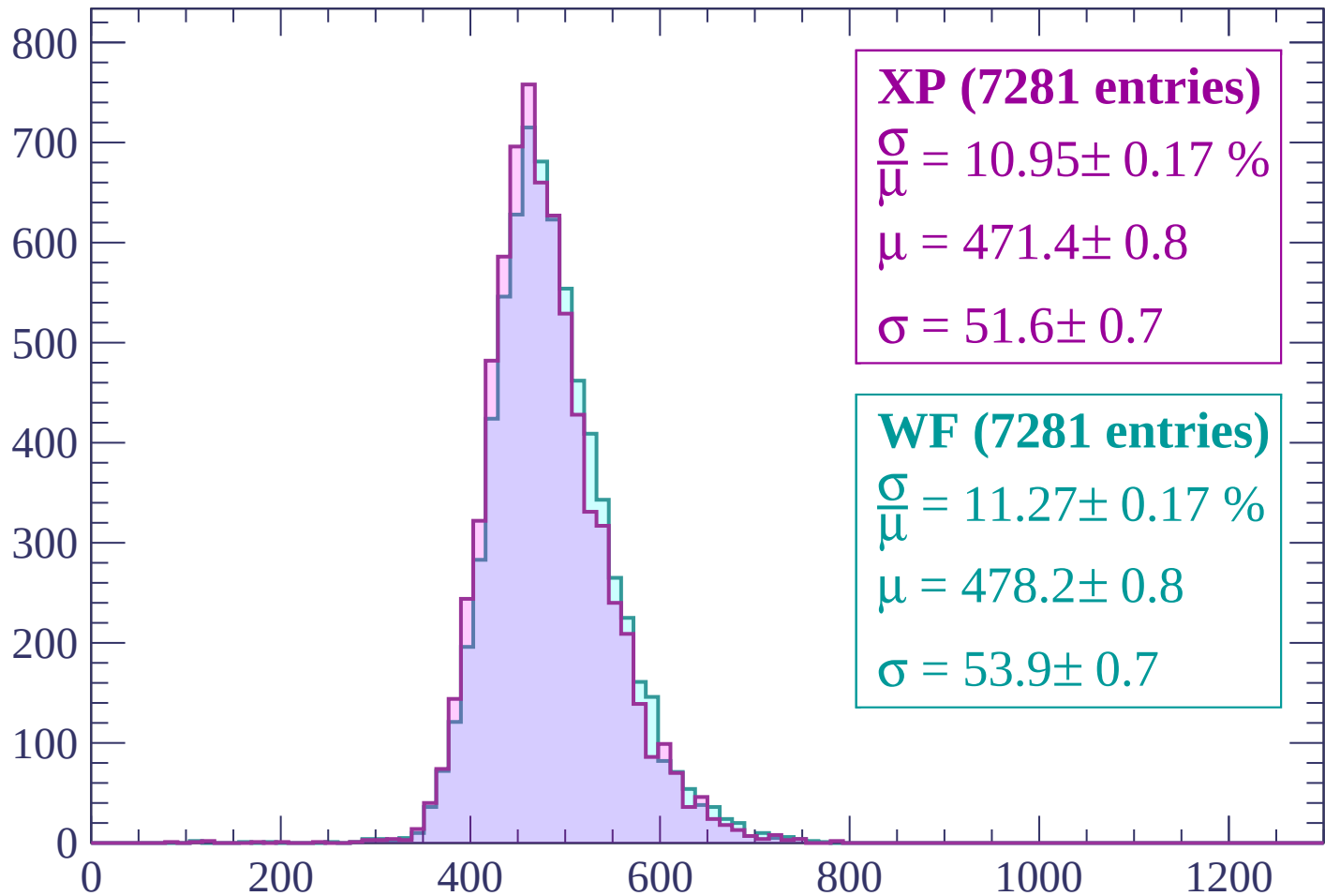
Energy loss |  $-1440 < p < -1380$

Count



# Energy loss | -1380 < p < -1320

Count



**XP (7281 entries)**

$$\frac{\sigma}{\mu} = 10.95 \pm 0.17 \%$$

$$\mu = 471.4 \pm 0.8$$

$$\sigma = 51.6 \pm 0.7$$

**WF (7281 entries)**

$$\frac{\sigma}{\mu} = 11.27 \pm 0.17 \%$$

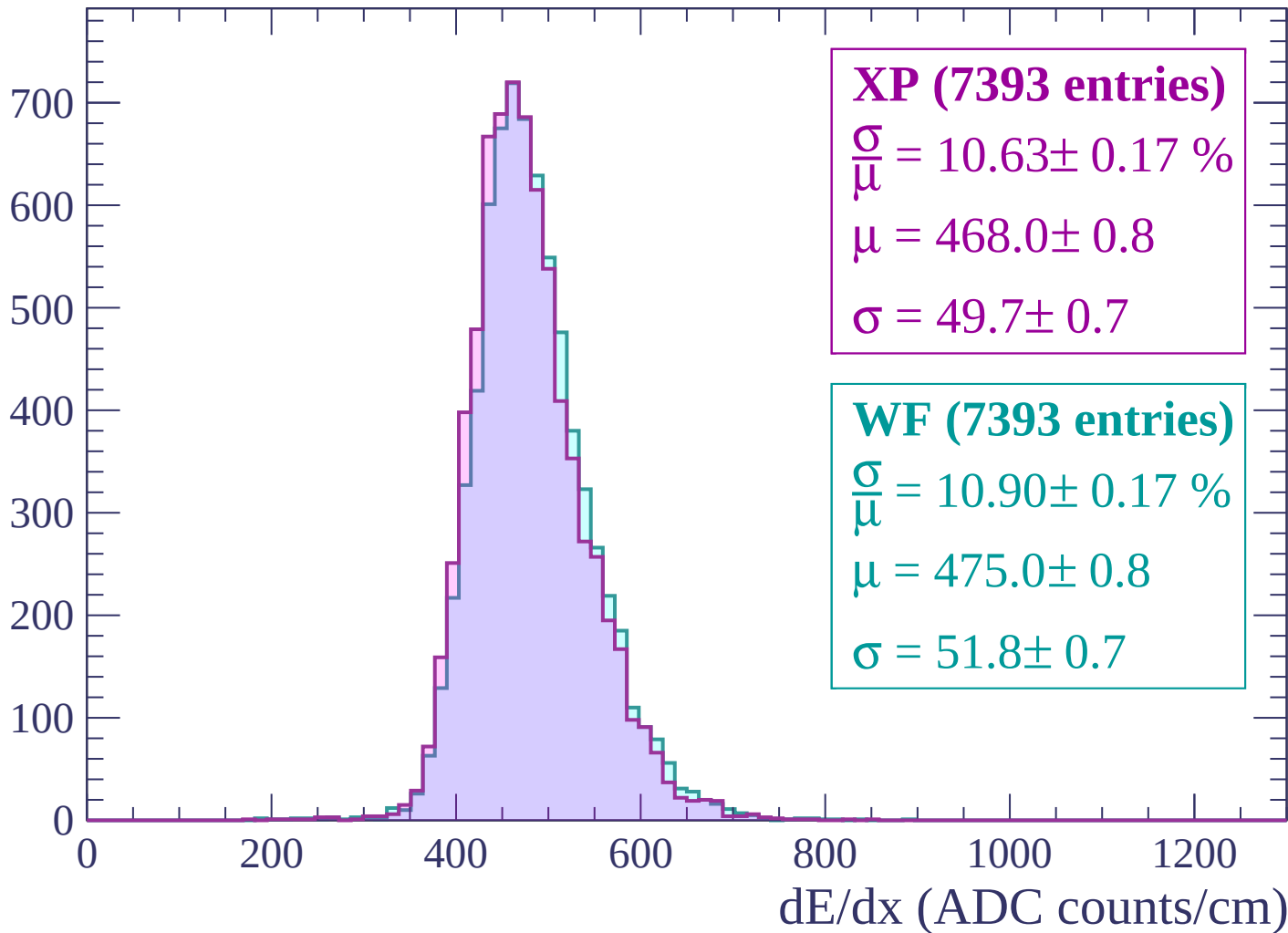
$$\mu = 478.2 \pm 0.8$$

$$\sigma = 53.9 \pm 0.7$$

dE/dx (ADC counts/cm)

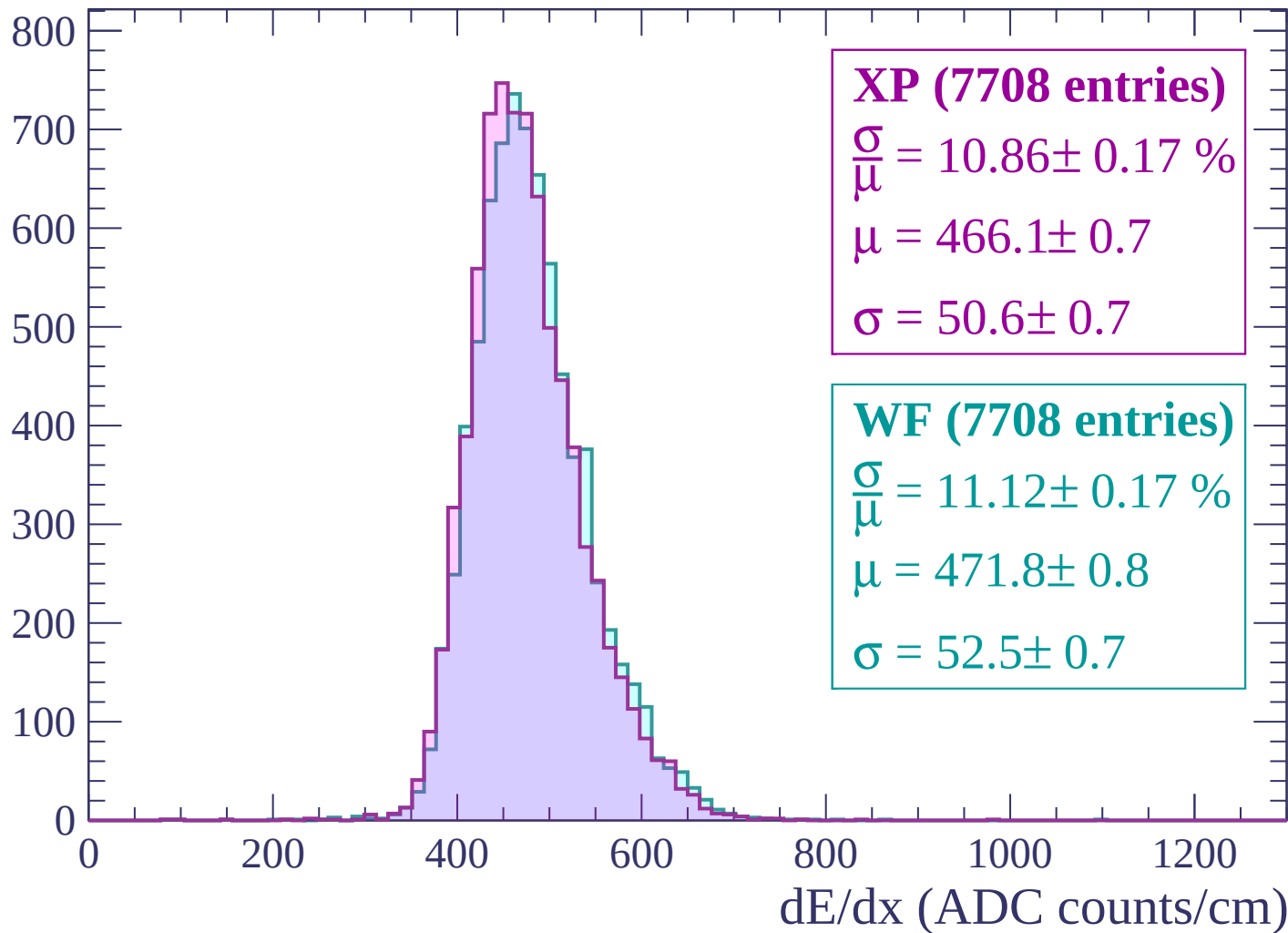
Energy loss |  $-1320 < p < -1260$

Count



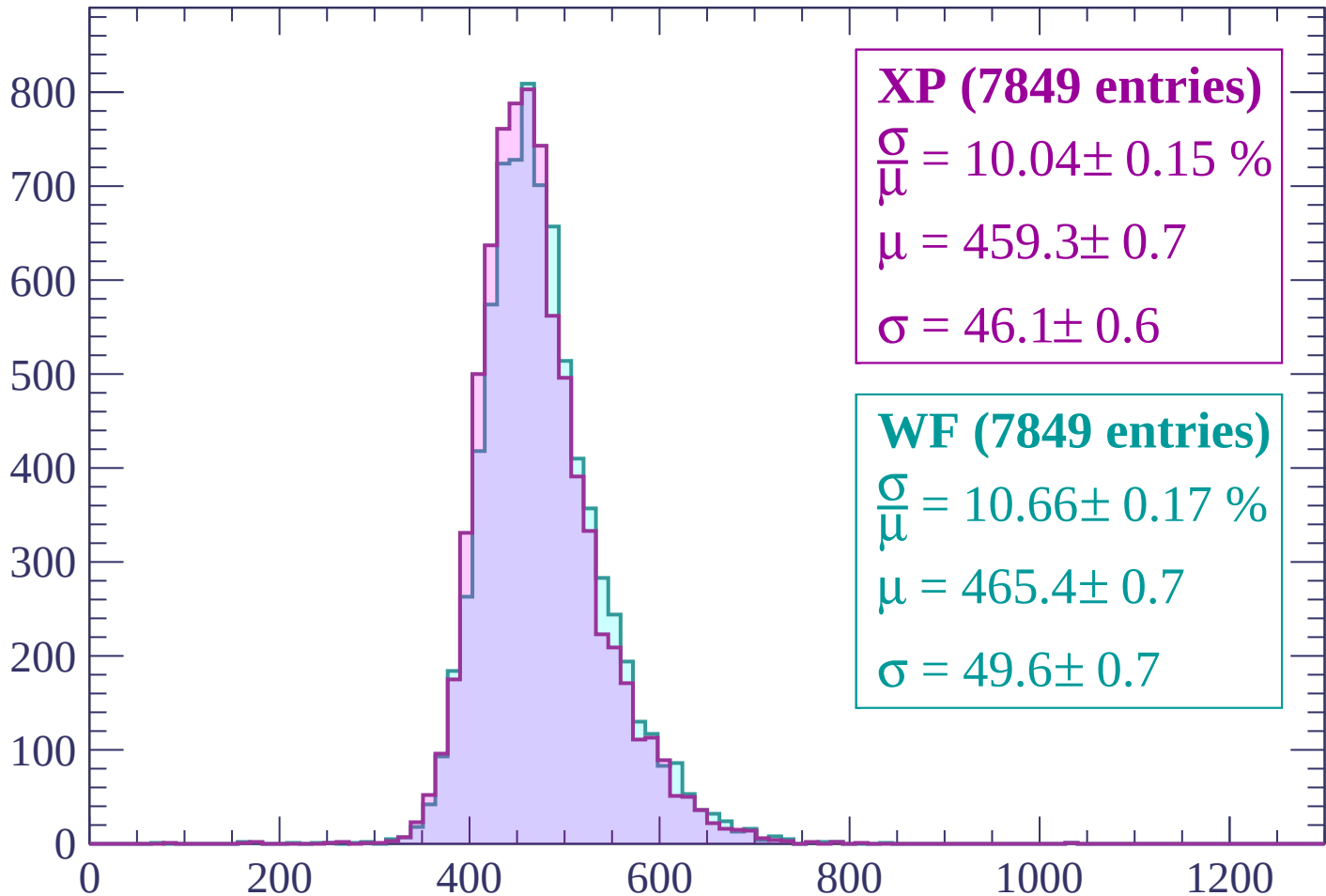
# Energy loss | $-1260 < p < -1200$

Count



# Energy loss | -1200 < p < -1140

Count



**XP (7849 entries)**

$$\frac{\sigma}{\mu} = 10.04 \pm 0.15 \%$$

$$\mu = 459.3 \pm 0.7$$

$$\sigma = 46.1 \pm 0.6$$

**WF (7849 entries)**

$$\frac{\sigma}{\mu} = 10.66 \pm 0.17 \%$$

$$\mu = 465.4 \pm 0.7$$

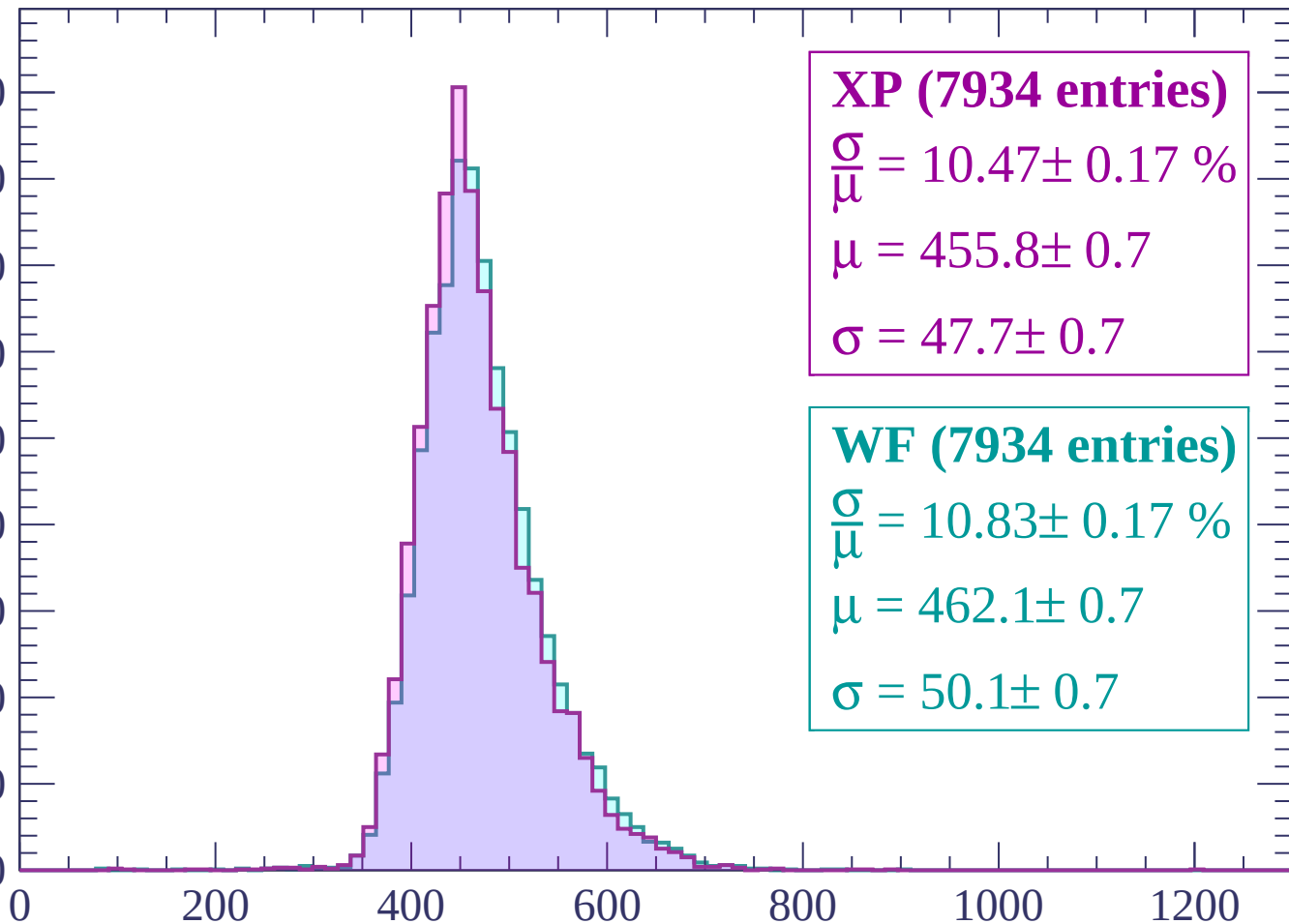
$$\sigma = 49.6 \pm 0.7$$

dE/dx (ADC counts/cm)

# Energy loss | -1140 < p < -1080

Count

900  
800  
700  
600  
500  
400  
300  
200  
100  
0



**XP (7934 entries)**

$\frac{\sigma}{\mu} = 10.47 \pm 0.17 \%$

$\mu = 455.8 \pm 0.7$

$\sigma = 47.7 \pm 0.7$

**WF (7934 entries)**

$\frac{\sigma}{\mu} = 10.83 \pm 0.17 \%$

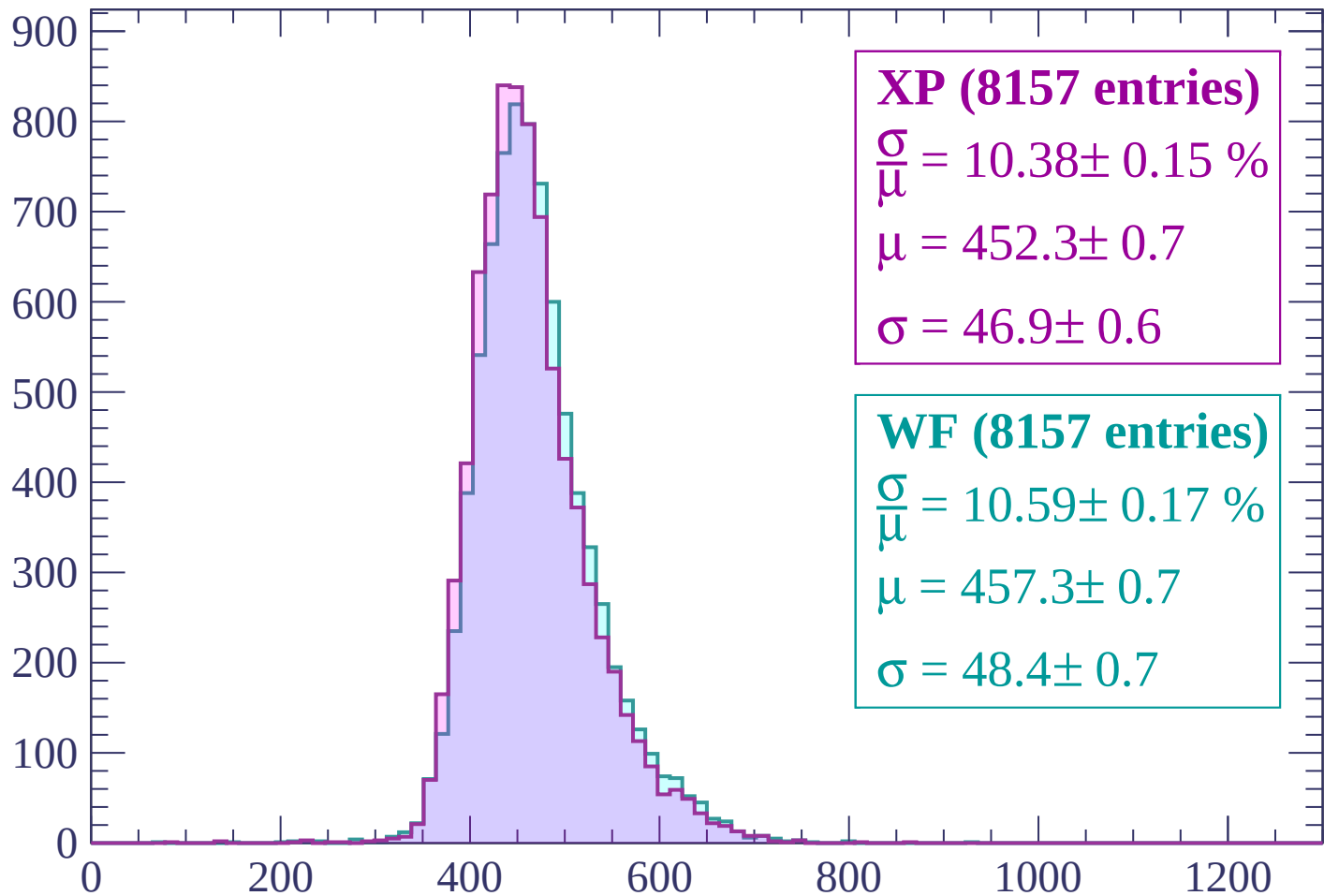
$\mu = 462.1 \pm 0.7$

$\sigma = 50.1 \pm 0.7$

dE/dx (ADC counts/cm)

# Energy loss | -1080 < p < -1020

Count



**XP (8157 entries)**

$\frac{\sigma}{\mu} = 10.38 \pm 0.15 \%$

$\mu = 452.3 \pm 0.7$

$\sigma = 46.9 \pm 0.6$

**WF (8157 entries)**

$\frac{\sigma}{\mu} = 10.59 \pm 0.17 \%$

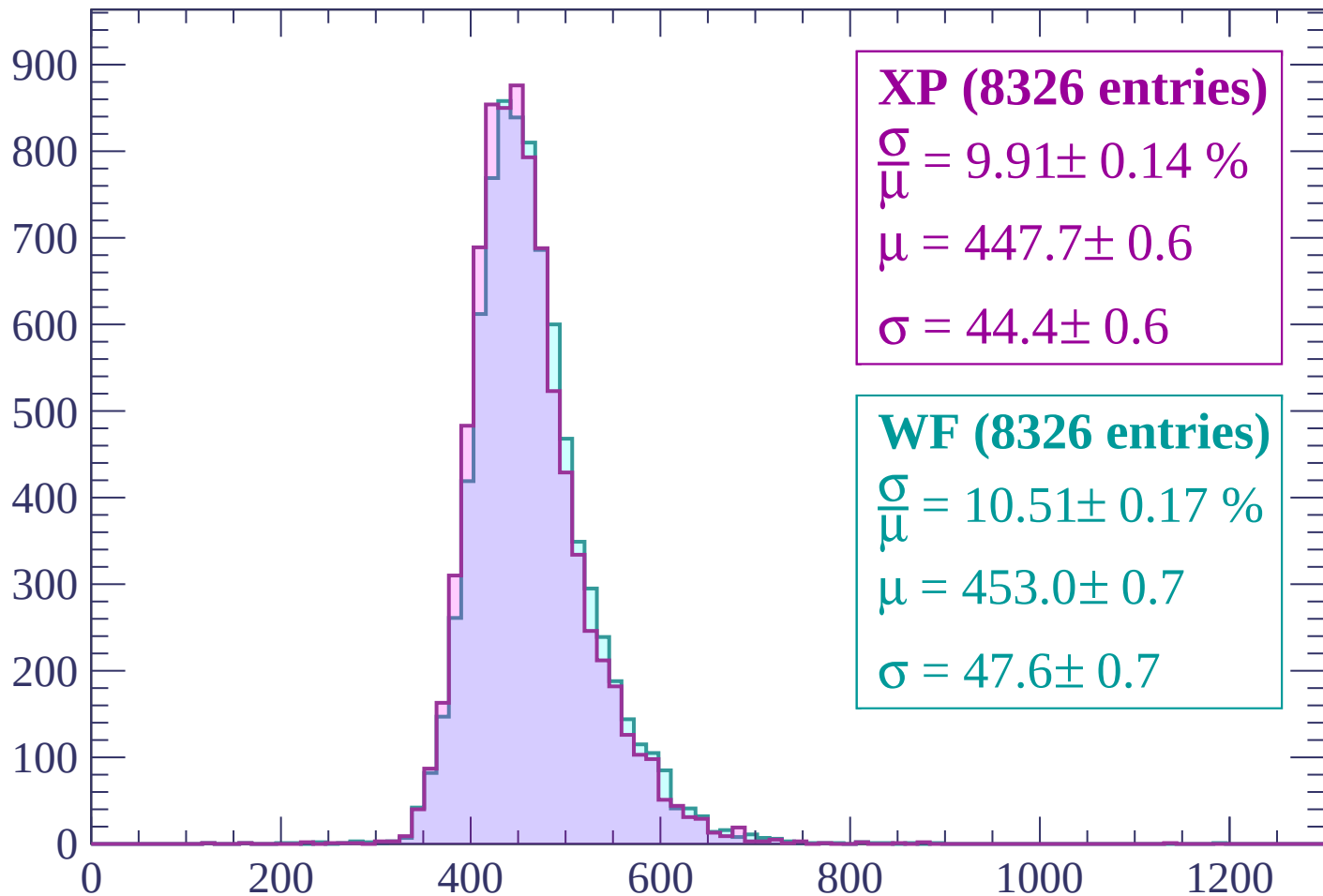
$\mu = 457.3 \pm 0.7$

$\sigma = 48.4 \pm 0.7$

dE/dx (ADC counts/cm)

Energy loss |  $-1020 < p < -960$

Count



**XP (8326 entries)**

$$\frac{\sigma}{\mu} = 9.91 \pm 0.14 \%$$

$$\mu = 447.7 \pm 0.6$$

$$\sigma = 44.4 \pm 0.6$$

**WF (8326 entries)**

$$\frac{\sigma}{\mu} = 10.51 \pm 0.17 \%$$

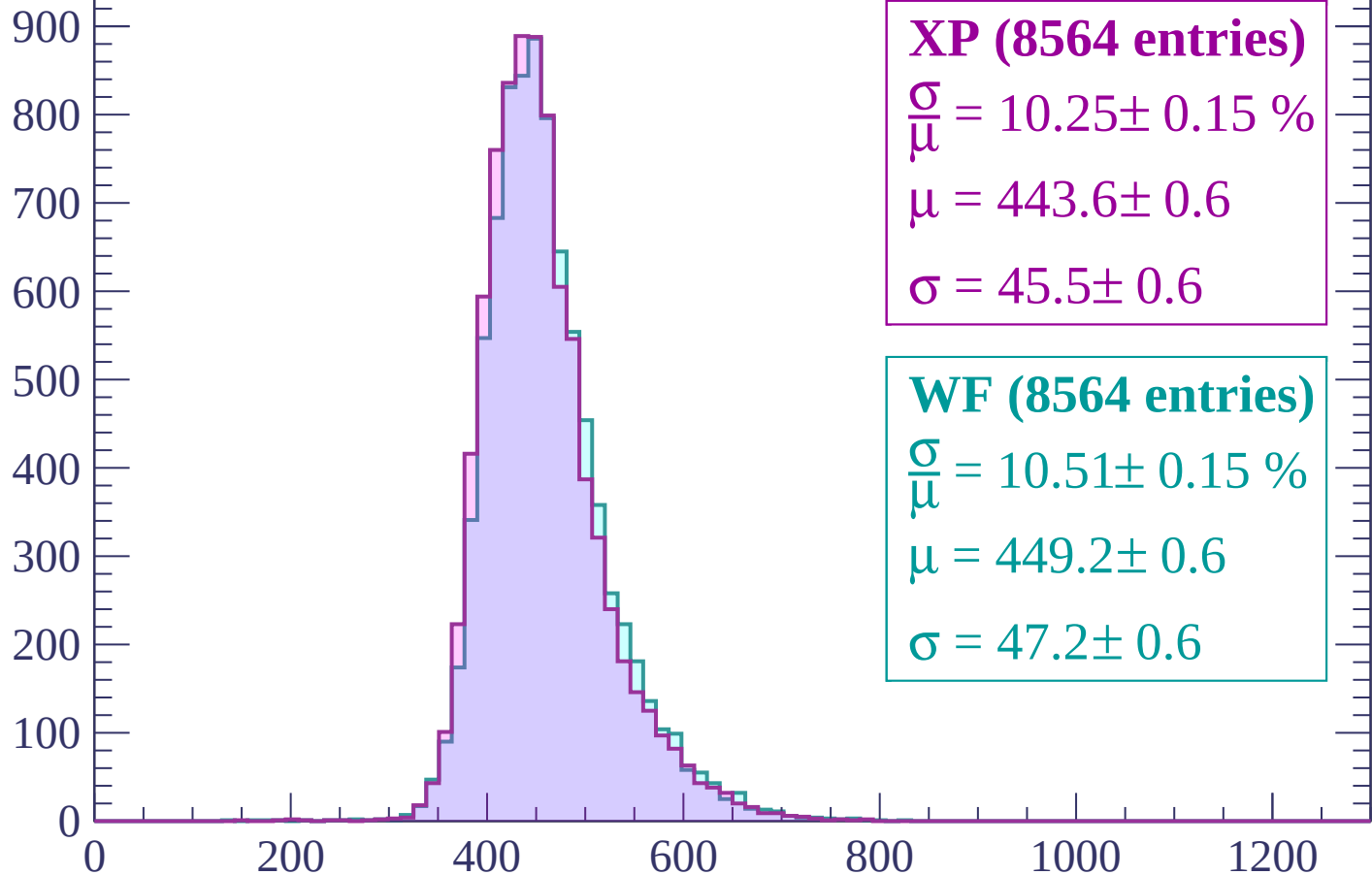
$$\mu = 453.0 \pm 0.7$$

$$\sigma = 47.6 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $-960 < p < -900$

Count



**XP (8564 entries)**

$\frac{\sigma}{\mu} = 10.25 \pm 0.15 \%$

$\mu = 443.6 \pm 0.6$

$\sigma = 45.5 \pm 0.6$

**WF (8564 entries)**

$\frac{\sigma}{\mu} = 10.51 \pm 0.15 \%$

$\mu = 449.2 \pm 0.6$

$\sigma = 47.2 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $-900 < p < -840$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP (8888 entries)**

$$\frac{\sigma}{\mu} = 9.88 \pm 0.14 \%$$

$$\mu = 438.5 \pm 0.6$$

$$\sigma = 43.3 \pm 0.6$$

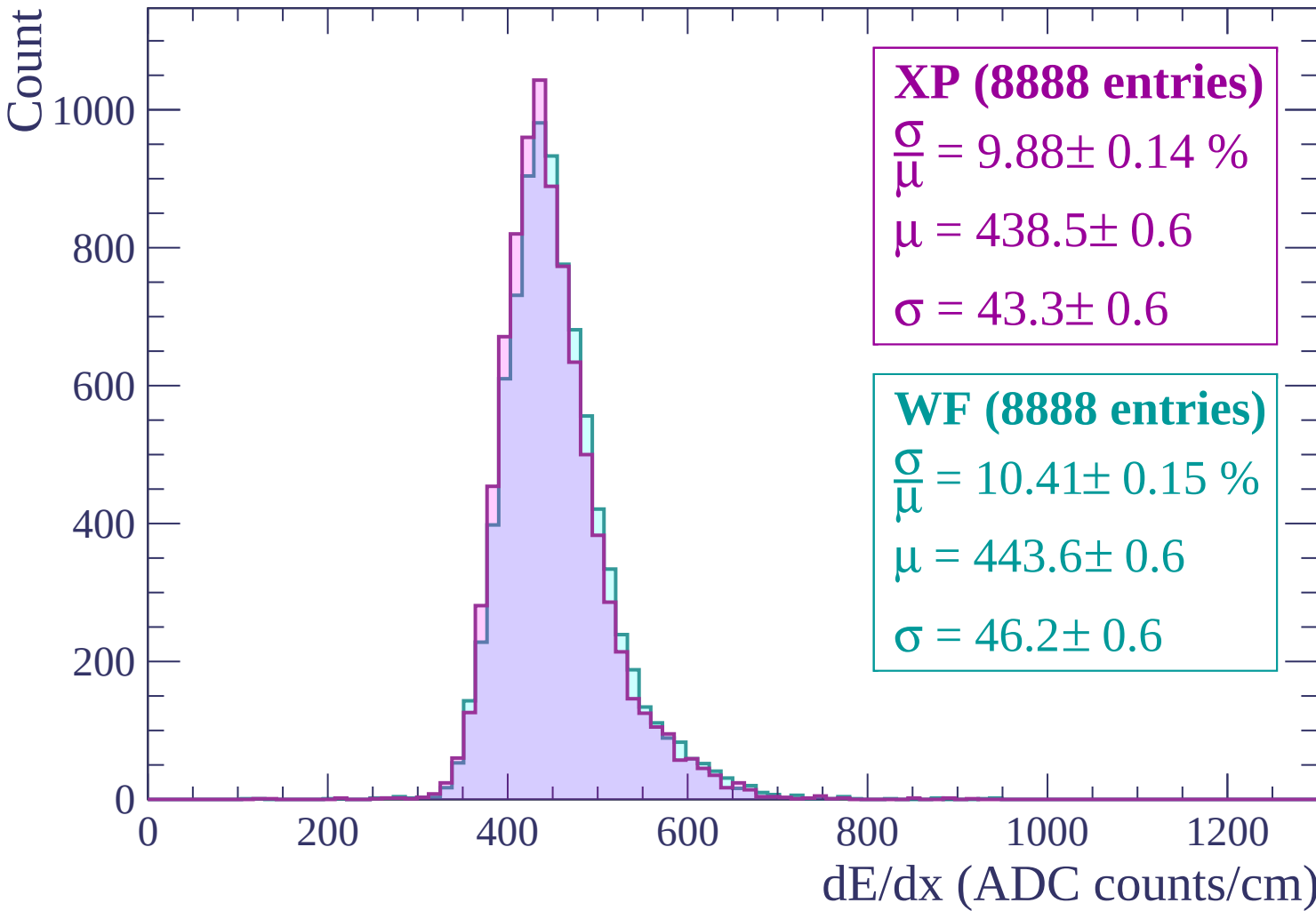
**WF (8888 entries)**

$$\frac{\sigma}{\mu} = 10.41 \pm 0.15 \%$$

$$\mu = 443.6 \pm 0.6$$

$$\sigma = 46.2 \pm 0.6$$

dE/dx (ADC counts/cm)



Energy loss |  $-840 < p < -780$

Count

1000

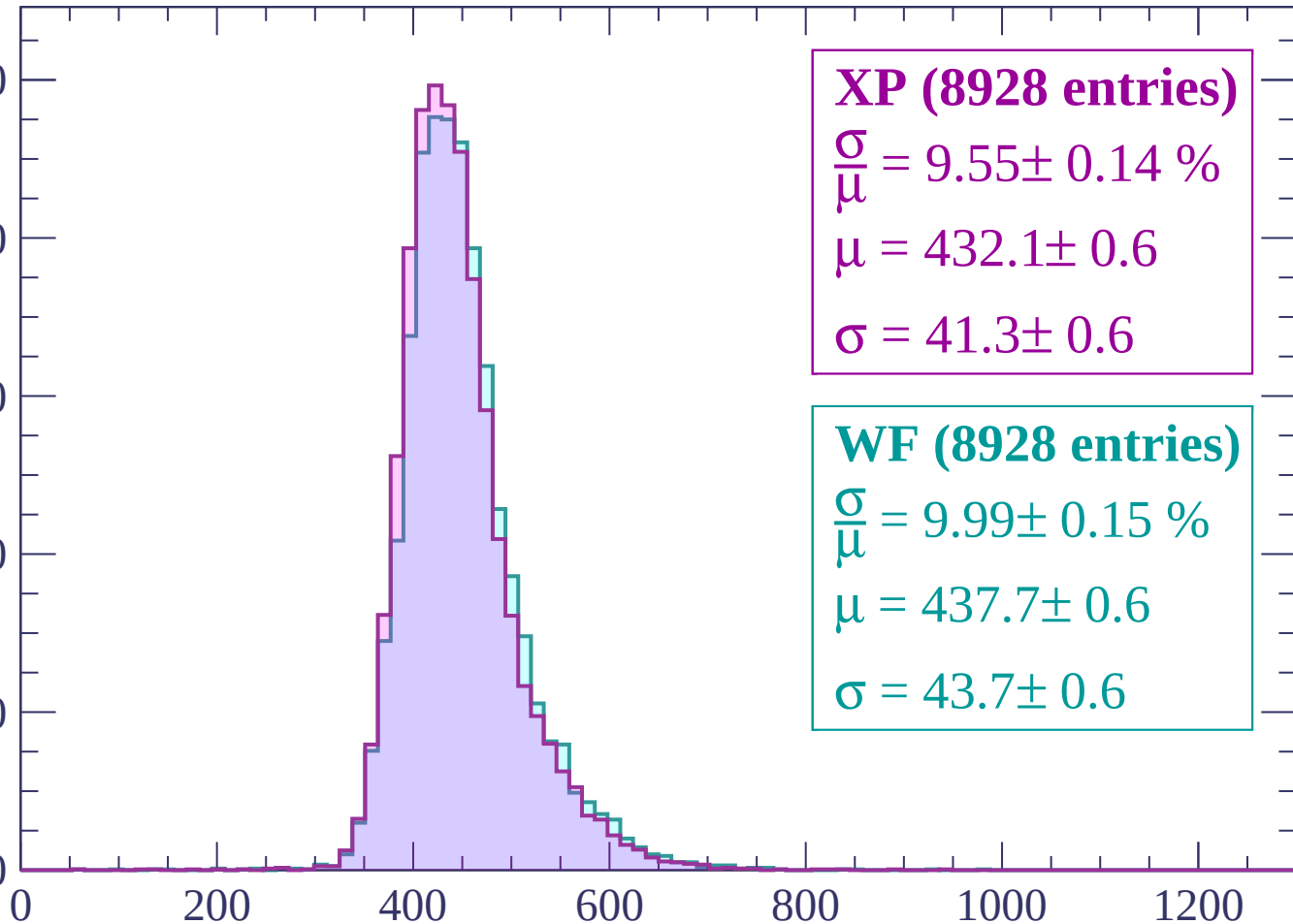
800

600

400

200

0



**XP (8928 entries)**

$\frac{\sigma}{\mu} = 9.55 \pm 0.14 \%$

$\mu = 432.1 \pm 0.6$

$\sigma = 41.3 \pm 0.6$

**WF (8928 entries)**

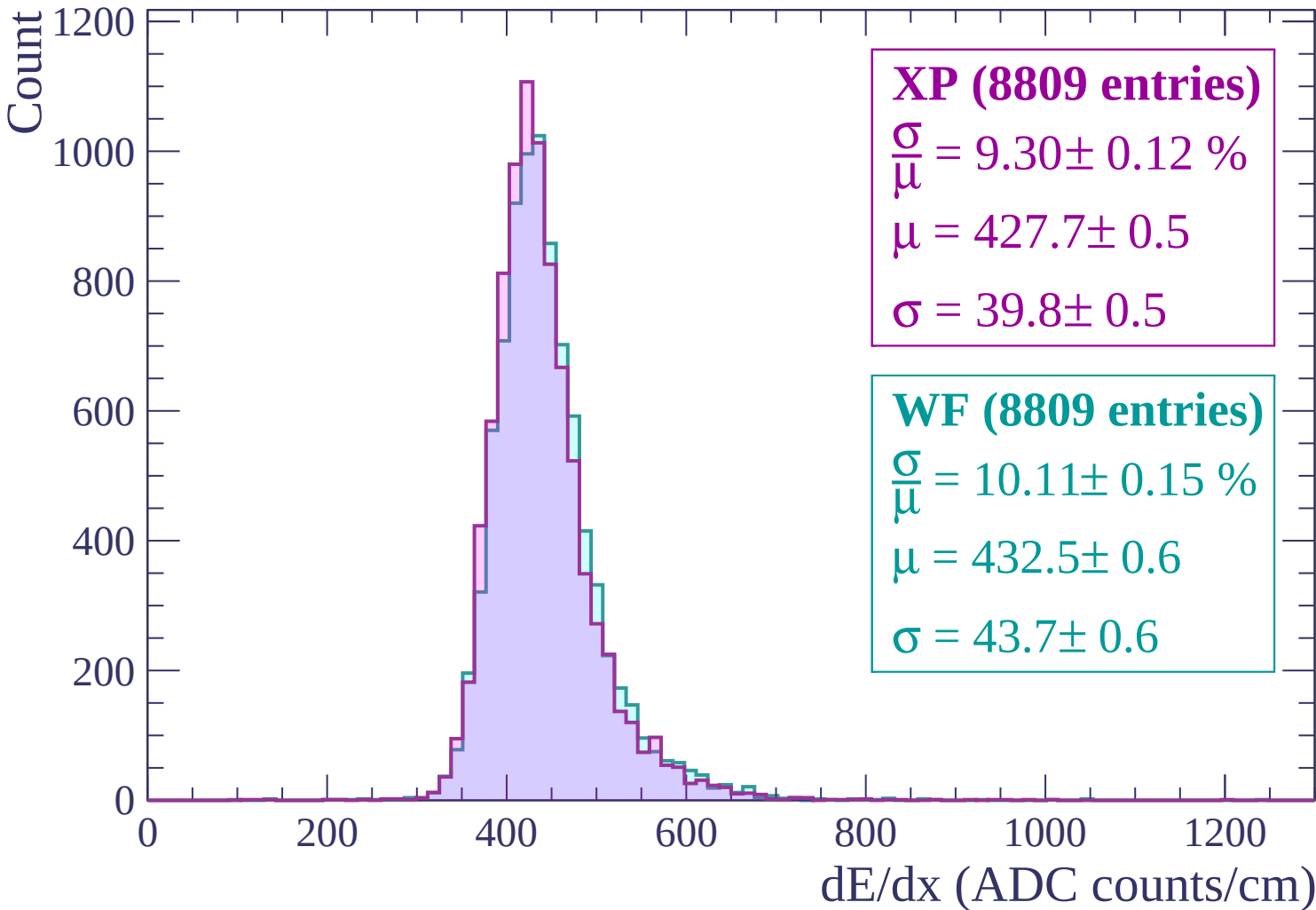
$\frac{\sigma}{\mu} = 9.99 \pm 0.15 \%$

$\mu = 437.7 \pm 0.6$

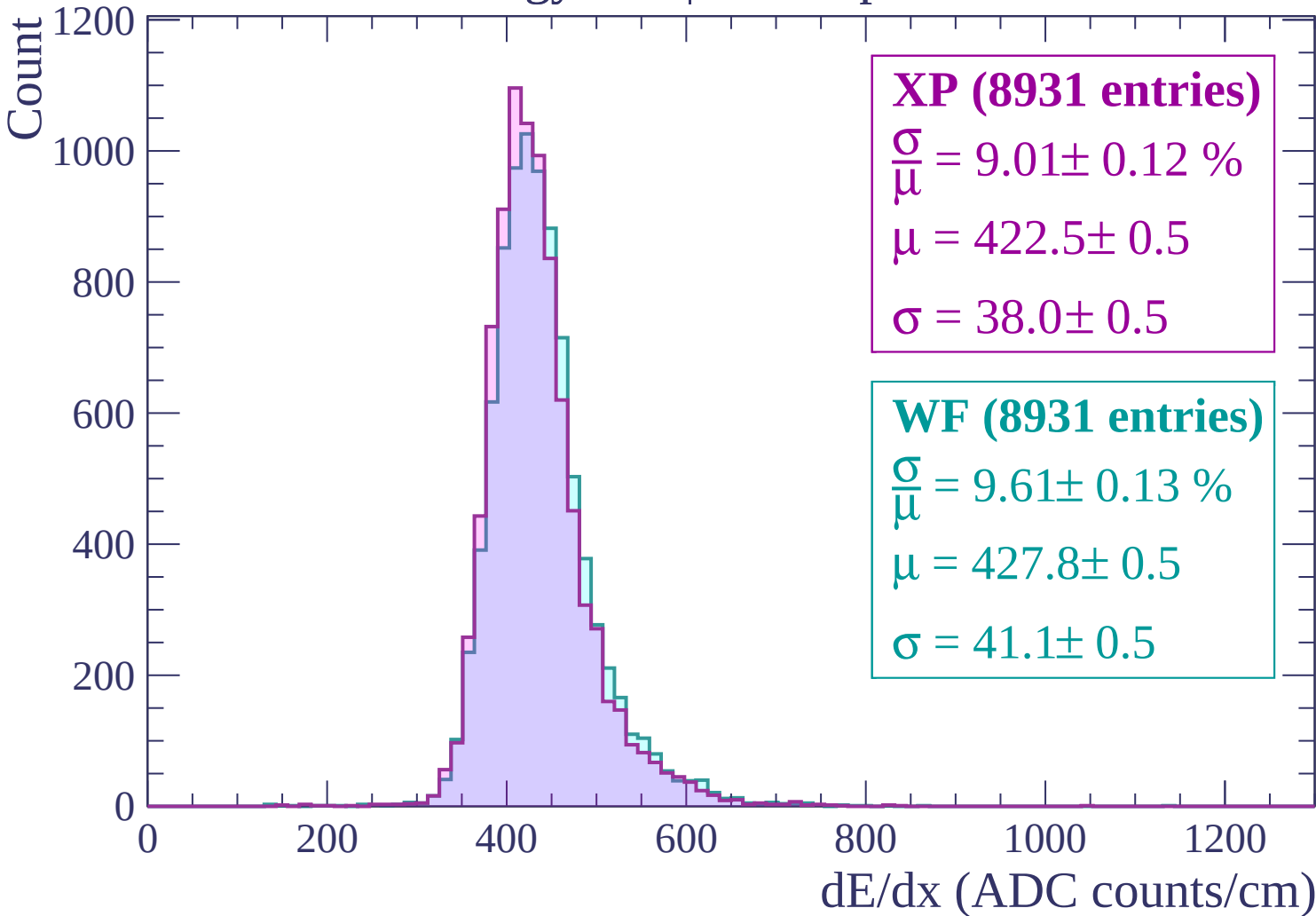
$\sigma = 43.7 \pm 0.6$

dE/dx (ADC counts/cm)

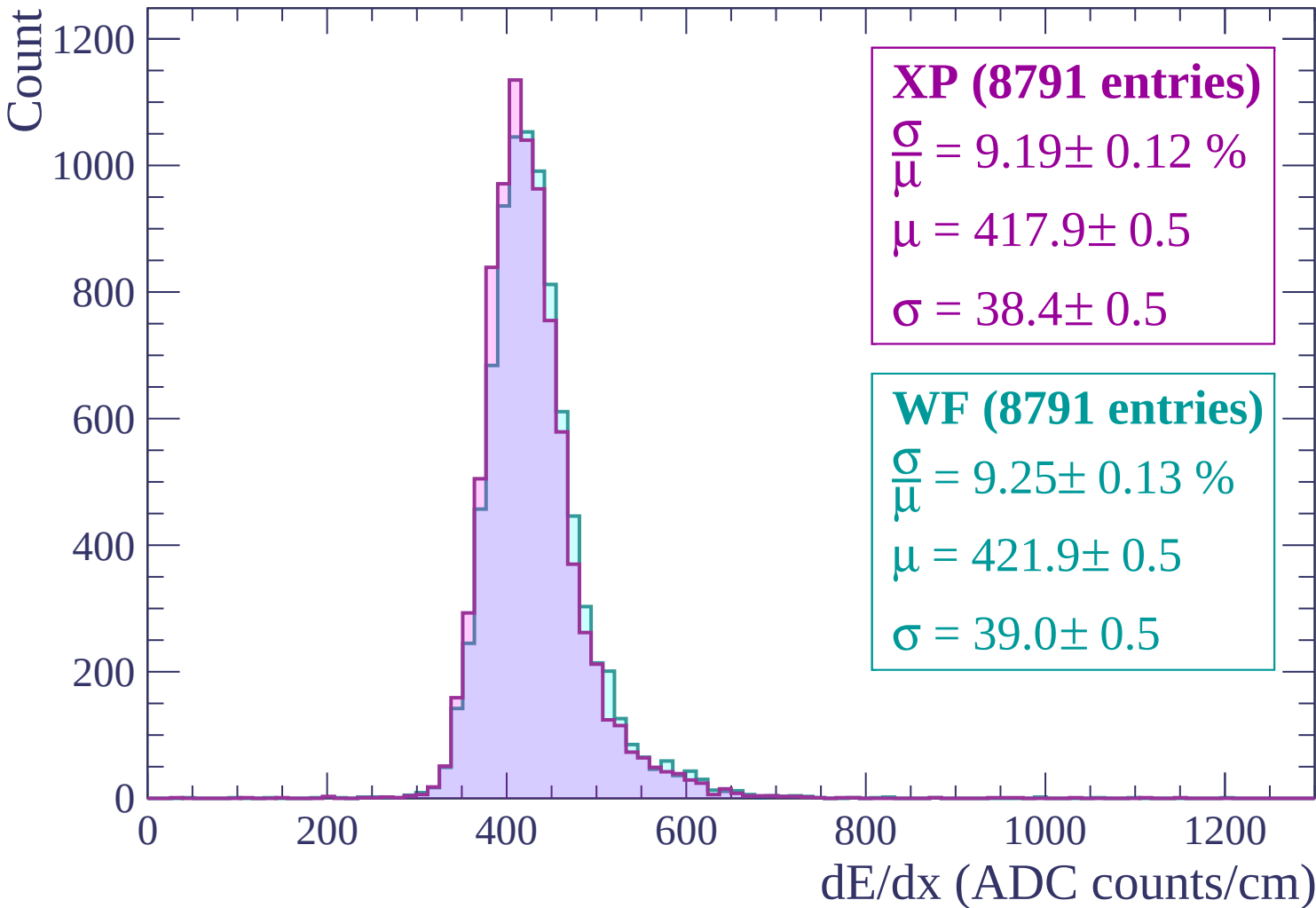
Energy loss |  $-780 < p < -720$



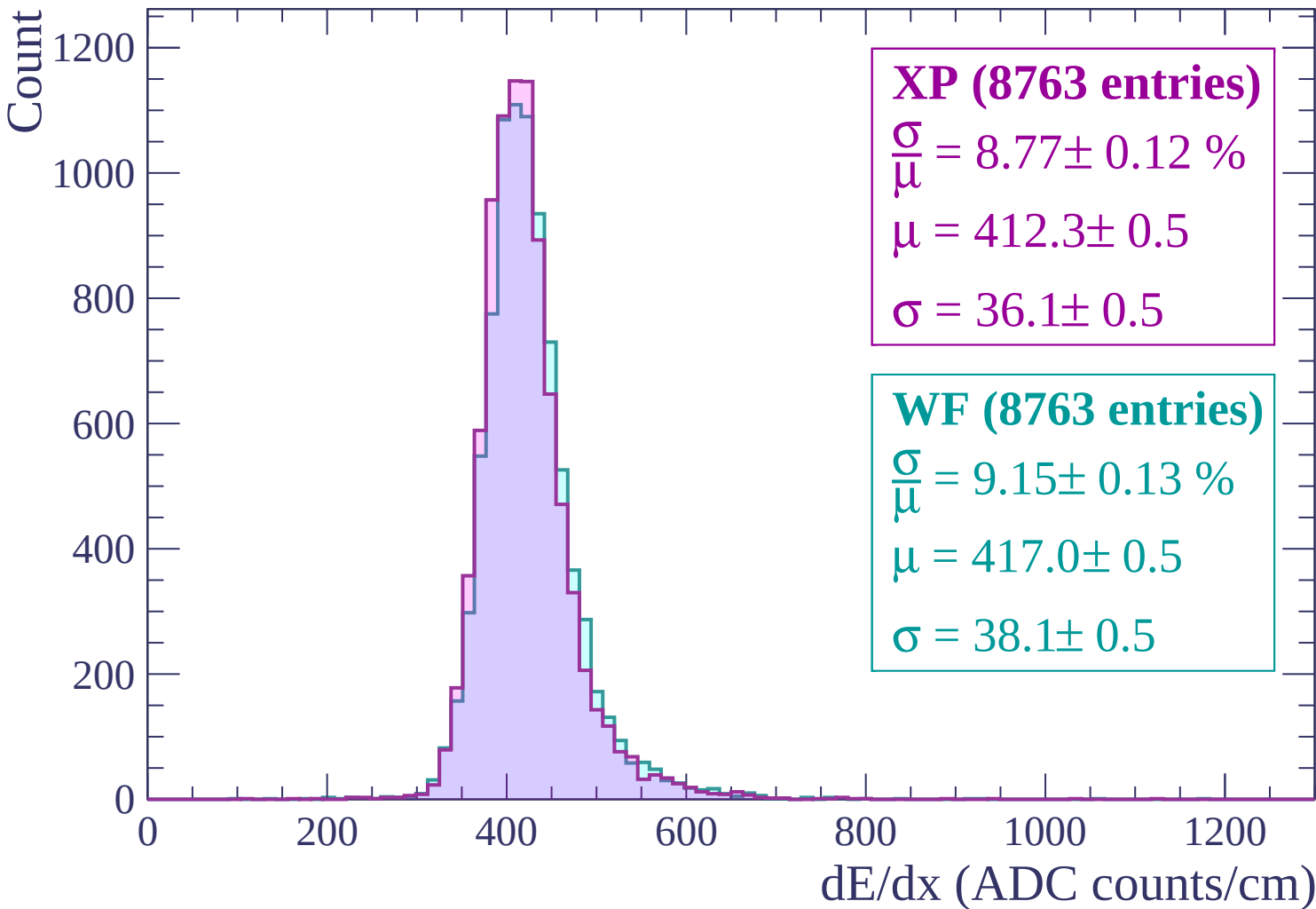
Energy loss |  $-720 < p < -660$



Energy loss |  $-660 < p < -600$



Energy loss |  $-600 < p < -540$



Energy loss |  $-540 < p < -480$

Count

1200

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

**XP (8819 entries)**

$$\frac{\sigma}{\mu} = 8.84 \pm 0.12 \%$$

$$\mu = 407.2 \pm 0.5$$

$$\sigma = 36.0 \pm 0.5$$

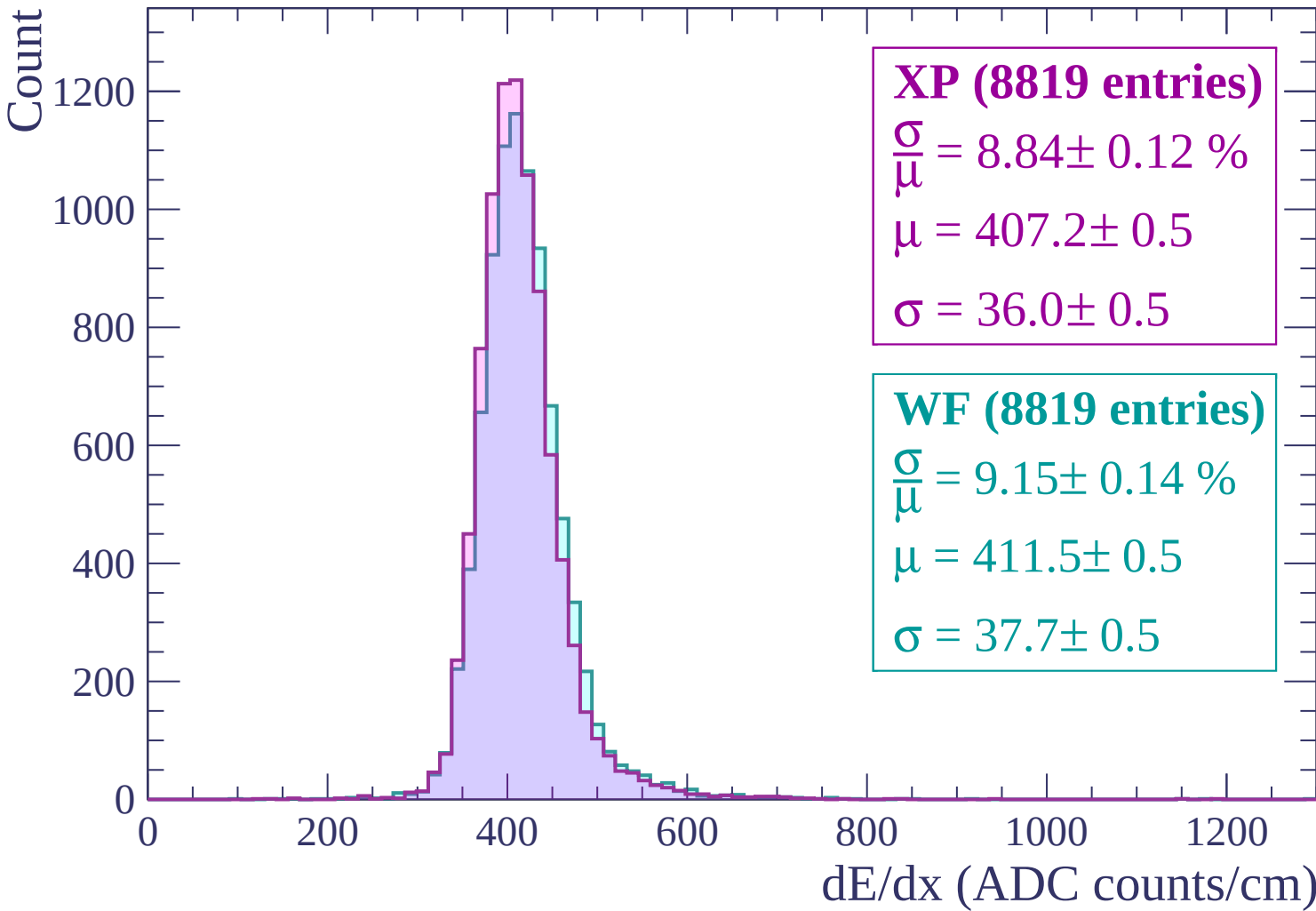
**WF (8819 entries)**

$$\frac{\sigma}{\mu} = 9.15 \pm 0.14 \%$$

$$\mu = 411.5 \pm 0.5$$

$$\sigma = 37.7 \pm 0.5$$

dE/dx (ADC counts/cm)



Energy loss |  $-480 < p < -420$

Count

1200  
1000  
800  
600  
400  
200  
0

**XP (8550 entries)**

$$\frac{\sigma}{\mu} = 8.49 \pm 0.13 \%$$

$$\mu = 403.3 \pm 0.5$$

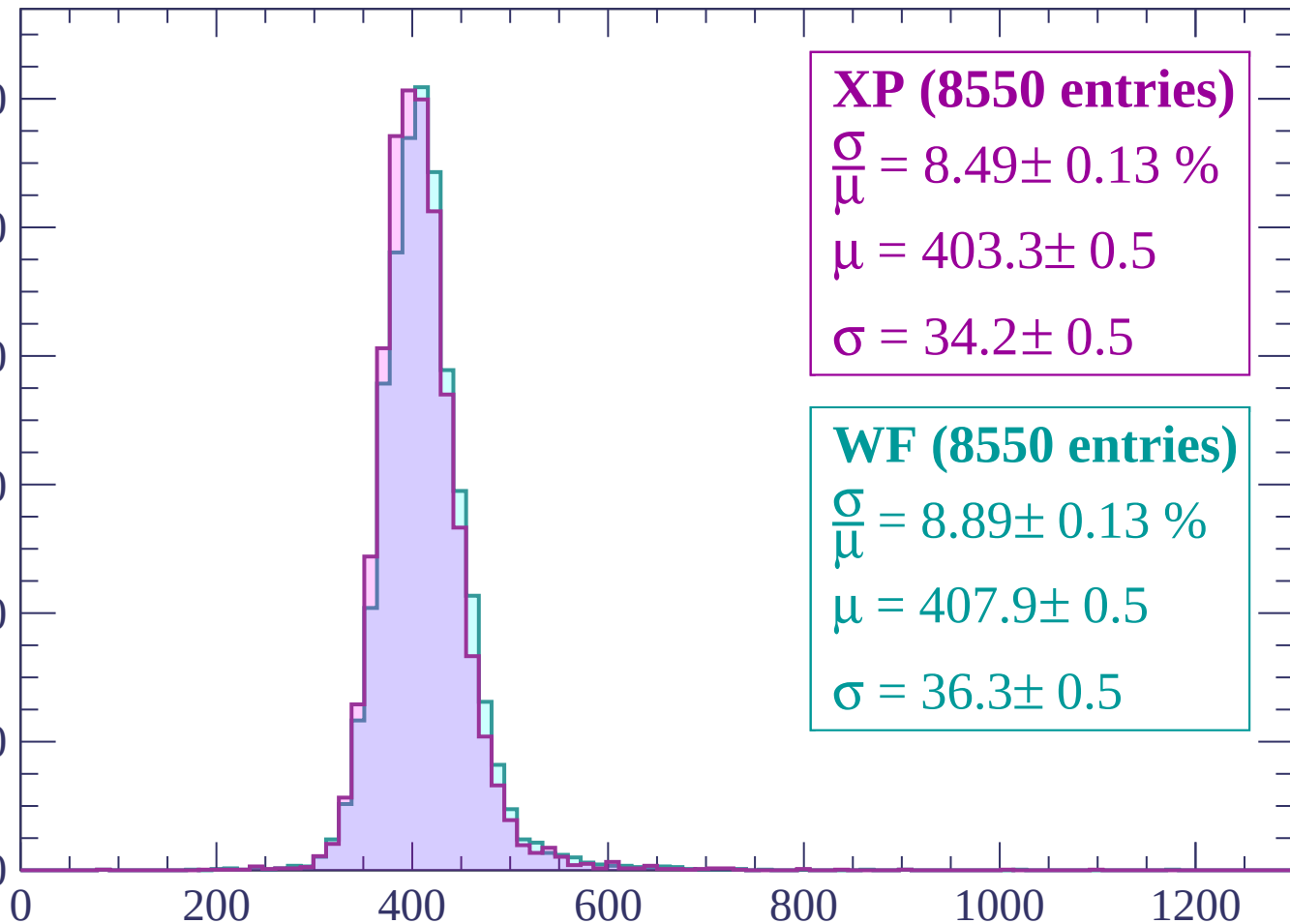
$$\sigma = 34.2 \pm 0.5$$

**WF (8550 entries)**

$$\frac{\sigma}{\mu} = 8.89 \pm 0.13 \%$$

$$\mu = 407.9 \pm 0.5$$

$$\sigma = 36.3 \pm 0.5$$



dE/dx (ADC counts/cm)

Energy loss |  $-420 < p < -360$

Count

1200  
1000  
800  
600  
400  
200  
0

**XP (8270 entries)**

$\frac{\sigma}{\mu} = 8.56 \pm 0.11 \%$

$\mu = 399.8 \pm 0.4$

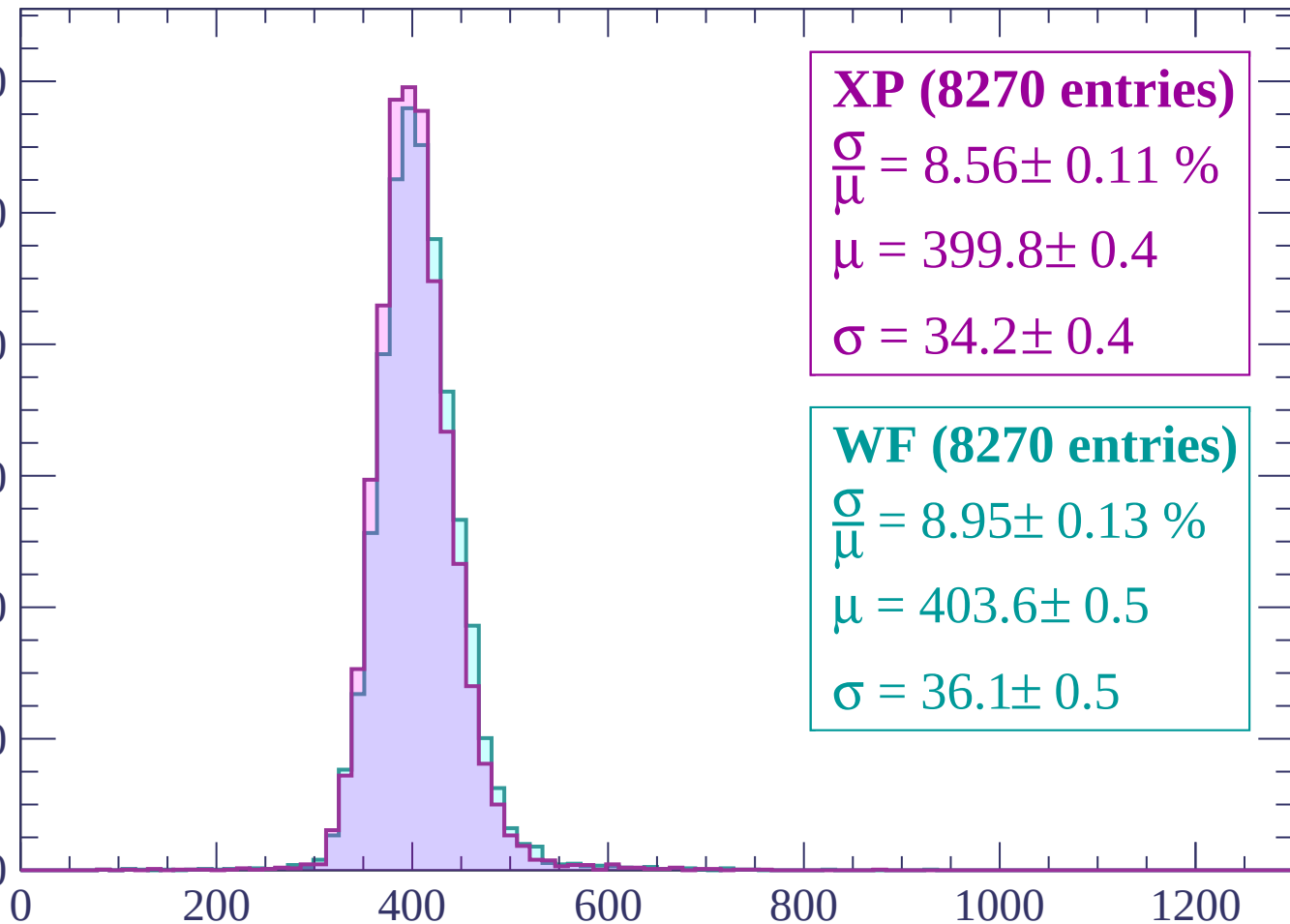
$\sigma = 34.2 \pm 0.4$

**WF (8270 entries)**

$\frac{\sigma}{\mu} = 8.95 \pm 0.13 \%$

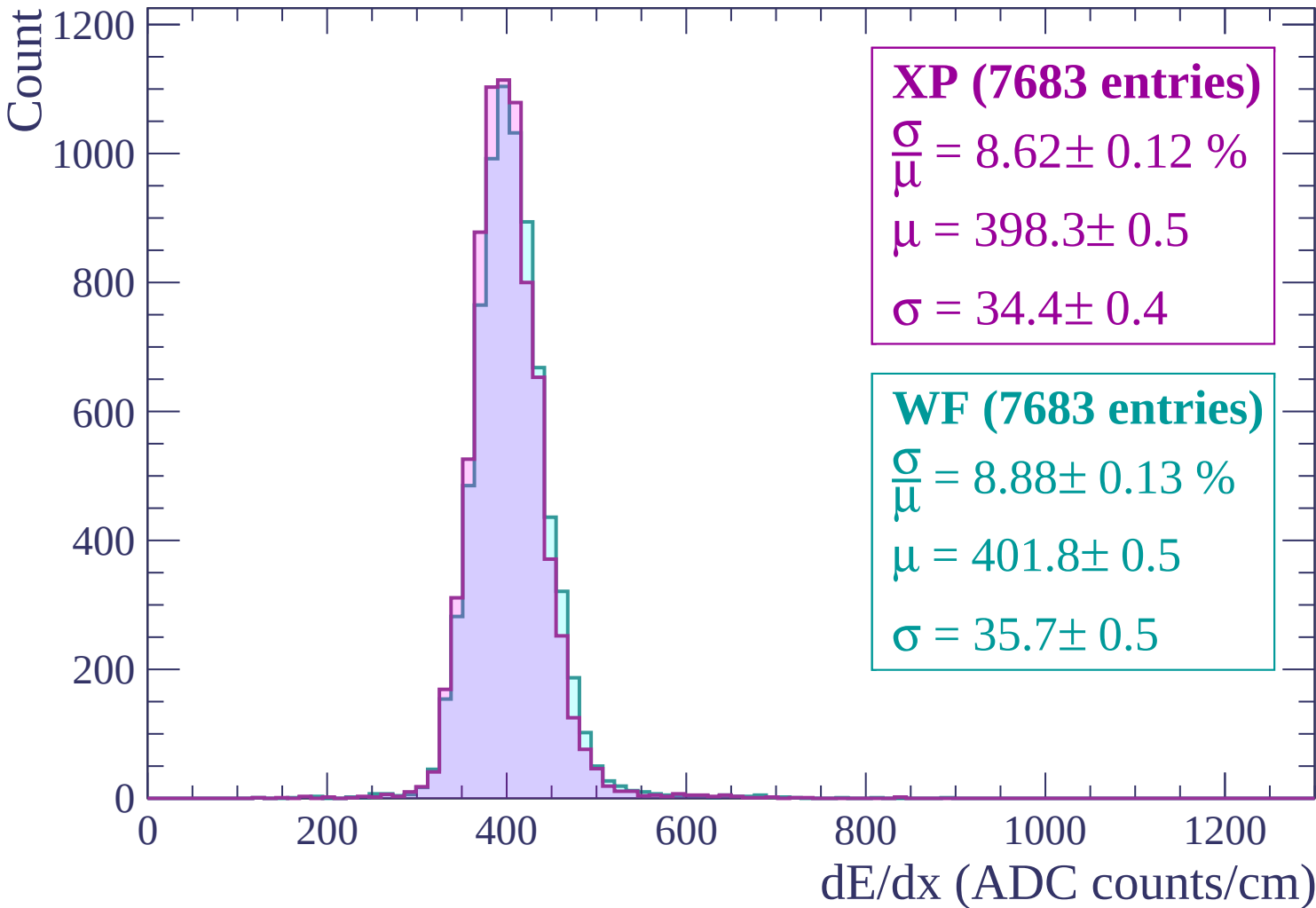
$\mu = 403.6 \pm 0.5$

$\sigma = 36.1 \pm 0.5$



dE/dx (ADC counts/cm)

Energy loss |  $-360 < p < -300$



Energy loss |  $-300 < p < -240$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (6796 entries)**

$\frac{\sigma}{\mu} = 8.41 \pm 0.12 \%$

$\mu = 399.8 \pm 0.5$

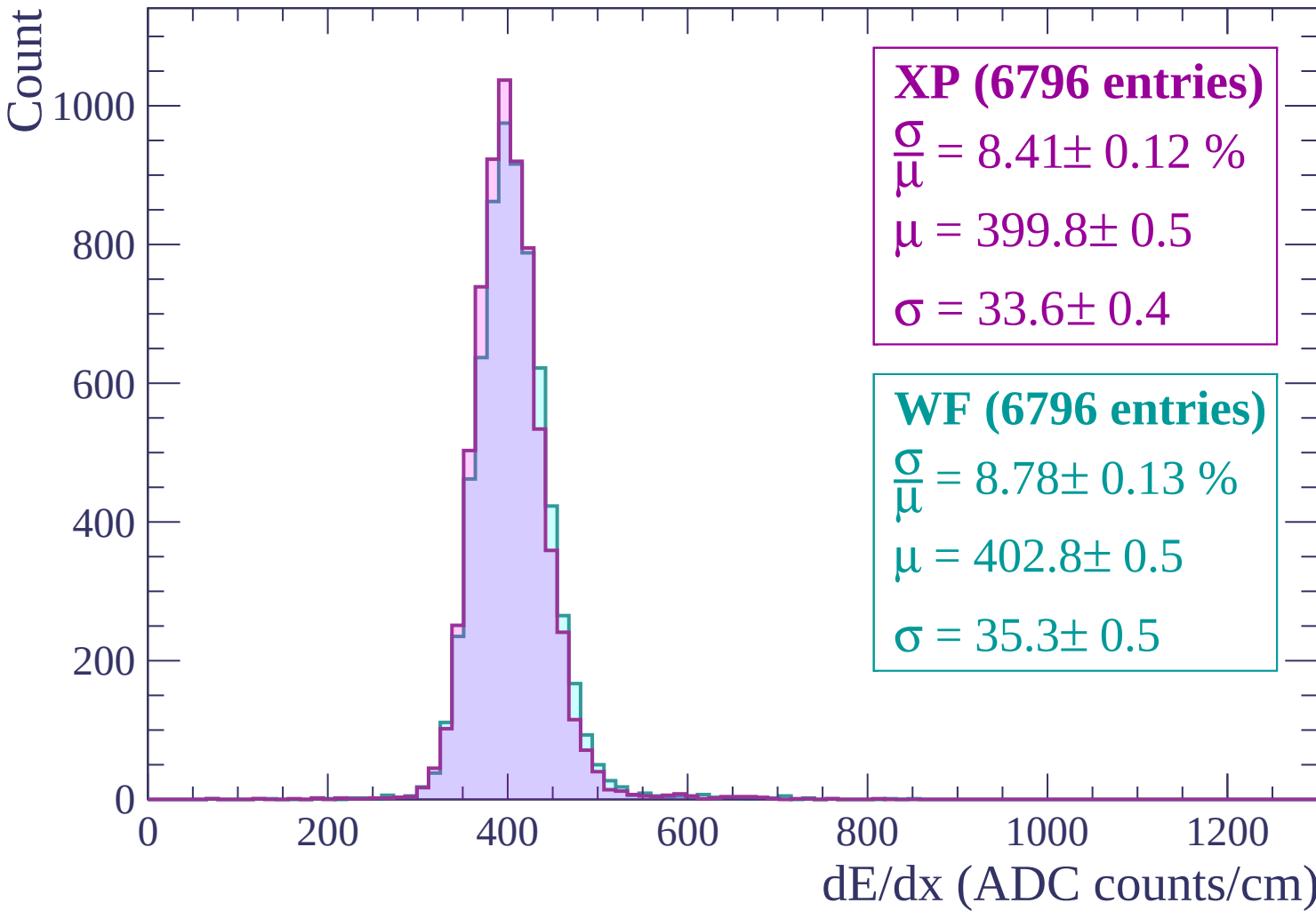
$\sigma = 33.6 \pm 0.4$

**WF (6796 entries)**

$\frac{\sigma}{\mu} = 8.78 \pm 0.13 \%$

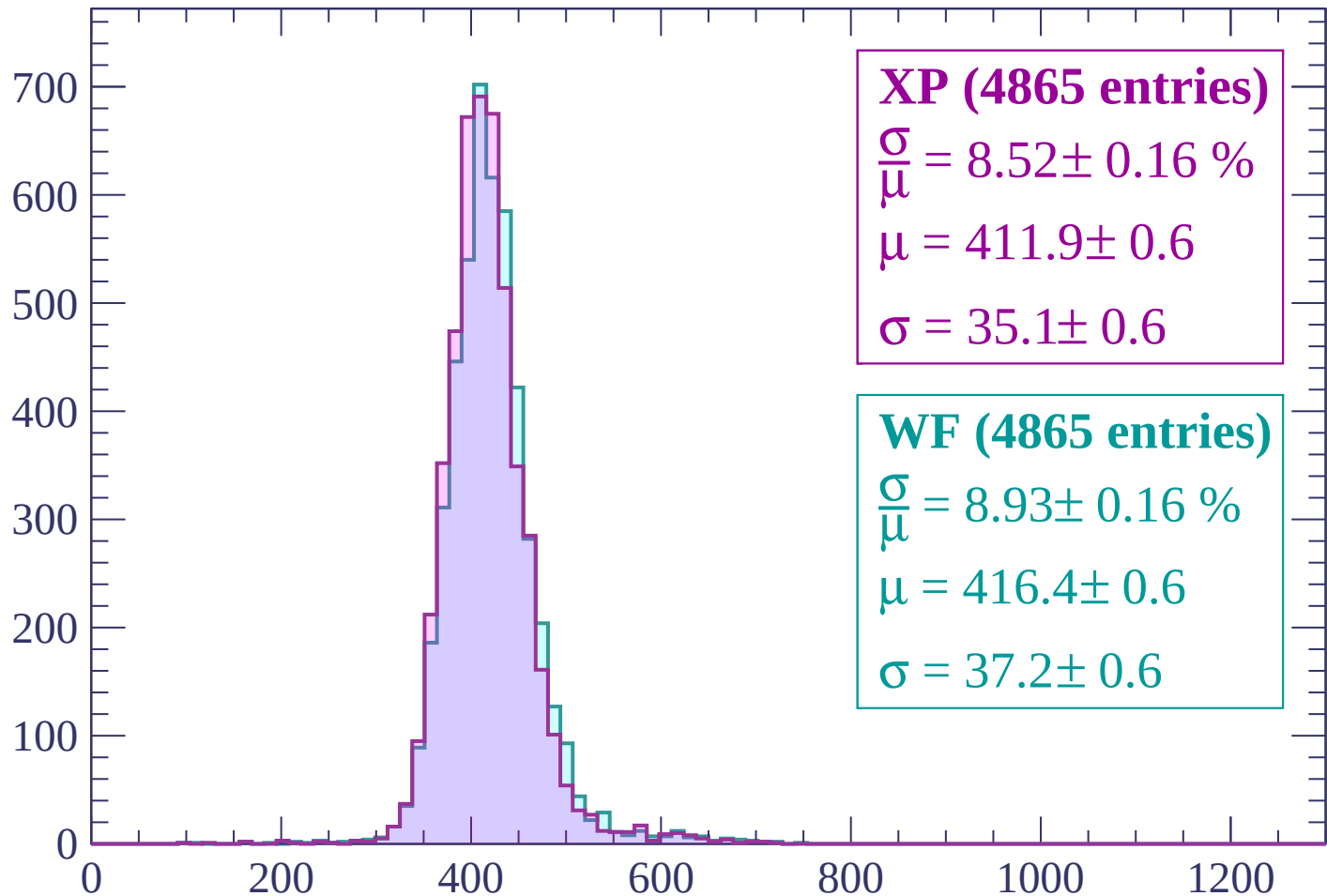
$\mu = 402.8 \pm 0.5$

$\sigma = 35.3 \pm 0.5$



Energy loss |  $-240 < p < -180$

Count



**XP (4865 entries)**

$$\frac{\sigma}{\mu} = 8.52 \pm 0.16 \%$$

$$\mu = 411.9 \pm 0.6$$

$$\sigma = 35.1 \pm 0.6$$

**WF (4865 entries)**

$$\frac{\sigma}{\mu} = 8.93 \pm 0.16 \%$$

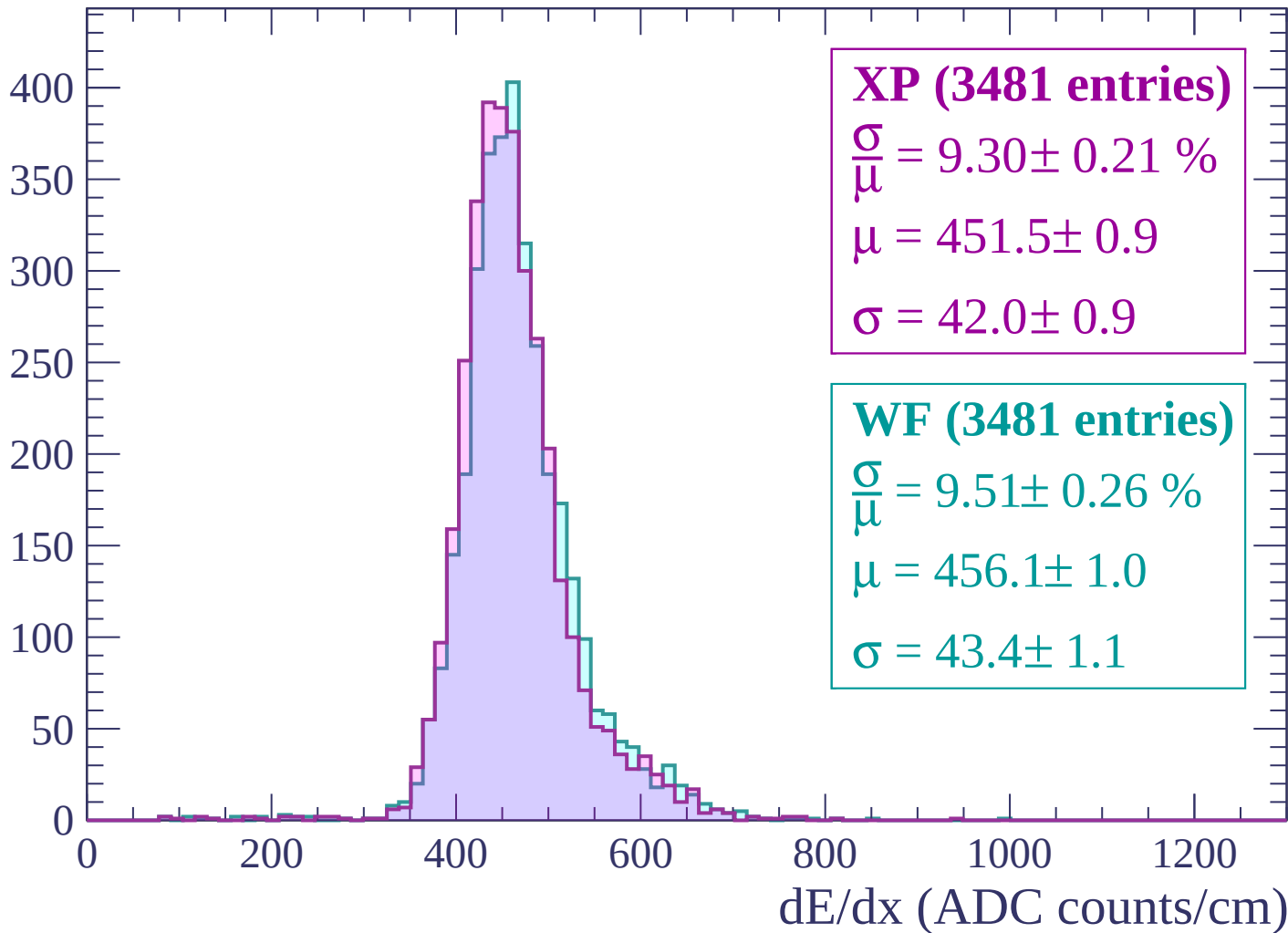
$$\mu = 416.4 \pm 0.6$$

$$\sigma = 37.2 \pm 0.6$$

dE/dx (ADC counts/cm)

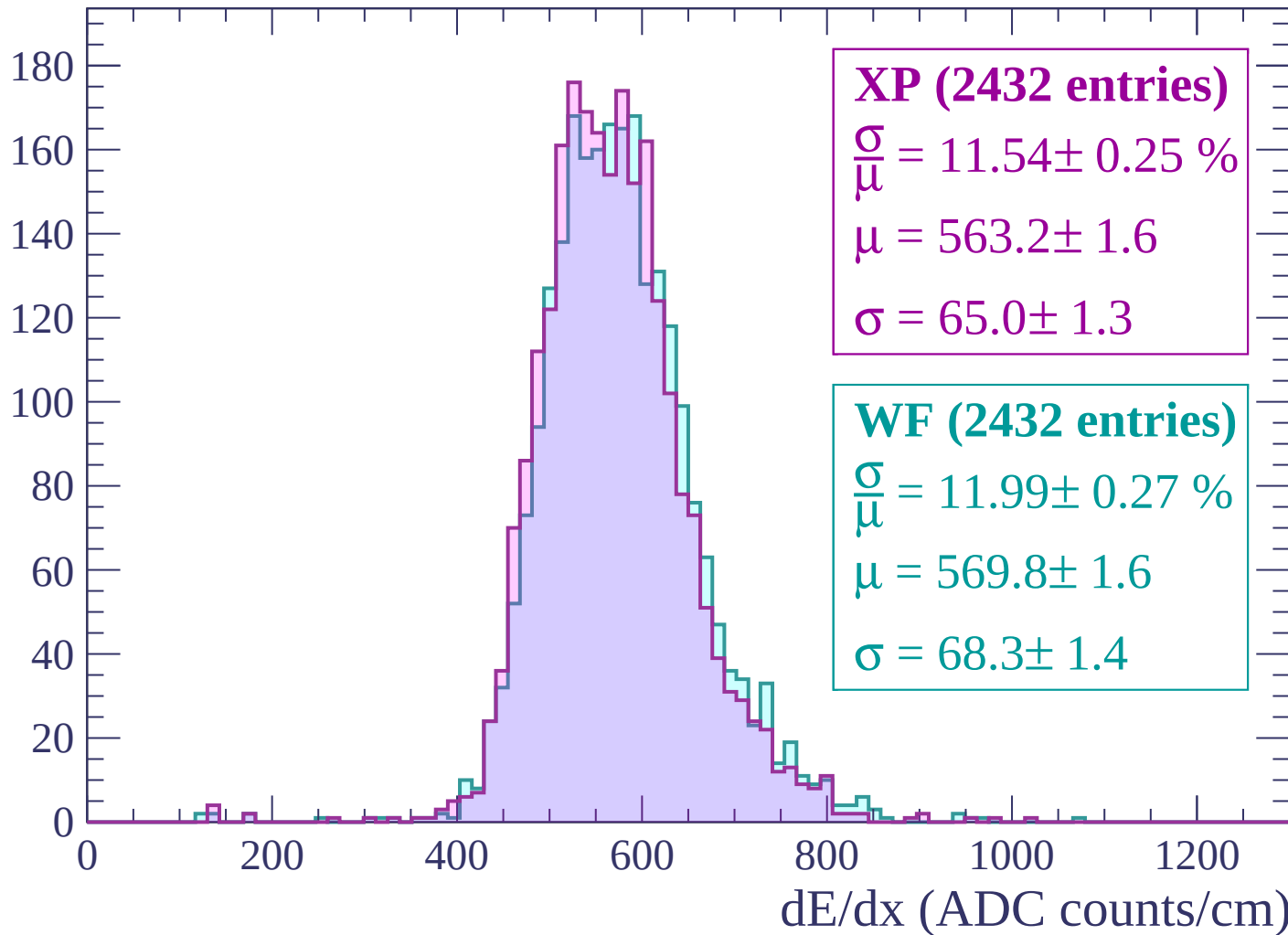
Energy loss |  $-180 < p < -120$

Count



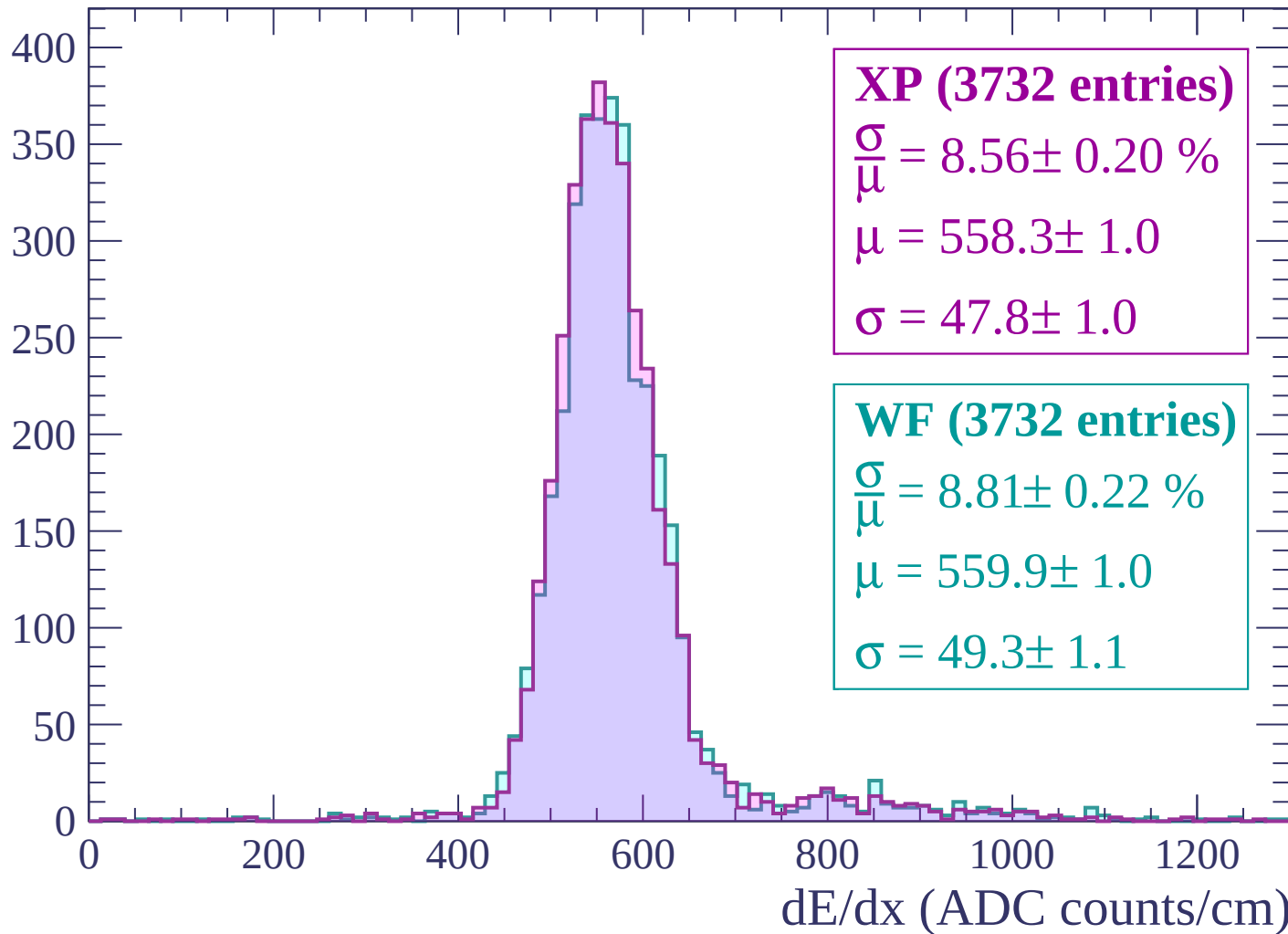
Energy loss |  $-120 < p < -60$

Count



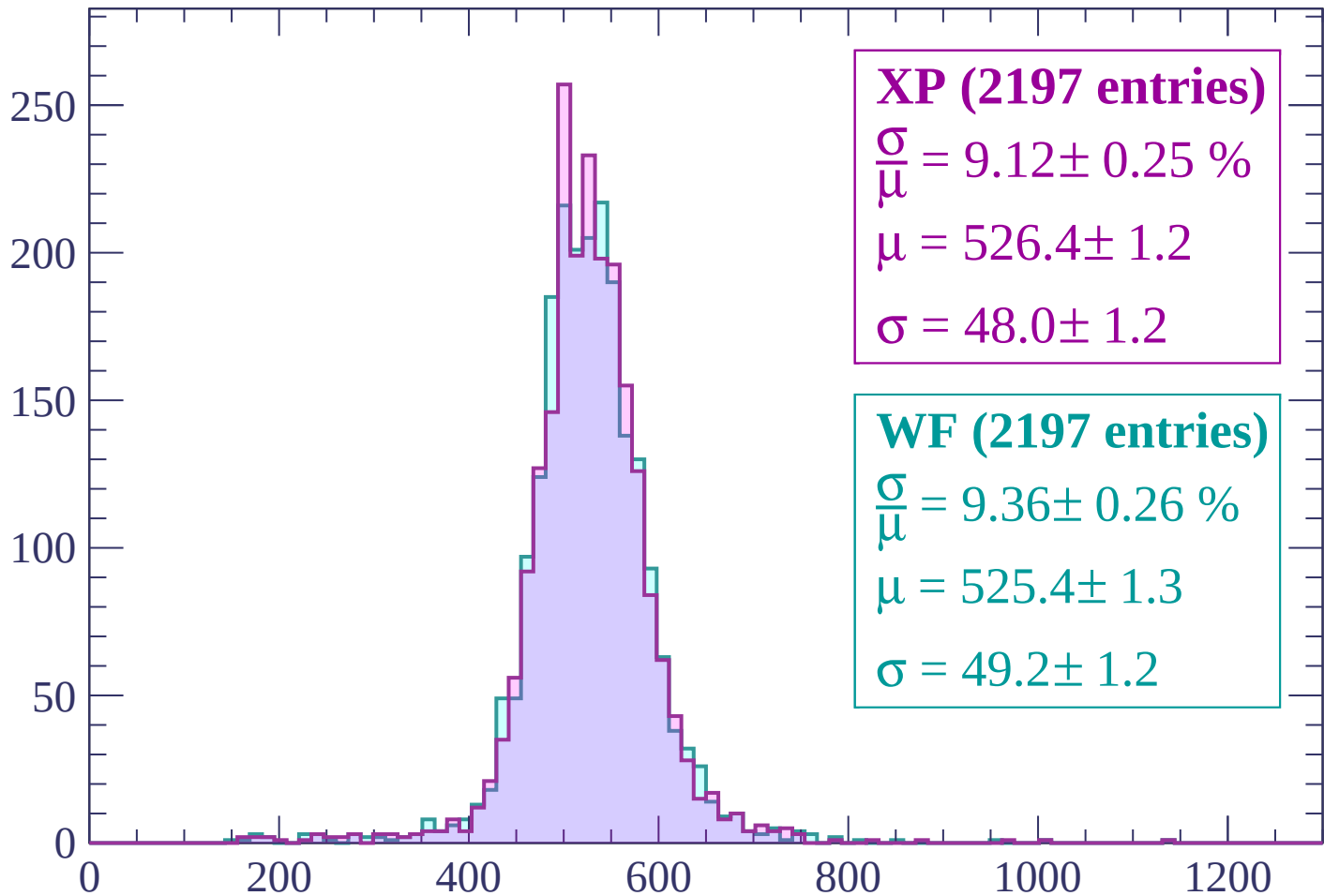
Energy loss |  $-60 < p < 0$

Count



Energy loss |  $0 < p < 60$

Count



**XP (2197 entries)**

$$\frac{\sigma}{\mu} = 9.12 \pm 0.25 \%$$

$$\mu = 526.4 \pm 1.2$$

$$\sigma = 48.0 \pm 1.2$$

**WF (2197 entries)**

$$\frac{\sigma}{\mu} = 9.36 \pm 0.26 \%$$

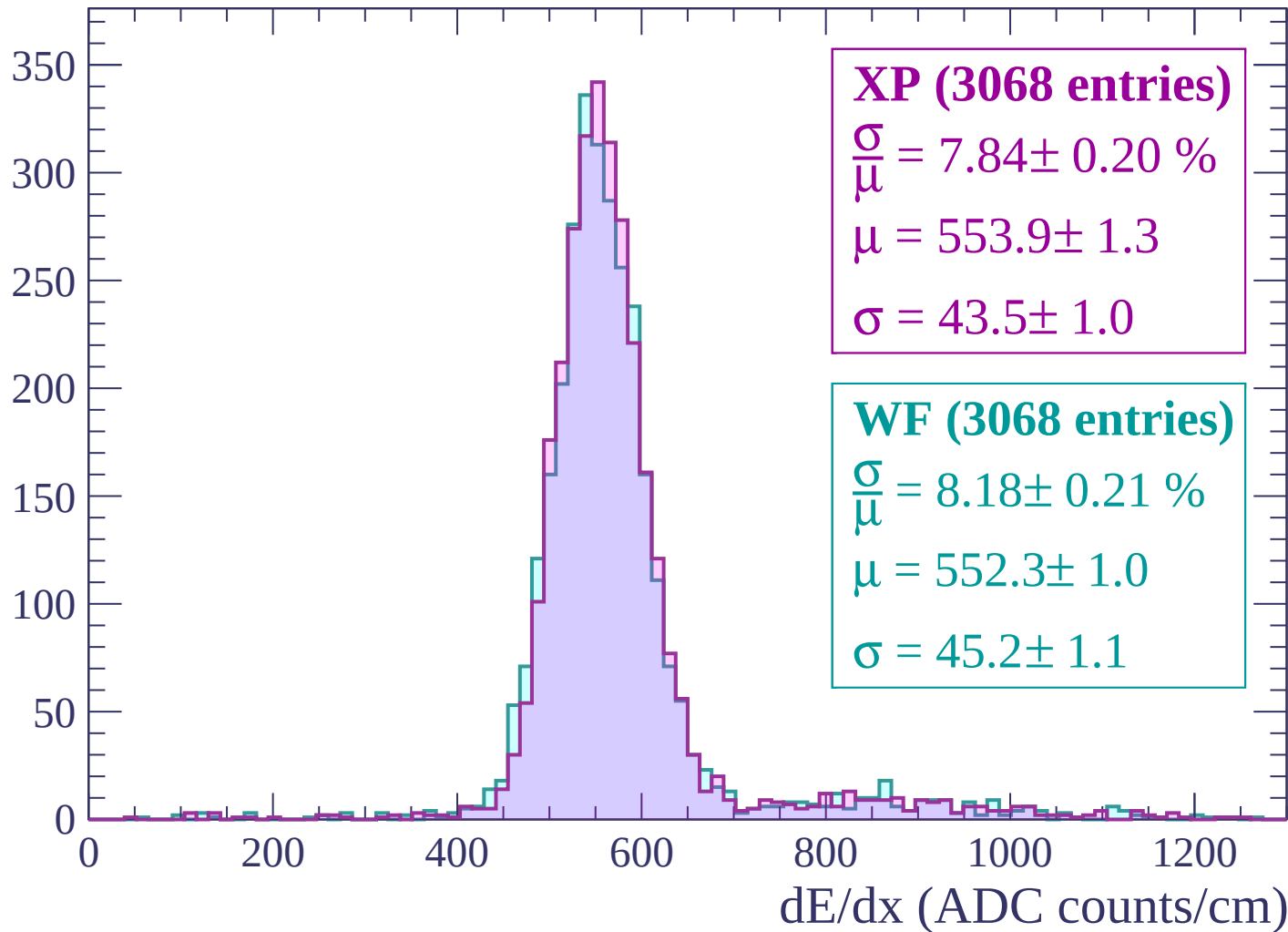
$$\mu = 525.4 \pm 1.3$$

$$\sigma = 49.2 \pm 1.2$$

$dE/dx$  (ADC counts/cm)

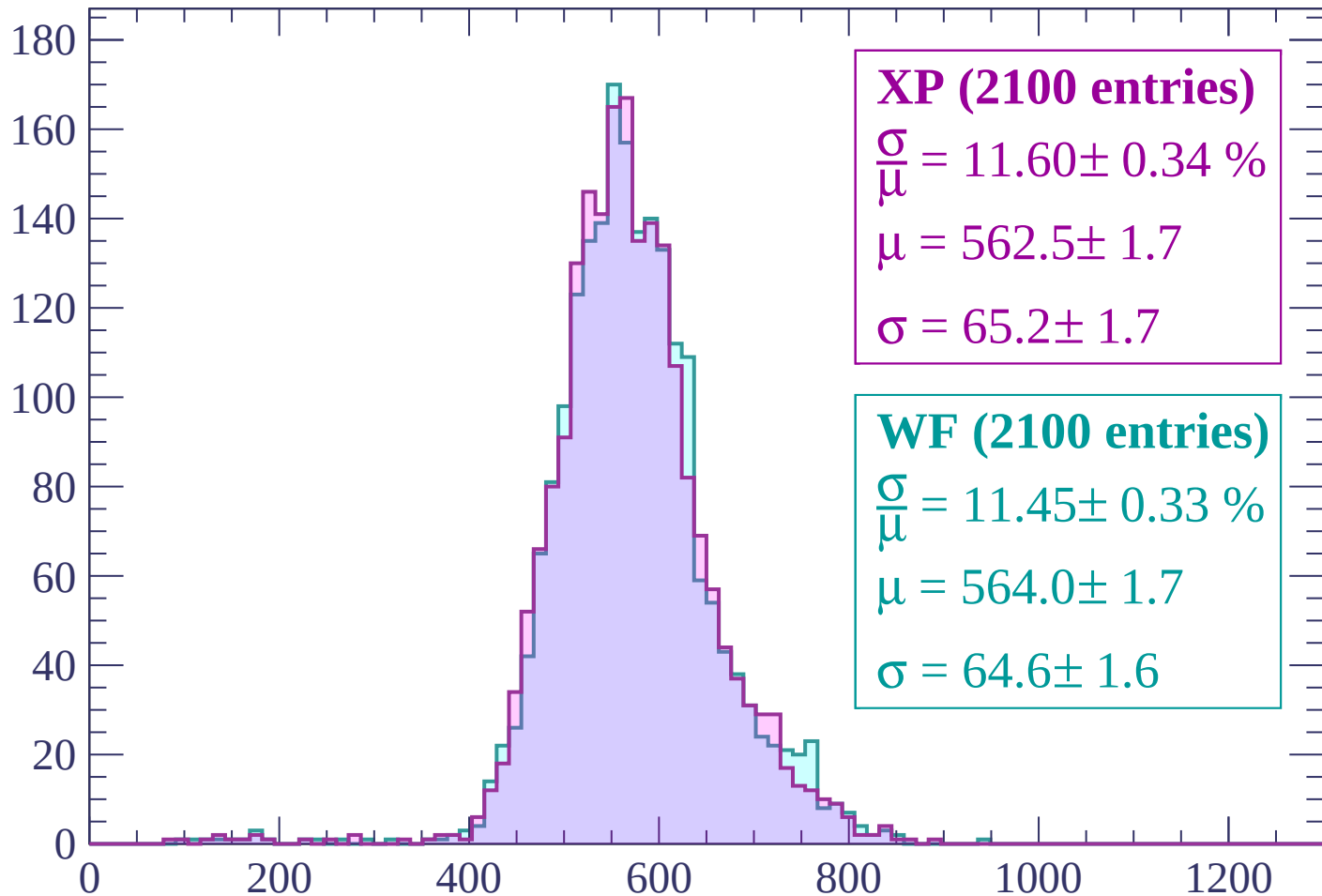
Energy loss |  $60 < p < 120$

Count



Energy loss |  $120 < p < 180$

Count



**XP (2100 entries)**

$$\frac{\sigma}{\mu} = 11.60 \pm 0.34 \%$$

$$\mu = 562.5 \pm 1.7$$

$$\sigma = 65.2 \pm 1.7$$

**WF (2100 entries)**

$$\frac{\sigma}{\mu} = 11.45 \pm 0.33 \%$$

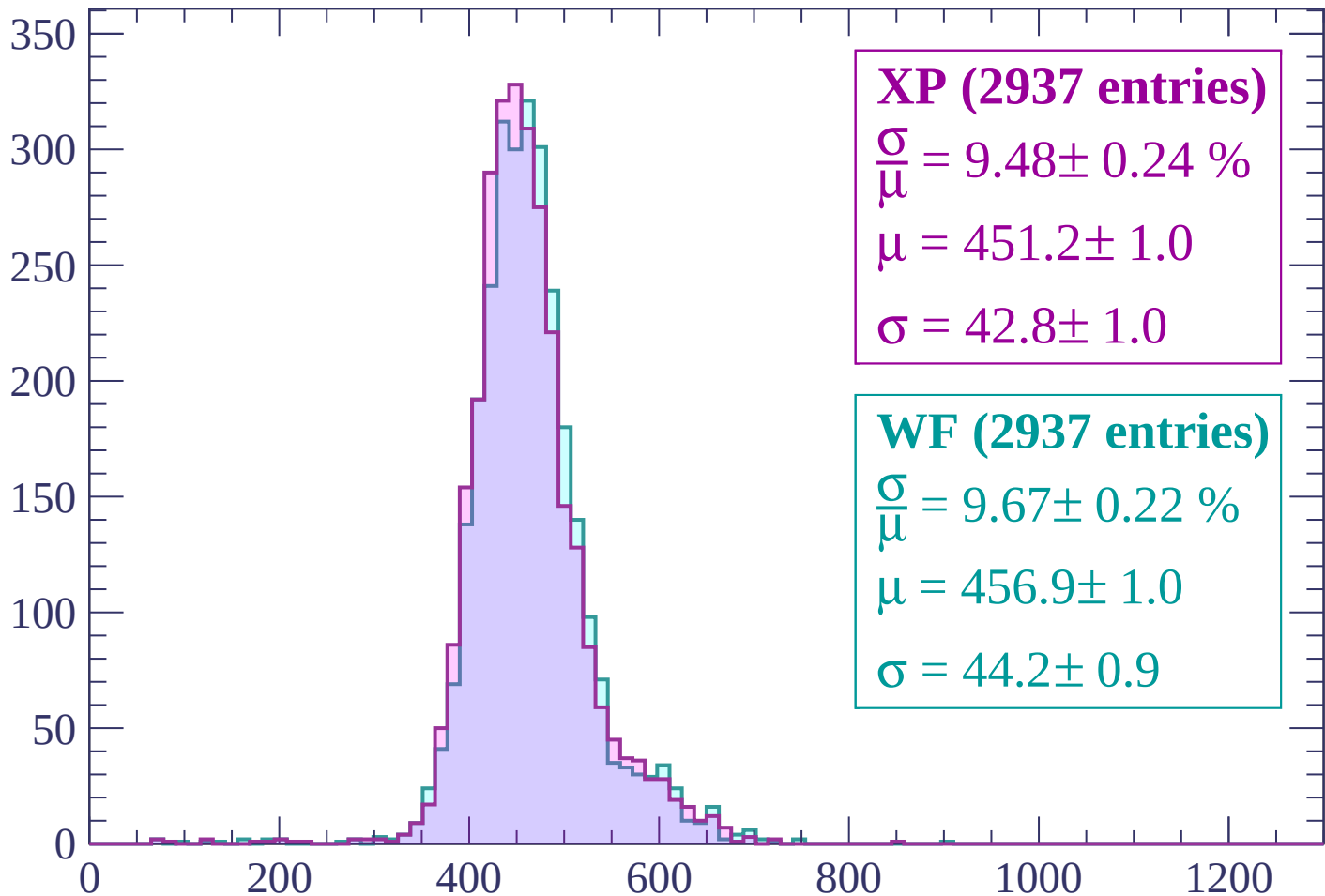
$$\mu = 564.0 \pm 1.7$$

$$\sigma = 64.6 \pm 1.6$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $180 < p < 240$

Count



**XP (2937 entries)**

$$\frac{\sigma}{\mu} = 9.48 \pm 0.24 \%$$

$$\mu = 451.2 \pm 1.0$$

$$\sigma = 42.8 \pm 1.0$$

**WF (2937 entries)**

$$\frac{\sigma}{\mu} = 9.67 \pm 0.22 \%$$

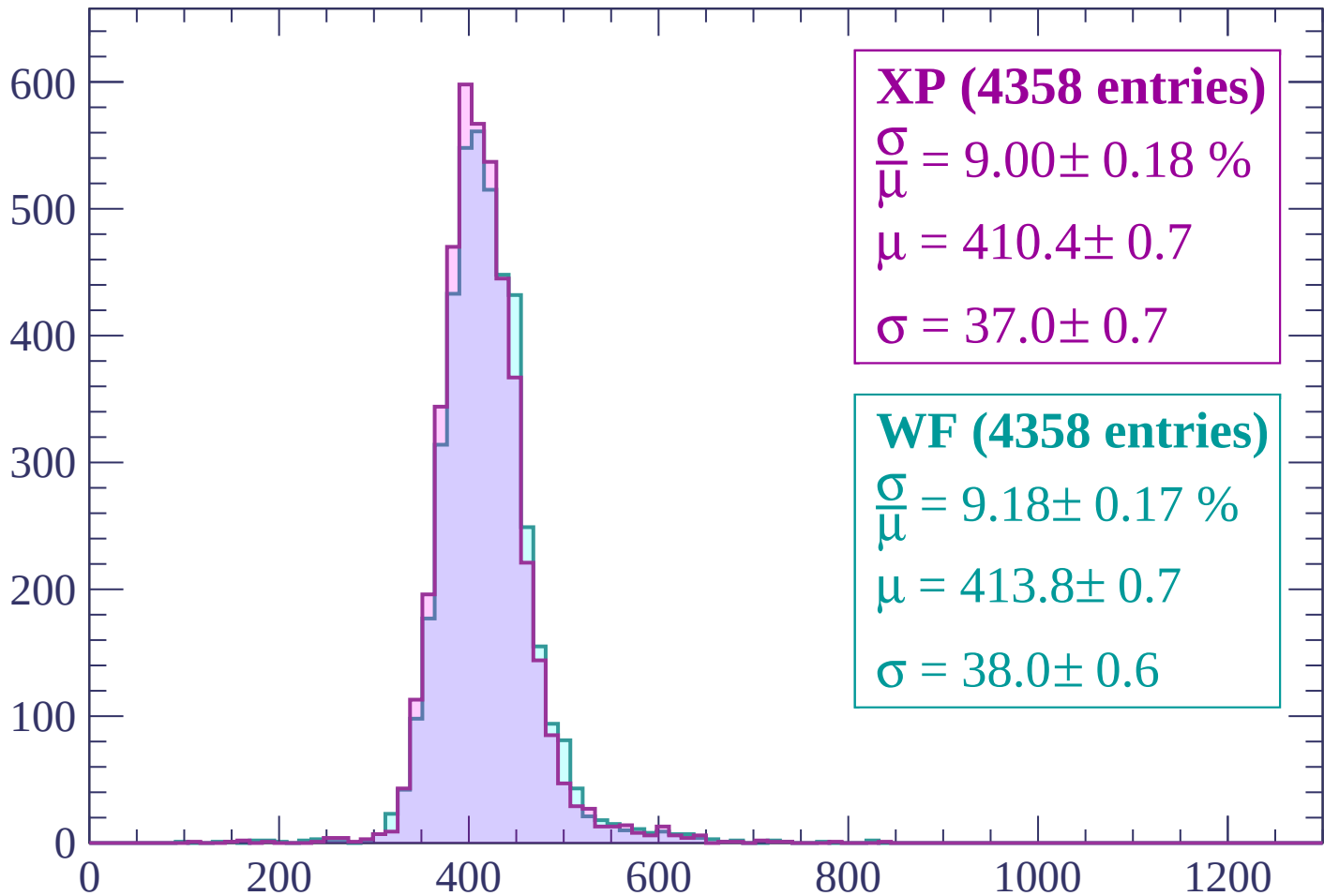
$$\mu = 456.9 \pm 1.0$$

$$\sigma = 44.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $240 < p < 300$

Count



**XP (4358 entries)**

$\frac{\sigma}{\mu} = 9.00 \pm 0.18 \%$

$\mu = 410.4 \pm 0.7$

$\sigma = 37.0 \pm 0.7$

**WF (4358 entries)**

$\frac{\sigma}{\mu} = 9.18 \pm 0.17 \%$

$\mu = 413.8 \pm 0.7$

$\sigma = 38.0 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $300 < p < 360$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (6673 entries)**

$\frac{\sigma}{\mu} = 8.61 \pm 0.12 \%$

$\mu = 397.6 \pm 0.5$

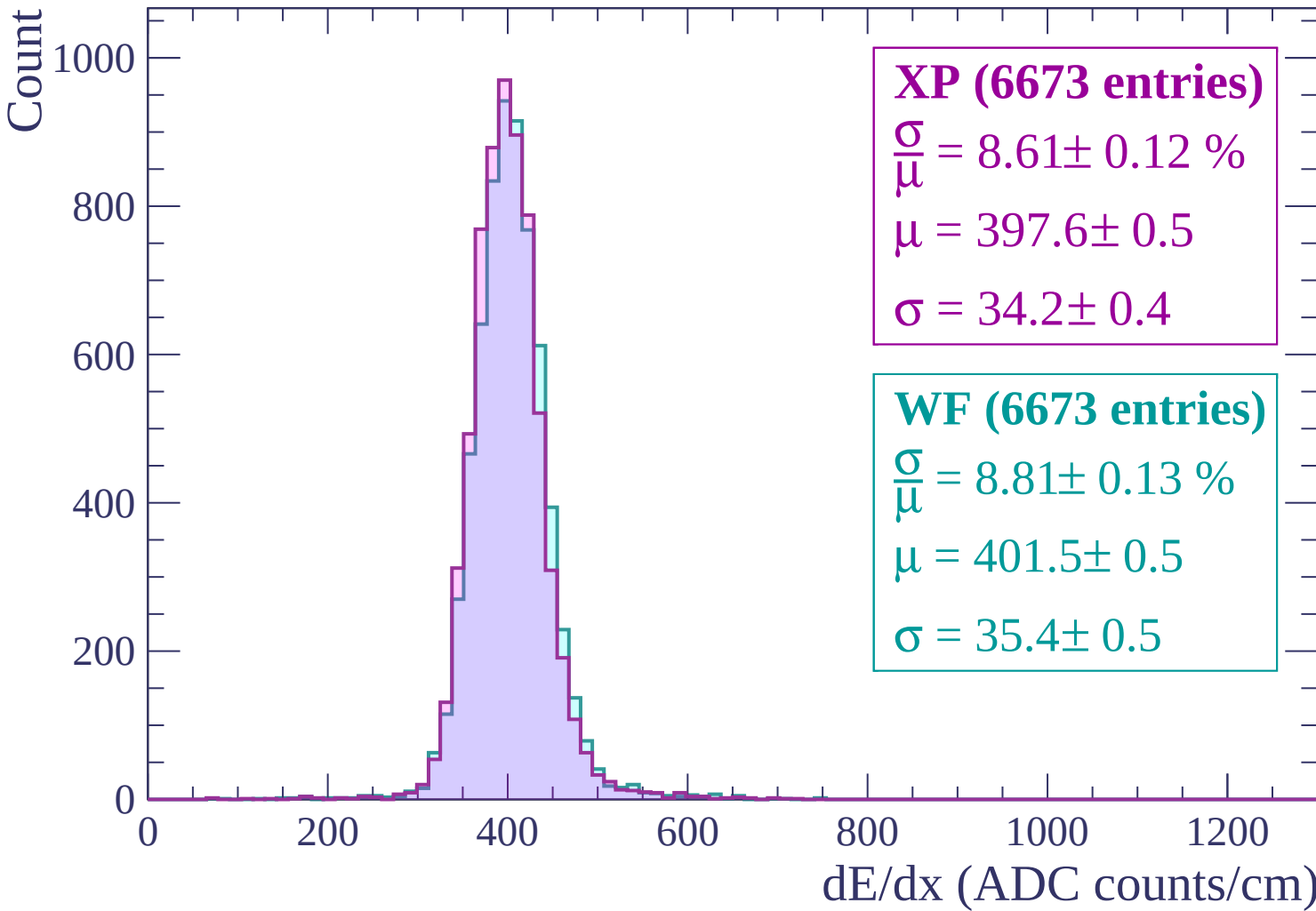
$\sigma = 34.2 \pm 0.4$

**WF (6673 entries)**

$\frac{\sigma}{\mu} = 8.81 \pm 0.13 \%$

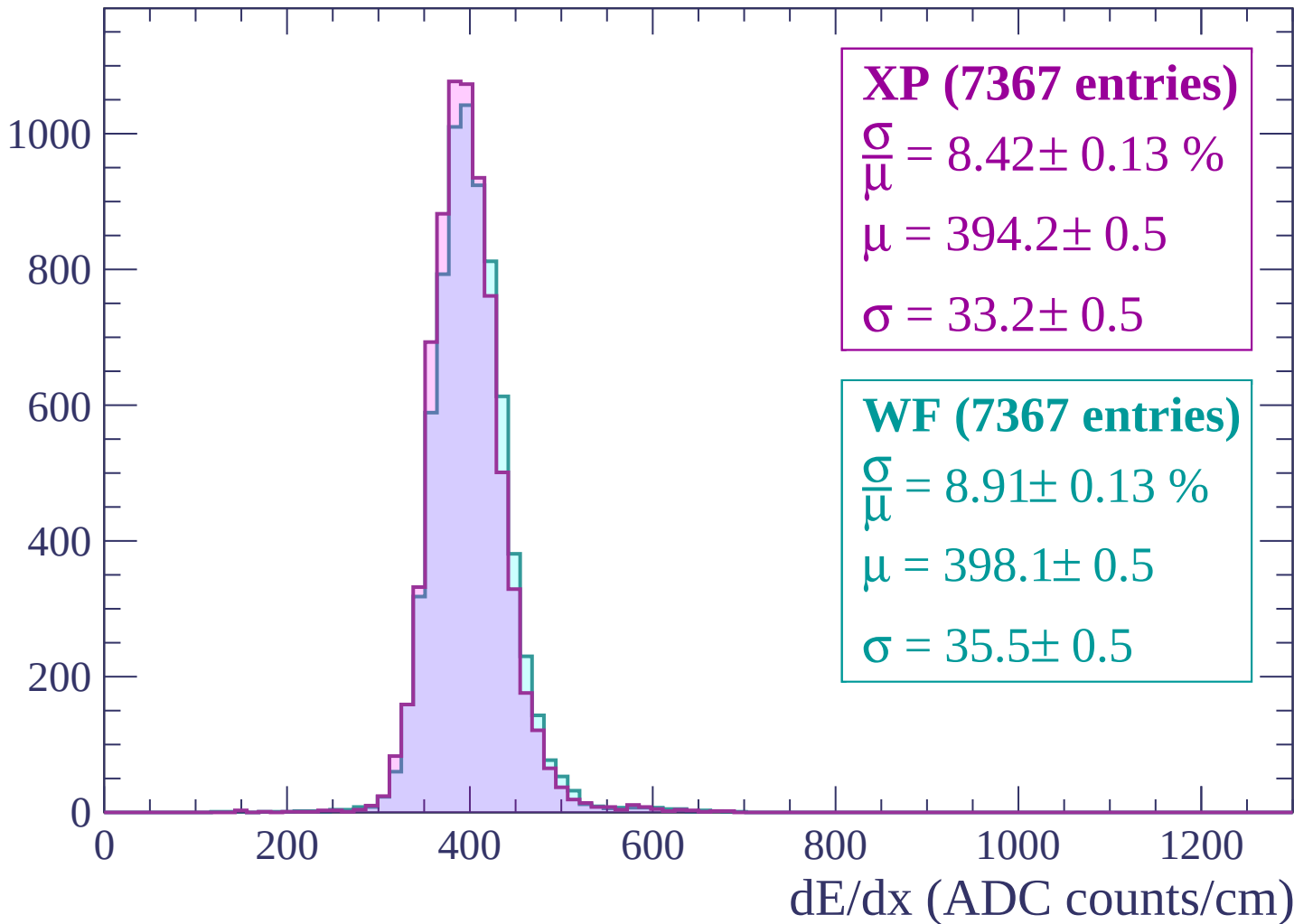
$\mu = 401.5 \pm 0.5$

$\sigma = 35.4 \pm 0.5$



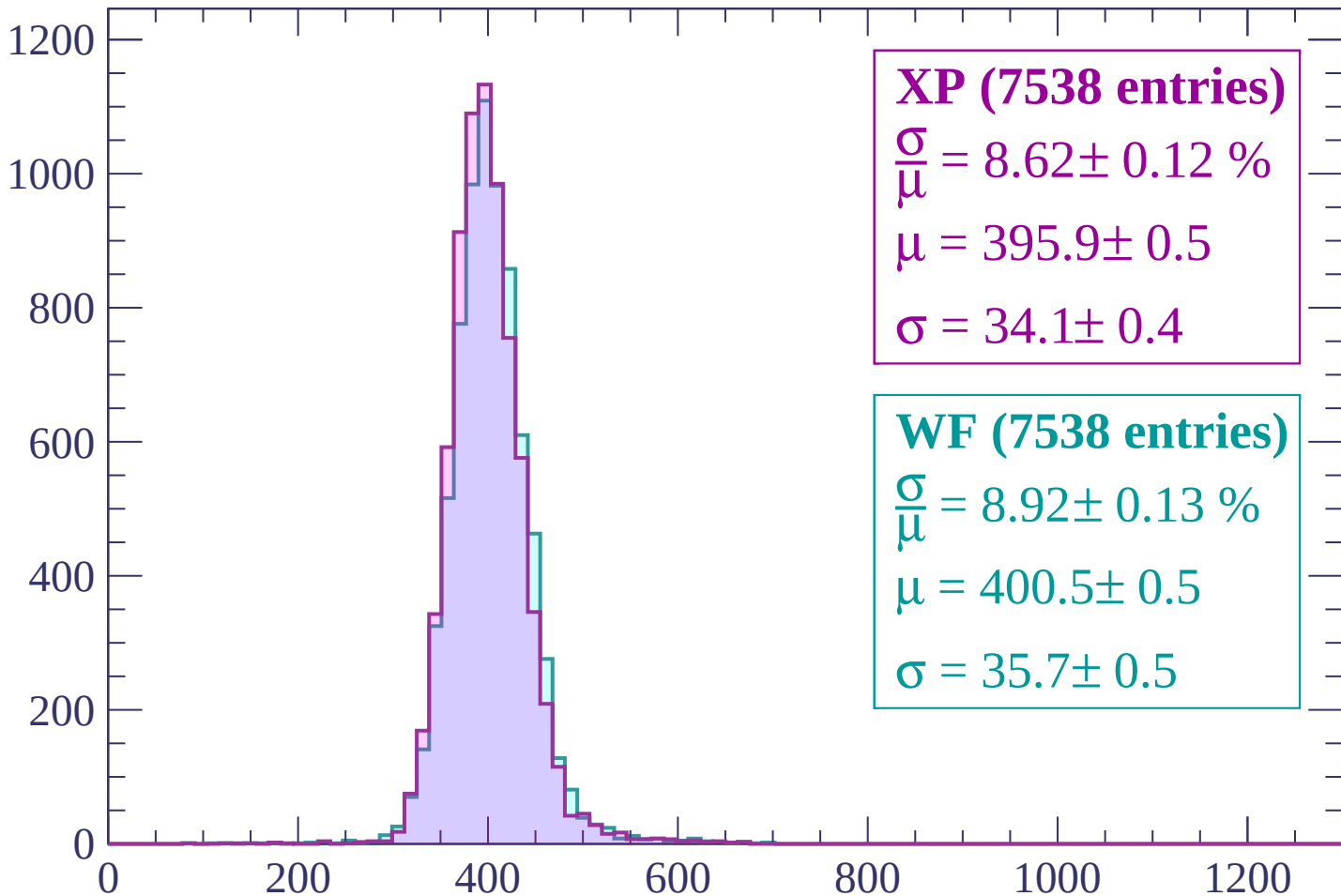
Energy loss |  $360 < p < 420$

Count



Energy loss |  $420 < p < 480$

Count



**XP (7538 entries)**

$$\frac{\sigma}{\mu} = 8.62 \pm 0.12 \%$$

$$\mu = 395.9 \pm 0.5$$

$$\sigma = 34.1 \pm 0.4$$

**WF (7538 entries)**

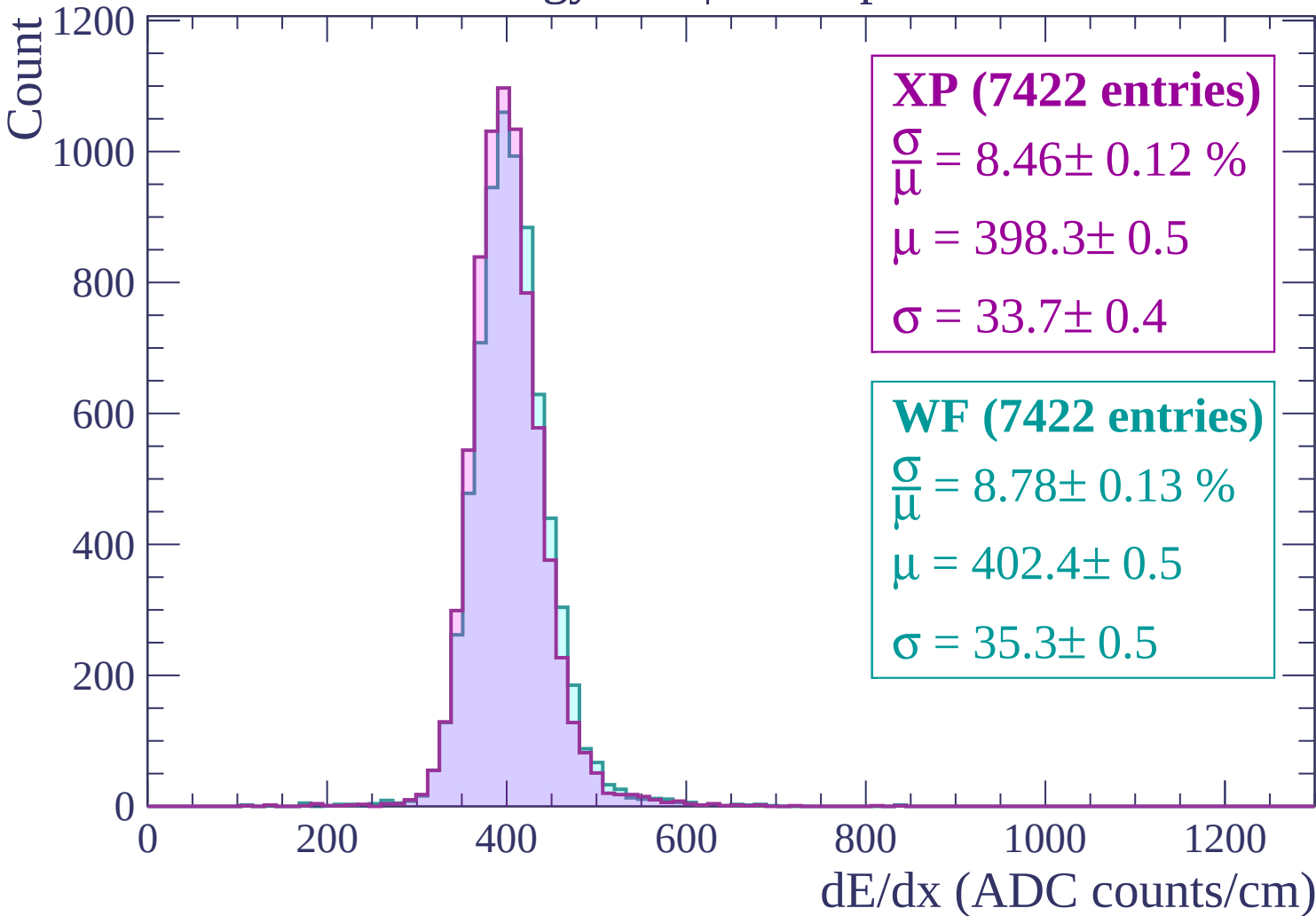
$$\frac{\sigma}{\mu} = 8.92 \pm 0.13 \%$$

$$\mu = 400.5 \pm 0.5$$

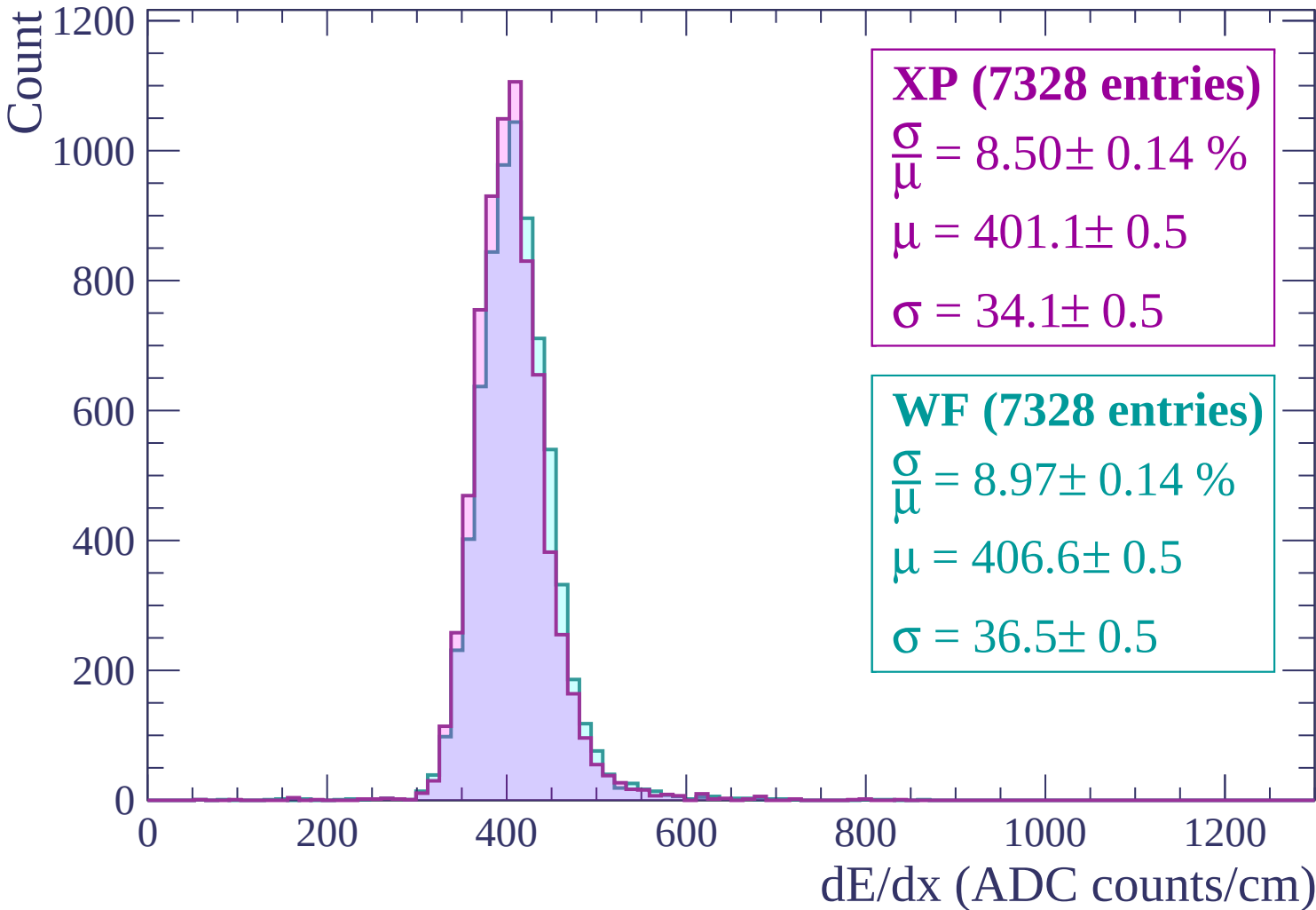
$$\sigma = 35.7 \pm 0.5$$

dE/dx (ADC counts/cm)

Energy loss |  $480 < p < 540$



Energy loss |  $540 < p < 600$



Energy loss |  $600 < p < 660$

Count

1000

800

600

400

200

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (7130 entries)**

$\frac{\sigma}{\mu} = 8.39 \pm 0.13 \%$

$\mu = 404.2 \pm 0.5$

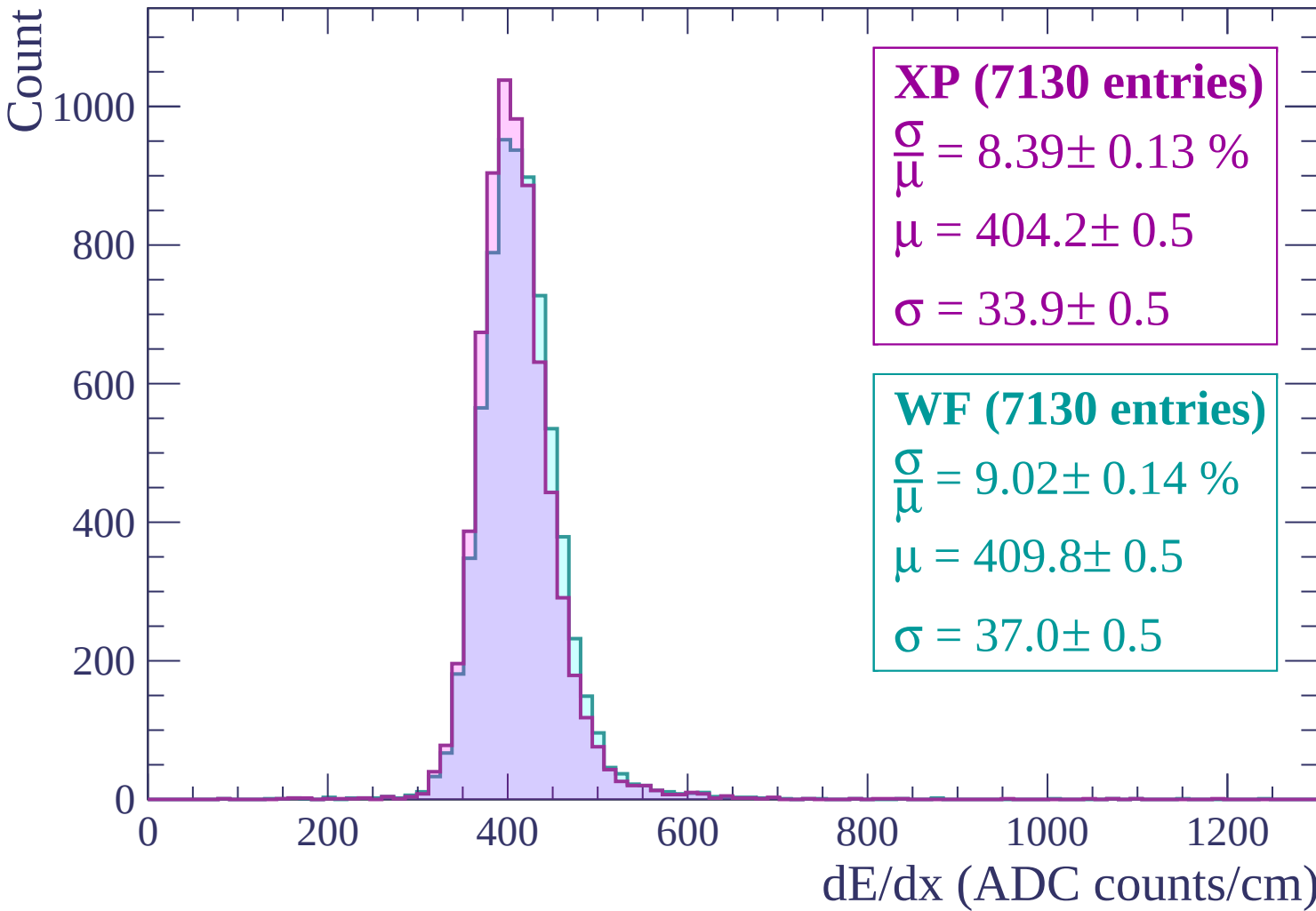
$\sigma = 33.9 \pm 0.5$

**WF (7130 entries)**

$\frac{\sigma}{\mu} = 9.02 \pm 0.14 \%$

$\mu = 409.8 \pm 0.5$

$\sigma = 37.0 \pm 0.5$



Energy loss |  $660 < p < 720$

Count

1000

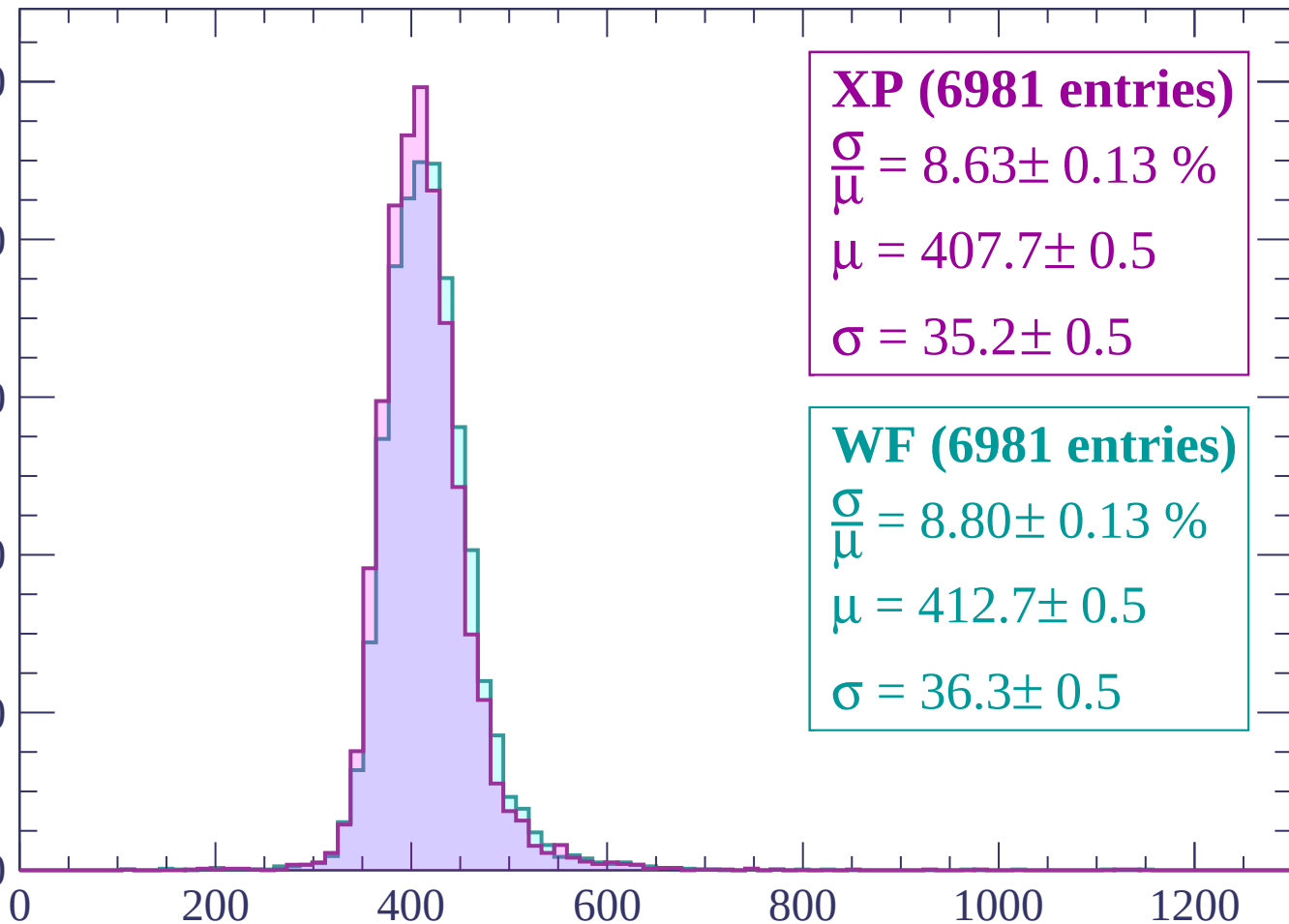
800

600

400

200

0



**XP (6981 entries)**

$\frac{\sigma}{\mu} = 8.63 \pm 0.13 \%$

$\mu = 407.7 \pm 0.5$

$\sigma = 35.2 \pm 0.5$

**WF (6981 entries)**

$\frac{\sigma}{\mu} = 8.80 \pm 0.13 \%$

$\mu = 412.7 \pm 0.5$

$\sigma = 36.3 \pm 0.5$

dE/dx (ADC counts/cm)

Energy loss |  $720 < p < 780$

Count

1000

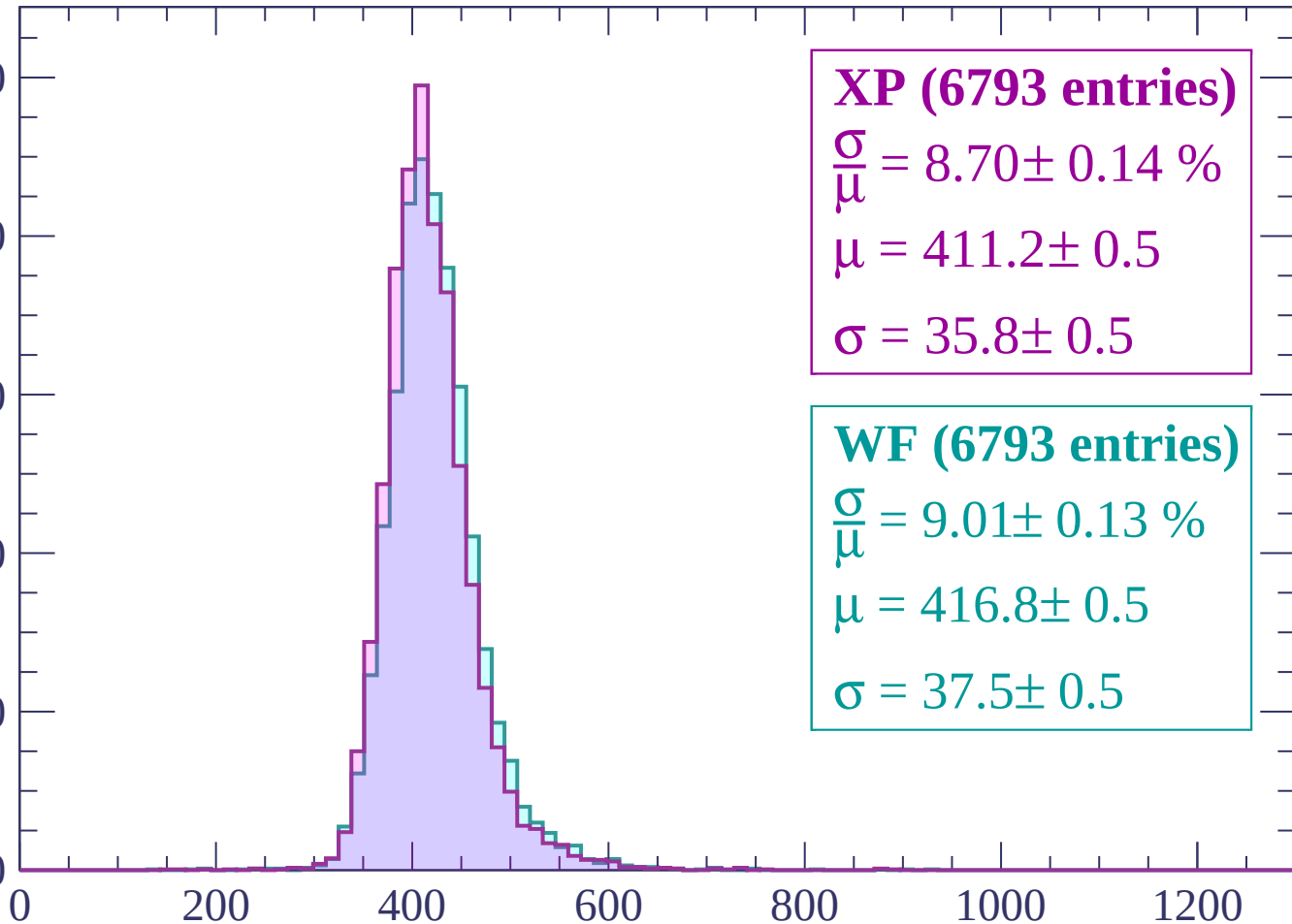
800

600

400

200

0



**XP (6793 entries)**

$\frac{\sigma}{\mu} = 8.70 \pm 0.14 \%$

$\mu = 411.2 \pm 0.5$

$\sigma = 35.8 \pm 0.5$

**WF (6793 entries)**

$\frac{\sigma}{\mu} = 9.01 \pm 0.13 \%$

$\mu = 416.8 \pm 0.5$

$\sigma = 37.5 \pm 0.5$

$dE/dx$  (ADC counts/cm)

Energy loss |  $780 < p < 840$

Count

1000

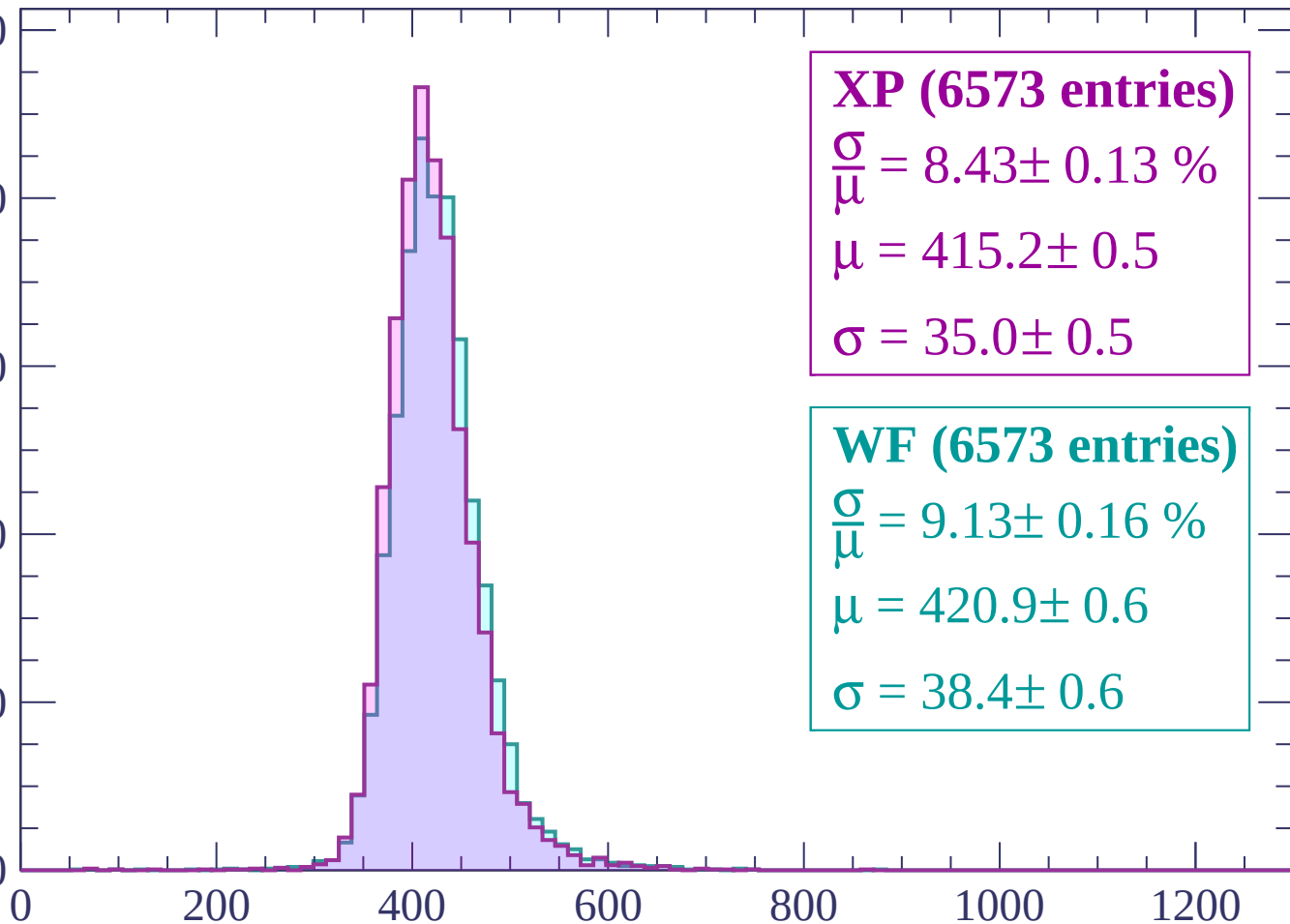
800

600

400

200

0



**XP (6573 entries)**

$\frac{\sigma}{\mu} = 8.43 \pm 0.13 \%$

$\mu = 415.2 \pm 0.5$

$\sigma = 35.0 \pm 0.5$

**WF (6573 entries)**

$\frac{\sigma}{\mu} = 9.13 \pm 0.16 \%$

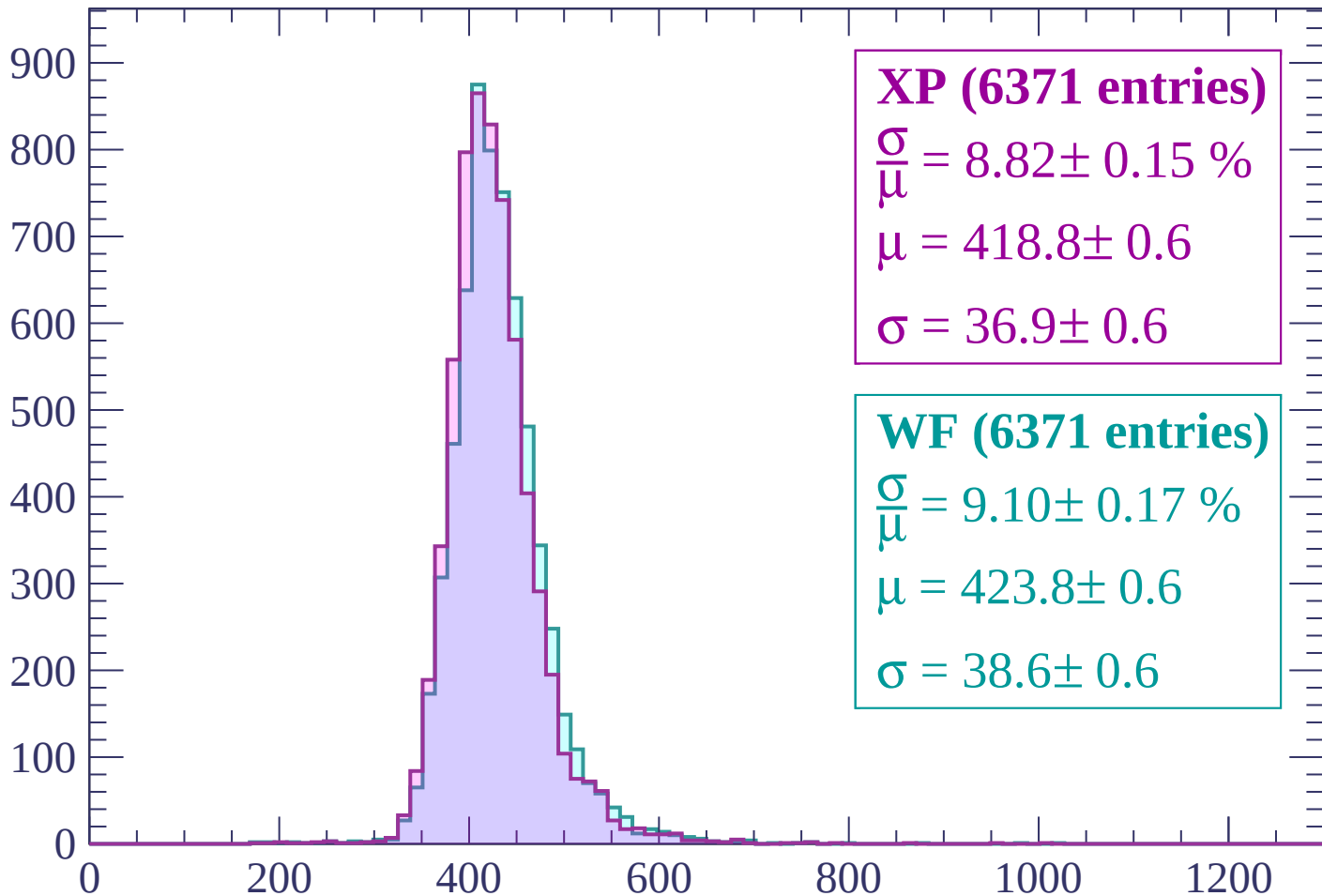
$\mu = 420.9 \pm 0.6$

$\sigma = 38.4 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $840 < p < 900$

Count



**XP (6371 entries)**

$$\frac{\sigma}{\mu} = 8.82 \pm 0.15 \%$$

$$\mu = 418.8 \pm 0.6$$

$$\sigma = 36.9 \pm 0.6$$

**WF (6371 entries)**

$$\frac{\sigma}{\mu} = 9.10 \pm 0.17 \%$$

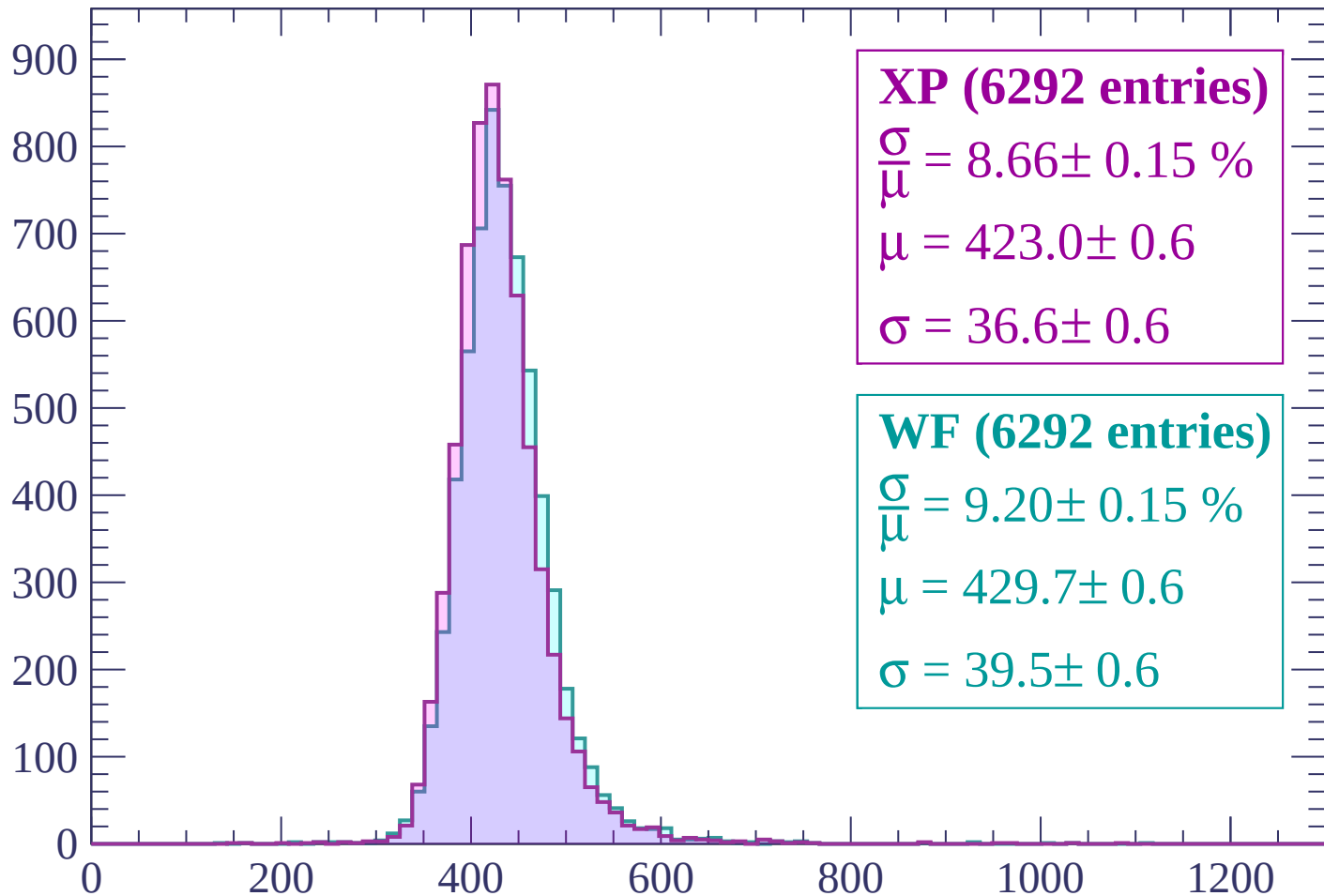
$$\mu = 423.8 \pm 0.6$$

$$\sigma = 38.6 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $900 < p < 960$

Count



**XP (6292 entries)**

$\frac{\sigma}{\mu} = 8.66 \pm 0.15 \%$

$\mu = 423.0 \pm 0.6$

$\sigma = 36.6 \pm 0.6$

**WF (6292 entries)**

$\frac{\sigma}{\mu} = 9.20 \pm 0.15 \%$

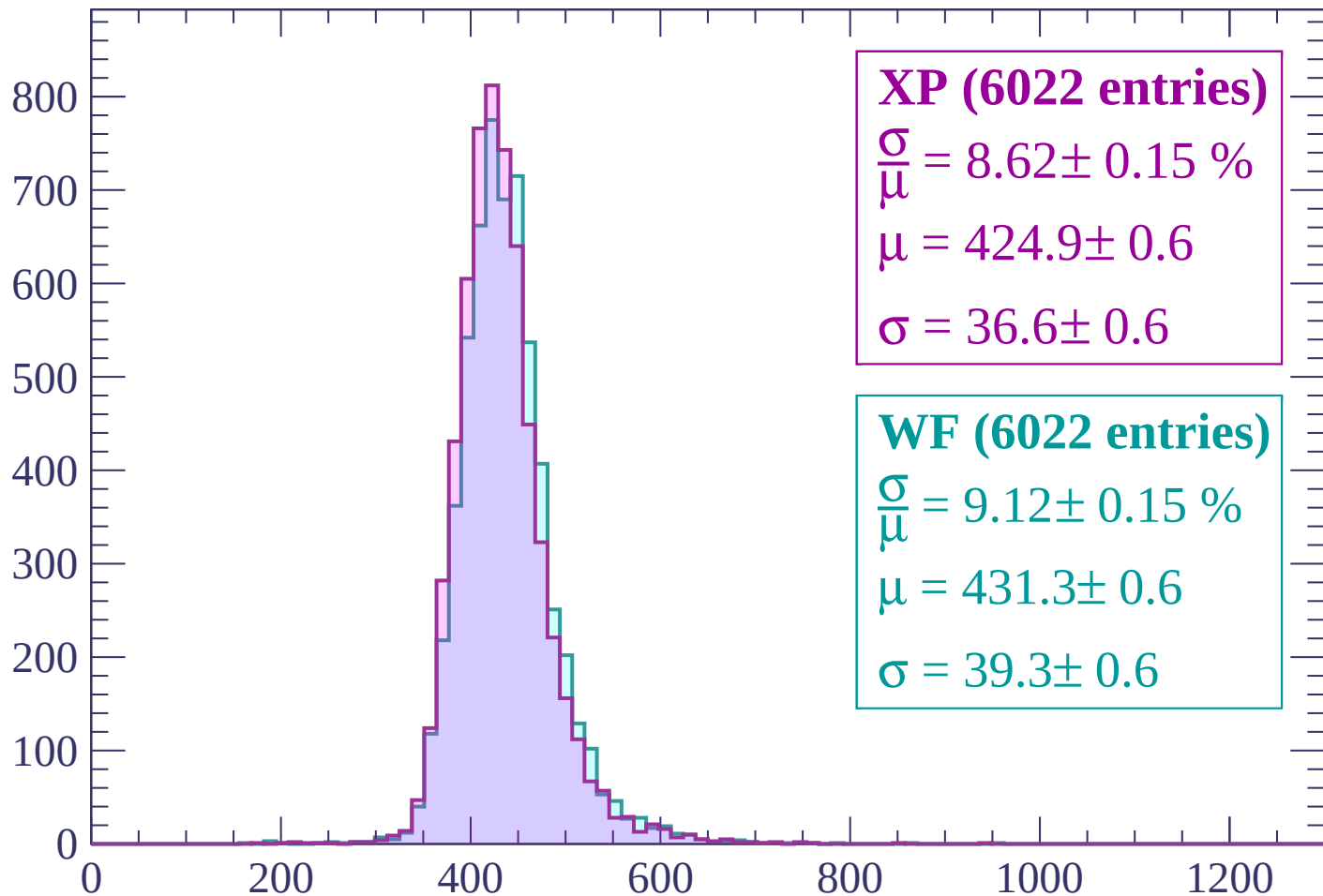
$\mu = 429.7 \pm 0.6$

$\sigma = 39.5 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $960 < p < 1020$

Count



**XP (6022 entries)**

$$\frac{\sigma}{\mu} = 8.62 \pm 0.15 \%$$

$$\mu = 424.9 \pm 0.6$$

$$\sigma = 36.6 \pm 0.6$$

**WF (6022 entries)**

$$\frac{\sigma}{\mu} = 9.12 \pm 0.15 \%$$

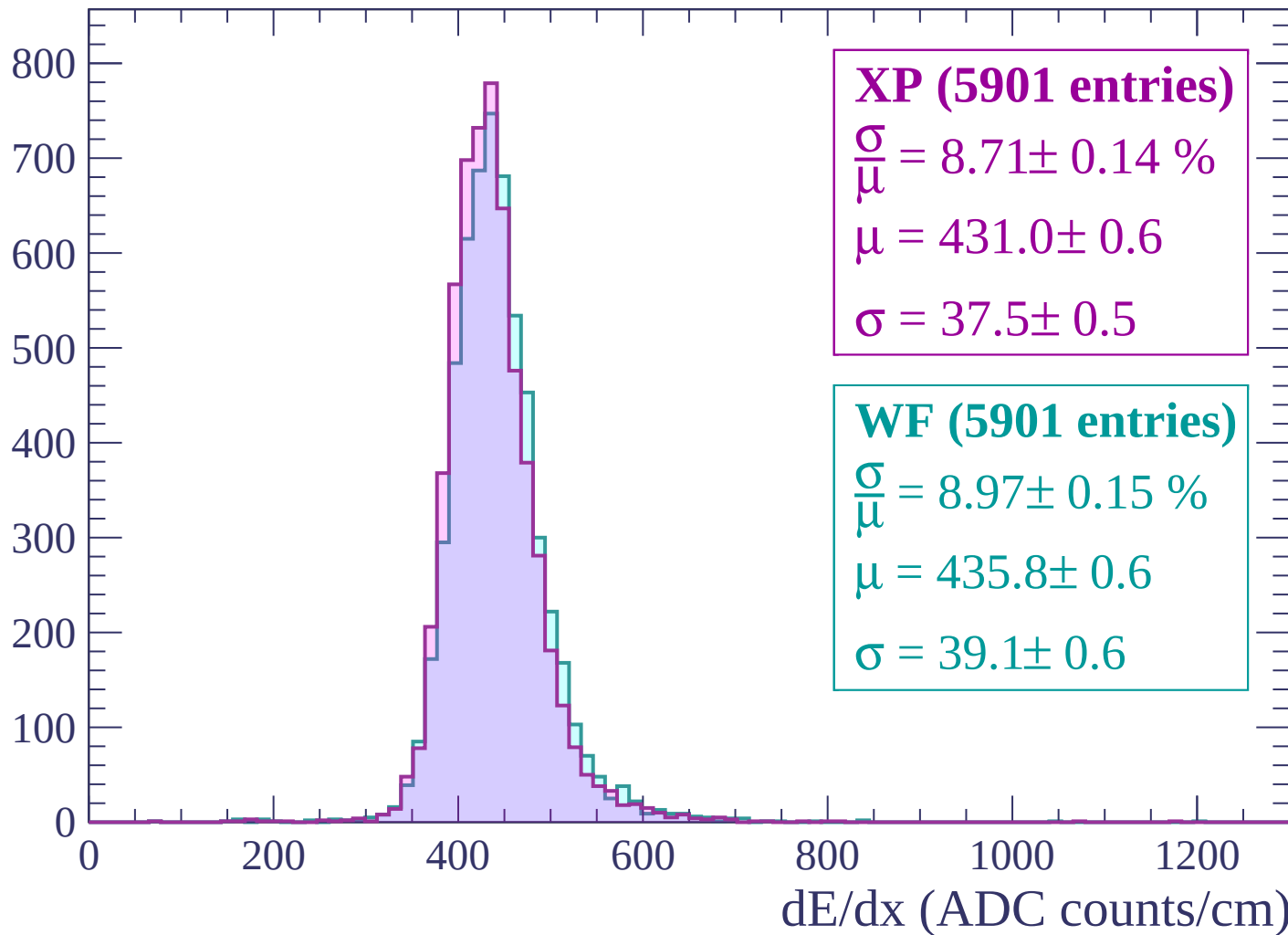
$$\mu = 431.3 \pm 0.6$$

$$\sigma = 39.3 \pm 0.6$$

dE/dx (ADC counts/cm)

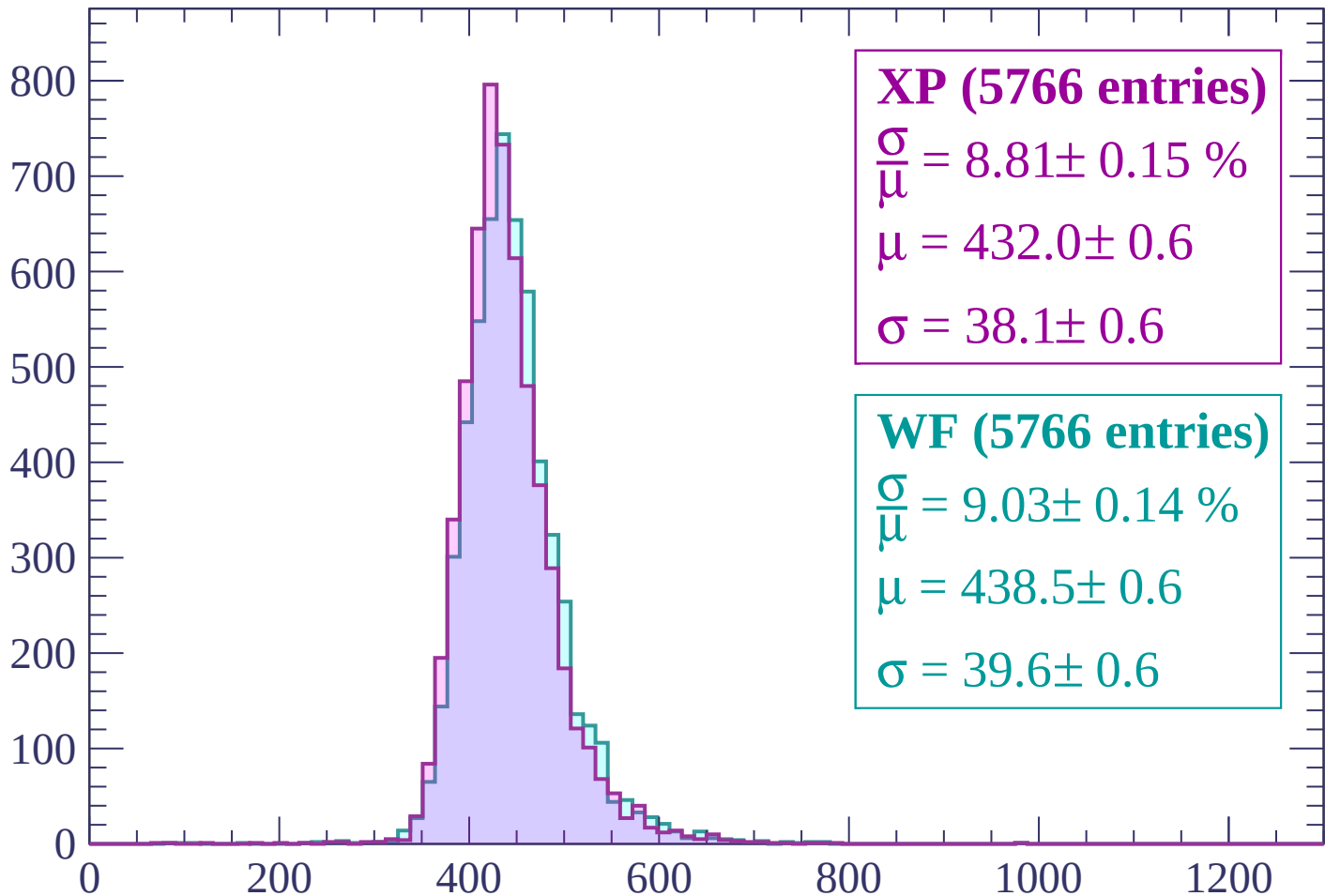
Energy loss |  $1020 < p < 1080$

Count



Energy loss |  $1080 < p < 1140$

Count



**XP (5766 entries)**

$$\frac{\sigma}{\mu} = 8.81 \pm 0.15 \%$$

$$\mu = 432.0 \pm 0.6$$

$$\sigma = 38.1 \pm 0.6$$

**WF (5766 entries)**

$$\frac{\sigma}{\mu} = 9.03 \pm 0.14 \%$$

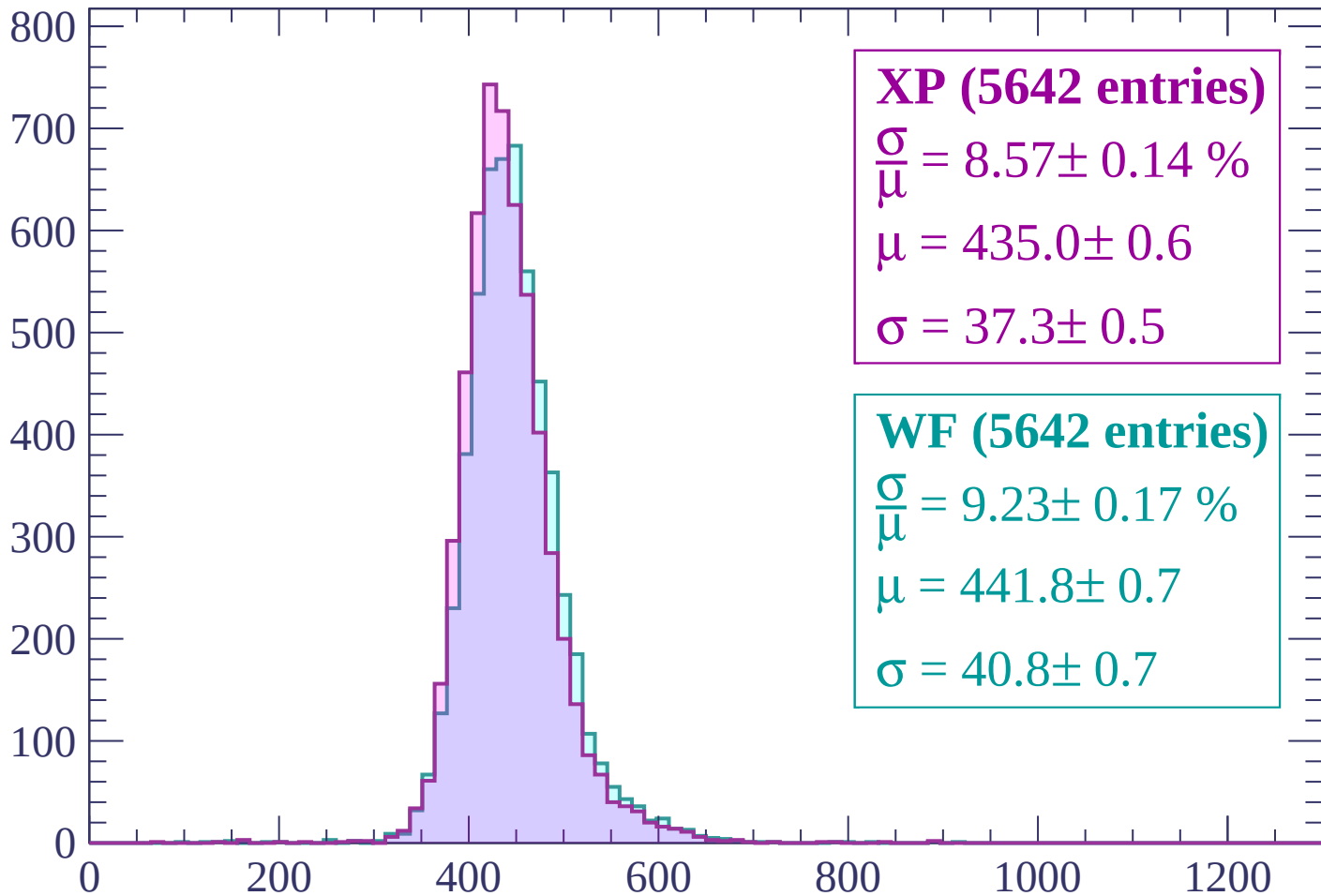
$$\mu = 438.5 \pm 0.6$$

$$\sigma = 39.6 \pm 0.6$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1140 < p < 1200$

Count



**XP (5642 entries)**

$$\frac{\sigma}{\mu} = 8.57 \pm 0.14 \%$$

$$\mu = 435.0 \pm 0.6$$

$$\sigma = 37.3 \pm 0.5$$

**WF (5642 entries)**

$$\frac{\sigma}{\mu} = 9.23 \pm 0.17 \%$$

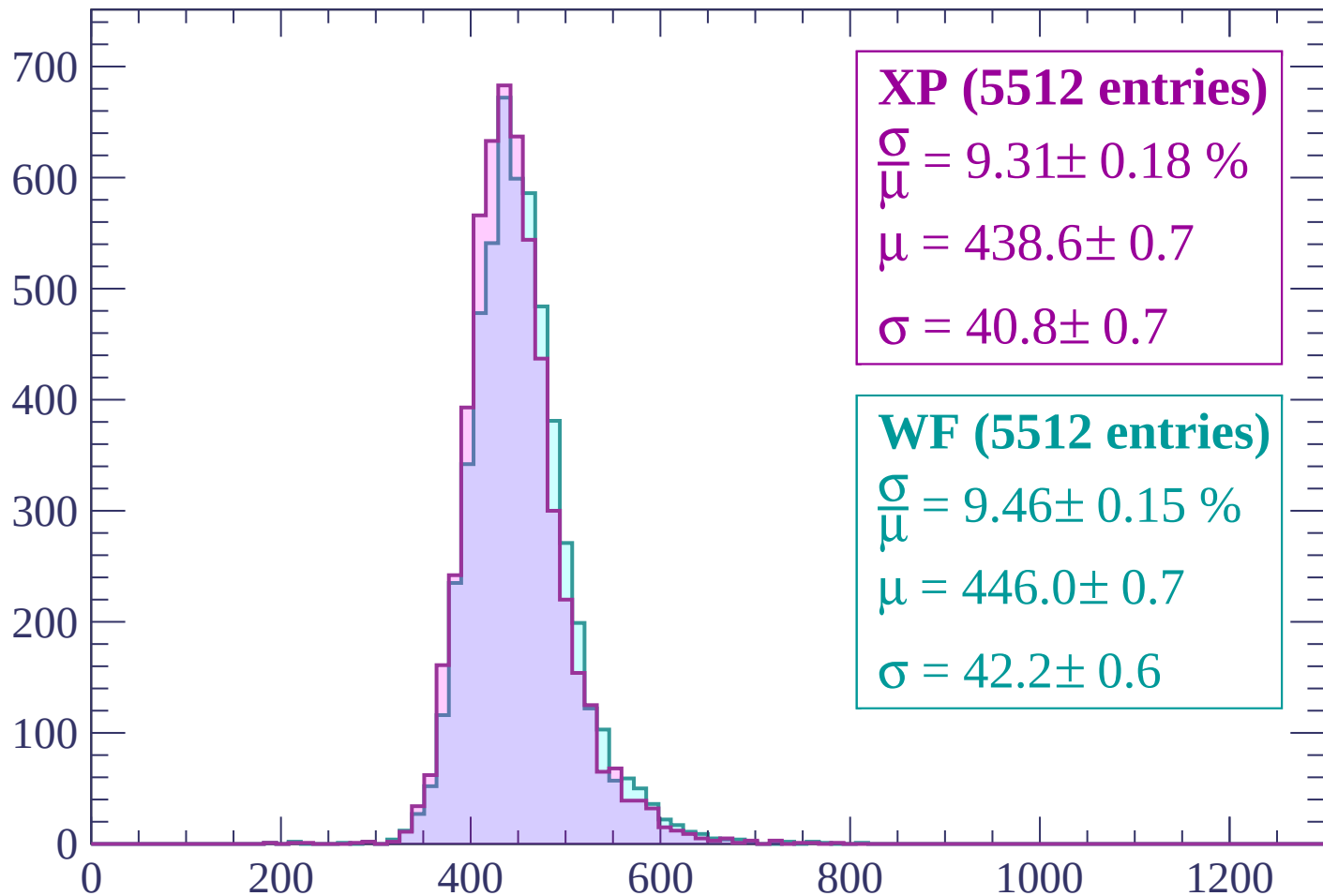
$$\mu = 441.8 \pm 0.7$$

$$\sigma = 40.8 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1200 < p < 1260$

Count



**XP (5512 entries)**

$\frac{\sigma}{\mu} = 9.31 \pm 0.18 \%$

$\mu = 438.6 \pm 0.7$

$\sigma = 40.8 \pm 0.7$

**WF (5512 entries)**

$\frac{\sigma}{\mu} = 9.46 \pm 0.15 \%$

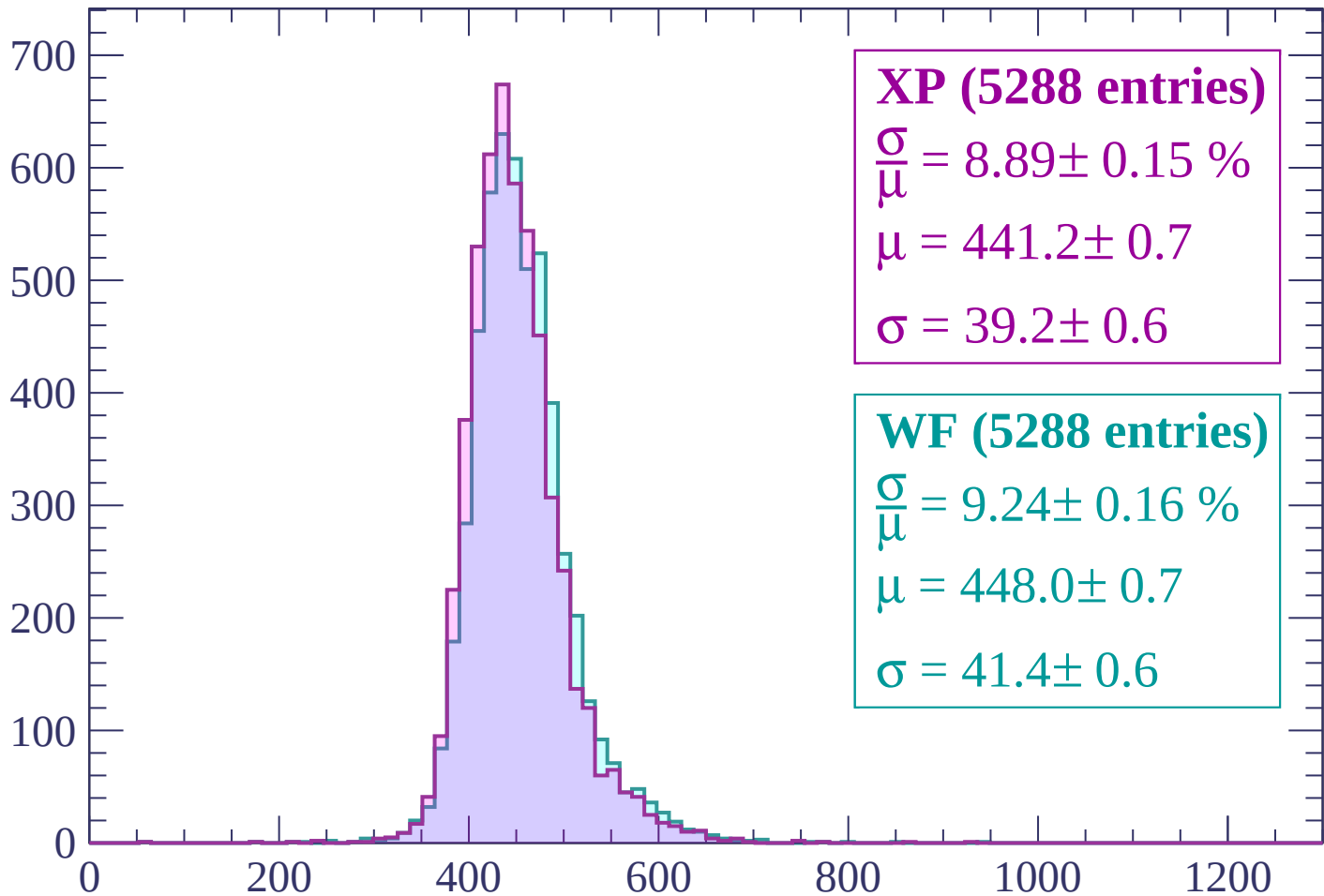
$\mu = 446.0 \pm 0.7$

$\sigma = 42.2 \pm 0.6$

dE/dx (ADC counts/cm)

Energy loss |  $1260 < p < 1320$

Count



**XP (5288 entries)**

$$\frac{\sigma}{\mu} = 8.89 \pm 0.15 \%$$

$$\mu = 441.2 \pm 0.7$$

$$\sigma = 39.2 \pm 0.6$$

**WF (5288 entries)**

$$\frac{\sigma}{\mu} = 9.24 \pm 0.16 \%$$

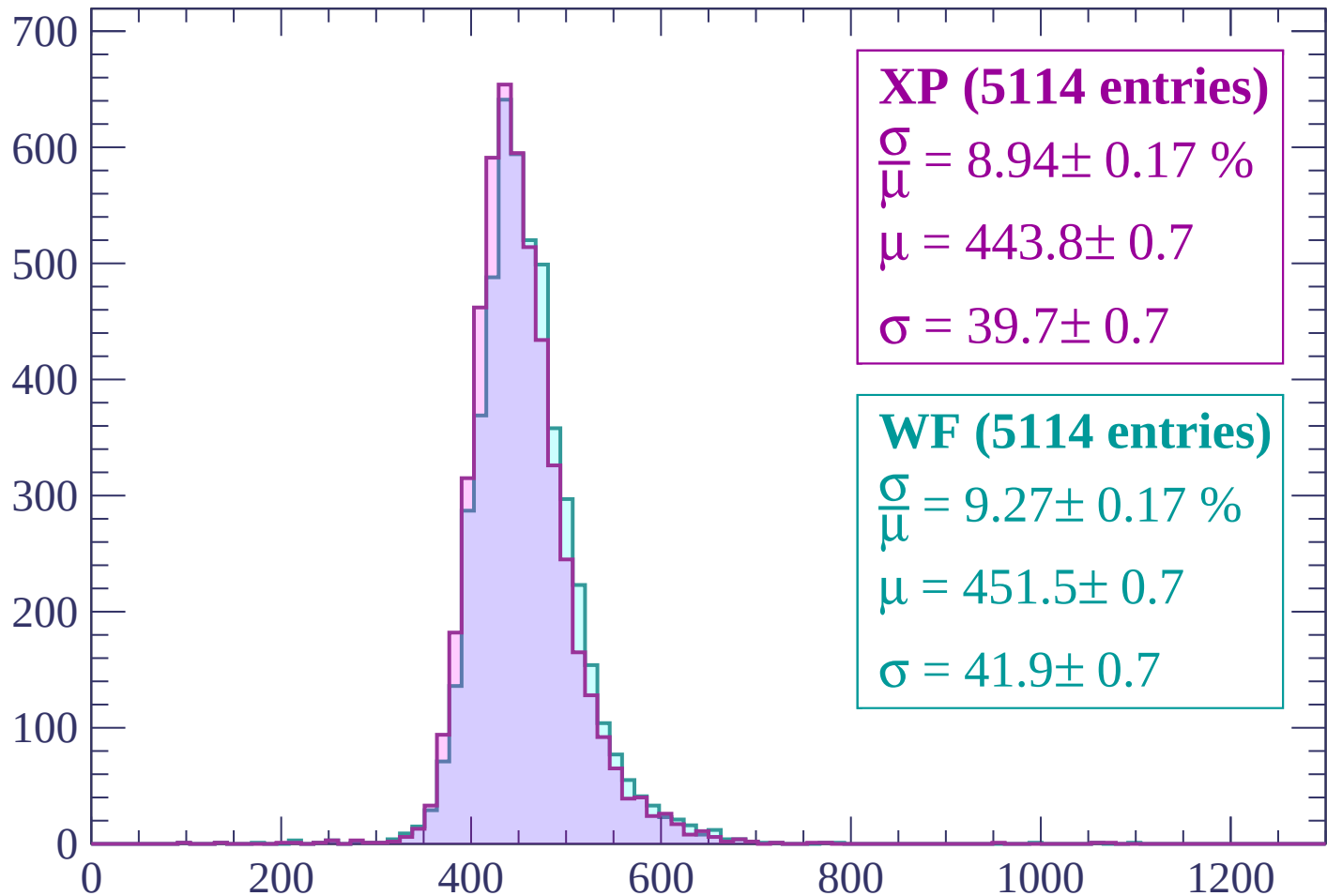
$$\mu = 448.0 \pm 0.7$$

$$\sigma = 41.4 \pm 0.6$$

dE/dx (ADC counts/cm)

Energy loss |  $1320 < p < 1380$

Count



**XP (5114 entries)**

$$\frac{\sigma}{\mu} = 8.94 \pm 0.17 \%$$

$$\mu = 443.8 \pm 0.7$$

$$\sigma = 39.7 \pm 0.7$$

**WF (5114 entries)**

$$\frac{\sigma}{\mu} = 9.27 \pm 0.17 \%$$

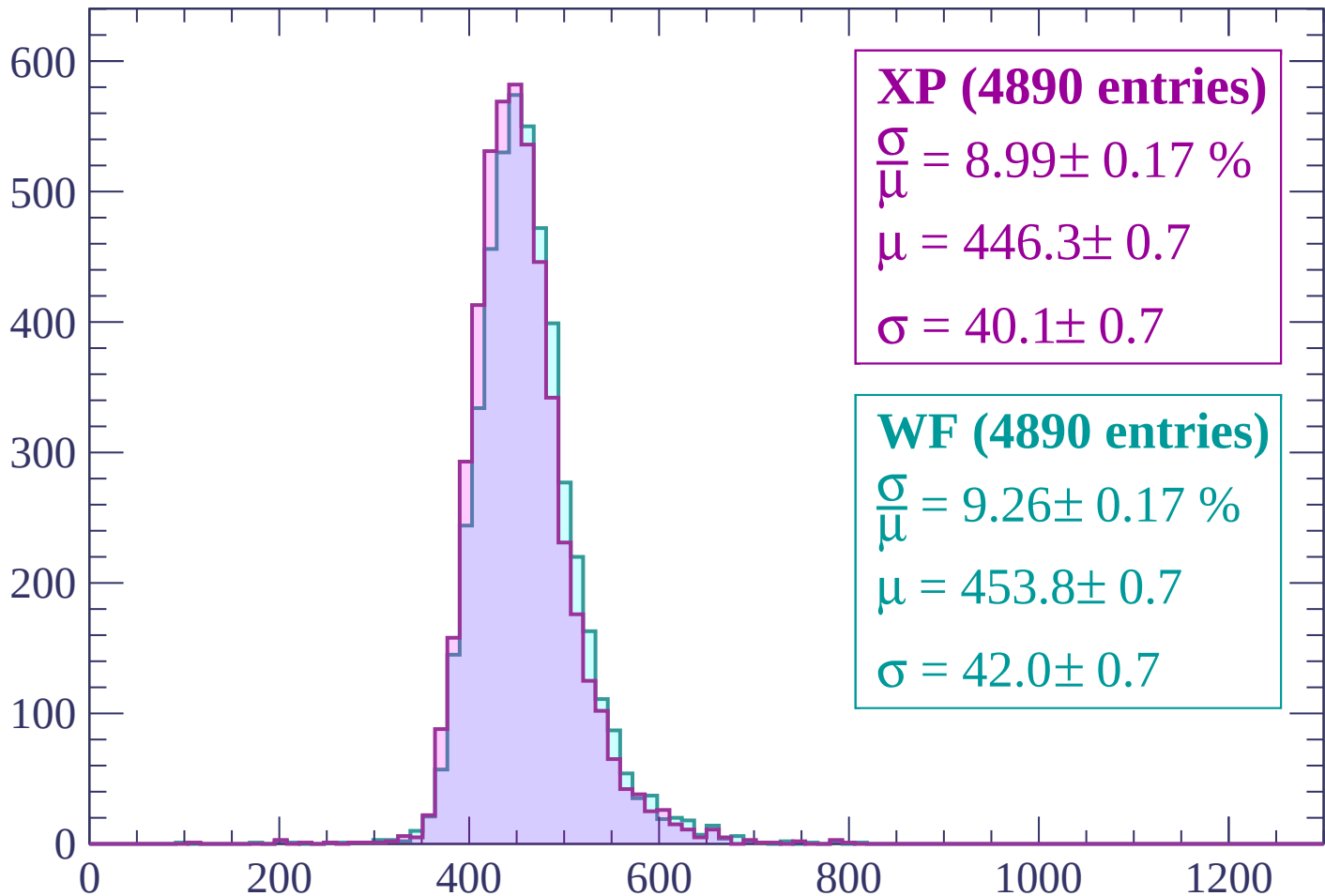
$$\mu = 451.5 \pm 0.7$$

$$\sigma = 41.9 \pm 0.7$$

dE/dx (ADC counts/cm)

Energy loss |  $1380 < p < 1440$

Count



**XP (4890 entries)**

$$\frac{\sigma}{\mu} = 8.99 \pm 0.17 \%$$

$$\mu = 446.3 \pm 0.7$$

$$\sigma = 40.1 \pm 0.7$$

**WF (4890 entries)**

$$\frac{\sigma}{\mu} = 9.26 \pm 0.17 \%$$

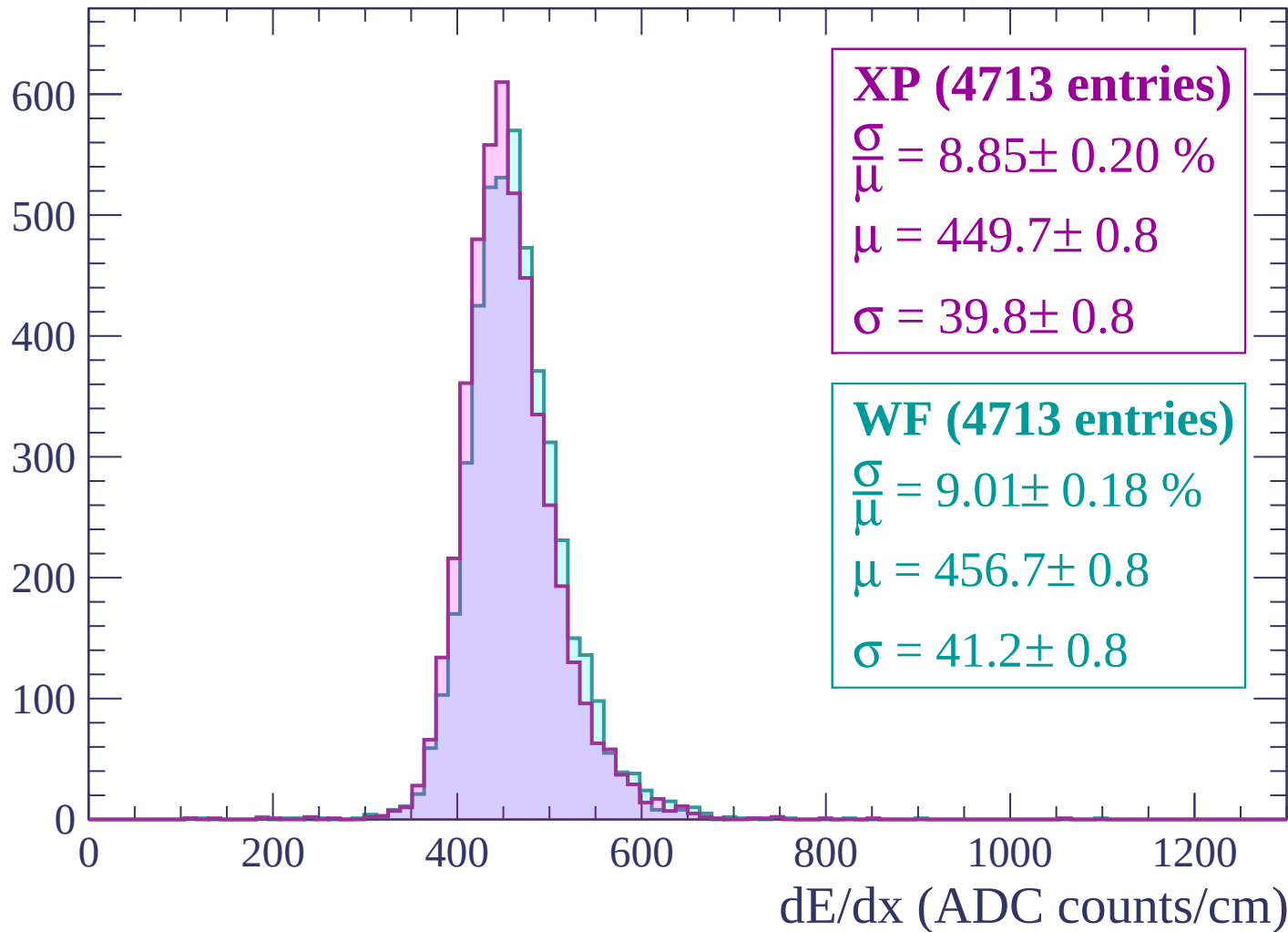
$$\mu = 453.8 \pm 0.7$$

$$\sigma = 42.0 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

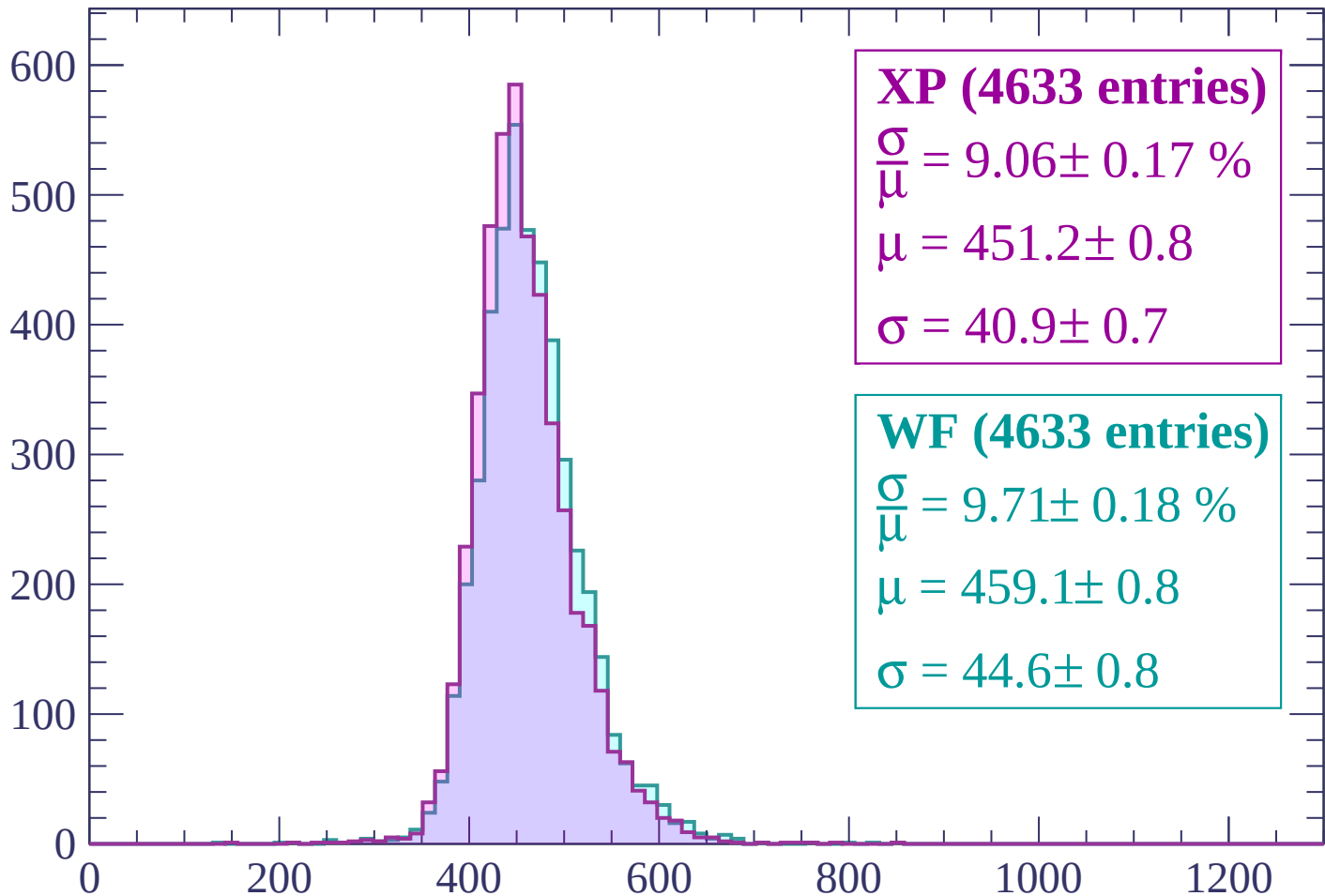
Energy loss |  $1440 < p < 1500$

Count



Energy loss |  $1500 < p < 1560$

Count



**XP (4633 entries)**

$$\frac{\sigma}{\mu} = 9.06 \pm 0.17 \%$$

$$\mu = 451.2 \pm 0.8$$

$$\sigma = 40.9 \pm 0.7$$

**WF (4633 entries)**

$$\frac{\sigma}{\mu} = 9.71 \pm 0.18 \%$$

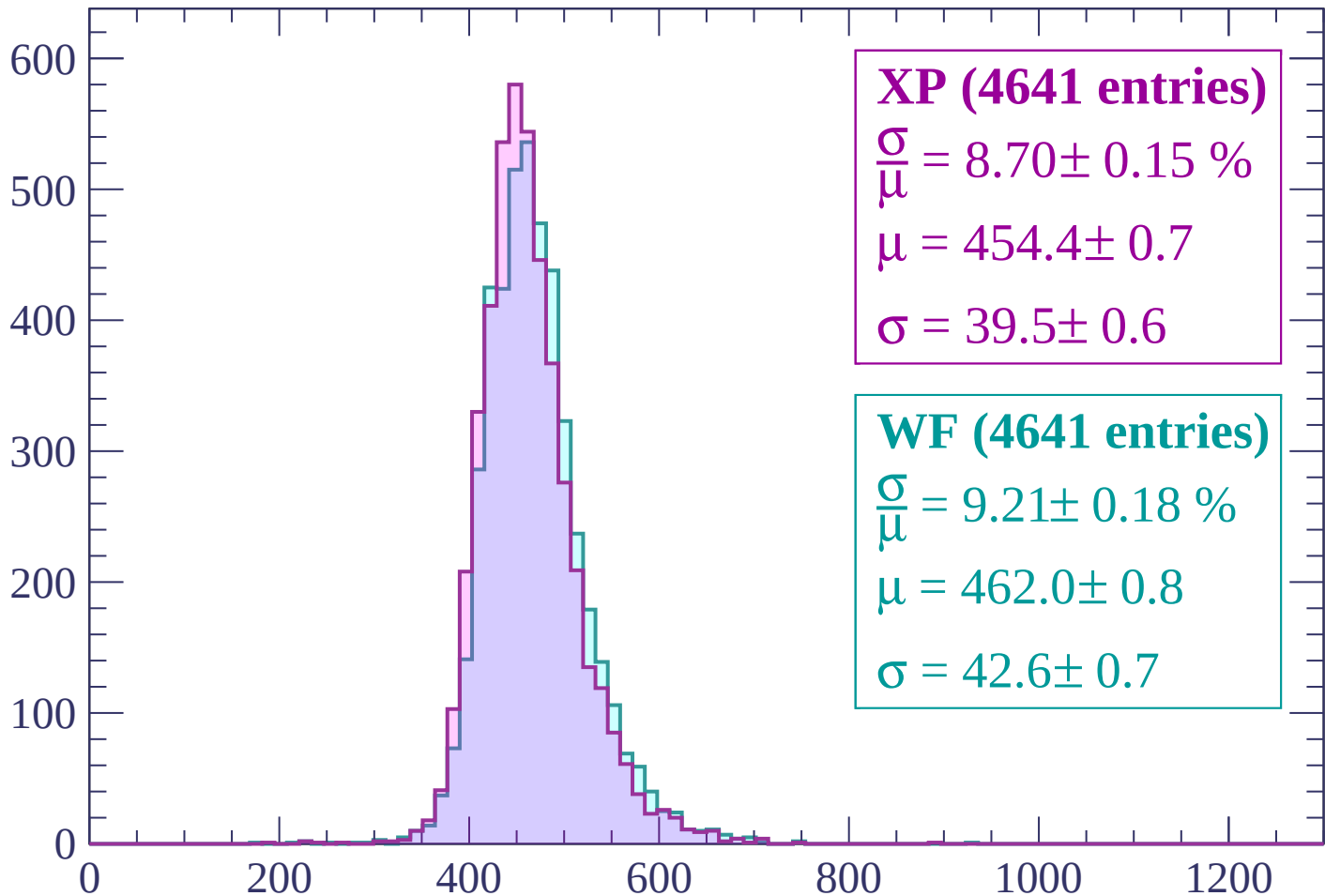
$$\mu = 459.1 \pm 0.8$$

$$\sigma = 44.6 \pm 0.8$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1560 < p < 1620$

Count



**XP (4641 entries)**

$$\frac{\sigma}{\mu} = 8.70 \pm 0.15 \%$$

$$\mu = 454.4 \pm 0.7$$

$$\sigma = 39.5 \pm 0.6$$

**WF (4641 entries)**

$$\frac{\sigma}{\mu} = 9.21 \pm 0.18 \%$$

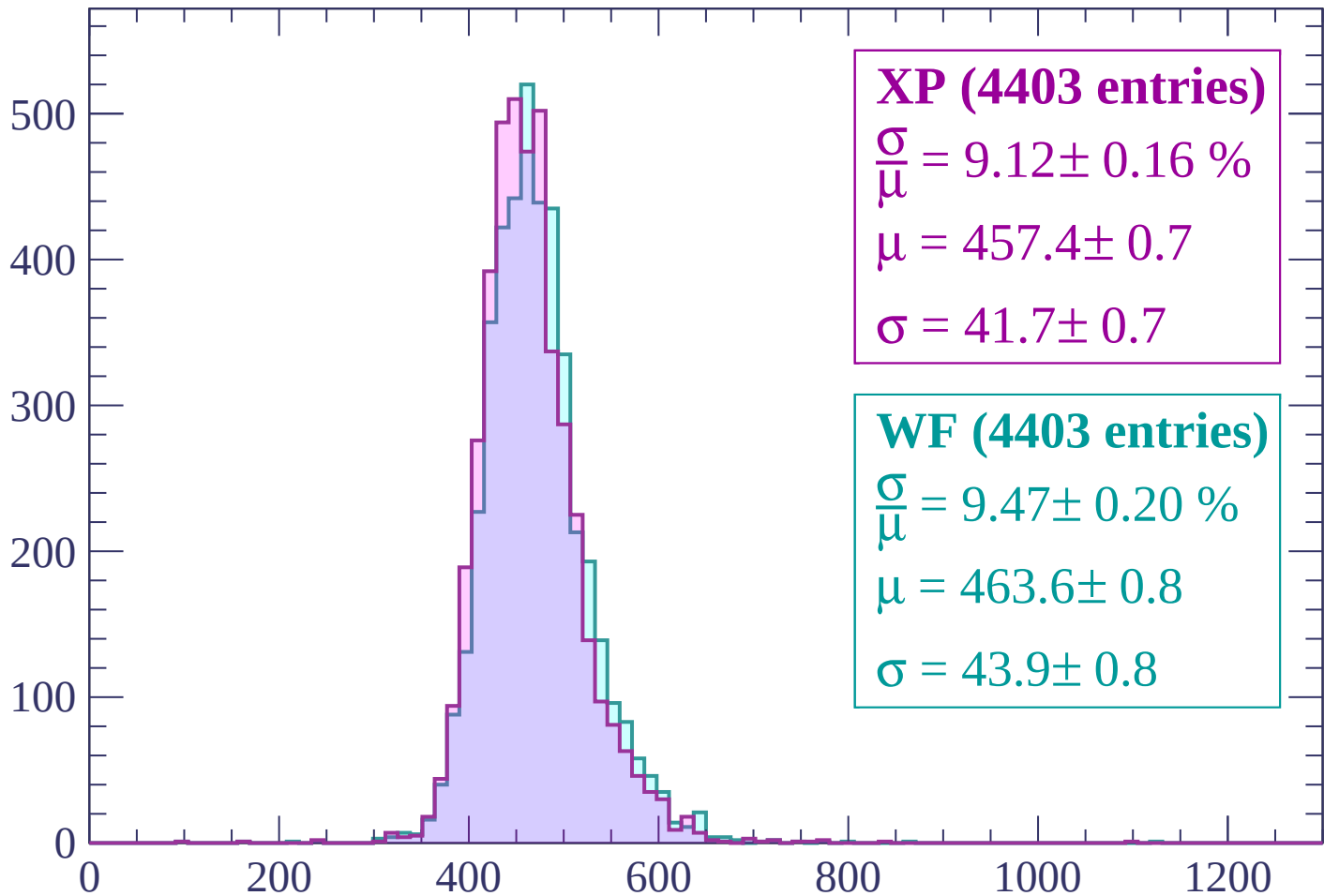
$$\mu = 462.0 \pm 0.8$$

$$\sigma = 42.6 \pm 0.7$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1620 < p < 1680$

Count



**XP (4403 entries)**

$\frac{\sigma}{\mu} = 9.12 \pm 0.16 \%$

$\mu = 457.4 \pm 0.7$

$\sigma = 41.7 \pm 0.7$

**WF (4403 entries)**

$\frac{\sigma}{\mu} = 9.47 \pm 0.20 \%$

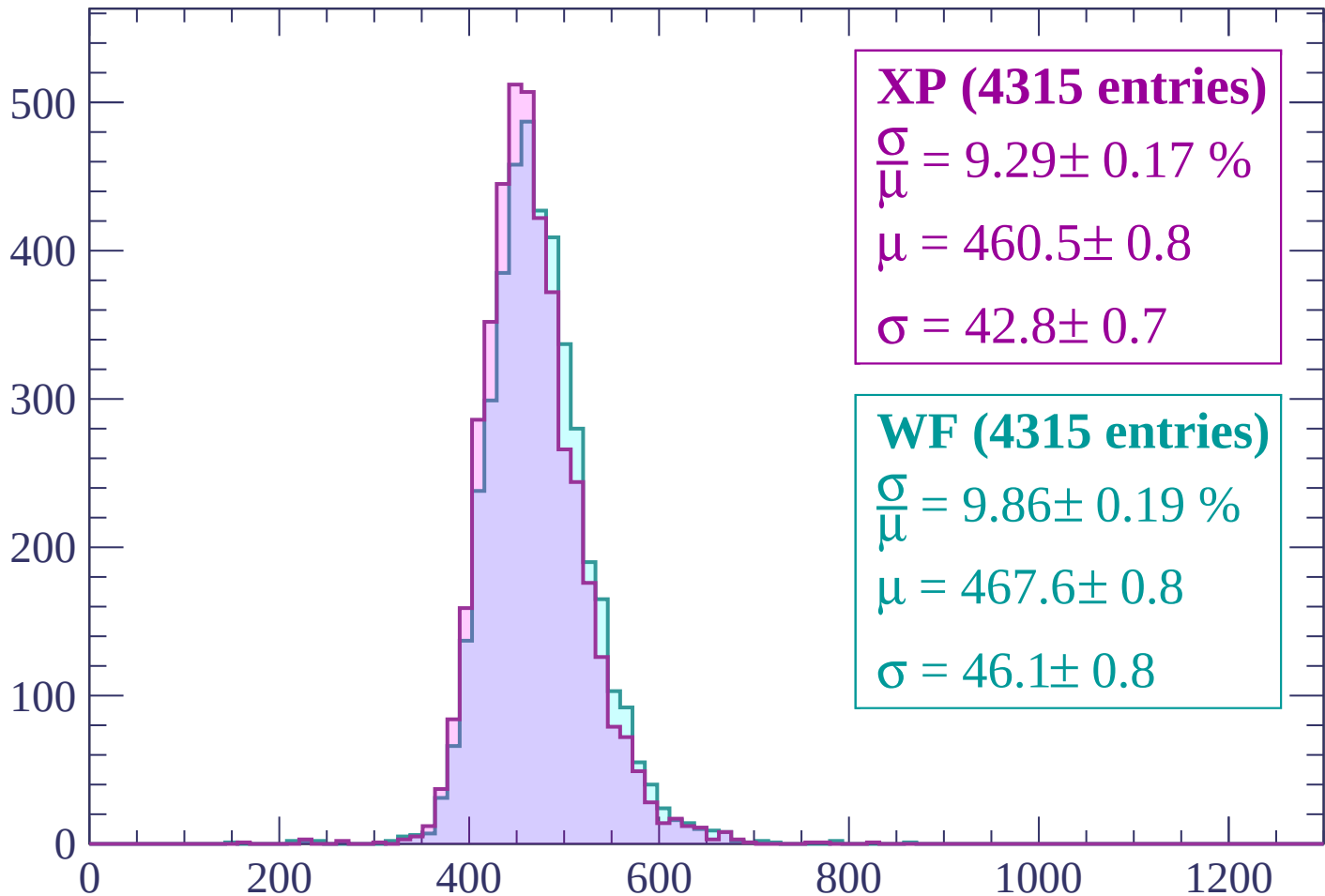
$\mu = 463.6 \pm 0.8$

$\sigma = 43.9 \pm 0.8$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1680 < p < 1740$

Count



**XP (4315 entries)**

$$\frac{\sigma}{\mu} = 9.29 \pm 0.17 \%$$

$$\mu = 460.5 \pm 0.8$$

$$\sigma = 42.8 \pm 0.7$$

**WF (4315 entries)**

$$\frac{\sigma}{\mu} = 9.86 \pm 0.19 \%$$

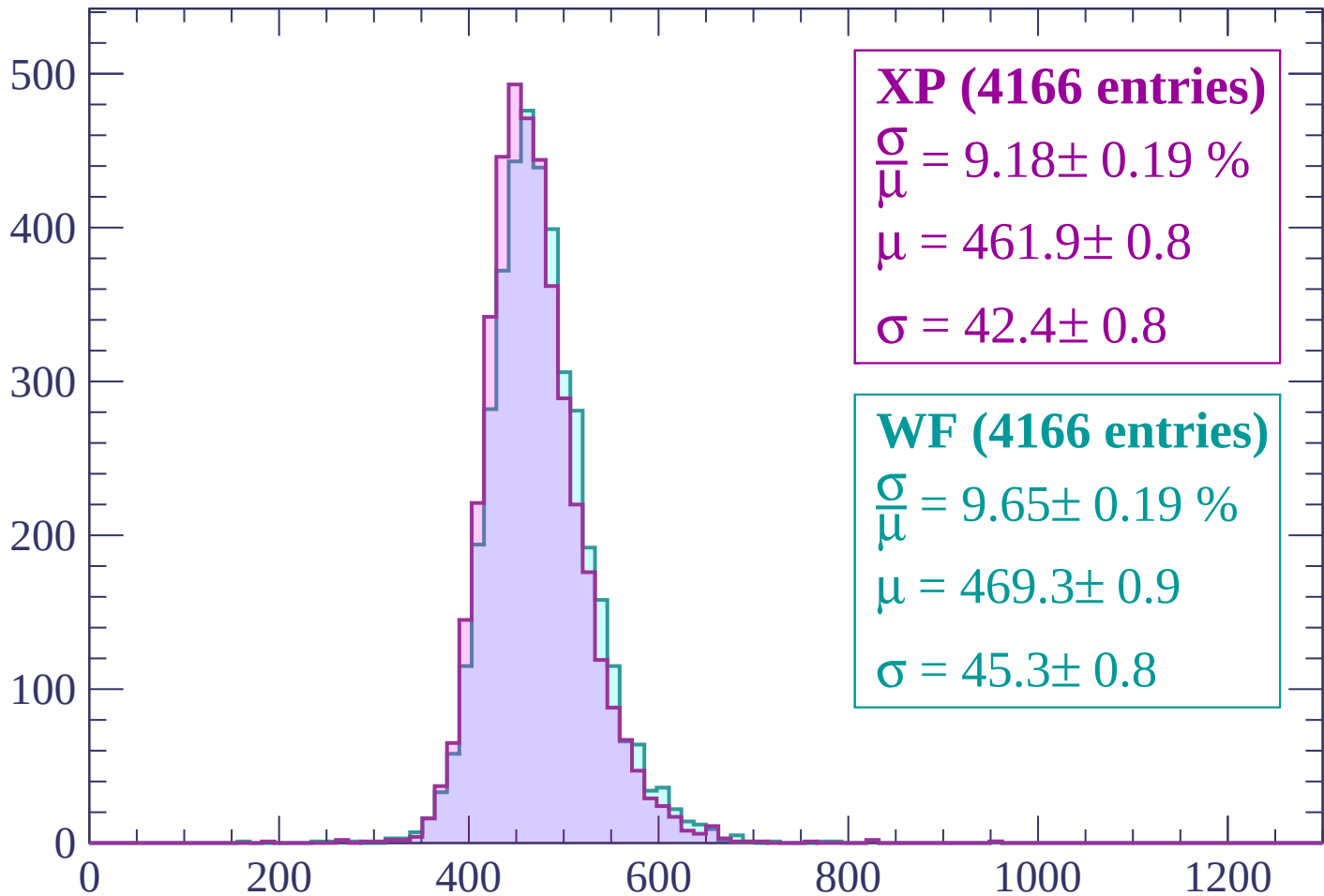
$$\mu = 467.6 \pm 0.8$$

$$\sigma = 46.1 \pm 0.8$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1740 < p < 1800$

Count



**XP (4166 entries)**

$$\frac{\sigma}{\mu} = 9.18 \pm 0.19 \%$$

$$\mu = 461.9 \pm 0.8$$

$$\sigma = 42.4 \pm 0.8$$

**WF (4166 entries)**

$$\frac{\sigma}{\mu} = 9.65 \pm 0.19 \%$$

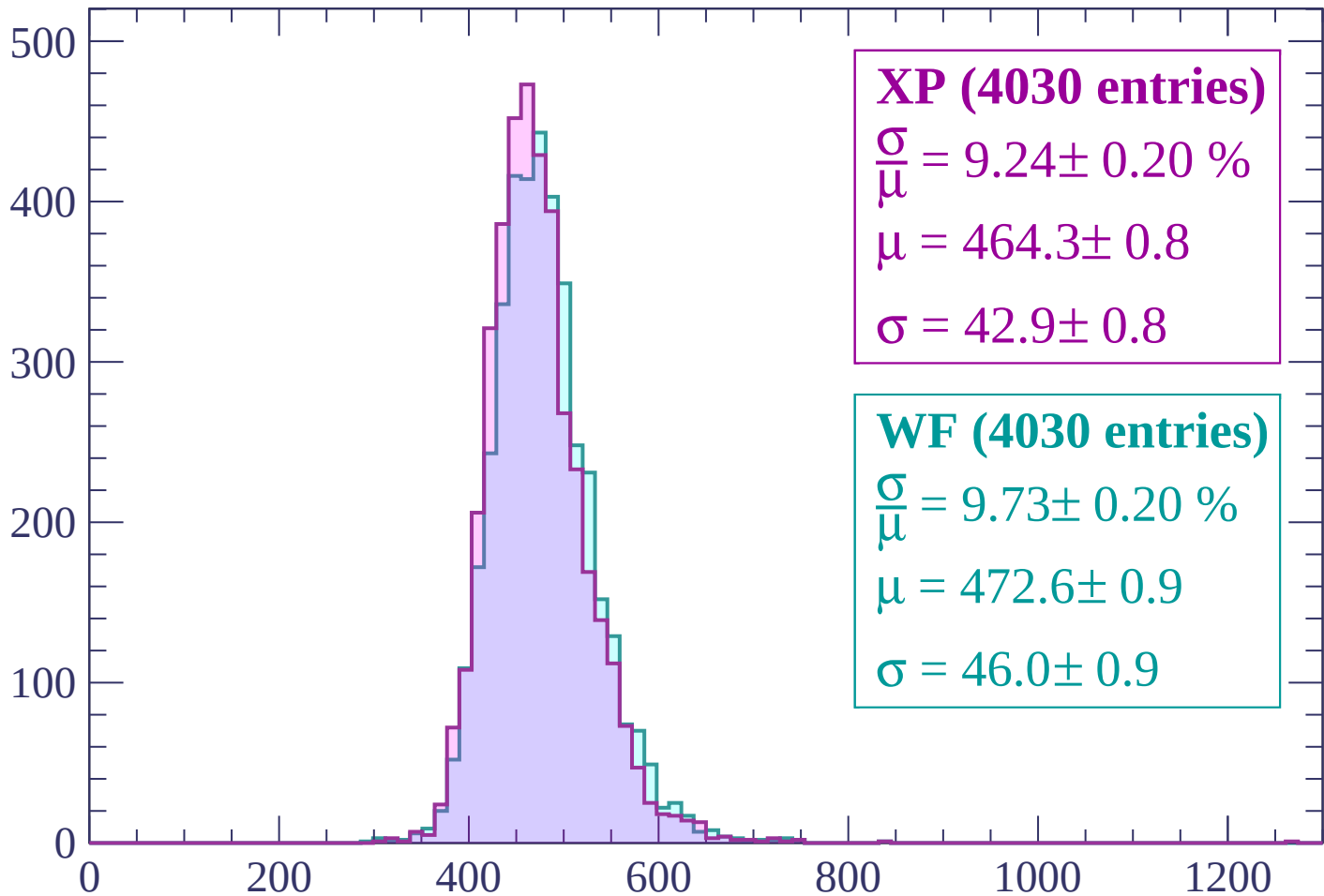
$$\mu = 469.3 \pm 0.9$$

$$\sigma = 45.3 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $1800 < p < 1860$

Count



**XP (4030 entries)**

$$\frac{\sigma}{\mu} = 9.24 \pm 0.20 \%$$

$$\mu = 464.3 \pm 0.8$$

$$\sigma = 42.9 \pm 0.8$$

**WF (4030 entries)**

$$\frac{\sigma}{\mu} = 9.73 \pm 0.20 \%$$

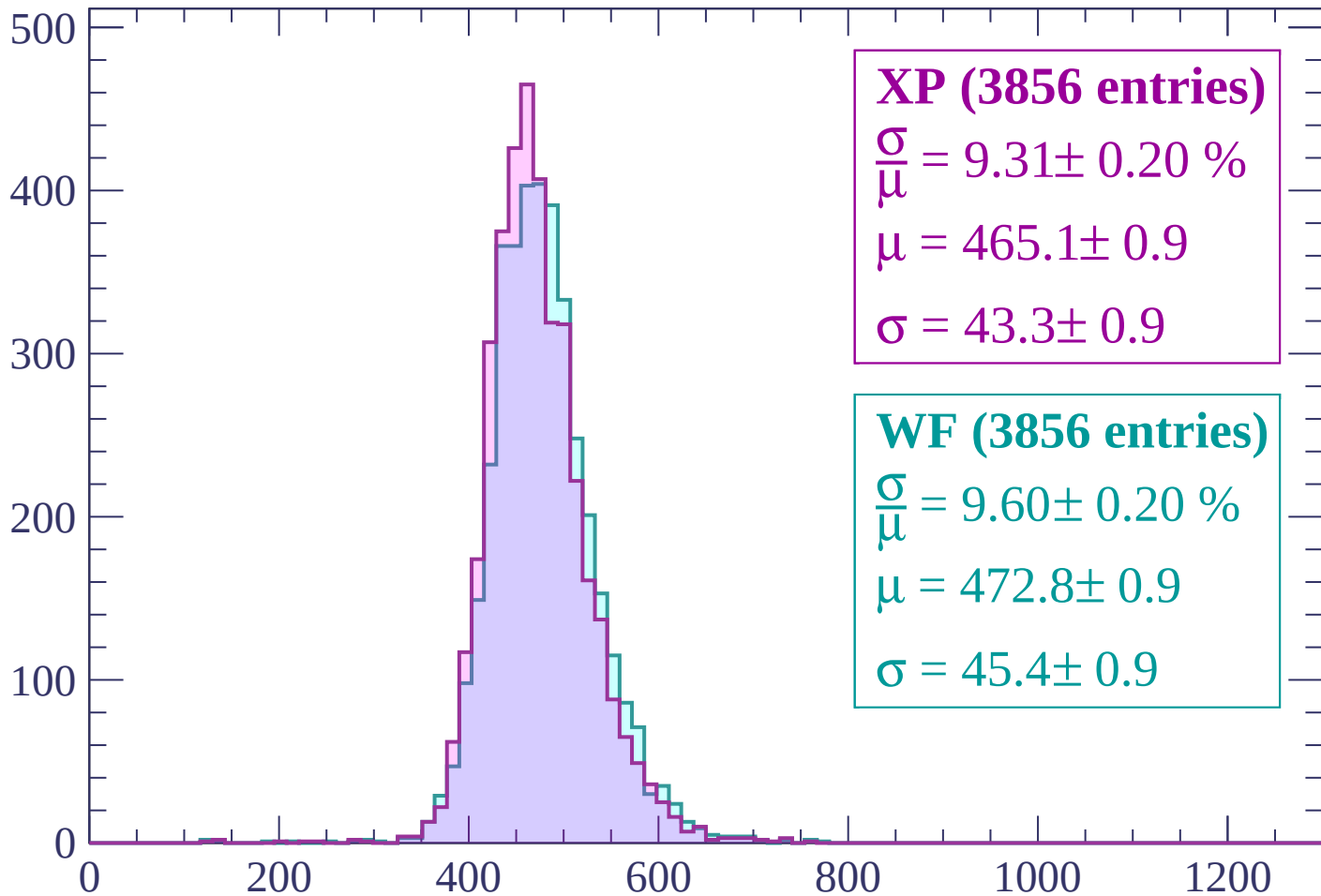
$$\mu = 472.6 \pm 0.9$$

$$\sigma = 46.0 \pm 0.9$$

$dE/dx$  (ADC counts/cm)

Energy loss |  $1860 < p < 1920$

Count



**XP (3856 entries)**

$$\frac{\sigma}{\mu} = 9.31 \pm 0.20 \%$$

$$\mu = 465.1 \pm 0.9$$

$$\sigma = 43.3 \pm 0.9$$

**WF (3856 entries)**

$$\frac{\sigma}{\mu} = 9.60 \pm 0.20 \%$$

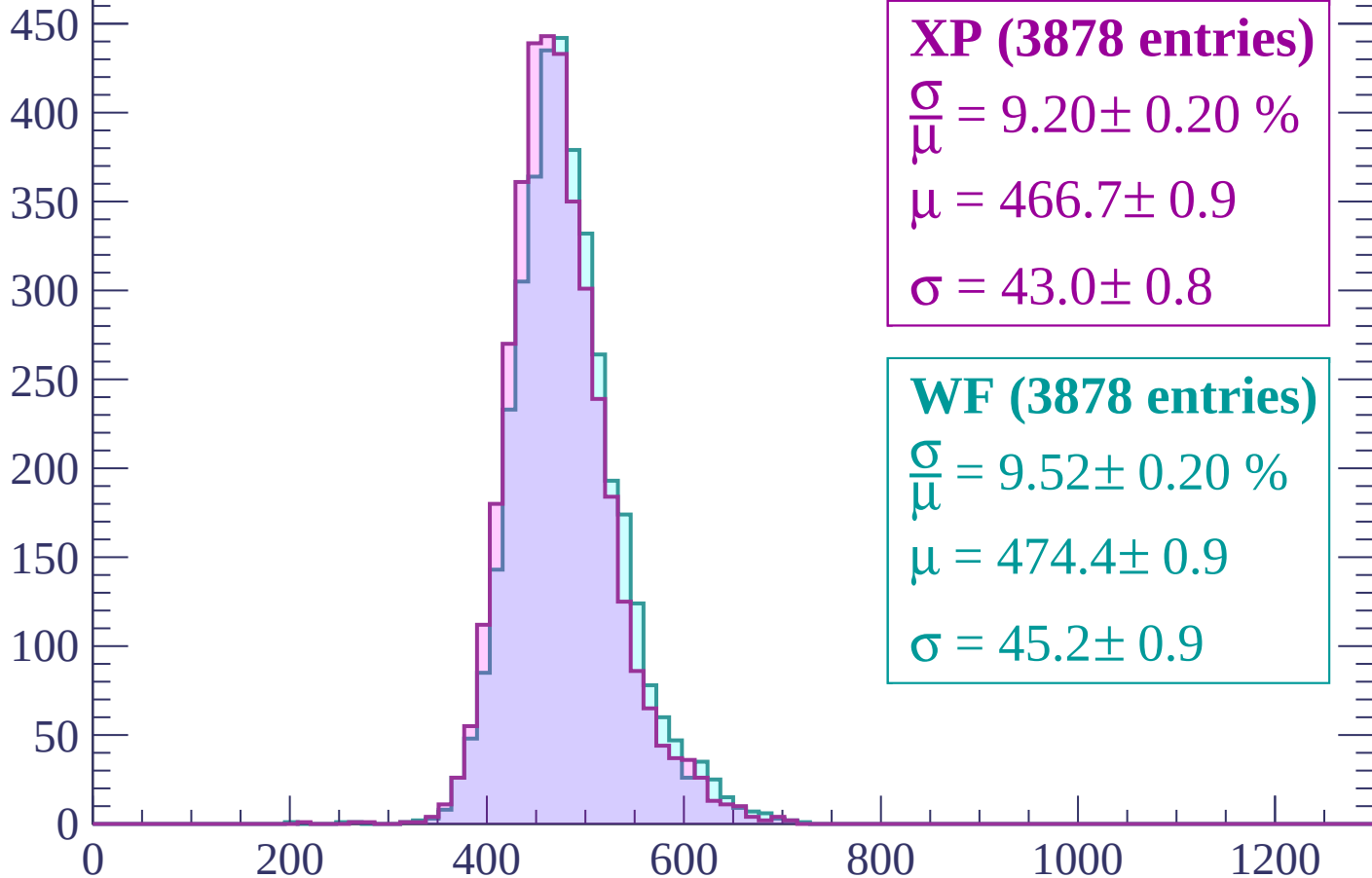
$$\mu = 472.8 \pm 0.9$$

$$\sigma = 45.4 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $1920 < p < 1980$

Count



**XP (3878 entries)**

$$\frac{\sigma}{\mu} = 9.20 \pm 0.20 \%$$

$$\mu = 466.7 \pm 0.9$$

$$\sigma = 43.0 \pm 0.8$$

**WF (3878 entries)**

$$\frac{\sigma}{\mu} = 9.52 \pm 0.20 \%$$

$$\mu = 474.4 \pm 0.9$$

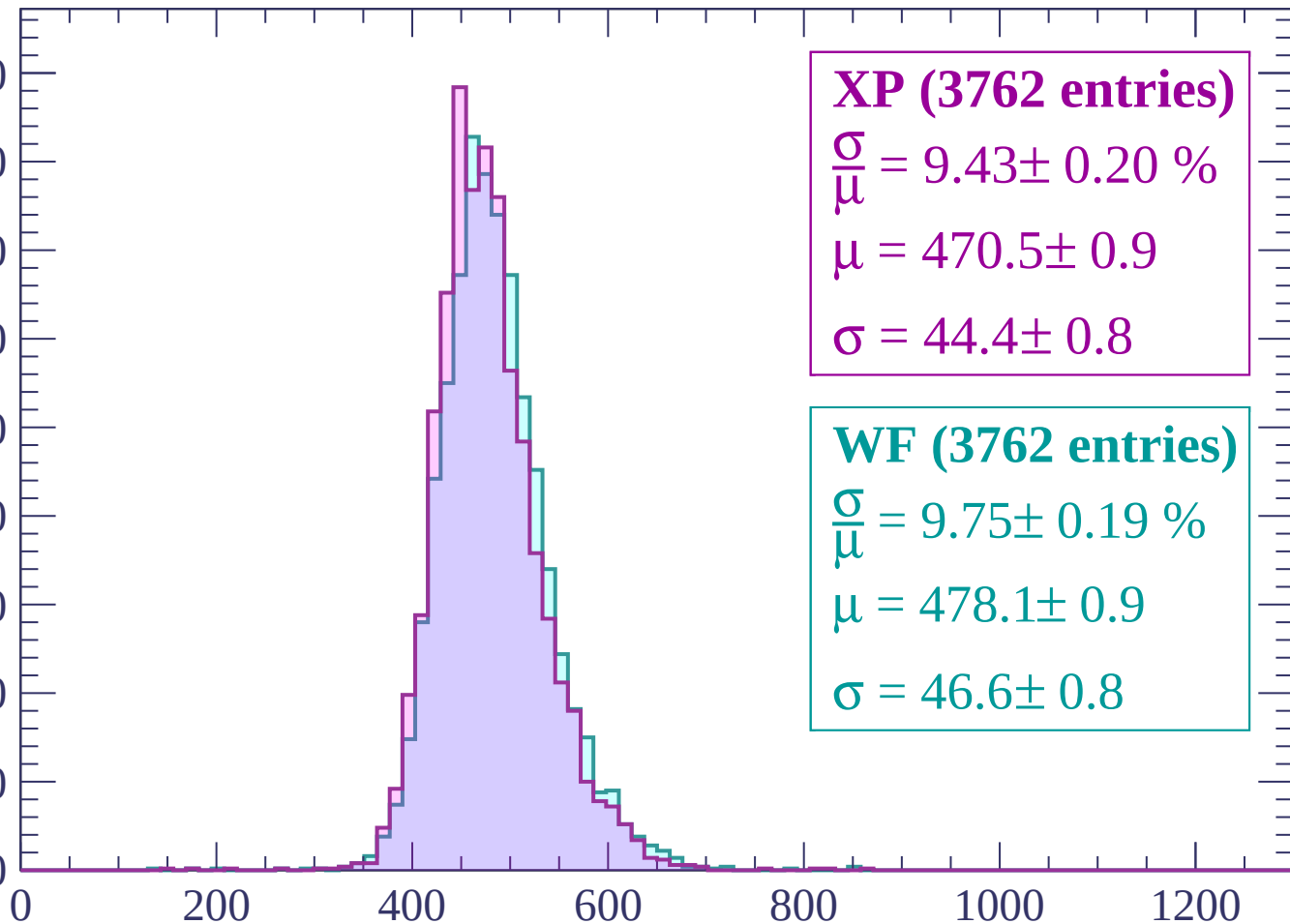
$$\sigma = 45.2 \pm 0.9$$

dE/dx (ADC counts/cm)

Energy loss |  $1980 < p < 2040$

Count

450  
400  
350  
300  
250  
200  
150  
100  
50  
0



**XP (3762 entries)**

$$\frac{\sigma}{\mu} = 9.43 \pm 0.20 \%$$

$$\mu = 470.5 \pm 0.9$$

$$\sigma = 44.4 \pm 0.8$$

**WF (3762 entries)**

$$\frac{\sigma}{\mu} = 9.75 \pm 0.19 \%$$

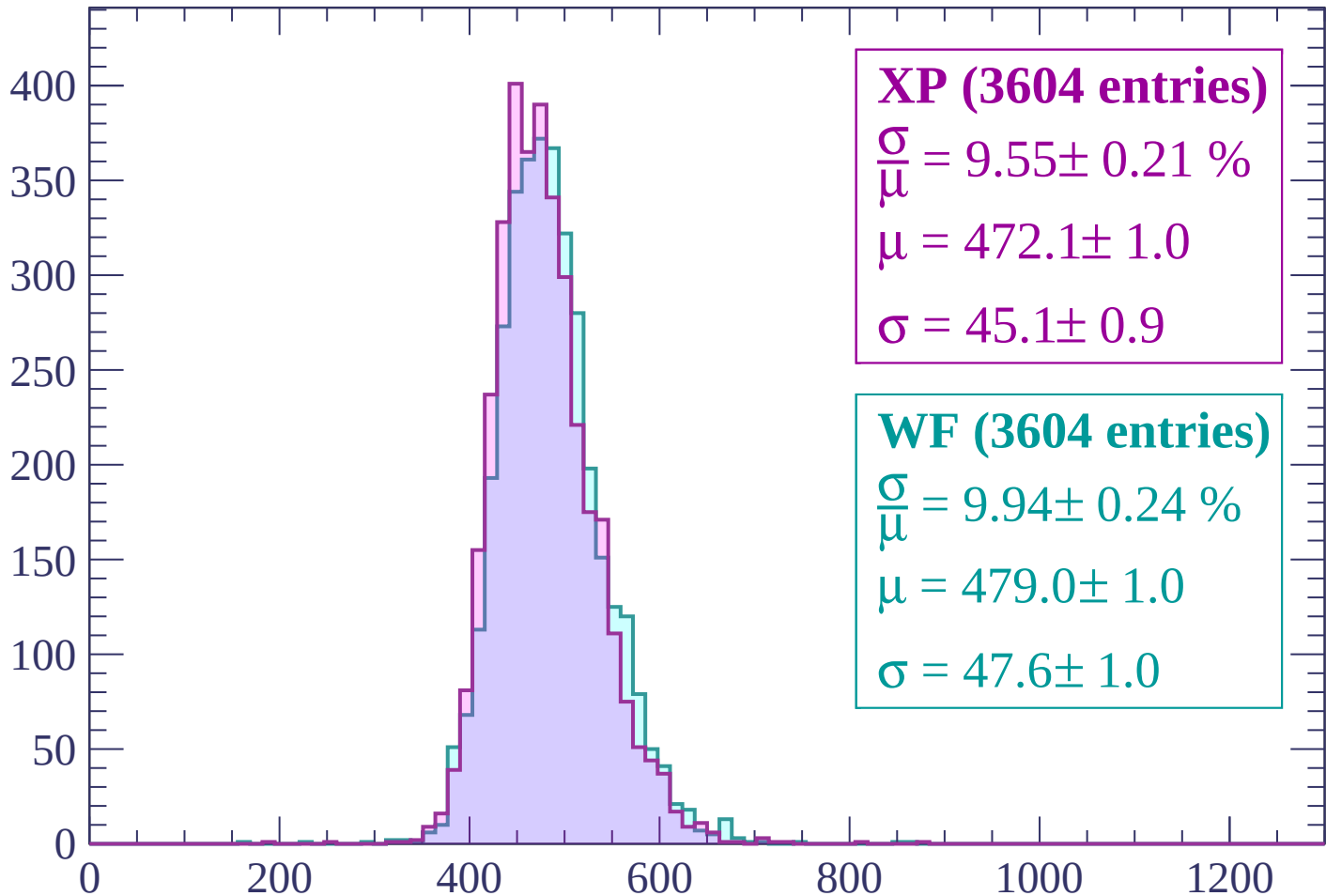
$$\mu = 478.1 \pm 0.9$$

$$\sigma = 46.6 \pm 0.8$$

dE/dx (ADC counts/cm)

Energy loss |  $2040 < p < 2100$

Count



**XP (3604 entries)**

$$\frac{\sigma}{\mu} = 9.55 \pm 0.21 \%$$

$$\mu = 472.1 \pm 1.0$$

$$\sigma = 45.1 \pm 0.9$$

**WF (3604 entries)**

$$\frac{\sigma}{\mu} = 9.94 \pm 0.24 \%$$

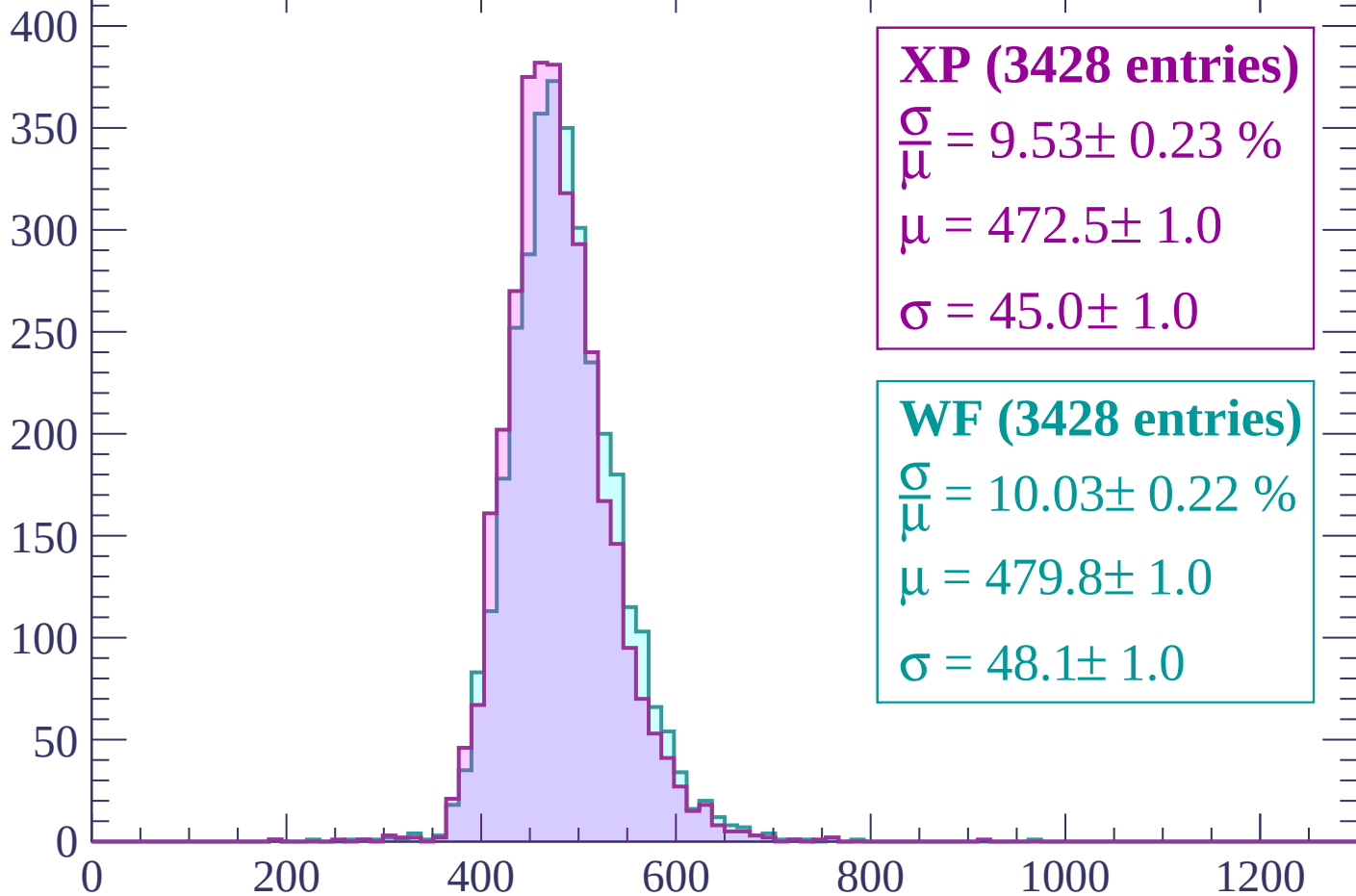
$$\mu = 479.0 \pm 1.0$$

$$\sigma = 47.6 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $2100 < p < 2160$

Count



**XP (3428 entries)**

$$\frac{\sigma}{\mu} = 9.53 \pm 0.23 \%$$

$$\mu = 472.5 \pm 1.0$$

$$\sigma = 45.0 \pm 1.0$$

**WF (3428 entries)**

$$\frac{\sigma}{\mu} = 10.03 \pm 0.22 \%$$

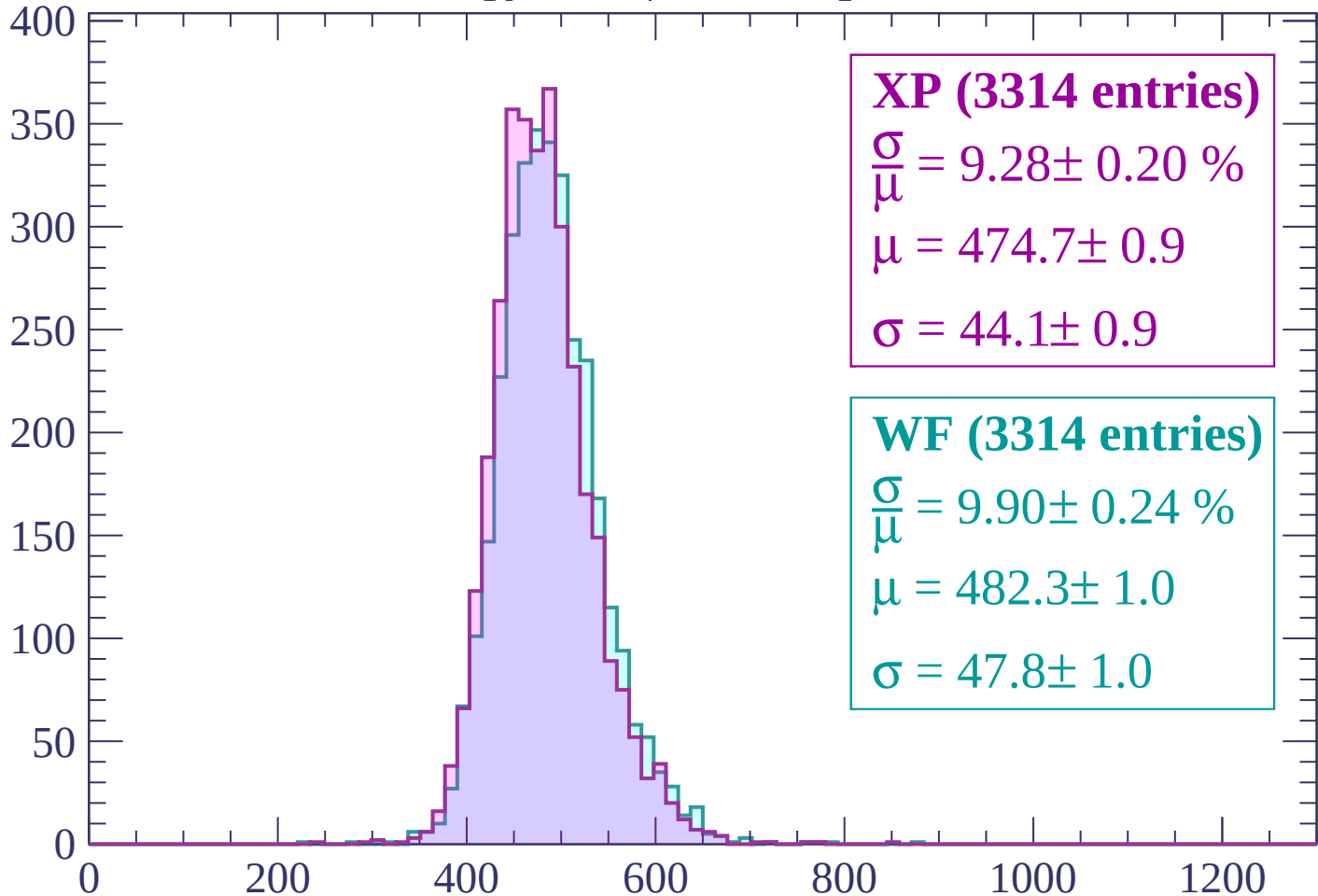
$$\mu = 479.8 \pm 1.0$$

$$\sigma = 48.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $2160 < p < 2220$

Count



**XP (3314 entries)**

$$\frac{\sigma}{\mu} = 9.28 \pm 0.20 \%$$

$$\mu = 474.7 \pm 0.9$$

$$\sigma = 44.1 \pm 0.9$$

**WF (3314 entries)**

$$\frac{\sigma}{\mu} = 9.90 \pm 0.24 \%$$

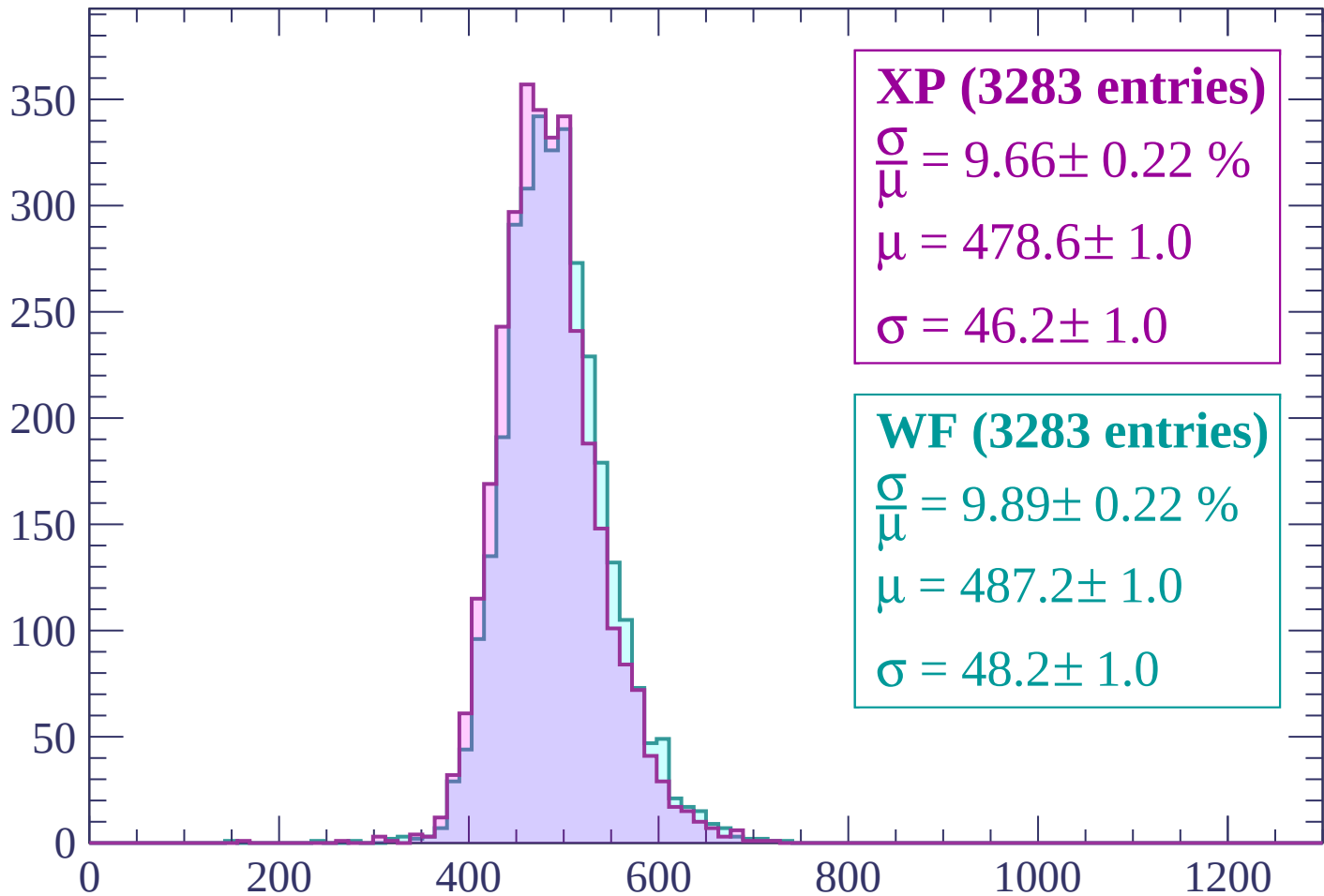
$$\mu = 482.3 \pm 1.0$$

$$\sigma = 47.8 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $2220 < p < 2280$

Count



**XP (3283 entries)**

$$\frac{\sigma}{\mu} = 9.66 \pm 0.22 \%$$

$$\mu = 478.6 \pm 1.0$$

$$\sigma = 46.2 \pm 1.0$$

**WF (3283 entries)**

$$\frac{\sigma}{\mu} = 9.89 \pm 0.22 \%$$

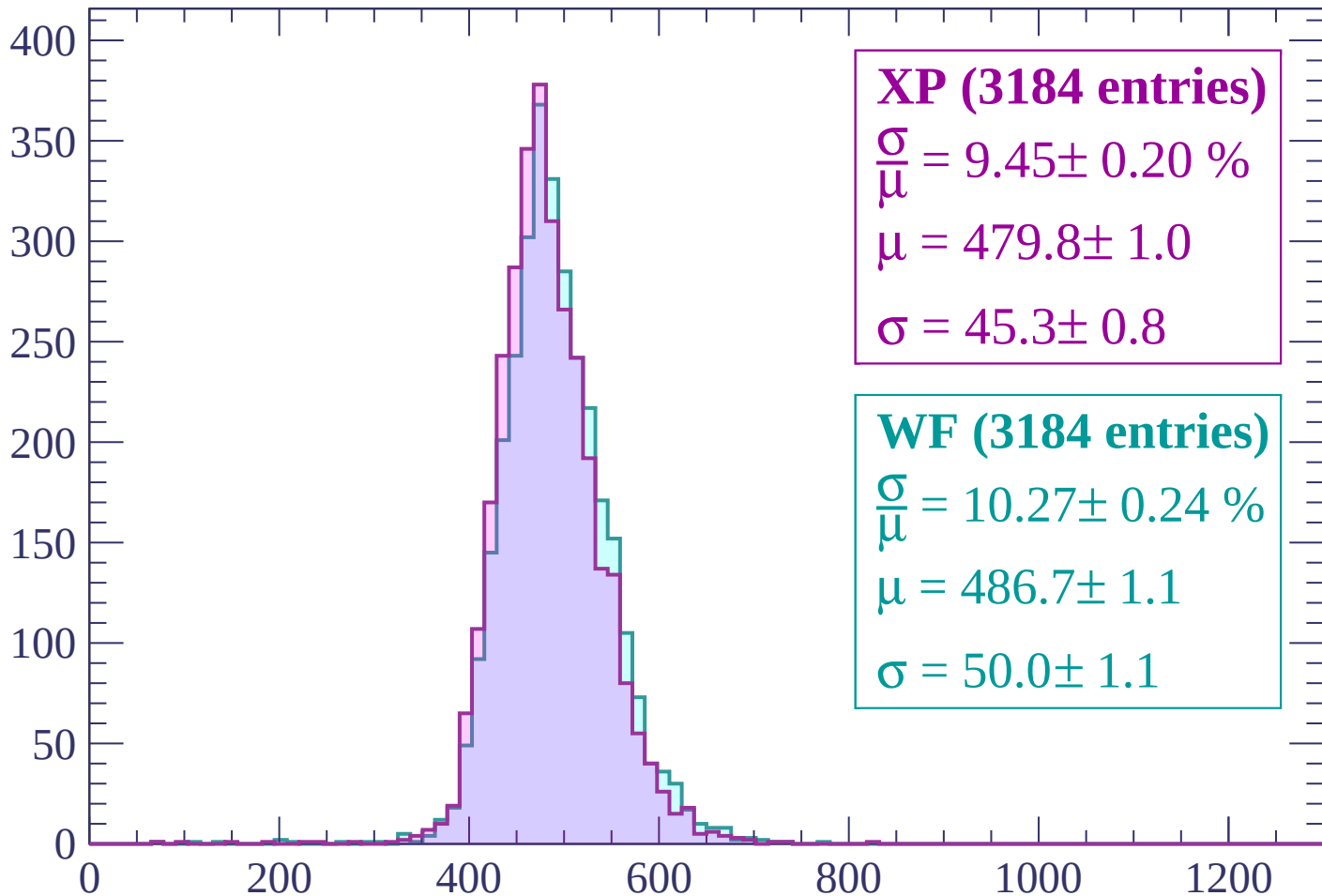
$$\mu = 487.2 \pm 1.0$$

$$\sigma = 48.2 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $2280 < p < 2340$

Count



**XP (3184 entries)**

$\frac{\sigma}{\mu} = 9.45 \pm 0.20 \%$

$\mu = 479.8 \pm 1.0$

$\sigma = 45.3 \pm 0.8$

**WF (3184 entries)**

$\frac{\sigma}{\mu} = 10.27 \pm 0.24 \%$

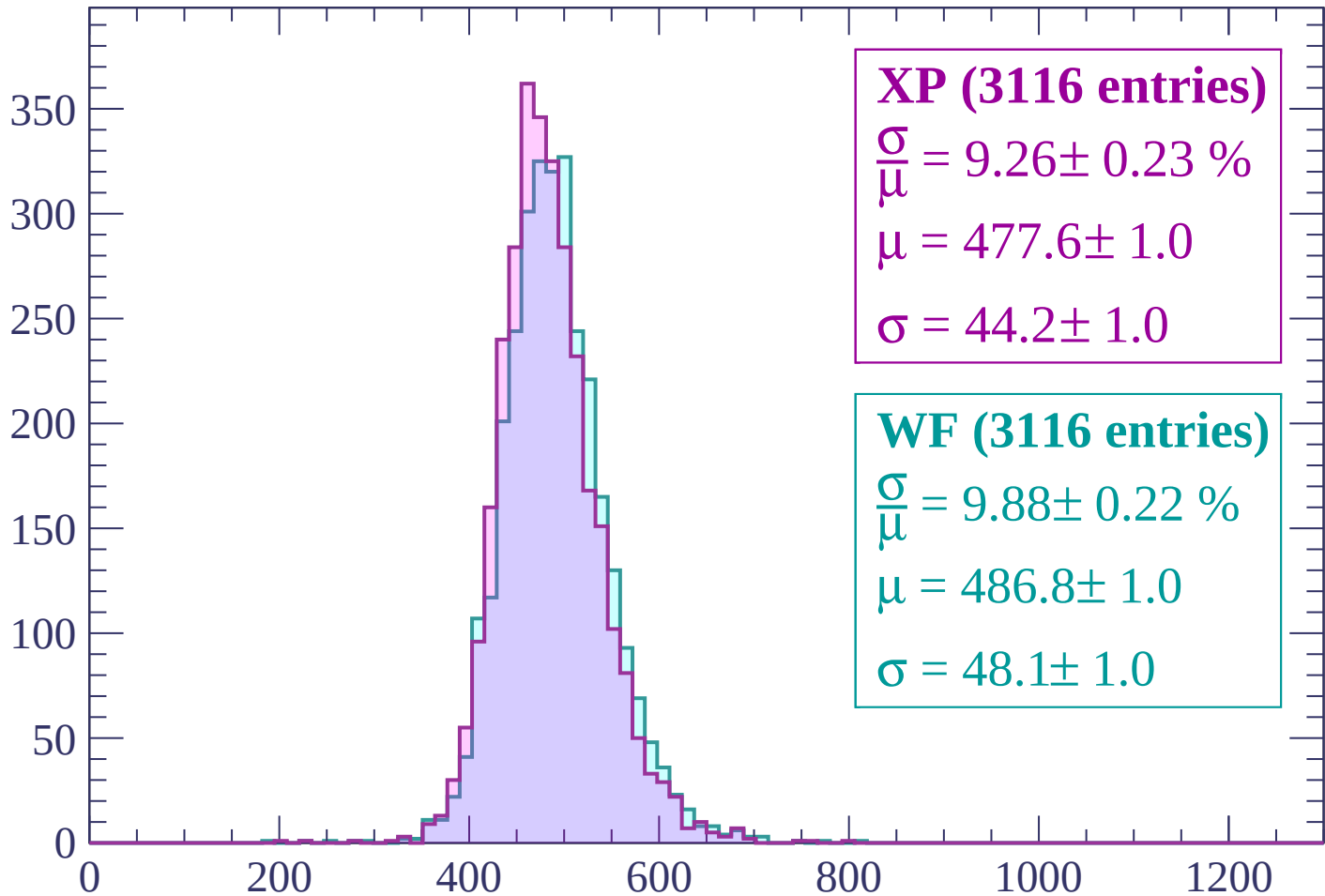
$\mu = 486.7 \pm 1.1$

$\sigma = 50.0 \pm 1.1$

dE/dx (ADC counts/cm)

Energy loss |  $2340 < p < 2400$

Count



**XP (3116 entries)**

$$\frac{\sigma}{\mu} = 9.26 \pm 0.23 \%$$

$$\mu = 477.6 \pm 1.0$$

$$\sigma = 44.2 \pm 1.0$$

**WF (3116 entries)**

$$\frac{\sigma}{\mu} = 9.88 \pm 0.22 \%$$

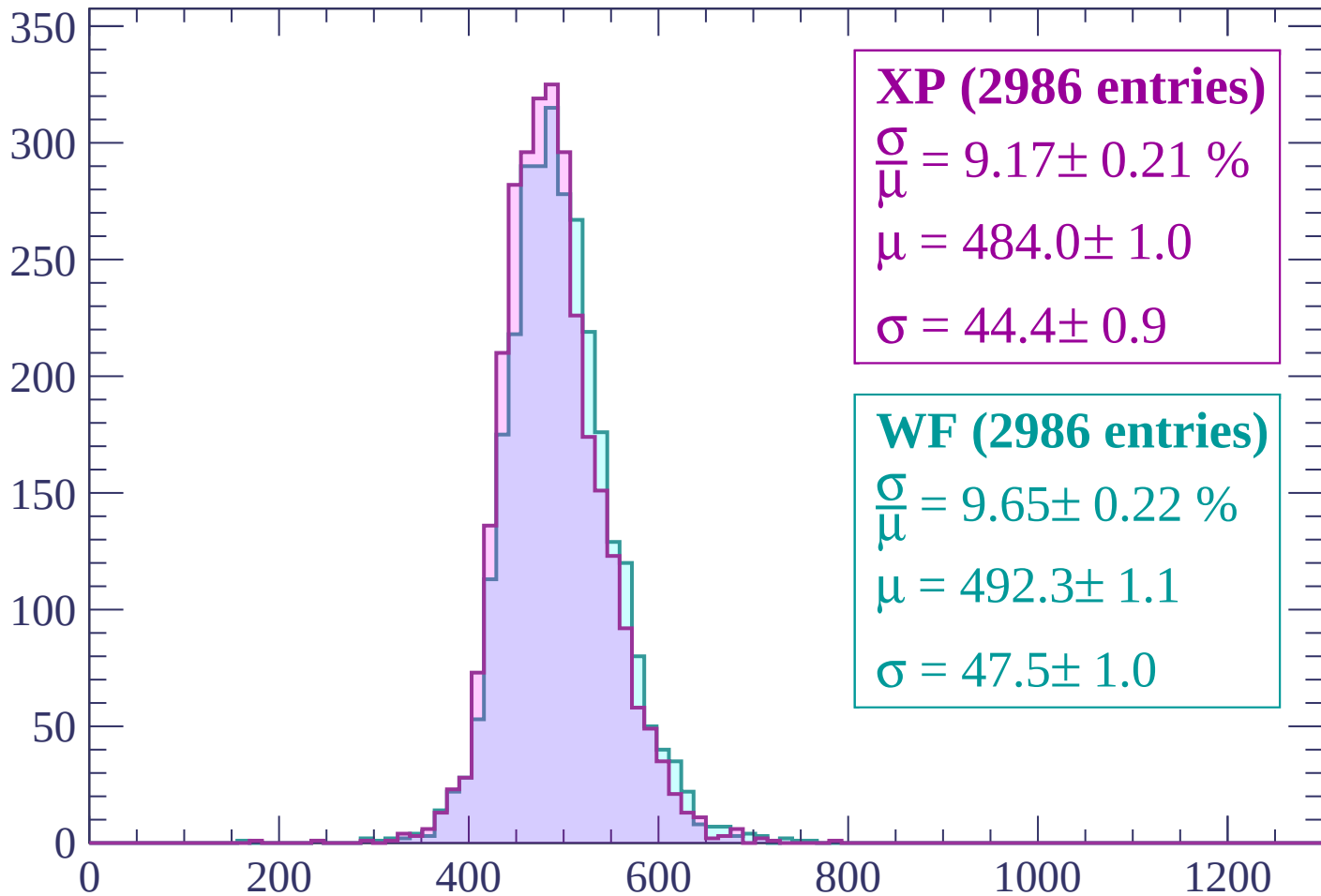
$$\mu = 486.8 \pm 1.0$$

$$\sigma = 48.1 \pm 1.0$$

dE/dx (ADC counts/cm)

Energy loss |  $2400 < p < 2460$

Count



**XP (2986 entries)**

$$\frac{\sigma}{\mu} = 9.17 \pm 0.21 \%$$

$$\mu = 484.0 \pm 1.0$$

$$\sigma = 44.4 \pm 0.9$$

**WF (2986 entries)**

$$\frac{\sigma}{\mu} = 9.65 \pm 0.22 \%$$

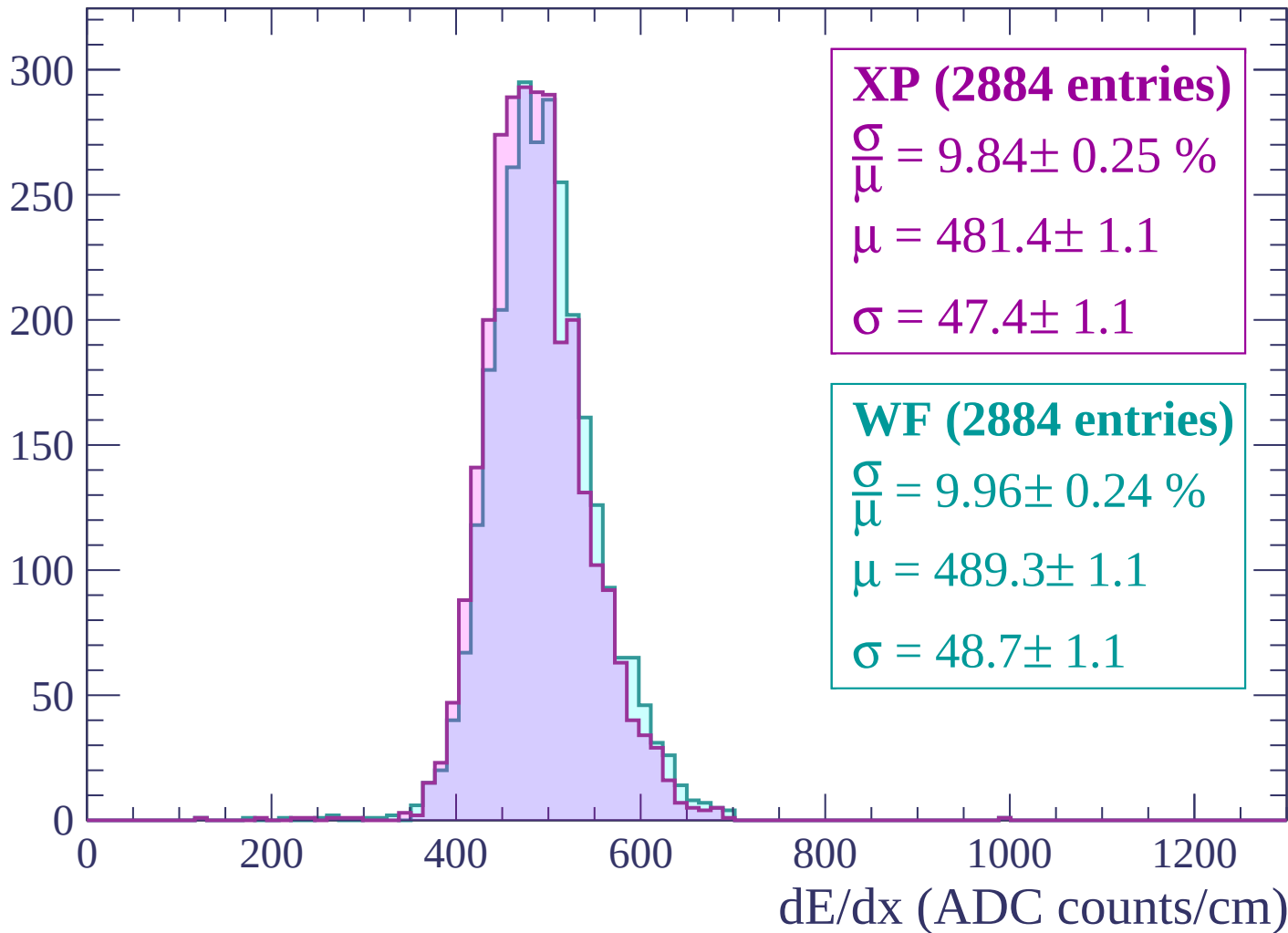
$$\mu = 492.3 \pm 1.1$$

$$\sigma = 47.5 \pm 1.0$$

dE/dx (ADC counts/cm)

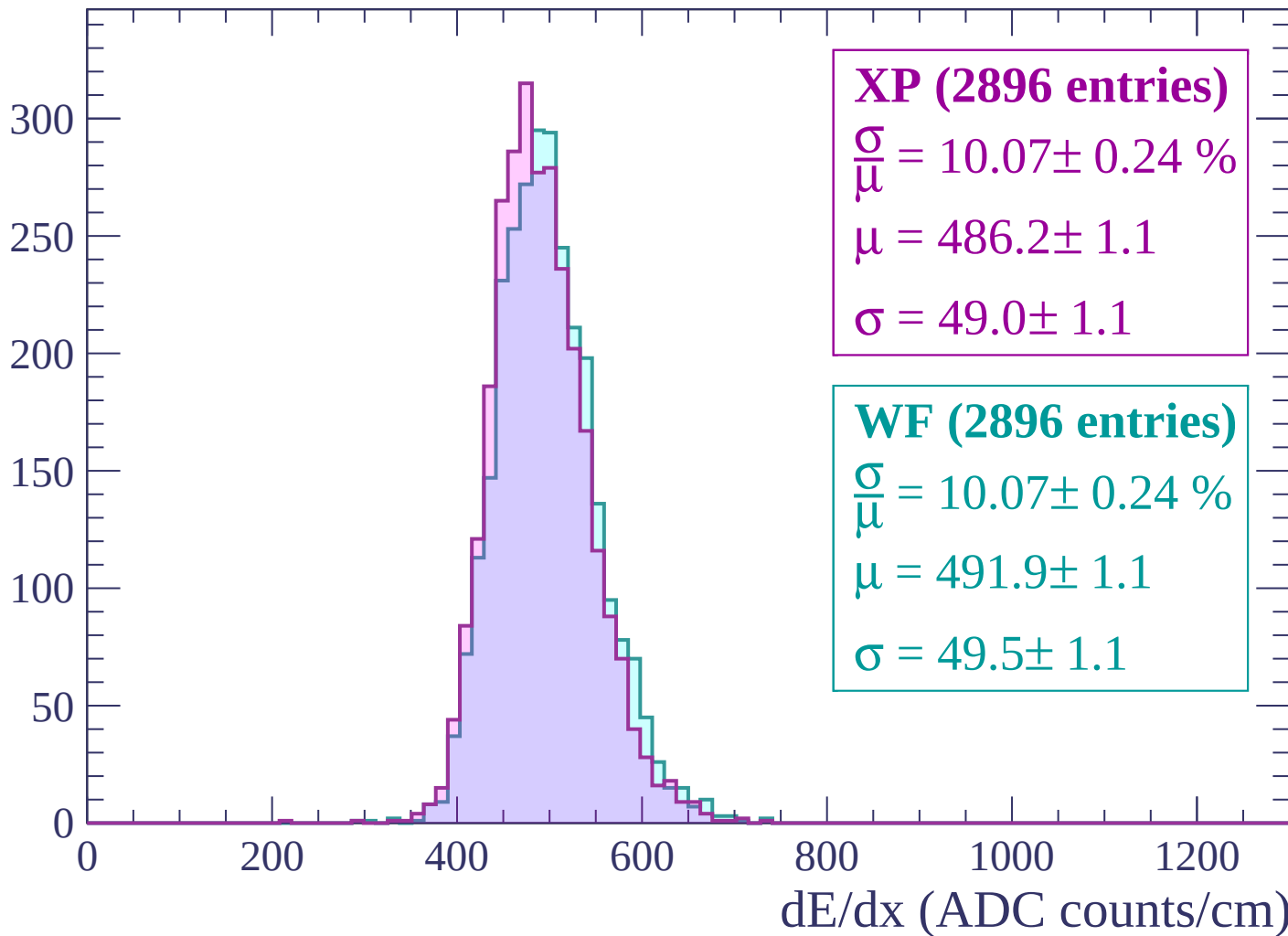
Energy loss |  $2460 < p < 2520$

Count



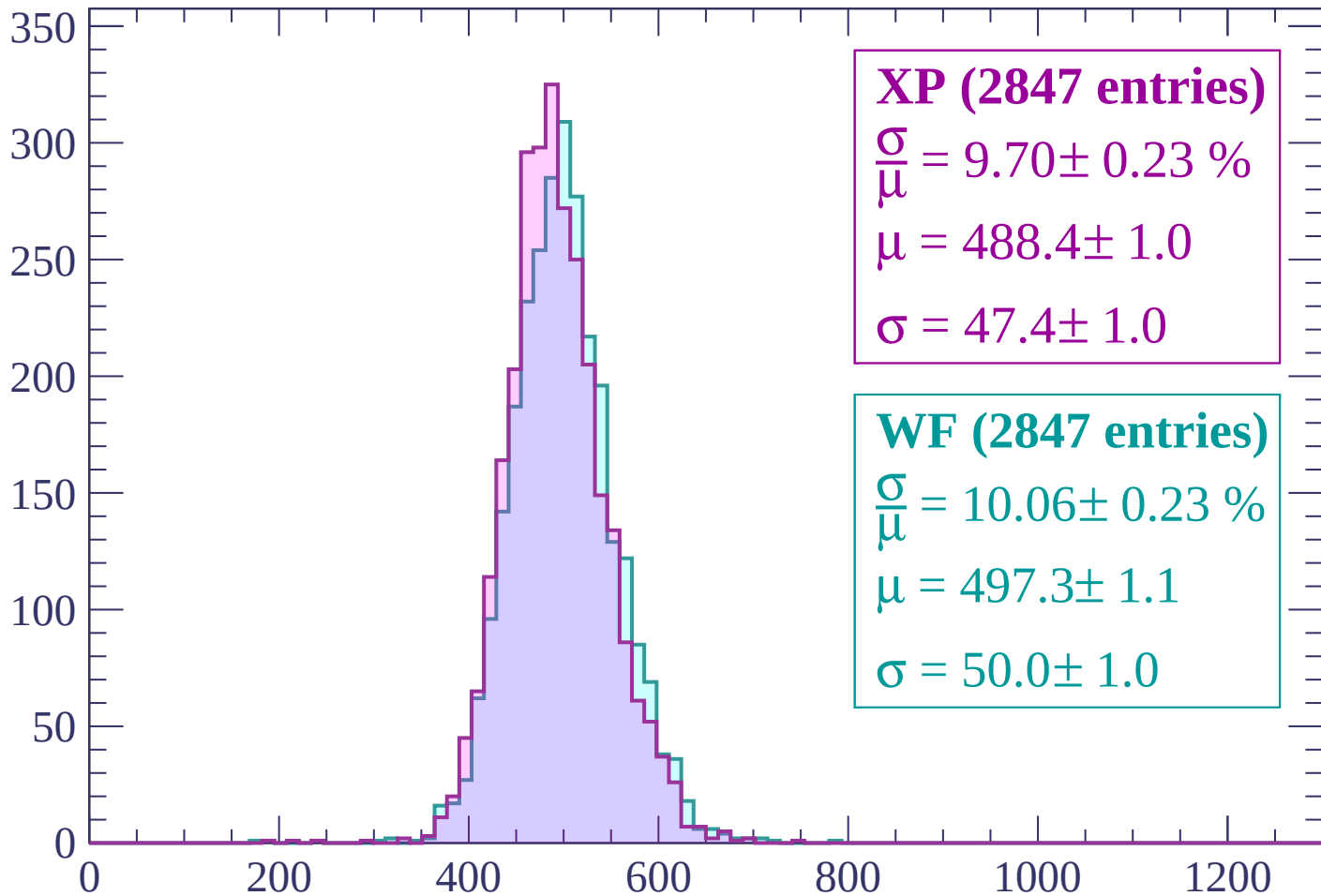
Energy loss |  $2520 < p < 2580$

Count



Energy loss |  $2580 < p < 2640$

Count



**XP (2847 entries)**

$$\frac{\sigma}{\mu} = 9.70 \pm 0.23 \%$$

$$\mu = 488.4 \pm 1.0$$

$$\sigma = 47.4 \pm 1.0$$

**WF (2847 entries)**

$$\frac{\sigma}{\mu} = 10.06 \pm 0.23 \%$$

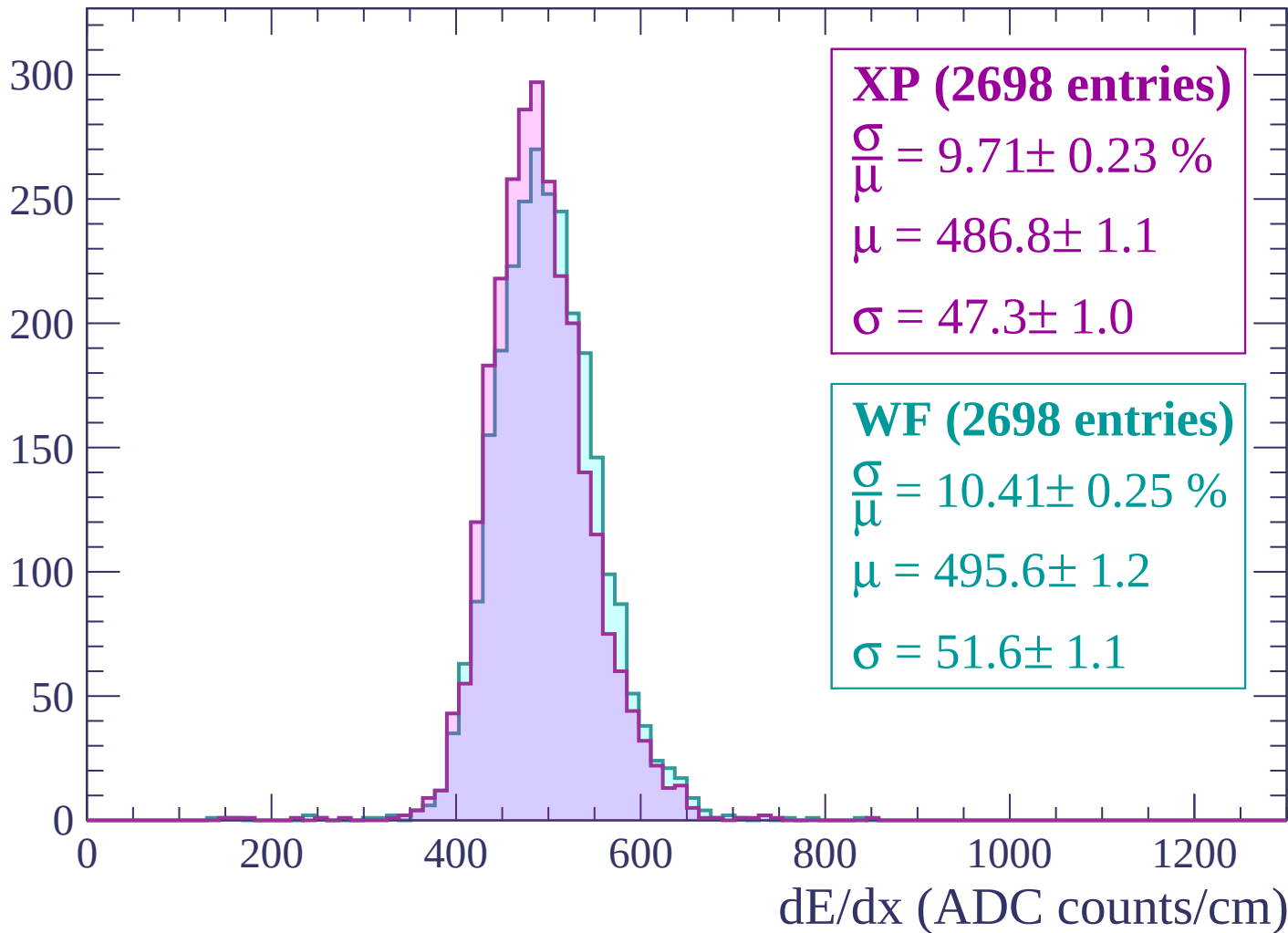
$$\mu = 497.3 \pm 1.1$$

$$\sigma = 50.0 \pm 1.0$$

dE/dx (ADC counts/cm)

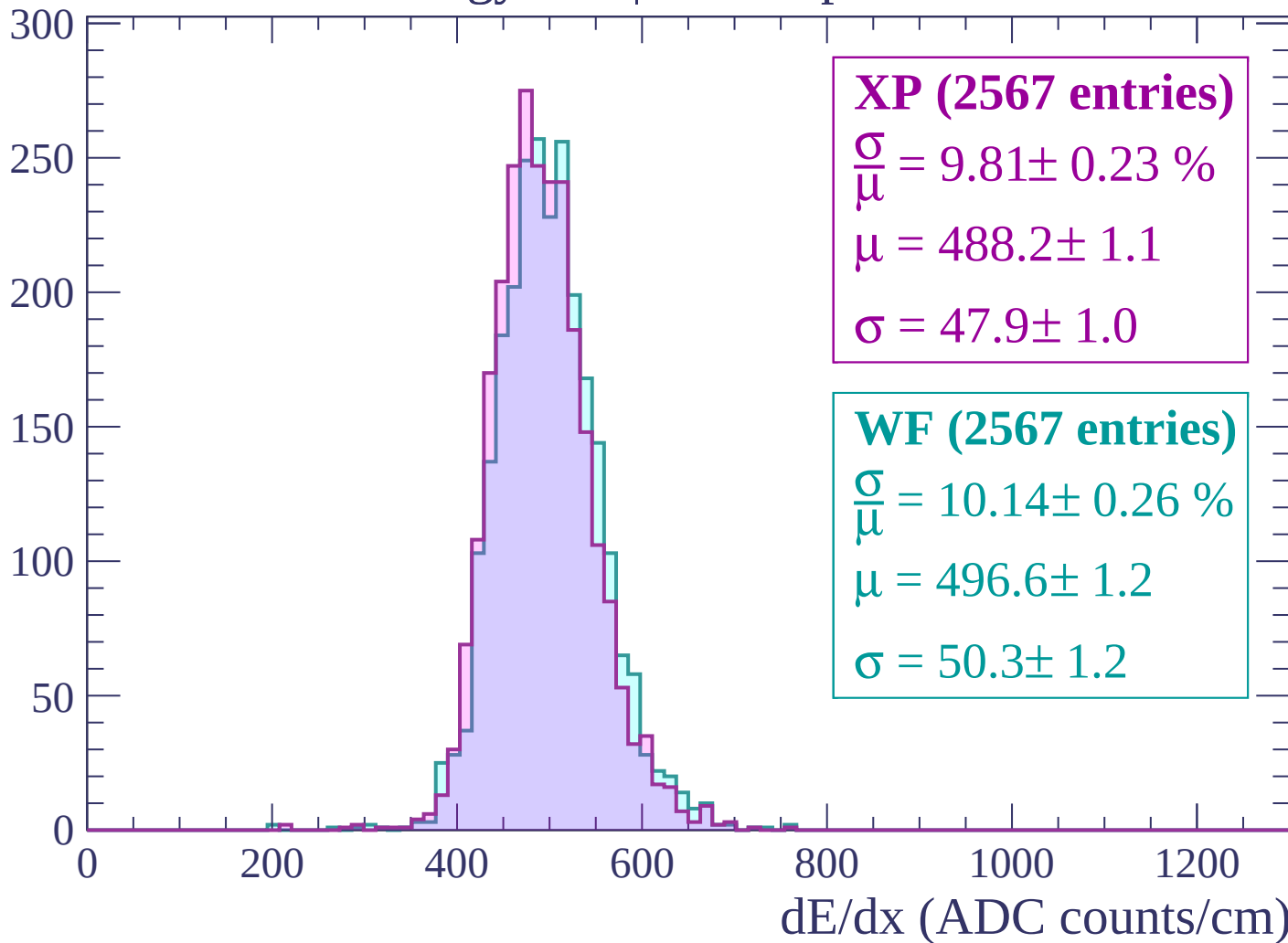
Energy loss |  $2640 < p < 2700$

Count



Energy loss |  $2700 < p < 2760$

Count



Energy loss |  $2760 < p < 2820$

Count

300

250

200

150

100

50

0

0

200

400

600

800

1000

1200

dE/dx (ADC counts/cm)

**XP (2491 entries)**

$\frac{\sigma}{\mu} = 9.53 \pm 0.26 \%$

$\mu = 490.5 \pm 1.1$

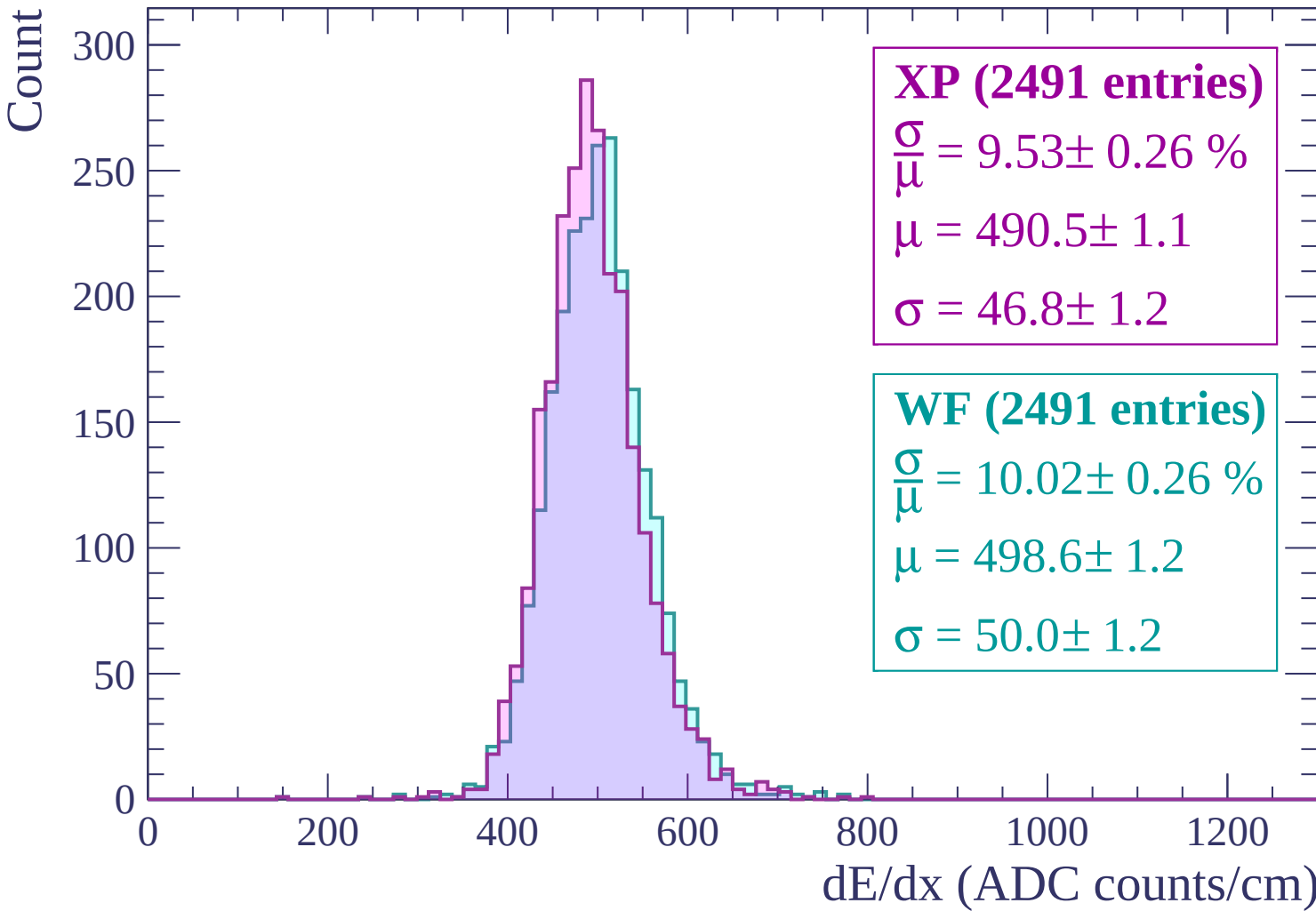
$\sigma = 46.8 \pm 1.2$

**WF (2491 entries)**

$\frac{\sigma}{\mu} = 10.02 \pm 0.26 \%$

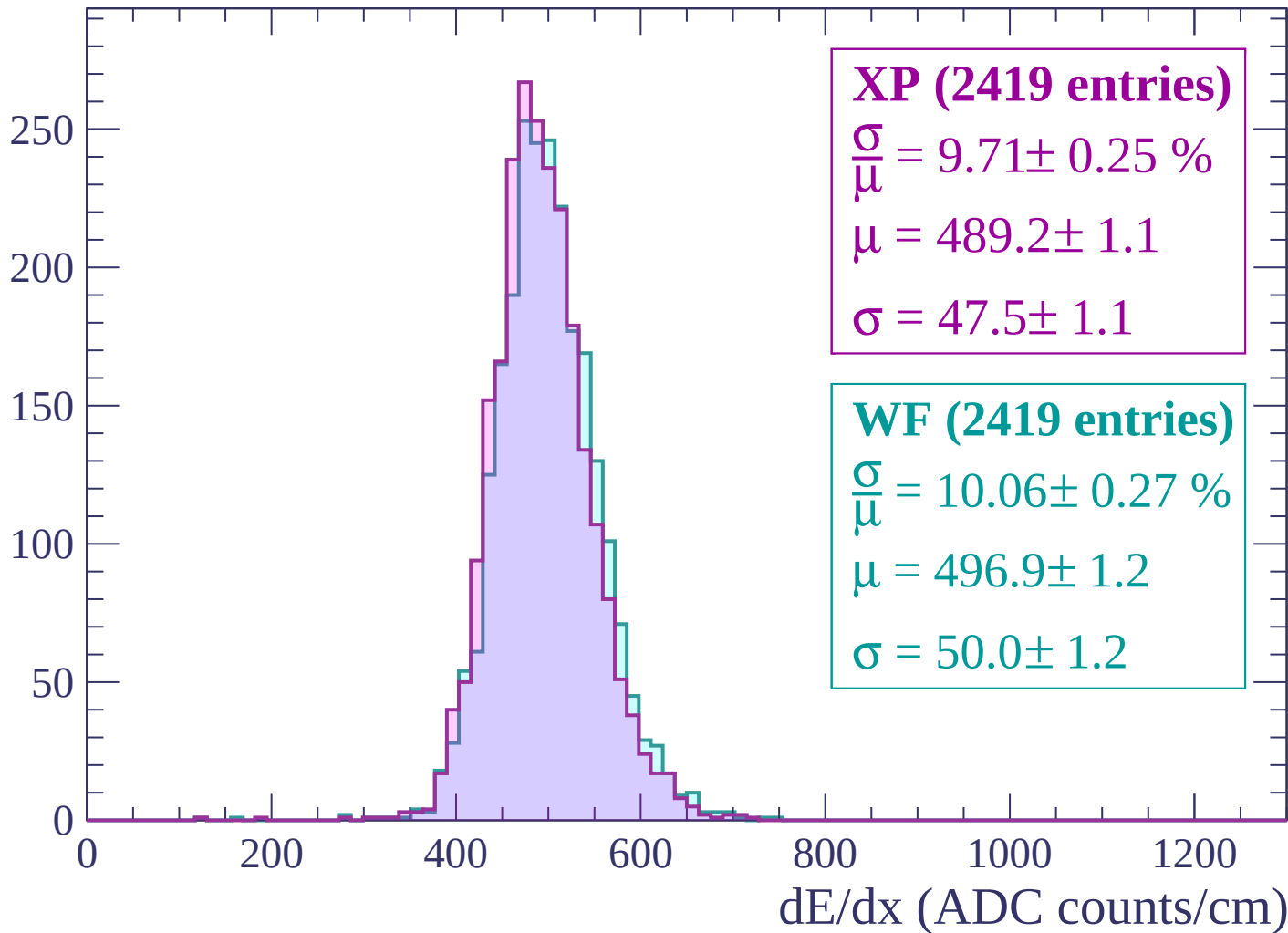
$\mu = 498.6 \pm 1.2$

$\sigma = 50.0 \pm 1.2$



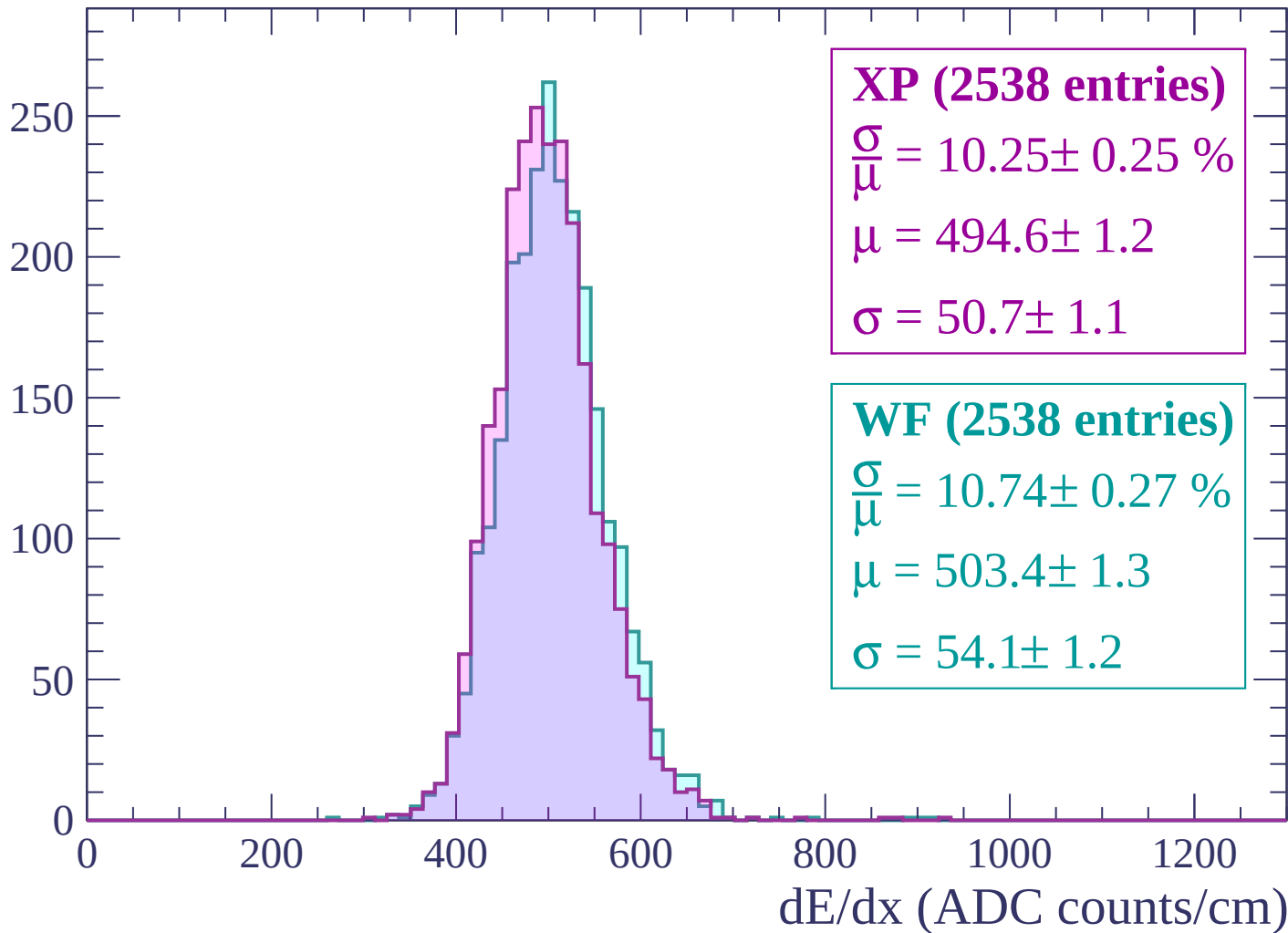
Energy loss |  $2820 < p < 2880$

Count



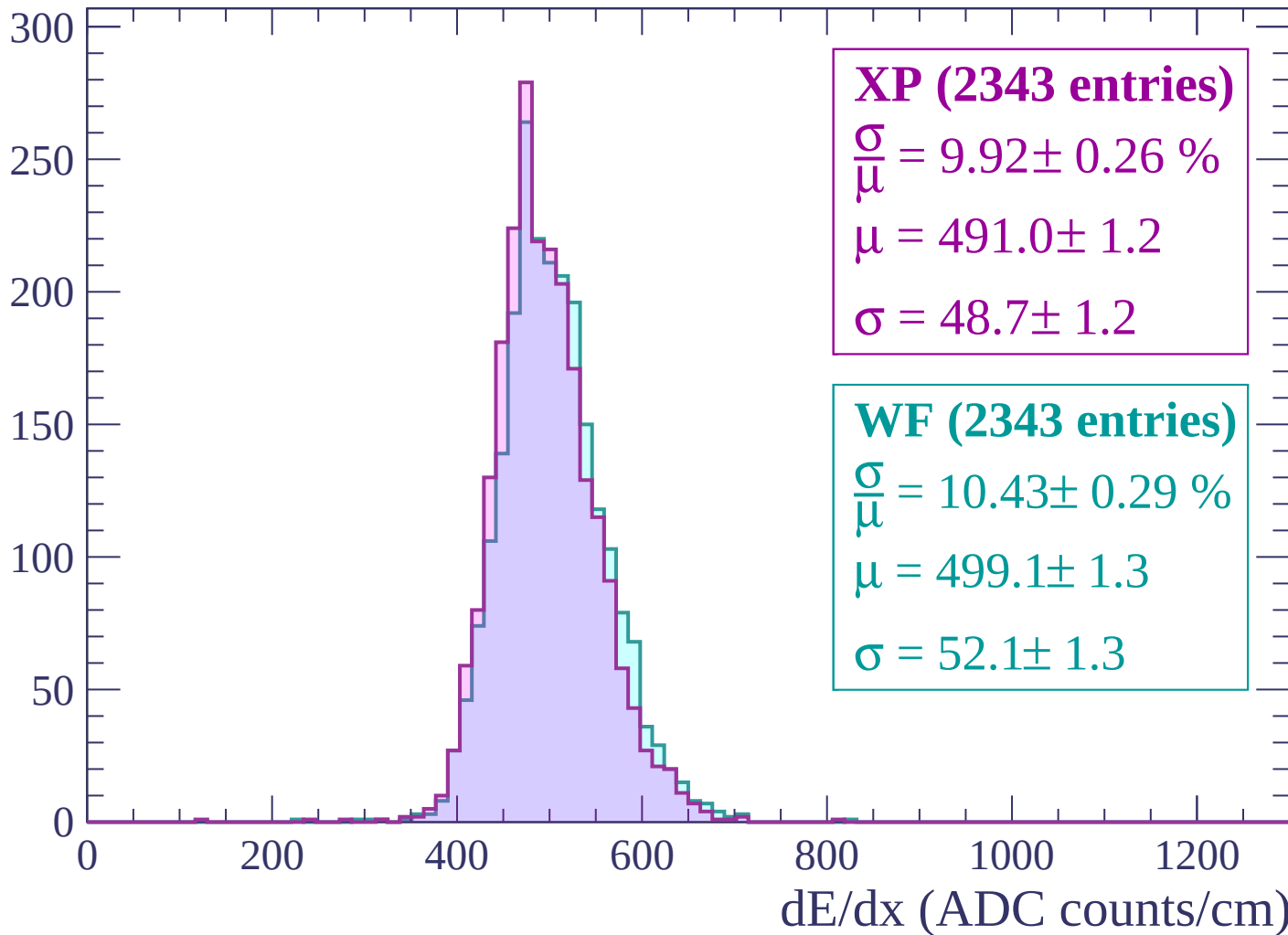
Energy loss |  $2880 < p < 2940$

Count



Energy loss |  $2940 < p < 3000$

Count



Energy loss |  $3000 < p < 3060$

Count

